

Exercise A

LITHIUM	BERYLLIUM
11 22.990	12 24.305
Na	Mg
SODIUM	MAGNESIUM
19 39.098	20 40.078

Calculate the effective nuclear charge on the outer shell electrons of Sodium.

Step 1: Write the electron configuration and separate the electrons in the the different sub-shells:



Step 2: Identify the electron of interest: $(1s^2) (2s^2 2p^6) 3s^1$

Step 3: Use the Slater's Rules:

$$\sigma = 0 \times 0.35 + 8 \times 0.85 + 2 \times 1 = 8.80$$

$$Z_{\text{eff}} = Z - \sigma = 11 - 8.80 = 2.20$$

s_i for Electrons in

Electron Group	All Higher Groups	Same Group	Groups with $n' = n - 1$	Groups with $n' < n - 1$
(1s)	0	0.30		
n = 3 (ns,p)	0	0.35	0.85	1.00
(nd), (nf)	0	0.35	1.00	1.00

Exercise B



7 minutes

N	FLUORINE	
65	17	35.453
	Cl	
R	CHLORINE	
96	35	79.904

Calculate the effective nuclear charge on the outer shell electrons of Chlorine.

Step 1: Write the electron configuration and separate the electrons in the the different sub-shells:



Step 2: Identify the electron of interest: $(1s^2) (2s^2 2p^6) (3s^2 3p^{4+1})$

Step 3: Use the Slater's Rules:

$$\sigma = 6 \times 0.35 + 8 \times 0.85 + 2 \times 1 = 10.9$$

$$Z_{eff} = Z - \sigma = 17 - 10.9 = 6.1$$

s_i for Electrons in

Electron Group	All Higher Groups	Same Group	Groups with $n' = n - 1$	Groups with $n' < n - 1$
(1s)	0	0.30		
n = 3 (ns,p)	0	0.35	0.85	1.00
(nd), (nf)	0	0.35	1.00	1.00

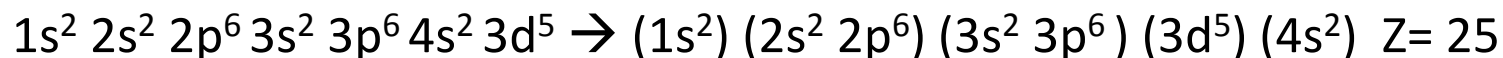
Exercise C



10 minutes

Using Slater's rules, calculate the effective nuclear charge on one of the $3d$ electrons compared with one of the $4s$ electrons for an atom of manganese. What is the electronic configuration of an hypothetical Mn^{1+} ion?

The ground state electron configuration of Manganese is:



$$4s) \sigma = 1 \times 0.35 + 13 \times 0.85 + 10 \times 1 = 21.4$$

$$Z_{\text{eff}} = 25 - 21.4 = \mathbf{3.6}$$

$$3d) \sigma = 4 \times 0.35 + 8 \times 1 + 8 \times 1 + 2 \times 1 = 19.4$$

$$Z_{\text{eff}} = 25 - 19.4 = \mathbf{5.6}$$

Mn^{1+} configuration: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$

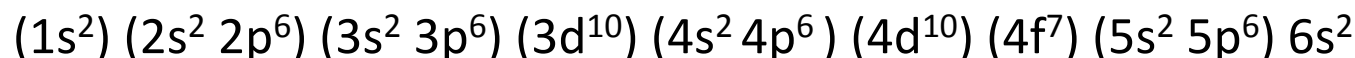
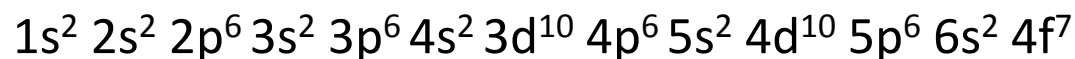
Exercise D



7 minutes

Use the Slater's rules to calculate the effective nuclear charge on one of the **4f** electrons and one of the **6s** electrons in a neutral Europium atom (**Z=63**)

The ground state electron configuration of Europium is:



$$4f) \sigma = 6 \times 0.35 + 46 \times 1 = 48.10$$

$$Z_{\text{eff}} = 63 - 48.1 = \mathbf{14.90}$$

$$6s) \sigma = 1 \times 0.35 + 8 \times 0.85 + 53 \times 1 = 60.15$$

$$Z_{\text{eff}} = 63 - 60.15 = \mathbf{2.85}$$

Electron Group	<i>s_i</i> for Electrons in			
	<i>All Higher Groups</i>	<i>Same Group</i>	<i>Groups with n' = n - 1</i>	<i>Groups with n' < n - 1</i>
(1s)	0	0.30		
(ns,p)	0	0.35	0.85	1.00
(nd), (nf)	0	0.35	1.00	1.00



5 minutes

Exercise E

If H and F had the same electronegativity, what would be the dissociation energy for the HCl molecule?

$$D(\text{Cl}_2) = 242 \text{ KJ mol}^{-1}$$

$$D(\text{H}_2) = 432 \text{ KJ mol}^{-1}$$

$$(242 * 432)^{0.5} = 323 \text{ KJ mol}^{-1}$$

Experimentally measured: 428 KJ mol⁻¹

105 KJ mol⁻¹ difference!!!!!!

Polar contribution

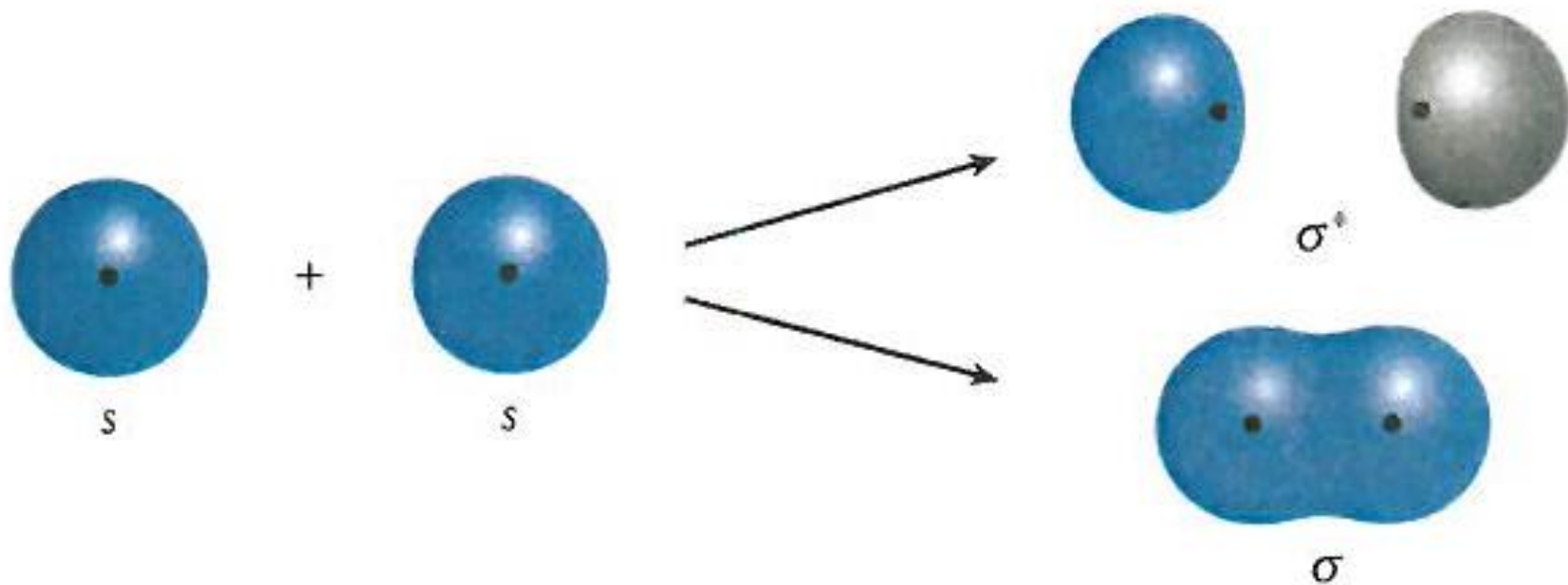
Exercise 3

Determine the bond order, the presence of lone pairs, and the geometry for an oxygen molecule O_2



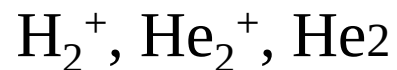
But, as we can see in this [movie](#), oxygen molecules are paramagnetic, which means that this structure cannot be correct!

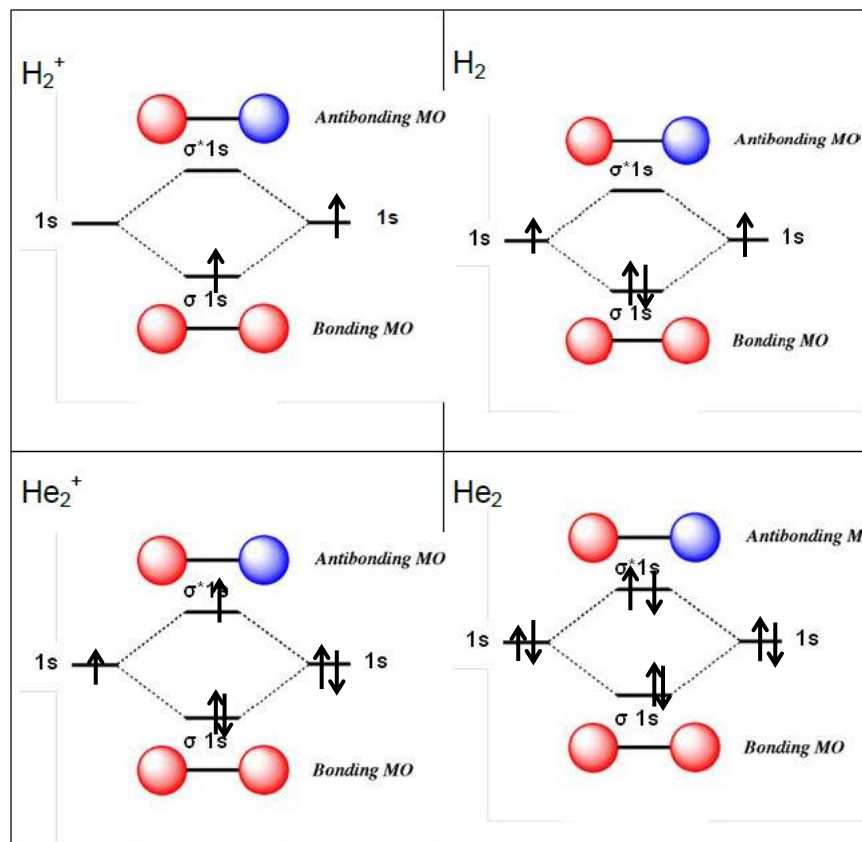
We have to think a little more carefully about the orbitals that are involved in chemical bonding.



Exercise 4

Draw the molecular orbital diagram for the following molecules:





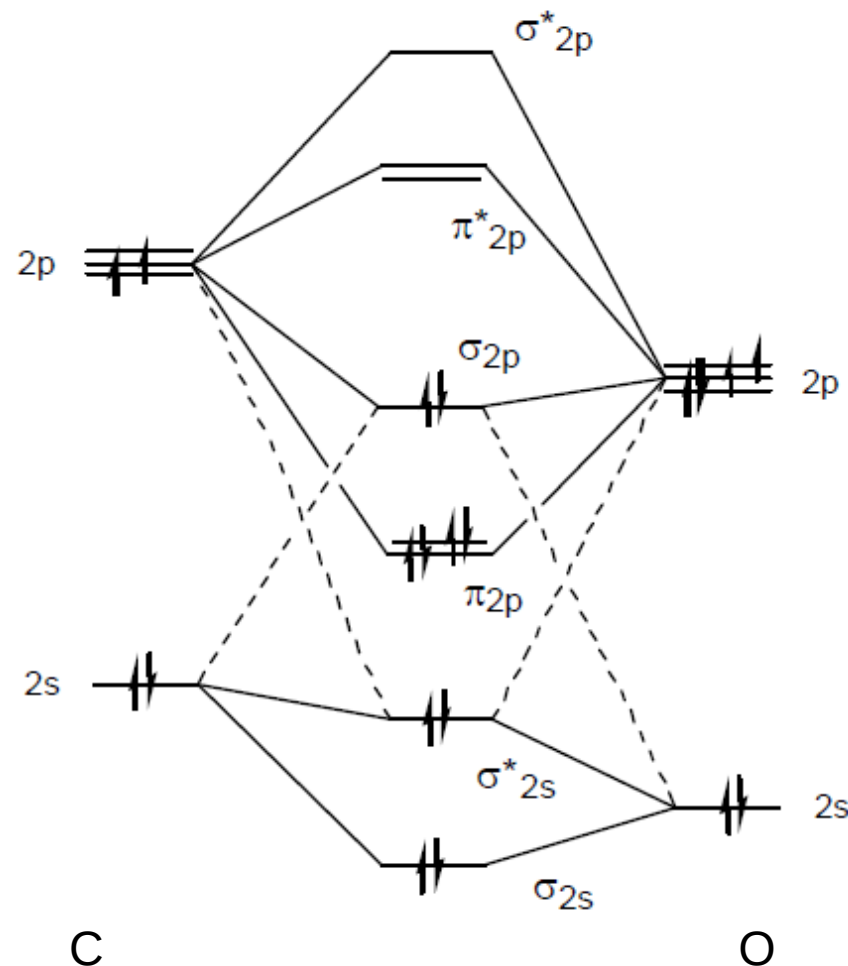
Molecule or Ion	Electron Configuration	Bond Order	Bond Length (pm)	Bond Energy (kJ/mol)
H_2^+	$(\sigma_{1s})^1$	$\frac{1}{2}$	106	269
H_2	$(\sigma_{1s})^2$	1	74	436
He_2^+	$(\sigma_{1s})^2(\sigma^*_{1s})^1$	$\frac{1}{2}$	108	251
He_2	$(\sigma_{1s})^2(\sigma^*_{1s})^2$	0	not observed	not observed

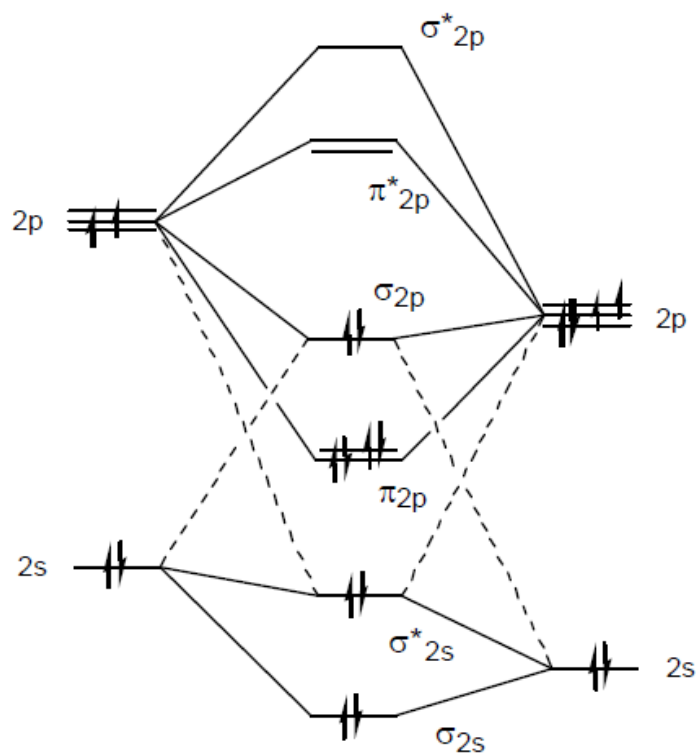
Exercise 1



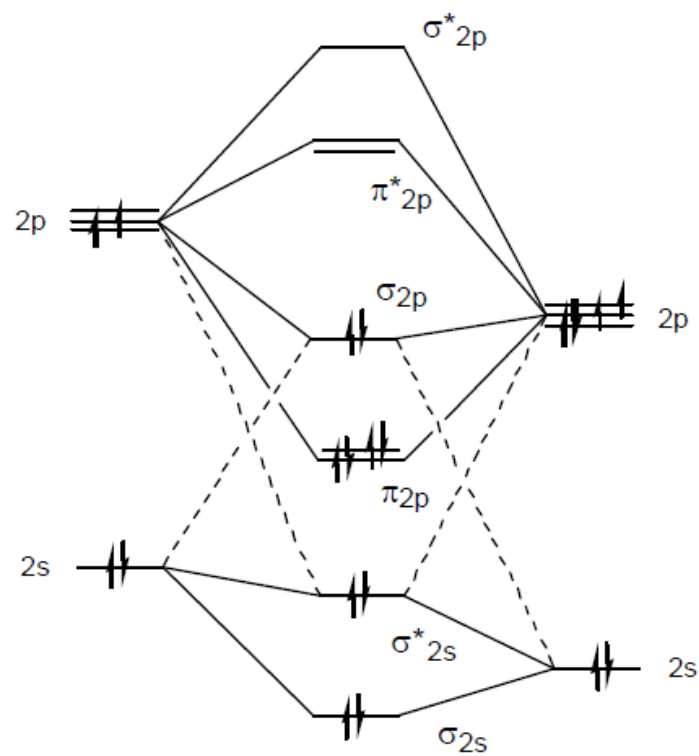
10 minutes

Using molecular orbital diagrams, predict which of the following free-radical gas-phase reactions is the more favored and give your reasoning.



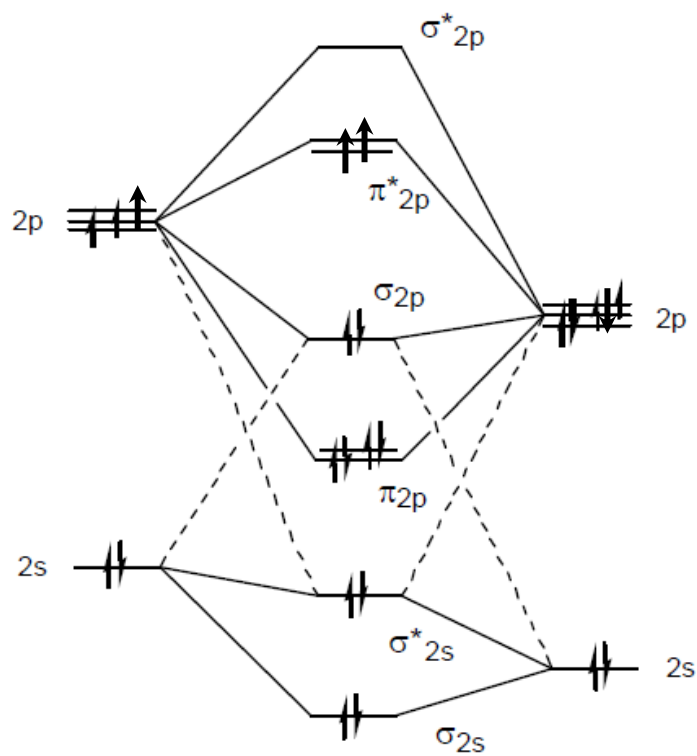


Bond order = 3



Bond order = 3

NO⁻

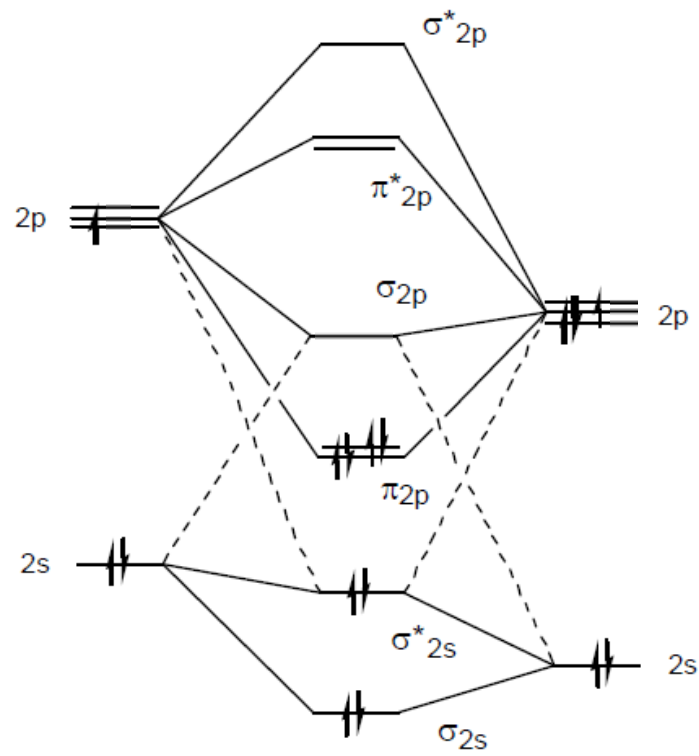


N

O⁻

Bond order = 2

CN⁺



C⁺

N

Bond order = 2

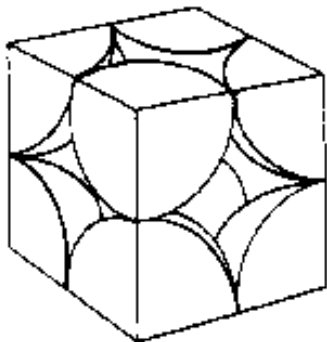
Worked Example

The edge length of the unit cell of alpha polonium is 336 pm (picometers).

(a) Determine the radius of a polonium (Po) atom (atomic mass = 209 u.m.a.).

(b) Determine the density of alpha polonium. Alpha polonium crystallizes in a simple cubic unit cell:

Simple cubic (sc)



(a) Two adjacent Po atoms contact each other, so the edge length of this cell is equal to two Po atomic radii: $l = 2r$. Therefore, the radius of Po is $336/2 = 168$ pm.

(b) Density is mass/volume.

The density of polonium can be found by determining the density of its unit cell (the mass contained within a unit cell divided by the volume of the unit cell). Since a Po unit cell contains one-eighth of a Po atom at each of its eight corners, a unit cell contains one Po atom.

The mass of a Po unit cell can be found by:

$$209(\text{g/mol}) / 6.022 \times 10^{23} (\text{atoms/mol}) = 3.47 \times 10^{-22} \text{ g/atom} = 3.47 \times 10^{-22} \text{ g/unit cell}$$

The volume of a Po unit cell is $(336)^3 = 3.79 \times 10^7 \text{ pm}^3 = 3.79 \times 10^{-23} \text{ cm}^3$ (Note that the edge length was converted from pm to cm to get the usual volume units for density: g/cm³).

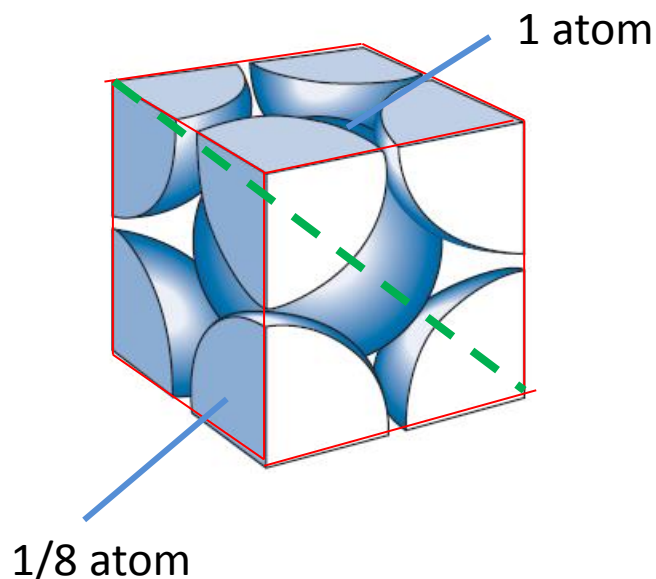
Therefore, the density of Polonium is $3.47 \times 10^{-22} \text{ g} / 3.79 \times 10^{-23} \text{ cm}^3 = 9.16 \text{ g/cm}^3$

Exercise 2



10 minutes

Chromium forms a body-centered cubic lattice in which the edge length of the unit cell is 288 pm. Calculate (a) the metallic radius of a chromium atom and (b) the density of chromium metal.



a) Cube diagonal = $288 \sqrt{3} = 499 \text{ pm}$
radius = $499/4 = 125 \text{ pm}$

(b) The density can be calculated from $\frac{\text{unit cell mass}}{\text{unit cell volume}}$.

The unit cell volume will be $(288 \times 10^{-10} \text{ cm})^3 = 2.39 \times 10^{-23} \text{ cm}^3$

Each unit cell contains two atoms, thus mass = $\frac{2 \times 52.0 \text{ g} \cdot \text{mol}^{-1}}{6.02 \times 10^{23} \text{ mol}^{-1}}$

$$= 1.73 \times 10^{-22} \text{ g.}$$

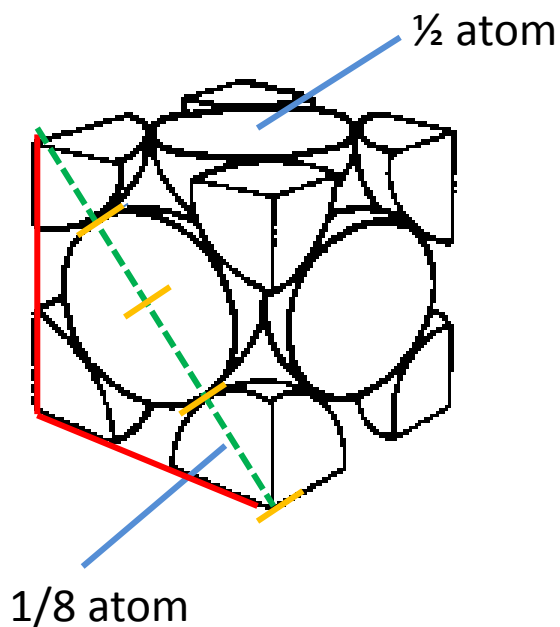
$$\text{Density} = \frac{1.73 \times 10^{-22} \text{ g}}{2.39 \times 10^{-23} \text{ cm}^3} = 7.24 \text{ g} \cdot \text{cm}^{-3}.$$

Exercise 3



10 minutes

The atoms in silver metal are arranged in a face centered cubic unit cell. Calculate the radius of a silver atom if the density of silver is 10.50 g cm^{-3} .



A face-centered cubic unit cell contains four atoms.

$$\text{Thus mass} = \frac{4 \times 107.9 \text{ g} \cdot \text{mol}^{-1}}{6.02 \times 10^{23} \text{ mol}^{-1}} = 7.17 \times 10^{-22} \text{ g}$$

$$\text{Volume} = \frac{7.17 \times 10^{-22} \text{ g}}{10.50 \text{ g} \cdot \text{cm}^{-3}} = 6.83 \times 10^{-23} \text{ cm}^3 = 6.83 \times 10^7 \text{ pm}^3$$

$$\text{Length of side} = \sqrt[3]{(6.83 \times 10^7 \text{ pm}^3)} = 409 \text{ pm}$$

$$\text{Square diagonal} = 409 \sqrt{2} = 578.4 \text{ pm}$$

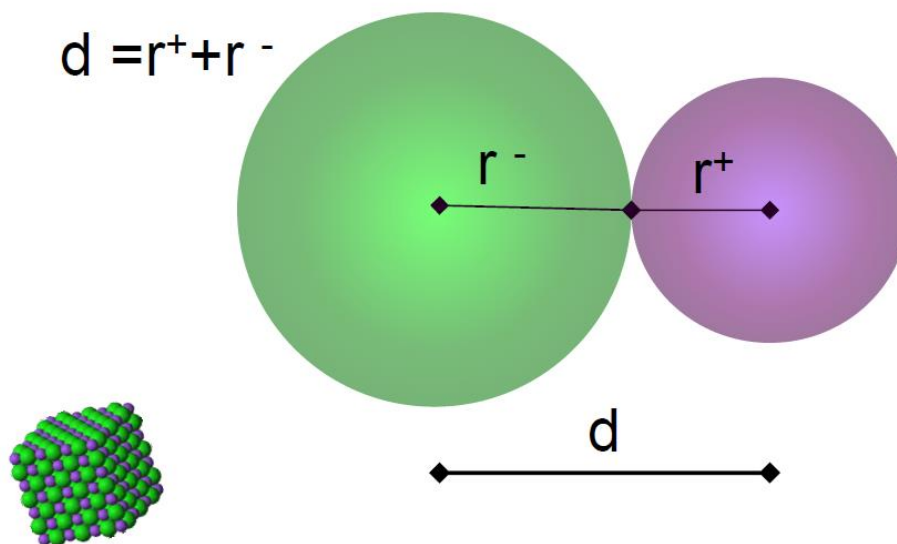
$$\text{radius} = 578.4 / 4 = 145 \text{ pm}$$

Ionic Radius

According to Pauling, for two ions with the same noble gas configuration, e.g. Na^+ and F^- , the ratio of the radii should be inversely proportional to the ratio of the “effective” nuclear charges, which can be calculated by Slater’s rules.

Exercise 1

Knowing that the interionic distance in NaF is $2.31 \text{ \AA}$, calculate the ionic radius of Na^+ and F^- .



According to Slater's rules, the shielding constant for Na^+ and F^- is
 $\sigma = 7 \times 0.35 + 2 \times 0.85 = 4.15$

Therefore, the effective nuclear charge is:

- For $\text{Na}^+ = 11 - 4.15 = 6.85$;
- For $\text{F}^- = 9 - 4.15 = 4.85$.

Thus, we have: $\frac{r_{\text{Na}^+}}{r_{\text{F}^-}} = \frac{4.85}{6.85} = 0.708$, that is, $r_{\text{Na}^+} = 0.708 r_{\text{F}^-}$

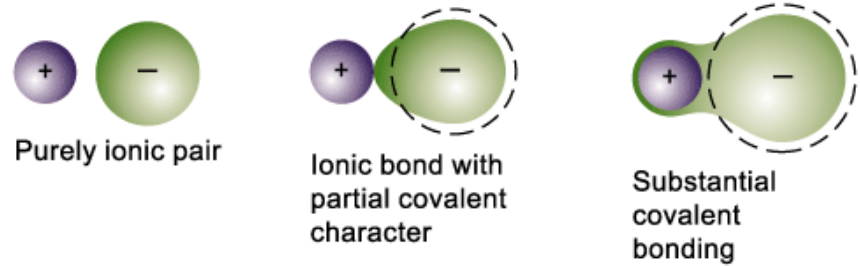
And the interionic distance in NaF is $2.31 \text{ \AA}$, i.e. $r_{\text{Na}^+} + r_{\text{F}^-} = 2.31 \text{ \AA}$

$$r_{\text{Na}^+} + 0.708 r_{\text{Na}^+} = 2.31 \text{ \AA}$$

1) We can conclude that:

$$r_{\text{F}^-} = 1.35 \text{ \AA}$$

$$r_{\text{Na}^+} = 0.96 \text{ \AA}$$



Increasing charge
Decreasing cation size



Increasing charge density



Exercise 2

Can charge density explain the different melting temperature of NaCl, MgCl₂, AlCl₃, and SiCl₄?

Calculate the charge density (Coulomb/mm³) of Na⁺ (ionic radius = 0,116 nm), Mg²⁺ (ionic radius = 0,086 nm), Al³⁺ (ionic radius = 0,068 nm), Si⁴⁺ (ionic radius = 0,041 nm).

Be aware: 1 proton has a charge of 1.60×10^{-19} C (Coulomb)

$$\text{Charge density of } Na^+ = \frac{1 \times (1.60 \times 10^{-19} C)}{\frac{4}{3} \pi (1.16 \times 10^{-7} mm)^3} = 24 C \cdot mm^{-3}$$

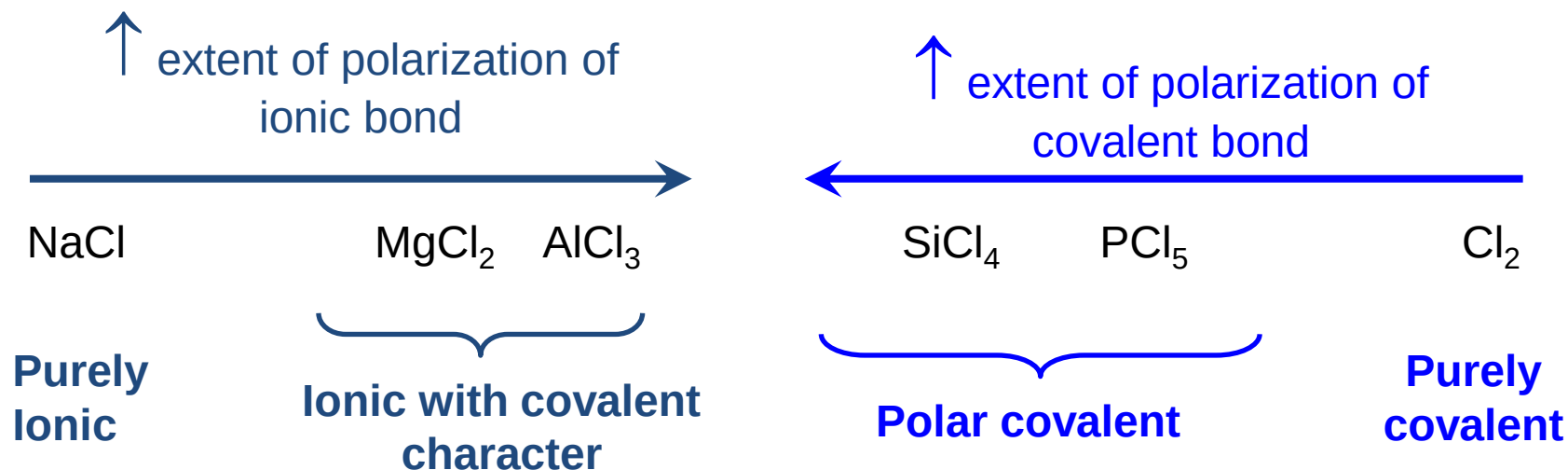
$$\text{Charge density of } Mg^{2+} = \frac{2 \times (1.60 \times 10^{-19} C)}{\frac{4}{3} \pi (8.6 \times 10^{-8} mm)^3} = 120 C \cdot mm^{-3}$$

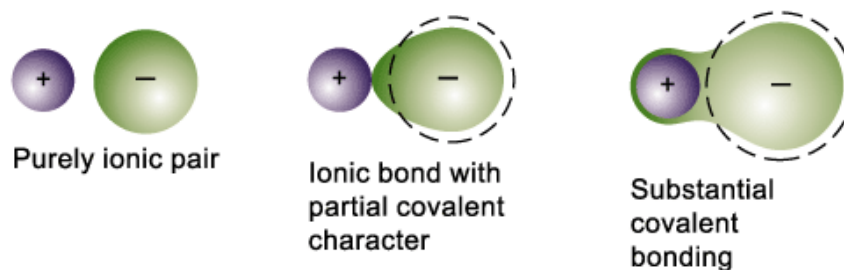
$$\text{Charge density of } Al^{3+} = \frac{3 \times (1.60 \times 10^{-19} C)}{\frac{4}{3} \pi (6.8 \times 10^{-8} mm)^3} = 365 C \cdot mm^{-3}$$

$$\text{Charge density of } Si^{4+} = \frac{4 \times (1.60 \times 10^{-19} C)}{\frac{4}{3} \pi (4.1 \times 10^{-8} mm)^3} = 2220 C \cdot mm^{-3}$$

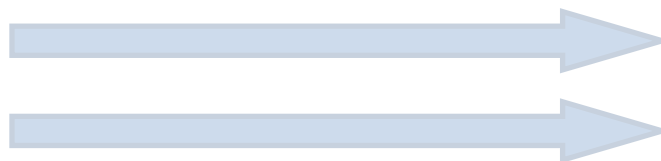
Chlorides can be used to demonstrate changes in bond type as the positive charge density increases due to higher charge (across Period 3) or larger size (down Group 1)

		Charge density (C/mm ³)	Chloride		Melting point /°C	water solubility
Period 3	Na ⁺	24	NaCl	GREATER POSITIVE CHARGE DENSITY ↓	808	soluble
	Mg ²⁺	120	MgCl ₂		714	soluble
	Al ³⁺	365	AlCl ₃		180	colloidal Al ₂ O ₃
	Si ⁴⁺	2220	SiCl ₄		-70	colloidal SiO ₄





Increasing charge
Decreasing cation size



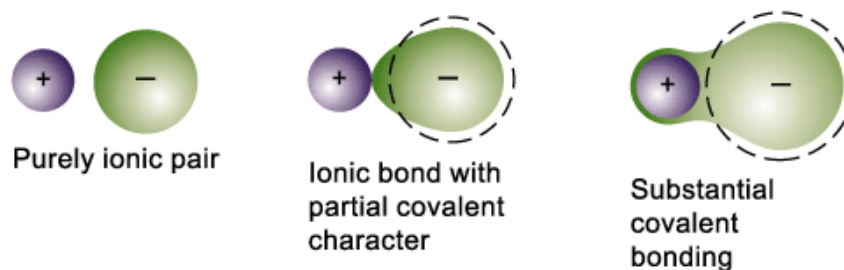
Increasing charge density



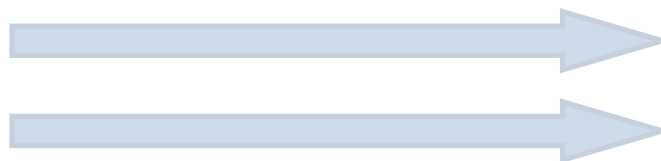
Increasing anion size



Ion	Radius (pm)	Compound	Melting point (°C)
F ⁻	119	KF	857
Cl ⁻	167	KCl	772
Br ⁻	182	KBr	735
I ⁻	206	KI	685



Increasing charge
Decreasing cation size



Increasing charge density



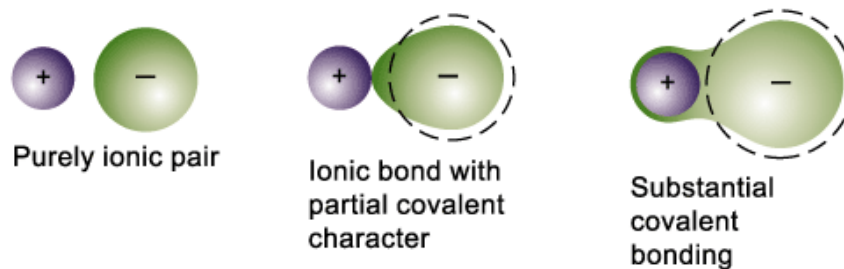
Increasing anion size



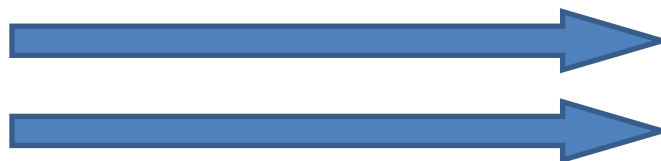
Incomplete valence shell



Compound	Melting point (°C)	Charge density ($\text{C} \cdot \text{mm}^{-3}$)	Bonding classification
KCl	770	11	Ionic
AgCl	455	15	Partially covalent



Increasing charge
Decreasing cation size



Increasing charge density

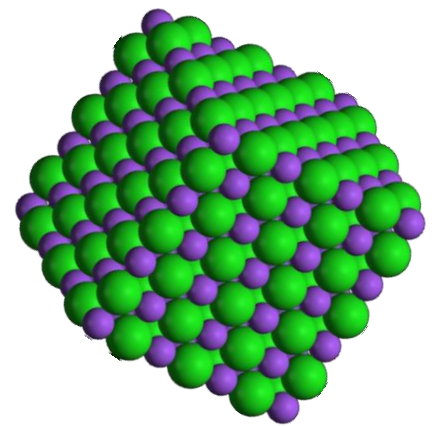


Increasing anion size



Incomplete valence shell





Ionic Solids

Inorganic solids: a 3D array of ions

Energy of this array of ions includes:

- 1) Electrostatic attractive and repulsive energies
- 2) Additional repulsive energy resulting from repulsion between the overlapping outer electron density of adjacent ions

Other contributions (e.g. van der Waals energy) are negligible

The lattice energy can be calculated by the Born-Landé equation:

$$U = - \frac{N A z^+ z^- e^2}{4\pi\epsilon_0 r_0} \left(1 - \frac{1}{n} \right)$$

Electrostatic interactions

Electron repulsion forces

N Avogadro constant = $6.022 \times 10^{23} \text{ mol}^{-1}$

A Madelung constant (it takes into account crystal geometry)

z_+ Charge of the cation

z_- Charge of the anion

e electron charge in coulomb = $-1.602 \times 10^{-19} \text{ C}$

ϵ_0 vacuum dielectric constant = $8.854... \times 10^{-12} \text{ C}^2 \cdot \text{m}^{-1} \cdot \text{J}^{-1}$

r_0 Inter-ionic distance [m]

The Born exponent, n , can be estimated for alkali halides from the correspondent noble gas electron configuration.

The Born exponent, n , can be estimated for alkali halides by using the average of the following numbers:

TABLE 6.4 Values for the Born exponent		
Electronic configuration of ion	Born exponent, n	Examples
He	5	Li^+
Ne	7	$\text{Na}^+, \text{Mg}^{2+}, \text{O}^{2-}, \text{F}^-$
Ar	9	$\text{K}^+, \text{Ca}^{2+}, \text{S}^{2-}, \text{Cl}^-$
Kr	10	Rb^+, Br^-
Xe	12	Cs^+, I^-

E.g. sodium chloride
$$U = - \frac{N A z^+ z^- e^2}{4\pi\epsilon_0 r_0} \left(1 - \frac{1}{n}\right)$$

Madelung constant, A , is 1.748

The Born exponent for NaCl can be approximated to $(7+9)/2=8$

$$U = - \frac{(6.02 \times 10^{23} \text{ mol}^{-1}) \times 1.748 \times 1 \times 1 \times (1.602 \times 10^{-19} \text{ C})^2}{4 \times 3.142 \times (8.854 \times 10^{-12} \text{ C}^2 \cdot \text{J}^{-1} \cdot \text{m}^{-1})(2.83 \times 10^{-10} \text{ m})} \left(1 - \frac{1}{8}\right)$$

$$= -751 \text{ kJ} \cdot \text{mol}^{-1}$$

The ionic model allowed us to calculate the enthalpy change in the formation of a lattice from a mole of gaseous isolated sodium and chloride ions, according to the equation:



This corresponds to 751 kJ/mol.

This is also the energy necessary to break the lattice compound into gaseous ions (reverse reaction).

To validate our model, the enthalpy of reaction 1 should be experimentally determined.

However this is not possible because NaCl does not vaporize to give Na^+ and Cl^- ions!

The Born-Haber Cycle

To circumvent this problem we will use the Born-Haber thermodynamic cycle.

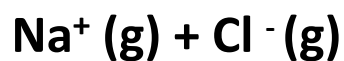
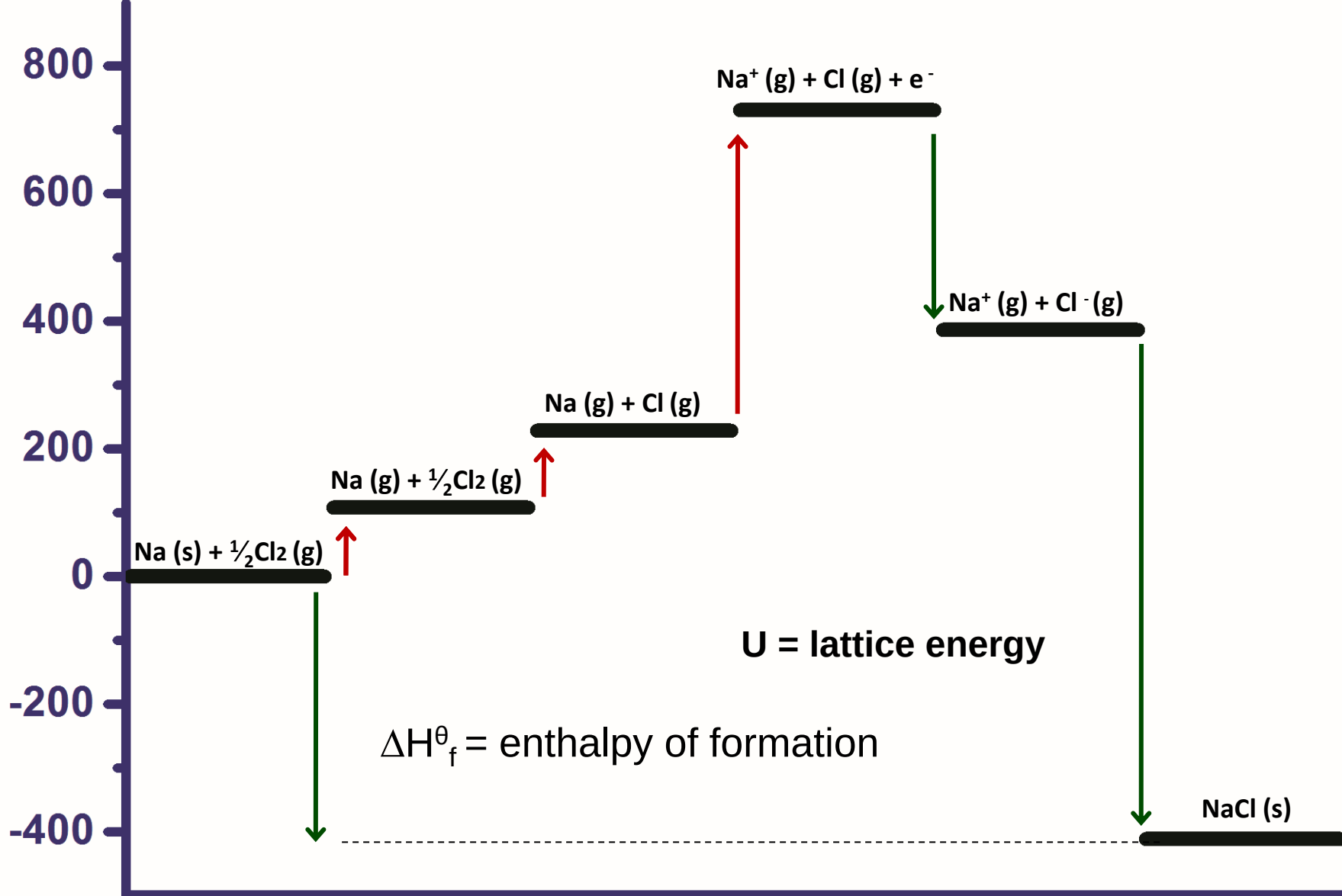
The reaction

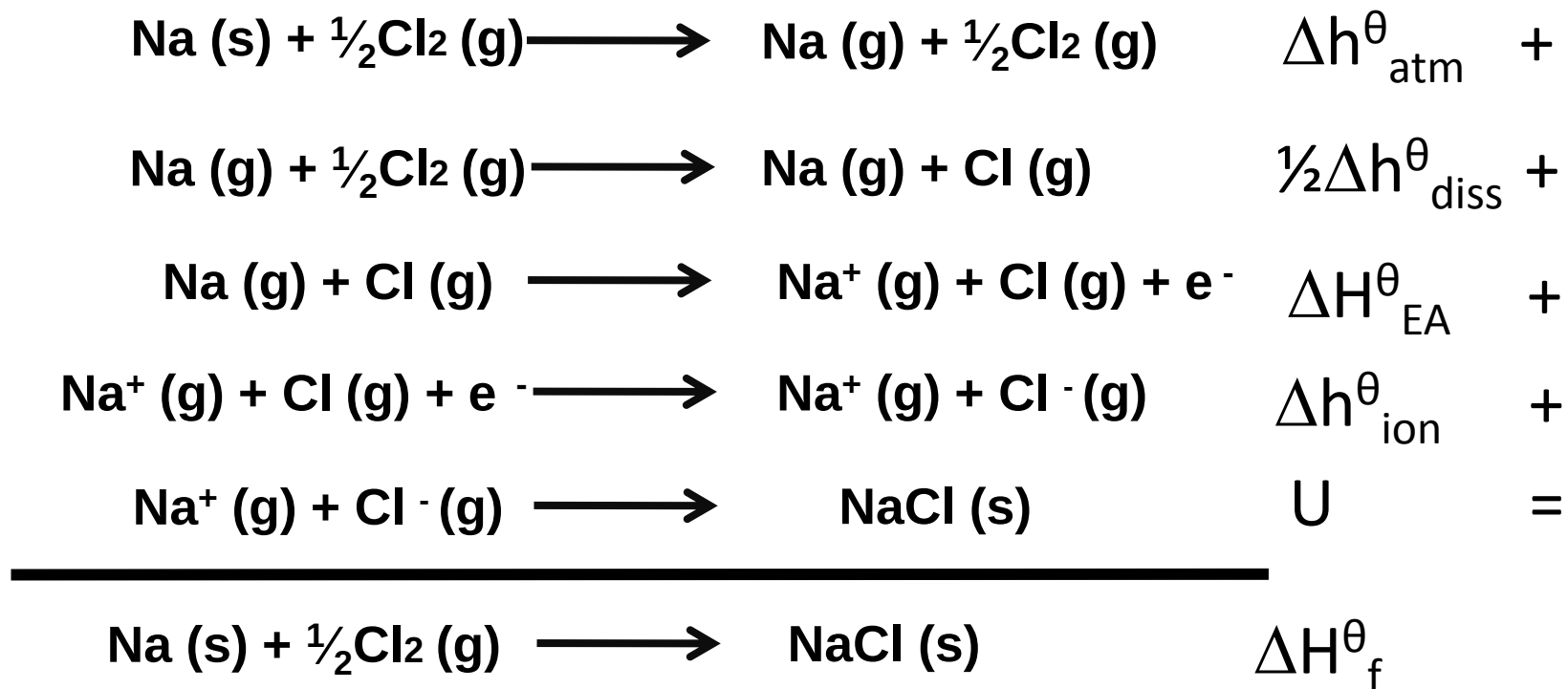


Can be “broken down” into a series of steps. The sum of the enthalpies of these steps is equal to the enthalpy for the overall reaction.

This is a consequence of the Law of Conservation of Energy (the First Law of Thermodynamics)

Enthalpy [kJ mol^{-1}]





$$\Delta H^{\theta}_{\text{atm}} + \frac{1}{2}\Delta H^{\theta}_{\text{diss}} + \Delta H^{\theta}_{\text{EA}} + \Delta H^{\theta}_{\text{ion}} + U = \Delta H^{\theta}_{\text{f}}$$

$$U = \Delta H^{\theta}_{\text{f}} - \Delta H^{\theta}_{\text{atm}} - \frac{1}{2}\Delta H^{\theta}_{\text{diss}} - \Delta H^{\theta}_{\text{EA}} - \Delta H^{\theta}_{\text{ion}}$$

$$U = \Delta H_f^\theta - \Delta H_{\text{atm}}^\theta - \frac{1}{2}\Delta H_{\text{diss}}^\theta - \Delta H_{\text{EA}}^\theta - \Delta H_{\text{ion}}^\theta$$

ΔH_f^θ	- 411
- $\Delta H_{\text{atm}}^\theta$	-108
- $0.5\Delta H_{\text{diss}}^\theta$	-121
- $\Delta H_{\text{EA}}^\theta$	+349
- $\Delta H_{\text{ion}}^\theta$	- 502
U	- 793 KJ mol ⁻¹

which is close to the value of U calculated by using the ionic model (-751 kJ mol⁻¹)!

Exercise 1



10 minutes

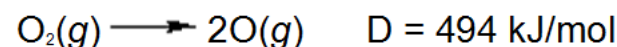
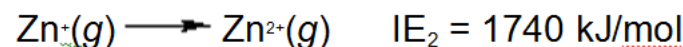
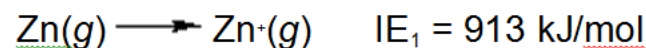
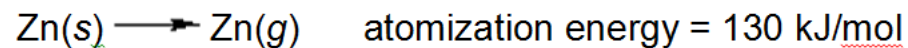
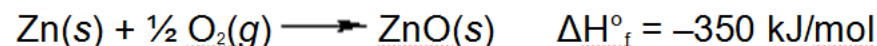
Construct a Born-Haber cycles for the formation of magnesium chloride and magnesium oxide. Do not perform any calculation.

Exercise 2



10 minutes

Construct a Born-Haber cycle for the formation of ZnO and calculate its lattice energy:



We can expect Lattice Energy to increase with the charge of the ions. In fact for the series MX, MX₂, MX₂, MX₃ the lattice energies are related in the ratios 1 : 3 : 6 : 10.

Question 1

Then, why we do not observe a prevalence of MX₃ salts?

Exercise 3



10 minutes

Construct the Born-Haber cycle for the theoretical compounds NaCl₂ and NaCl₃ using the numerical values from the NaCl cycle and using the following values:

Lattice Energy for NaCl₂ = -2500 KJ/mol;

Lattice Energy for NaCl₃ = -5400 KJ/mol;

Na second ionization energy IE₂ = 4569 KJ/mol;

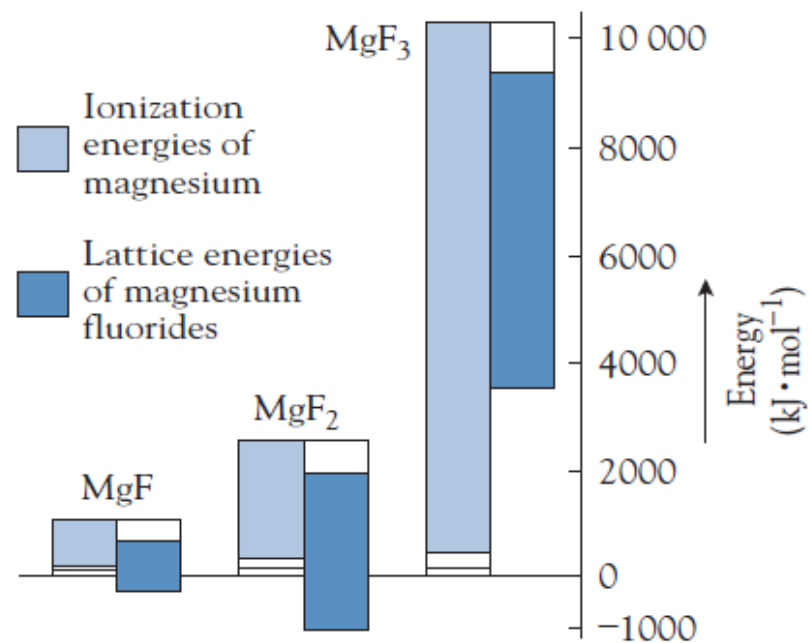
Na third ionization energy IE₃ = 6919 KJ/mol;

Why NaCl is the preferred product?

MgF MgF₂ MgF₃

TABLE 6.6 Thermodynamic factors in the formation of three possible magnesium fluorides

Enthalpy factors (kJ·mol ⁻¹)	MgF	MgF ₂	MgF ₃
Mg atomization	+150	+150	+150
F—F bond energy	+80	+160	+240
Mg ionization (total)	+740	+2190	+9930
F electron affinity	-330	-660	-990
Lattice energy	≈ -900	-2880	≈ -5900
$\Delta H_f^\ominus$ (estimated)	-260	-1040	+3430



Entropy

Entropy is often related to the degree of disorder of the substance (although the concept of entropy is really more complex).

Entropy is discussed on an absolute basis. Thus, even elements have a listed value of entropy. The zero point is taken to be that of a perfect crystal of a substance at the absolute zero of temperature.

We can calculate the standard entropy change for a reaction in the same way as that of the enthalpy change.

$$\Delta S^{\ominus}(\text{reaction}) = \Sigma S^{\ominus}(\text{products}) - \Sigma S^{\ominus}(\text{reactants})$$

Standard Entropies

- These are molar entropy values of substances in their standard states.
- Standard entropies tend to increase with increasing molar mass.

Substance	S° , J/mol-K
Gases	
H ₂ (g)	130.6
N ₂ (g)	191.5
O ₂ (g)	205.0
H ₂ O(g)	188.8
NH ₃ (g)	192.5
CH ₃ OH(g)	237.6
C ₆ H ₆ (g)	269.2
Liquids	
H ₂ O(l)	69.9
CH ₃ OH(l)	126.8
C ₆ H ₆ (l)	172.8
Solids	
Li(s)	29.1
Na(s)	51.4
K(s)	64.7
Fe(s)	27.23
FeCl ₃ (s)	142.3
NaCl(s)	72.3

Free Energy

The Driving Force of a Reaction.

For a spontaneous reaction, there must be an increase in entropy overall of the universe (reaction system + its surroundings).

We can determine whether a reaction is spontaneous from the entropy and enthalpy changes of the reaction itself.

$$\Delta G^{\theta} = \Delta H_{sys}^{\theta} - T \Delta S_{system}^{\theta}$$

ΔH	ΔS	$-T\Delta S$	$\Delta G = \Delta H - T\Delta S$	Reaction Characteristics	Example
–	+	–	–	Spontaneous at all temperatures	$2 \text{ O}_3(g) \longrightarrow 3 \text{ O}_2(g)$
+	–	+	+	Nonspontaneous at all temperatures	$3 \text{ O}_2(g) \longrightarrow 2 \text{ O}_3(g)$
–	–	+	+ or –	Spontaneous at low T ; nonspontaneous at high T	$\text{H}_2\text{O}(l) \longrightarrow \text{H}_2\text{O}(s)$
+	+	–	+ or –	Spontaneous at high T ; nonspontaneous at low T	$\text{H}_2\text{O}(s) \longrightarrow \text{H}_2\text{O}(l)$

$$\Delta G^{\theta}(\text{reaction}) = \Sigma \Delta G_f^{\theta}(\text{products}) - \Sigma \Delta G_f^{\theta}(\text{reactants})$$

Exercise 4



10 minutes

Calculate the free energy of formation of sodium chloride under standard conditions (298K):

$$S^\circ_{(\text{Na})} = 51 \text{ J/(mol K)};$$

$$S^\circ_{(\text{Cl}_2)} = 222 \text{ J/(mol K)};$$

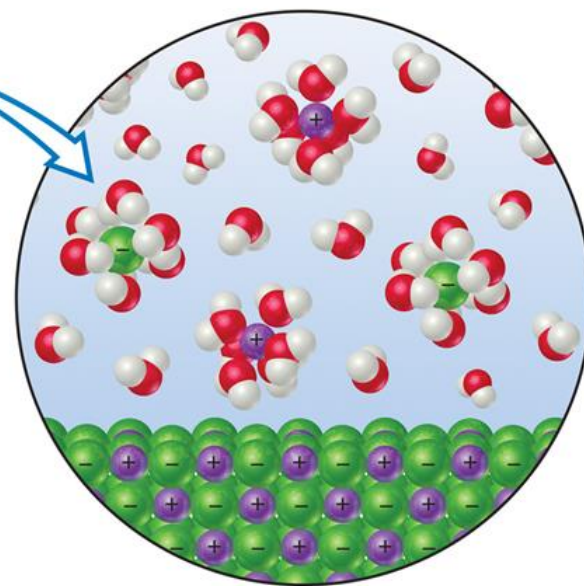
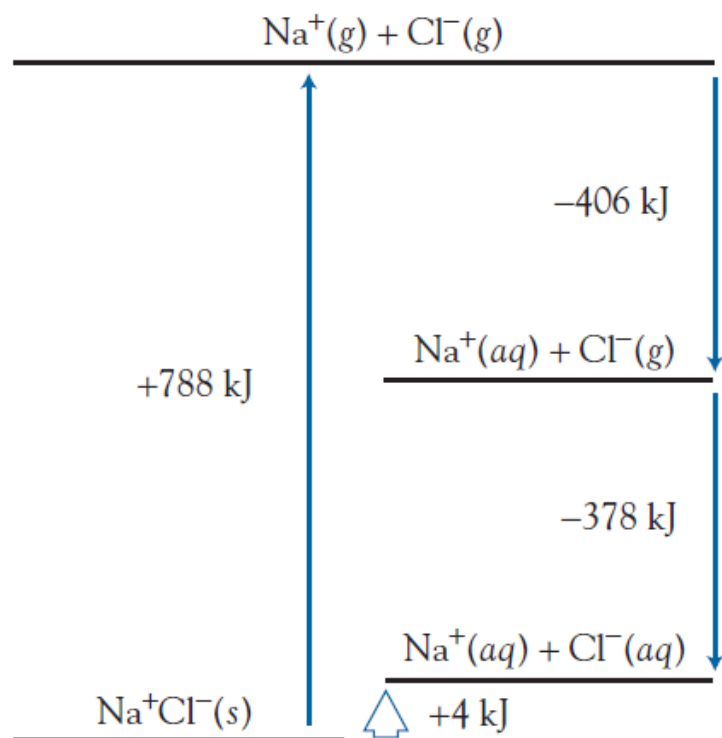
$$S^\circ_{(\text{NaCl})} = 72 \text{ J/(mol K)};$$

$$\Delta H^\circ_{\text{f}(\text{NaCl})} = - 411 \text{ kJ/mol};$$

Energy of hydration

Question 3:

Why does sodium chloride dissolve in water?

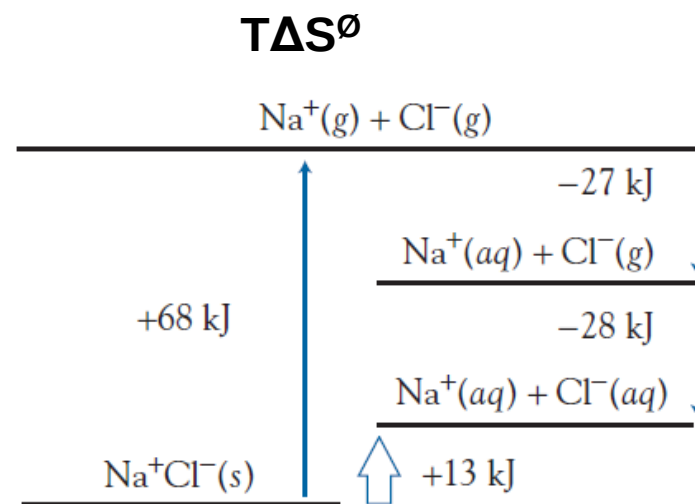
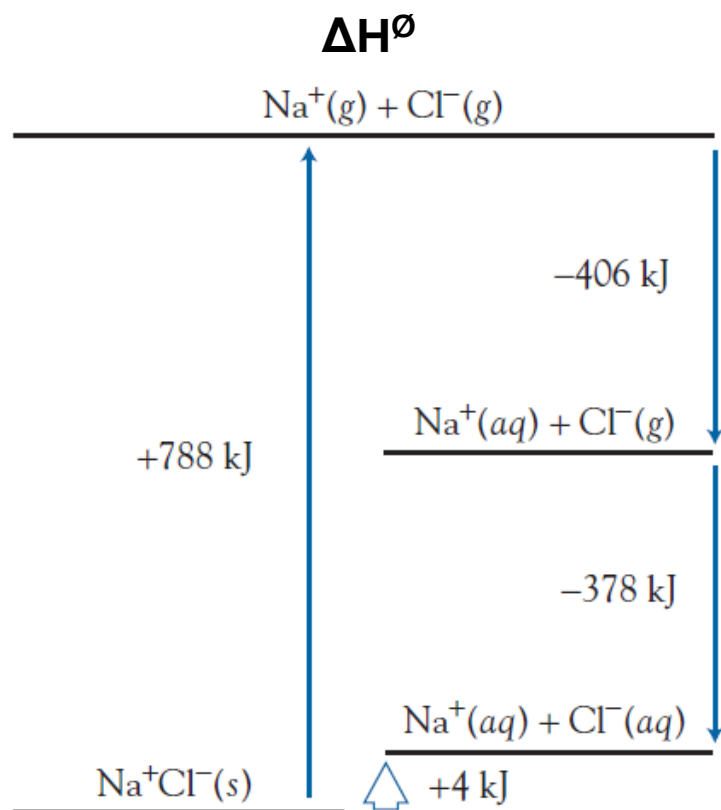


Lattice energy vs. energy of hydration

Energy of hydration

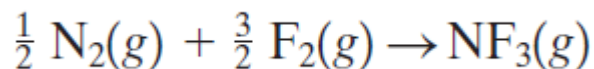
Question 3:

Why does sodium chloride dissolve in water?

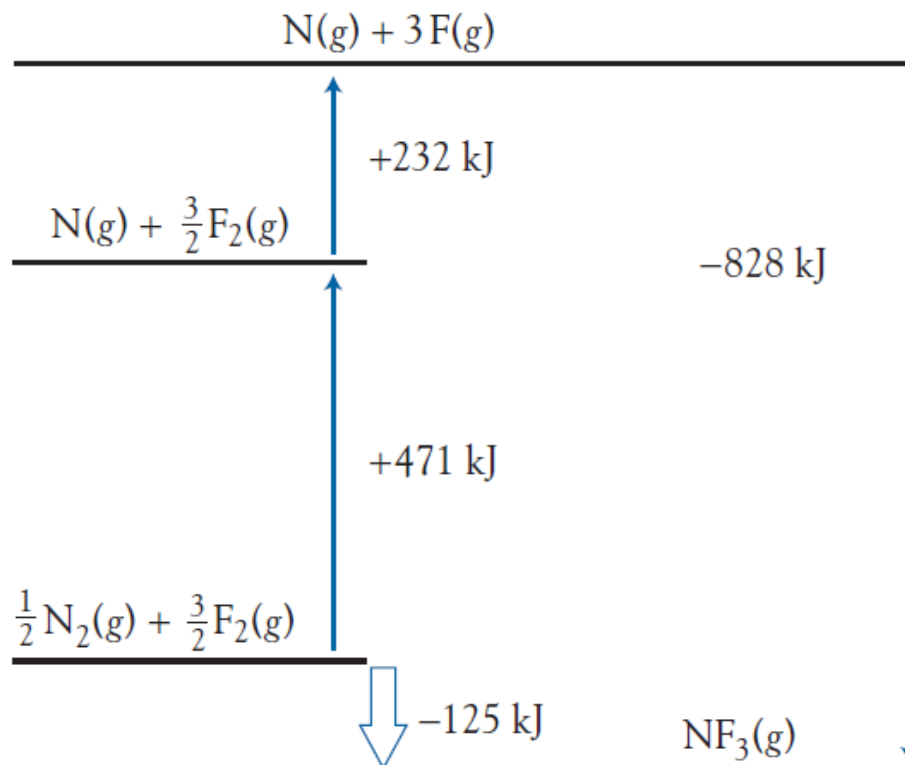


$$\begin{aligned}\Delta G^\ominus &= \Delta H^\ominus - T\Delta S^\ominus \\ &= (+4 \text{ kJ}\cdot\text{mol}^{-1}) - (+13 \text{ kJ}\cdot\text{mol}^{-1}) \\ &= -9 \text{ kJ}\cdot\text{mol}^{-1}\end{aligned}$$

Formation of covalent compounds



1) Born-Haber cycle:
enthalpy



2) Formation entropy: $\Delta S^\circ_f = S^\circ_{(\text{NF}_3)} - \frac{1}{2} S^\circ_{(\text{N}_2)} - \frac{3}{2} S^\circ_{(\text{F}_2)} = -140 \text{ J / (mol K)}$

3) Free energy of formation = $\Delta H^\circ_f - T = 125 - 298 \times (-0.140) = -83 \text{ kJ/mol}$

Exercise 5



10 minutes

Ammonia synthesis

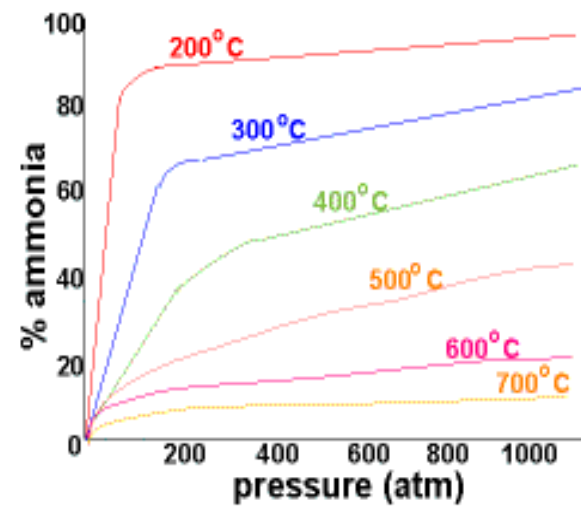
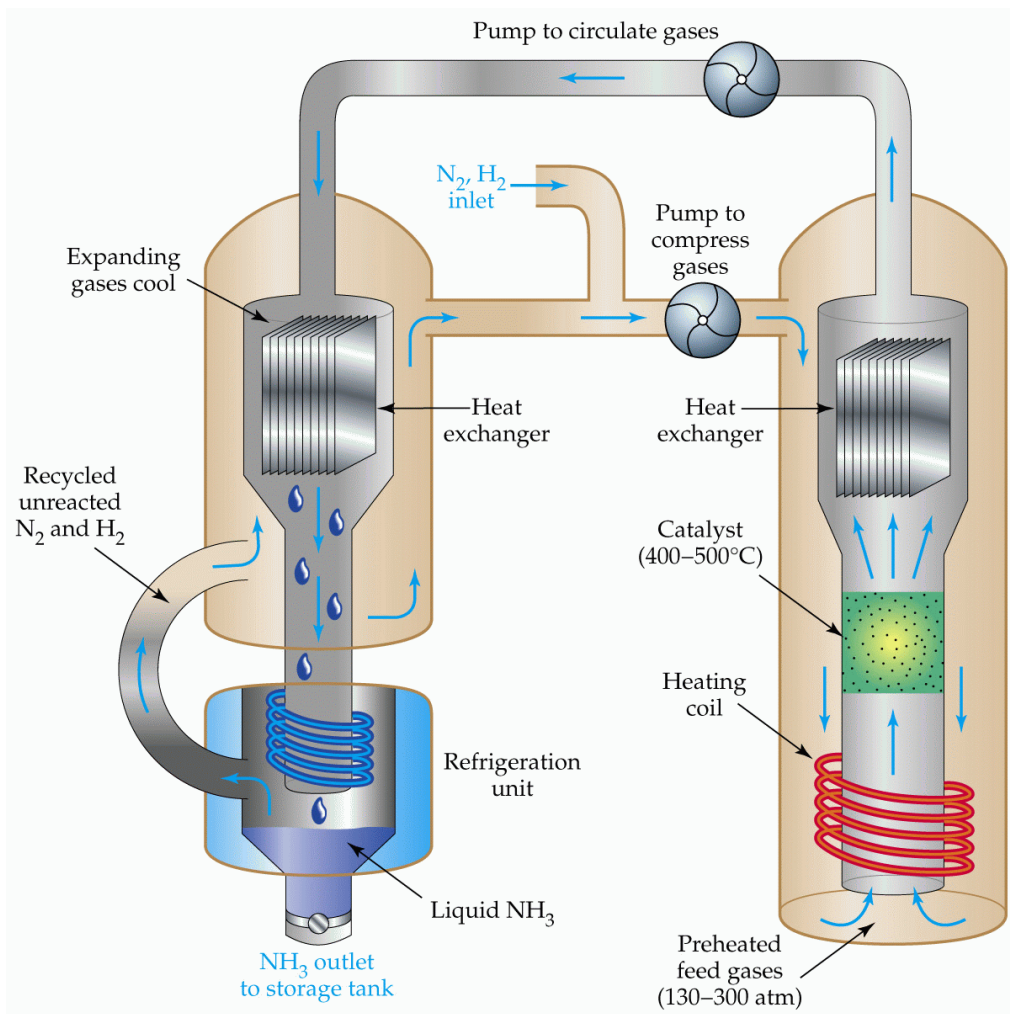
At the beginning of the 20th century there was a shortage of naturally occurring, nitrogen-rich fertilizers, which prompted the German Chemist Fritz Haber, and others, to look for ways of combining nitrogen (obtained by air distillation) with hydrogen to form ammonia

$$\Delta H^\circ_{f(\text{NH}_3)} = -46 \text{ kJ/mol}$$

$$S^\circ_{(\text{NH}_3)} = 193 \text{ J/(mol K)}, S^\circ_{(\text{N}_2)} = 192 \text{ J/(mol K)}, \text{ and } S^\circ_{(\text{H}_2)} = 131 \text{ J/(mol K)},$$

- 1) Calculate the free energy variation to form a mole of ammonia from gaseous nitrogen and hydrogen.
- 2) Ammonia is typically synthesized at temperature of 500°C and pressure of 250 atm in presence of a catalyst. Can you suggest a reason for that?

Thermodynamic vs Kinetic



Lecture 6

Solvent systems and Acid-Base Behaviour

Chapter 7

Content:

- **Solvents**
 - **Acid/base definitions**
 - **Hard-Soft Acid-Base Concepts**
- 

Solvent Properties

- 1) The temperature and pressure range over which it is a liquid.
- 2) Its dielectric constant.
- 3) Its donor and acceptor (Lewis acid-base) properties.
- 4) Its protonic (Brønsted) acidity or basicity.
- 5) The nature and extent of autodissociation.

From last week...

The Born exponent, n , can be estimated for alkali halides by using the average of the following numbers:

TABLE 6.4 Values for the Born exponent		
Electronic configuration of ion	Born exponent, n	Examples
He	5	Li^+
Ne	7	$\text{Na}^+, \text{Mg}^{2+}, \text{O}^{2-}, \text{F}^-$
Ar	9	$\text{K}^+, \text{Ca}^{2+}, \text{S}^{2-}, \text{Cl}^-$
Kr	10	Rb^+, Br^-
Xe	12	Cs^+, I^-

E.g. **sodium chloride**

$$U = - \frac{N A z^+ z^- e^2}{4 \pi \epsilon_0 r_0} \left(1 - \frac{1}{n} \right)$$

Madelung constant, A , is 1.748

The Born exponent for NaCl can be approximated to $(7+9)/2=8$

$$\begin{aligned} U &= - \frac{(6.02 \times 10^{23} \text{ mol}^{-1}) \times 1.748 \times 1 \times 1 \times (1.602 \times 10^{-19} \text{ C})^2}{4 \times 3.142 \times (8.854 \times 10^{-12} \text{ C}^2 \cdot \text{J}^{-1} \cdot \text{m}^{-1})(2.83 \times 10^{-10} \text{ m})} \left(1 - \frac{1}{8} \right) \\ &= -751 \text{ kJ} \cdot \text{mol}^{-1} \end{aligned}$$

Solvents

Common name	Formula	Dielectric constant
Hydrofluoric acid	HF	84
Water	H ₂ O	78
Dimethylsulfoxide (DMSO)	(CH ₃) ₂ SO	47
<i>N,N</i> -dimethylformamide (DMF)	(CH ₃) ₂ NCHO	38
Acetonitrile	CH ₃ CN	37
Methanol	CH ₃ OH	32
Ammonia	NH ₃	27
Acetone	(CH ₃) ₂ CO	21
Dichloromethane	CH ₂ Cl ₂	9
Tetrahydrofuran (THF)	C ₄ H ₈ O	8
Diethylether	(C ₂ H ₅) ₂ O	4
Toluene	C ₆ H ₅ CH ₃	2
Hexane	C ₆ H ₁₄	2

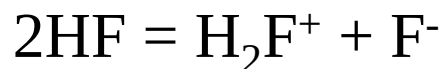
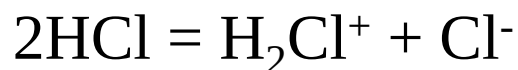
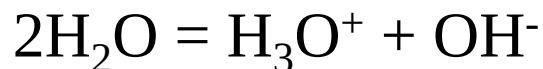
Dielectric constant (ϵ): A measure of a substance's ability to insulate charges from each other. Higher ϵ means higher polarity, and greater ability to stabilize charges. Notice ϵ is related, but is not the same as dipole moment.

1) Polar Protic Solvents

The solvents containing ionizable protons are called protic solvents, e.g. H_2O , HCl , HF , and H_2SO_4 . They are more or less acidic.

Autodissociation of Protic Solvents

Several examples of autodissociation reaction:



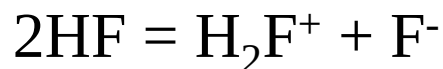
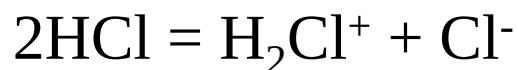
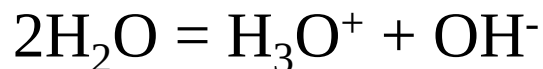
Feature: proton transfer between two solvent molecules to give the **protonated** cation and the **deprotonated** anion.

1) Polar Protic Solvents

The solvents containing ionizable protons are called protic solvents, e.g. H_2O , HCl , HF , and H_2SO_4 . They are more or less acidic.

Autodissociation of Protic Solvents

Several examples of autodissociation reaction:



Feature: proton transfer between two solvent molecules to give the **protonated** cation and the **deprotonated** anion.

2) Polar aprotic solvents.

For instance, acetonitrile (CH_3CN), acetone ($\text{CH}_3)_2\text{CO}$ N,N-dimethylformamide (DMF). They have the features:

aprotic and no autodissociation equilibria , but they strongly solvates ions.

3) Nonpolar, or very weakly polar, nondissociated liquids, which do not solvate strongly

For instance, carbon tetrachloride and hydrocarbons *are not powerful solvents due to their low polarity, low dielectric constants, and poor donor power. They can be used when a inert solvent is needed.*

Exercise 1



10 minutes

Identify acids and bases in the following reactions:

1. $\text{H}_2\text{SO}_4 + \text{H}_2\text{O} = \text{H}_3\text{O}^+ + \text{HSO}_4^-$
2. $\text{NaOH} = \text{Na}^+ + \text{OH}^-$
3. $\text{PF}_5 + 2\text{HF (solvent)} = \text{H}_2\text{F}^+ + \text{PF}_6^-$
4. $\text{SiO}_2 + \text{Na}_2\text{O} = \text{Na}_2\text{SiO}_3$
5. $\text{B(OR)}_3 + \text{NaH} = \text{Na[HB(OR)}_3]$
6. $\text{Li}_3\text{N} + 2\text{NH}_3 \text{ (solvent)} = 3\text{Li}^+ + 3\text{NH}_2^-$
7. $0.5\text{Al}_2\text{Cl}_6 + \text{PF}_3 = \text{Cl}_3\text{Al-PF}_3$
8. $\text{PCl}_5 + \text{ICl (solvent)} = \text{PCl}_4^+ + \text{ICl}_2^-$

Brønsted-Lowry Definition

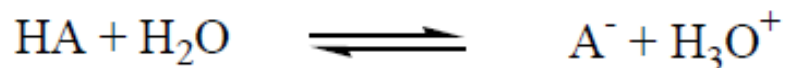
“An acid is a substance that supplies protons and a base is a proton acceptor.”

Acid strength

A protonic acid gives H^+ to water and generates the oxonium ion H_3O^+ . The strength of an acid in a dilute aqueous solution is estimated from the equilibrium constant K_a

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

for the dissociation equilibrium,



but it is more convenient to use,

$$\text{p}K_a = -\log K_a$$

Acid	HA	A ⁻	K_a (at 25°C)	p <i>K</i> _a
Perchloric acid	HClO ₄	ClO ₄ ⁻	10 ¹⁰	-10
Hydrochloric acid	HCl	Cl ⁻	10 ²	-2
Hydrofluoric acid	HF	F ⁻	3.5 × 10 ⁻⁴	3.45
Ammonium ion	NH ₄ ⁺	NH ₃	5.5 × 10 ⁻¹⁰	9.26

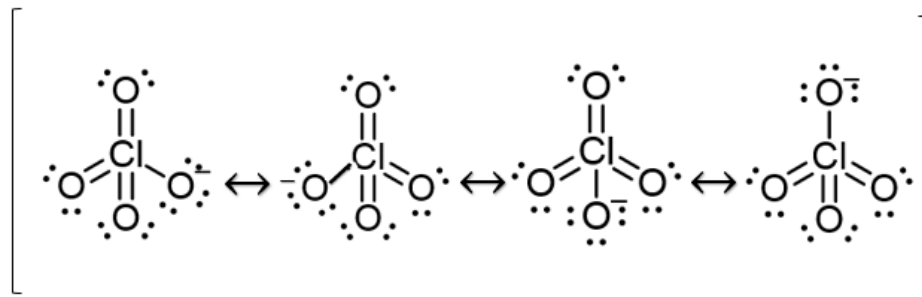
Question

- Which is the strongest acid among the substances listed in the table ?

Acid	HA	A ⁻	K _a (at 25°C)	pK _a
Perchloric acid	HClO ₄	ClO ₄ ⁻	↑ 10 ¹⁰	↓ -10
Hydrochloric acid	HCl	Cl ⁻	10 ²	-2
Hydrofluoric acid	HF	F ⁻	3.5 × 10 ⁻⁴	3.45
Ammonium ion	NH ₄ ⁺	NH ₃	5.5 × 10 ⁻¹⁰	↓ 9.26



$$\uparrow K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} \quad \text{or} \quad K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

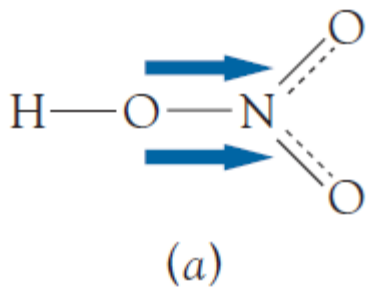


Hydrohalic acids

Acid	pK_a	H—X bond energy (kJ·mol ⁻¹)	
HF(<i>aq</i>)	+3	565	OH = 459 kJ/mol
HCl(<i>aq</i>)	-7	428	
HBr(<i>aq</i>)	-9	362	
HI(<i>aq</i>)	-10	295	



Oxyacids

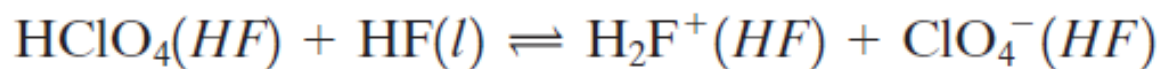


Nitric acid is stronger than nitrous acid



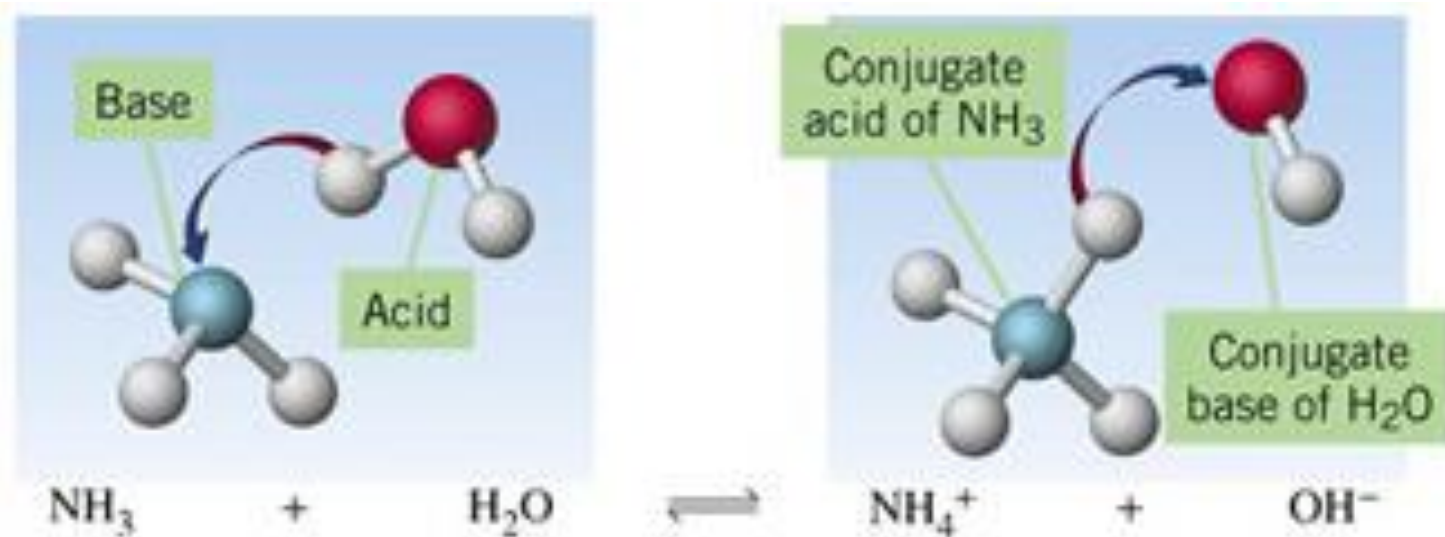
Acid	HA	A ⁻	K _a (at 25°C)	pK _a
Perchloric acid	HClO ₄	ClO ₄ ⁻	10 ¹⁰	-10
Hydrochloric acid	HCl	Cl ⁻	10 ²	-2
Hydrofluoric acid	HF	F ⁻	3.5 × 10 ⁻⁴	3.45
Ammonium ion	NH ₄ ⁺	NH ₃	5.5 × 10 ⁻¹⁰	9.26

Strength of acids depends on the solvent



in the Brønsted–Lowry theory, an acid is a proton donor and a base is a proton acceptor. **Every acid has a conjugate base**, which is the acid minus a proton, and **every base has a conjugate acid**, which is the base plus a proton.

In an acid–base reaction, the forward reaction is between an acid and a base and the reverse reaction is between the conjugate base and conjugate acid.



For reversible acid–base reactions, the equilibrium constants are called the acid ionization constant (K_a), and the base ionization constant (K_b), respectively.

The K_b values can be obtained, given the K_a values:

$$K_a \cdot K_b = K_w$$

If a Brønsted–Lowry acid is strong, its conjugate base is weak;

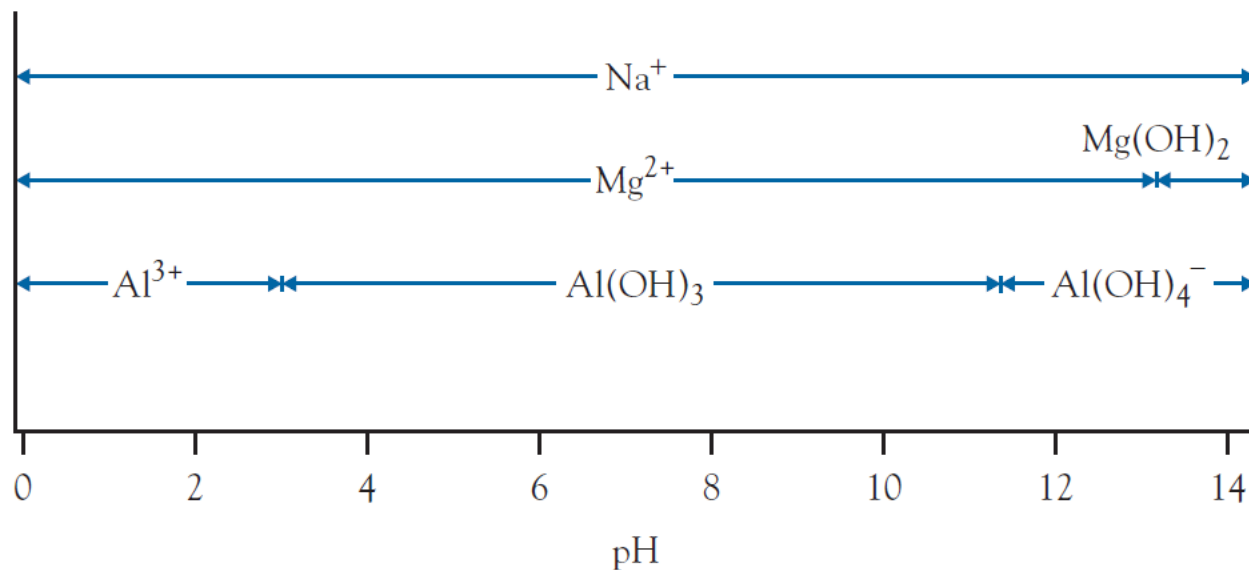
if a Brønsted–Lowry base is strong, its conjugate acid is weak.

TABLE 7.4 Base ionization constants of various inorganic bases

Base	A^-	HA	K_b (at 25°C)	pK_b
Phosphate ion	PO_4^{3-}	HPO_4^{2-}	4.7×10^{-2}	1.33
Cyanide ion	CN^-	HCN	1.6×10^{-5}	4.79
Ammonia	NH_3	NH_4^+	1.8×10^{-5}	4.74
Hydrazine	N_2H_4	$N_2H_5^+$	8.5×10^{-7}	6.07

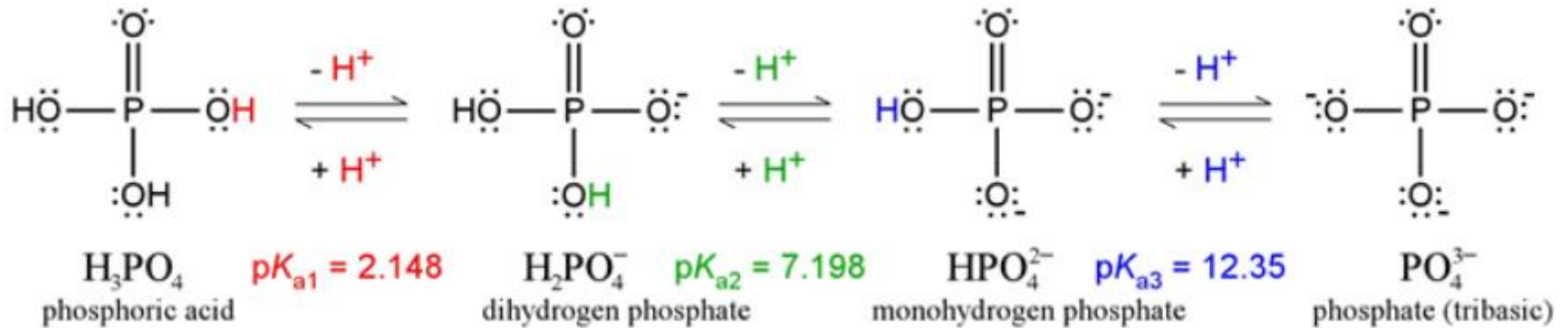
Acidity of metal ions

Li^+ Slightly acidic	Be^{2+} Weakly acidic		
Na^+ Neutral	Mg^{2+} Weakly acidic	Al^{3+} Acidic	
K^+ Neutral	Ca^{2+} Slightly acidic	Sc^{3+} Acidic	Ti^{4+} Very acidic



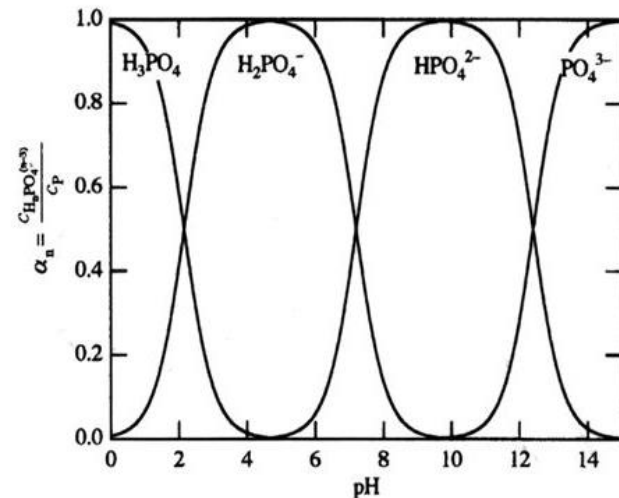
Order the following cations in terms of increasing Brønsted acidity in water:
 Sr^{2+} , Ba^{2+} , and Hg^{2+} .

Polyprotic Acids



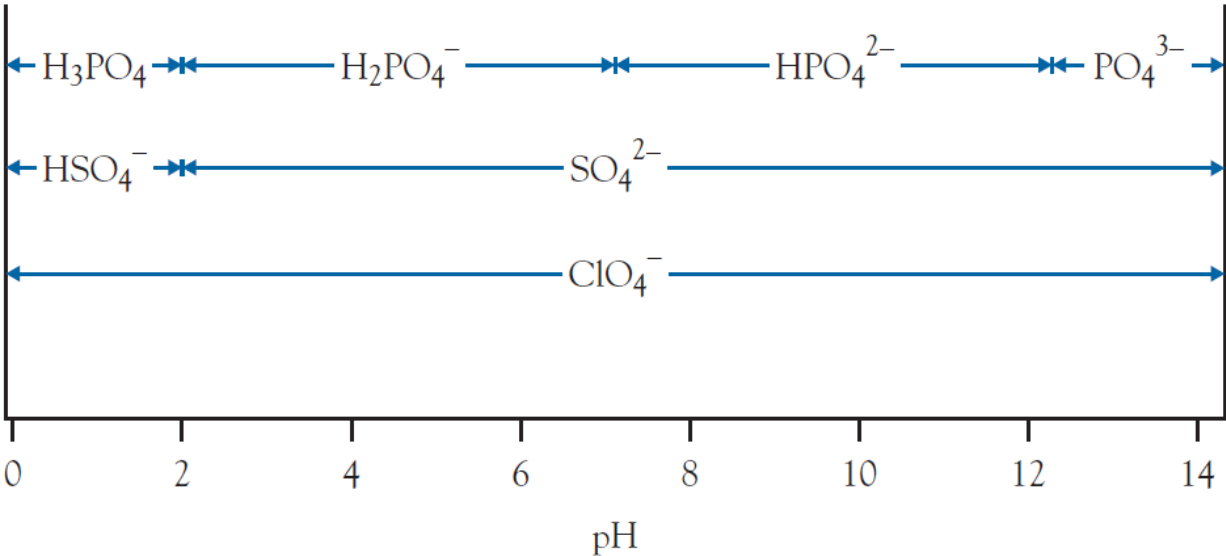
The first acid dissociation is that of the strongest acidic species of the series; K_{a2} goes with the next strongest acidic species, and so on.

Also in this case acid and basic dissociation constant are related each other.



Basicity trends

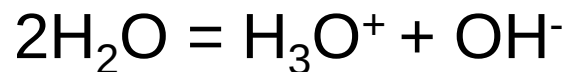
Classification	Type	Examples
Neutral	XO_4^-	ClO_4^- , MnO_4^-
Weakly basic	XO_4^{2-}	SO_4^{2-} , CrO_4^{2-} , MoO_4^{2-}
Moderately basic	XO_4^{3-}	PO_4^{3-} , VO_4^{3-}
Strongly basic	XO_4^{4-}	SiO_4^{4-}



Classification	Type	Examples
Neutral	XO_4^-	ClO_4^- , MnO_4^-
	XO_3^-	NO_3^- , ClO_3^-
Weakly basic	XO_2^-	NO_2^- , ClO_2^-
Moderately basic	XO^-	ClO^-

Solvent System Definition

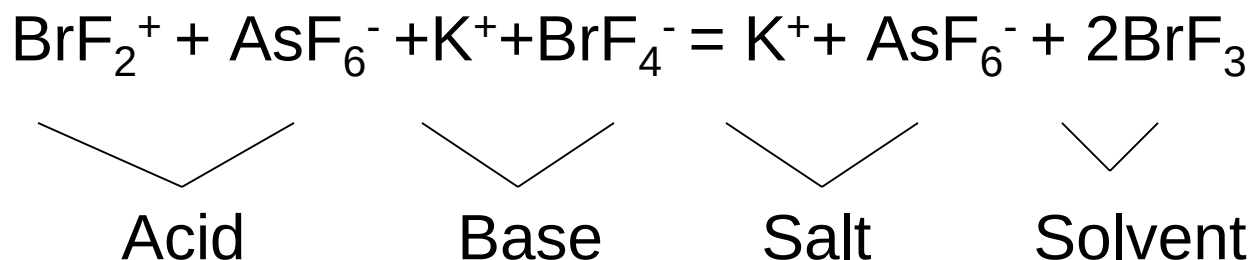
“A solute that increases the cationic species to the solvent is an acid; one that increases the anionic is a base.” This definition is much broader, since it can be applied in all cases, where the solvent has a significant autoionization reaction, e.g.



For the BrF_3 , is the compound BrF_2AsF_6 an acid or a base?

The answer: it is an acid. Because the following reaction occurs: $\text{BrF}_2\text{AsF}_6 = \text{BrF}_2^+ + \text{AsF}_6^-$, which increases the cationic species (BrF_2^+) of the solvent (BrF_3). In contrast, KBrF_4 is a base.

If the two above mentioned compounds are mixed, the following neutralisation reaction takes place:



Note: Acid or base is not an absolute property of the solute, since it is related to the solvent.

The Lux-Flood Definition

“An acid can be defined as an oxide ion acceptor and a base as an oxide ion donor.”

For example, $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
where CaO is a base and SiO₂ is an acid.

Acids

p-block oxides
Early transition
metal oxides

Bases

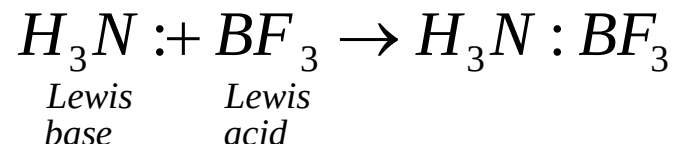
s-block oxides
Later transition
metal oxides

The Lewis Definition

“Acid is defined as an electron-pair acceptor and a base as an electron-pair donor.”

A very general, solid and useful definition!

This definition covers a great many systems where protons are not involved at all. For example,



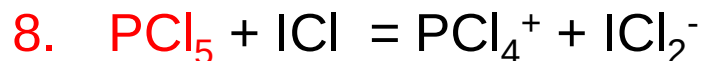
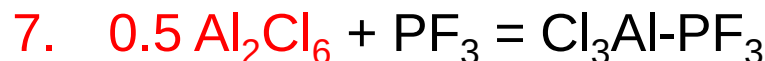
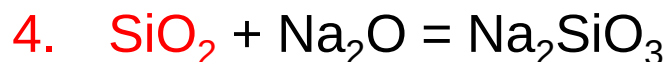
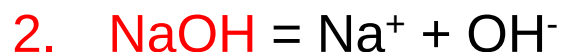
- 1) All the usual *ligands* can be regarded as bases and all metal ions as acids.
- 2) *Lewis acidity*: the degree of affinity of a metal ion for ligands. *Lewis basicity*: tendency of a ligand to be bound to a metal ion.

Exercise 1



4 minutes

Check your answers:



Exercise 2



8 minutes

Taken the reaction

... $\text{H}_3\text{PO}_4 = \text{H}^+ + \text{H}_2\text{PO}_4^-$, is H_3PO_4 a base or an acid? Why?

... $\text{CaO} + \text{SiO}_2 = \text{CaSiO}_3$, is SiO_2 a base or an acid? Why?

... $\text{AlCl}_3 + \text{PF}_3 = \text{Cl}_3\text{Al}:\text{PF}_3$, is PF_3 a base or an acid? Why?

... $3\text{Na}_2\text{O} + \text{P}_2\text{O}_5 = 2 \text{Na}_3\text{PO}_4$, is Na_2O a base or an acid? Why?

... $10 \text{S} + 4 \text{NH}_3 (\text{solvent}) = \text{S}_6^{2-} + \text{S}_4\text{N}^- + 3 \text{NH}_4^+$, is S a base or an acid? Why?

Pearson Hard-Soft Acid-Base:

Hard: metal cations (Lewis acids) small, compact and not very polarizable, prefer bases that are also small and less polarizable anions (Lewis bases).

Soft: metal ions (Lewis acids) with low charge density and not so low electronegativity. These acids tend to form complexes with larger and more polarizable Lewis bases.

The empirical relationship: *hard acids prefer hard bases and soft acids prefer soft bases.*

Hard, borderline (intermediate), and soft acids

1 H																	13 Al	14 Si		
3 Li	4 Be																			
11 Na	12 Mg																			
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As						
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb						
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	76 Re	77 Os	78 Ir	79 Pt	80 Au	81 Hg	82 Tl	83 Pb	84 Bi						
87 Fr	88 Ra	89 Ac																		

hard soft intermediate

TABLE 7.11 Common hard, borderline, and soft bases	
Hard	Soft
Li ⁺ Be ²⁺ B ³⁺ C ⁴⁺ N ⁵⁺ O ⁶⁺ F ⁷⁺ Ne ⁸⁺ Na ⁺ Mg ²⁺ Al ³⁺ Si ⁴⁺ P ⁵⁺ S ⁶⁺ Cl ⁷⁺ Ar ⁸⁺ K ⁺ Ca ²⁺ Sc ³⁺ Ti ⁴⁺ V ⁵⁺ Cr ⁶⁺ Mn ⁷⁺ Fe ⁸⁺ Co ⁹⁺ Ni ¹⁰⁺ Cu ¹¹⁺ Zn ¹²⁺ Ga ¹³⁺ Ge ¹⁴⁺ As ¹⁵⁺ Se ¹⁶⁺ Br ¹⁷⁺ Kr ¹⁸⁺ Rb ⁺ Sr ²⁺ Y ³⁺ Zr ⁴⁺ Nb ⁵⁺ Mo ⁶⁺ Tc ⁷⁺ Ru ⁸⁺ Rh ⁹⁺ Pd ¹⁰⁺ Ag ¹¹⁺ Hg ¹²⁺ Tl ¹³⁺ Pb ¹⁴⁺ Bi ¹⁵⁺ Po ¹⁶⁺ At ¹⁷⁺ Rn ¹⁸⁺ Cs ⁺ Ba ²⁺ La ³⁺ Ce ⁴⁺ Pr ⁵⁺ Nd ⁶⁺ Pm ⁷⁺ Sm ⁸⁺ Eu ⁹⁺ Gd ¹⁰⁺ Tb ¹¹⁺ Dy ¹²⁺ Ho ¹³⁺ Er ¹⁴⁺ Tm ¹⁵⁺ Yb ¹⁶⁺ Lu ¹⁷⁺ Hf ⁴⁺ Ta ⁵⁺ W ⁶⁺ Re ⁷⁺ Os ⁸⁺ Ir ⁹⁺ Pt ¹⁰⁺ Au ¹¹⁺ Hg ¹²⁺ Tl ¹³⁺ Pb ¹⁴⁺ Bi ¹⁵⁺ Po ¹⁶⁺ At ¹⁷⁺ Rn ¹⁸⁺ Fr ⁺ Ra ²⁺ Ac ³⁺ Th ⁴⁺ Pa ⁵⁺ U ⁶⁺ Np ⁷⁺ Pu ⁸⁺ Am ⁹⁺ Cm ¹⁰⁺ Bk ¹¹⁺ Cf ¹²⁺ Es ¹³⁺ Fm ¹⁴⁺ Md ¹⁵⁺ No ¹⁶⁺ Lr ¹⁷⁺	H ⁺ Li ⁺ Na ⁺ K ⁺ Rb ⁺ Cs ⁺ Fr ⁺ Be ²⁺ Mg ²⁺ Ca ²⁺ Sr ²⁺ Ba ²⁺ Ra ²⁺ Sc ³⁺ Y ³⁺ La ³⁺ Ce ³⁺ Pr ³⁺ Nd ³⁺ Sm ³⁺ Eu ³⁺ Gd ³⁺ Tb ³⁺ Dy ³⁺ Ho ³⁺ Er ³⁺ Tm ³⁺ Yb ³⁺ Lu ³⁺ Ac ³⁺ Th ³⁺ Pa ³⁺ U ³⁺ Np ³⁺ Pu ³⁺ Am ³⁺ Cm ³⁺ Bk ³⁺ Cf ³⁺ Es ³⁺ Fm ³⁺ Md ³⁺ No ³⁺ Lr ³⁺ H ⁻ Li ⁻ Na ⁻ K ⁻ Rb ⁻ Cs ⁻ Fr ⁻ Be ⁻ Mg ⁻ Ca ⁻ Sr ⁻ Ba ⁻ Ra ⁻ Sc ⁻ Y ⁻ La ⁻ Ce ⁻ Pr ⁻ Nd ⁻ Sm ⁻ Eu ⁻ Gd ⁻ Tb ⁻ Dy ⁻ Ho ⁻ Er ⁻ Tm ⁻ Yb ⁻ Lu ⁻ Ac ⁻ Th ⁻ Pa ⁻ U ⁻ Np ⁻ Pu ⁻ Am ⁻ Cm ⁻ Bk ⁻ Cf ⁻ Es ⁻ Fm ⁻ Md ⁻ No ⁻ Lr ⁻ B ³⁺ Al ³⁺ Ga ³⁺ In ³⁺ Tl ³⁺ Pb ³⁺ Bi ³⁺ Po ³⁺ At ³⁺ Rn ³⁺ Fr ³⁺ Ra ³⁺ Ac ³⁺ Th ³⁺ Pa ³⁺ U ³⁺ Np ³⁺ Pu ³⁺ Am ³⁺ Cm ³⁺ Bk ³⁺ Cf ³⁺ Es ³⁺ Fm ³⁺ Md ³⁺ No ³⁺ Lr ³⁺ C ⁴⁺ Si ⁴⁺ Ge ⁴⁺ Sn ⁴⁺ Pb ⁴⁺ Bi ⁴⁺ Po ⁴⁺ At ⁴⁺ Rn ⁴⁺ Fr ⁴⁺ Ra ⁴⁺ Ac ⁴⁺ Th ⁴⁺ Pa ⁴⁺ U ⁴⁺ Np ⁴⁺ Pu ⁴⁺ Am ⁴⁺ Cm ⁴⁺ Bk ⁴⁺ Cf ⁴⁺ Es ⁴⁺ Fm ⁴⁺ Md ⁴⁺ No ⁴⁺ Lr ⁴⁺ N ⁵⁺ P ⁵⁺ As ⁵⁺ Sb ⁵⁺ Bi ⁵⁺ Po ⁵⁺ At ⁵⁺ Rn ⁵⁺ Fr ⁵⁺ Ra ⁵⁺ Ac ⁵⁺ Th ⁵⁺ Pa ⁵⁺ U ⁵⁺ Np ⁵⁺ Pu ⁵⁺ Am ⁵⁺ Cm ⁵⁺ Bk ⁵⁺ Cf ⁵⁺ Es ⁵⁺ Fm ⁵⁺ Md ⁵⁺ No ⁵⁺ Lr ⁵⁺ O ⁶⁺ S ⁶⁺ Se ⁶⁺ Te ⁶⁺ Po ⁶⁺ At ⁶⁺ Rn ⁶⁺ Fr ⁶⁺ Ra ⁶⁺ Ac ⁶⁺ Th ⁶⁺ Pa ⁶⁺ U ⁶⁺ Np ⁶⁺ Pu ⁶⁺ Am ⁶⁺ Cm ⁶⁺ Bk ⁶⁺ Cf ⁶⁺ Es ⁶⁺ Fm ⁶⁺ Md ⁶⁺ No ⁶⁺ Lr ⁶⁺ F ⁷⁺ Cl ⁷⁺ Br ⁷⁺ I ⁷⁺ At ⁷⁺ Rn ⁷⁺ Fr ⁷⁺ Ra ⁷⁺ Ac ⁷⁺ Th ⁷⁺ Pa ⁷⁺ U ⁷⁺ Np ⁷⁺ Pu ⁷⁺ Am ⁷⁺ Cm ⁷⁺ Bk ⁷⁺ Cf ⁷⁺ Es ⁷⁺ Fm ⁷⁺ Md ⁷⁺ No ⁷⁺ Lr ⁷⁺ Ne ⁸⁺ Ar ⁸⁺ Kr ⁸⁺ Xe ⁸⁺ Rn ⁸⁺ Fr ⁸⁺ Ra ⁸⁺ Ac ⁸⁺ Th ⁸⁺ Pa ⁸⁺ U ⁸⁺ Np ⁸⁺ Pu ⁸⁺ Am ⁸⁺ Cm ⁸⁺ Bk ⁸⁺ Cf ⁸⁺ <

Hard	Borderline	Soft
F^- , O^{2-} , OH^- , H_2O , CO_3^{2-} , NH_3 , NO_3^- , SO_4^{2-} , ClO_4^- , PO_4^{3-} , (Cl^-)	Br^- , N_3^- , NCS^-	I^- , S^{2-} , P^{3-} , H^- , CN^- , CO , SCN^- , $S_2O_3^{2-}$

9	F	Fluorine
17	Cl	Chlorine
35	Br	Bromine
53	I	Iodine
85	At	Astatine
117	Ts	Tennessine

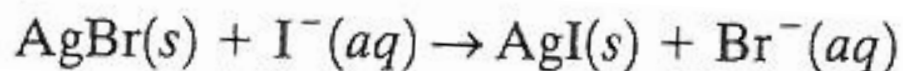
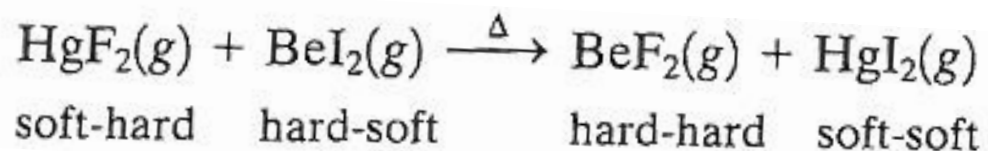


TABLE 7.12 Solubilities of sodium and silver halides (mol·L ⁻¹)				
	Fluoride	Chloride	Bromide	Iodide
Sodium	1.0	6.1	11.3	12.3
Silver	14.3	1.3×10^{-5}	7.2×10^{-7}	9.1×10^{-9}

Lecture 7

Coordination compounds (part A)

Chapter 19

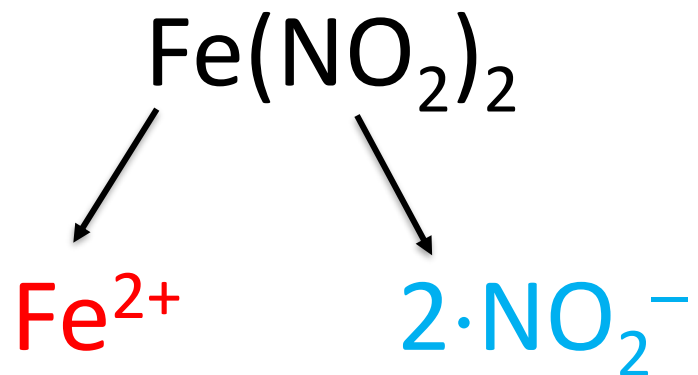
Content:

- **Introduction to Coordination Compounds**
- **Application of the Cristal Field Theory to the octahedral, tetrahedral and square planar geometries**
- **Colors of the coordination complexes**
- **Use of the CFT to understand color trends, to predict normal and inverse spinel structures, and to explain magnetic properties of coordination compounds**

Coordination chemistry: the branch of inorganic chemistry concerned with the combined behavior of cations and their ligands.

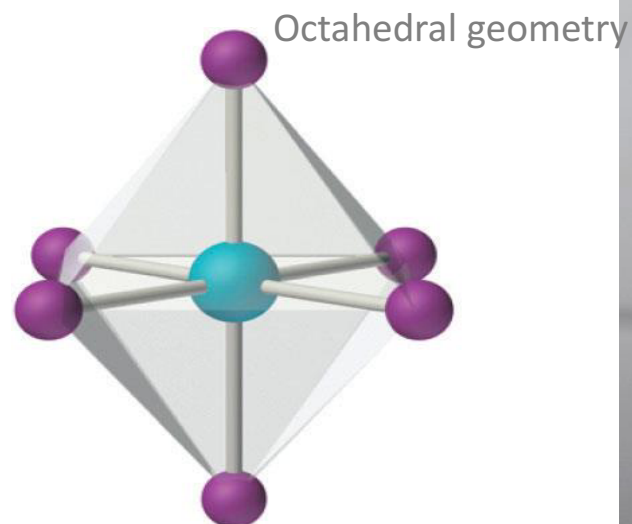
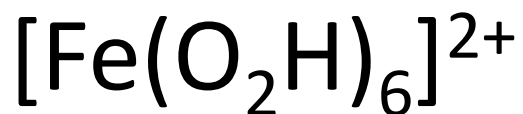
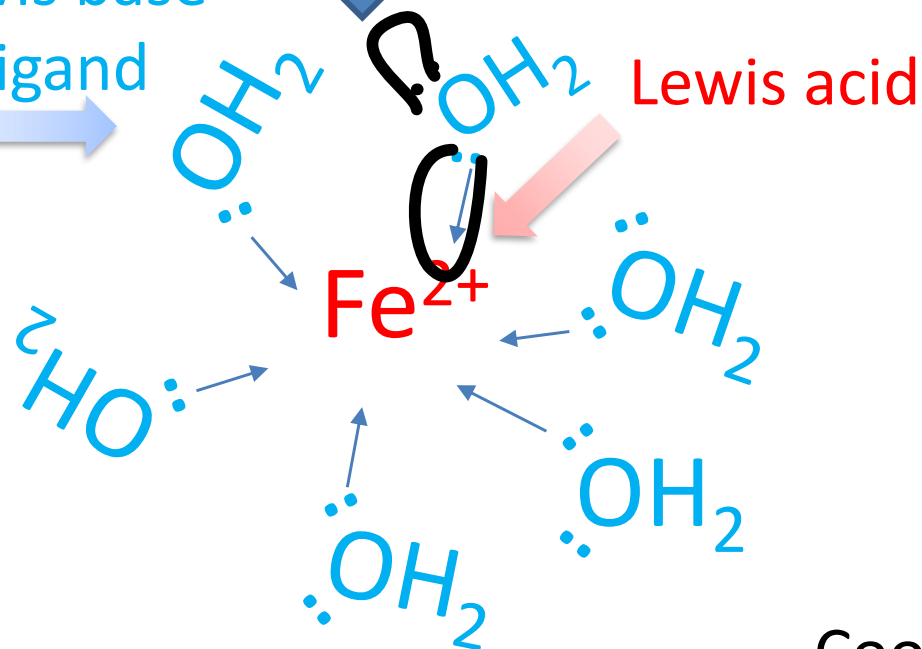
There is **no sharp dividing line** between coordination chemistry and the chemistry of covalent molecules on the one hand and that of ionic solids on the other.

Ligand: In a compound, a cation is surrounded by anions or neutral molecules. The groups immediately surrounding a cation are called ligands.



In water

Lewis base
or ligand



Coordination number = 6

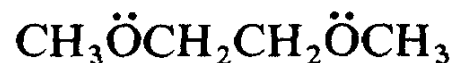
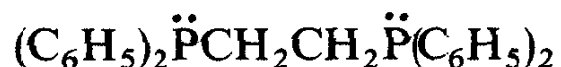
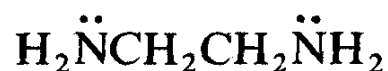
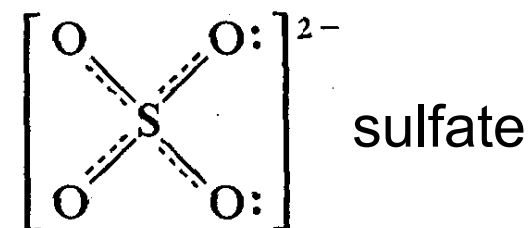
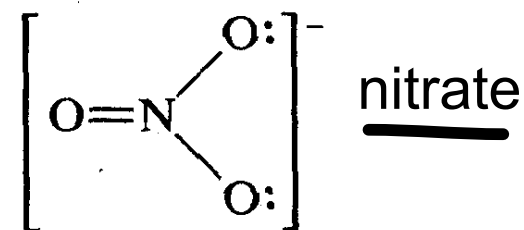
Oxidation state of Iron = 2+

Monodentate and polydentate

Monodentate

Chloride ion (Cl^-),
Hydroxide ion (OH^-),
Water (H_2O),
Ammonia (NH_3).

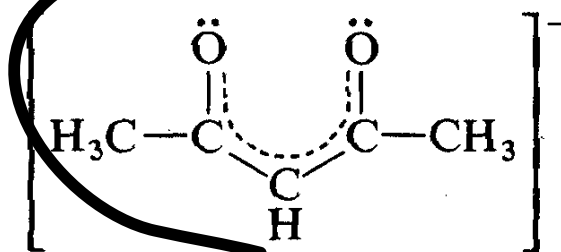
Bidentate Ligands



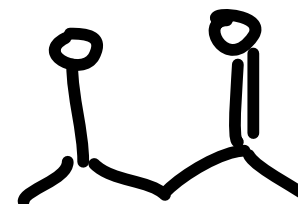
Ethylenediamine (en)

Bis(diphenylphosphino)ethane (diphos)

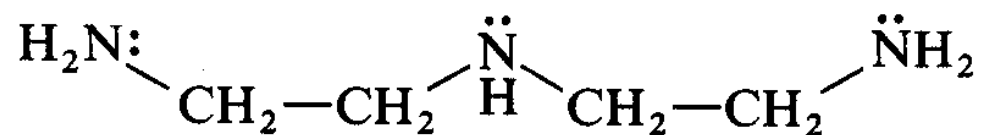
1,2 dimethoxyethane (glyme)



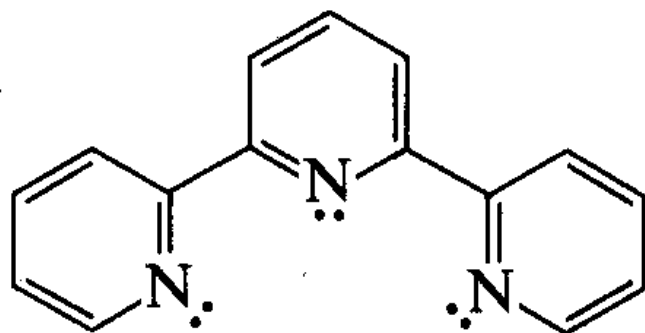
acetylacetonate, acac



tridentate ligands:

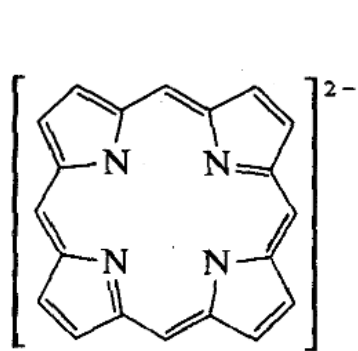


diethylene, dien

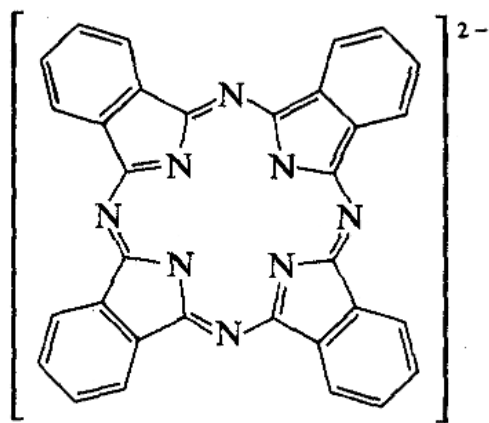


terpyridyl, terpy

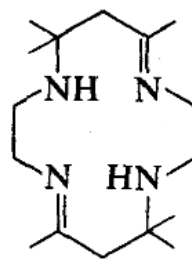
Quadridentate ligands:



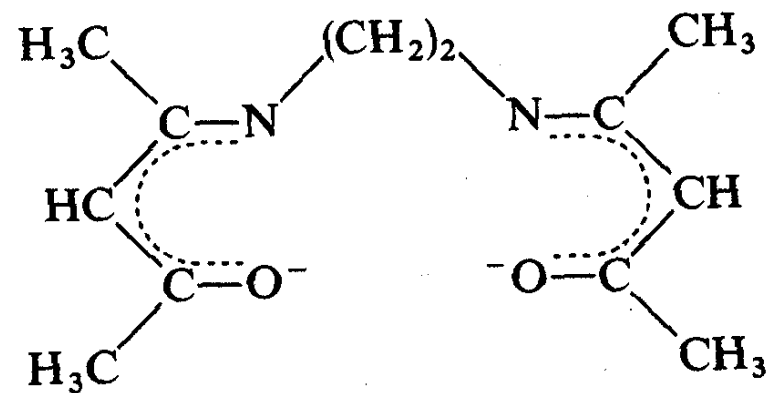
6-XII

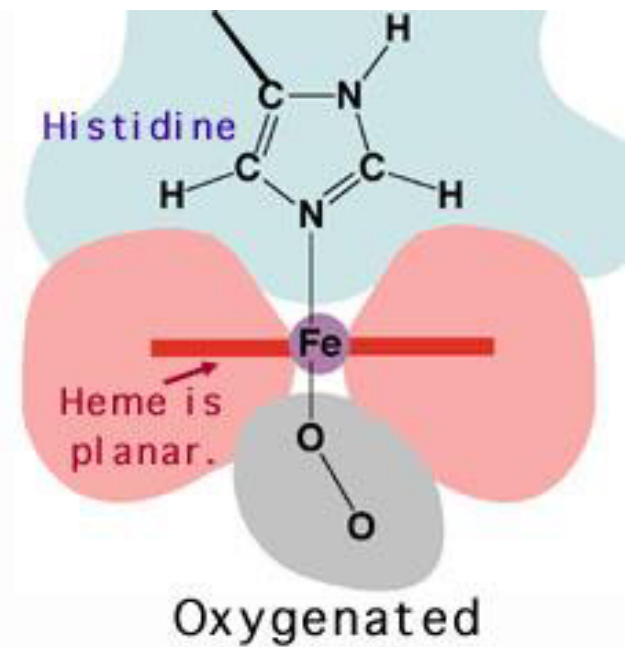
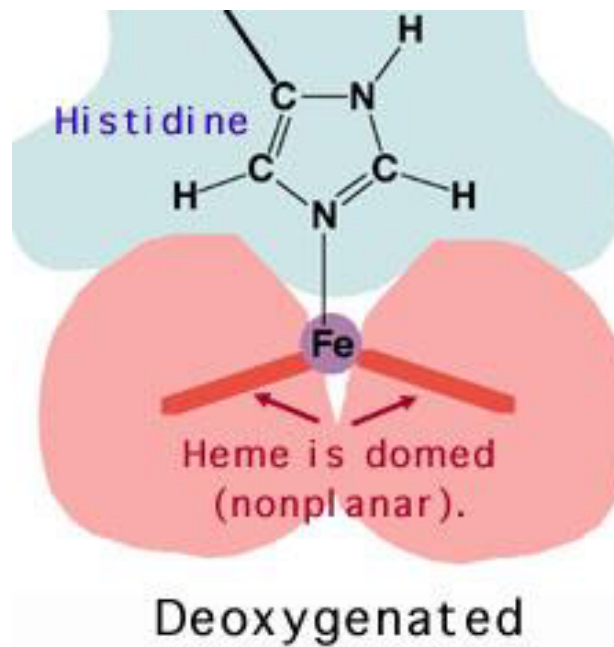
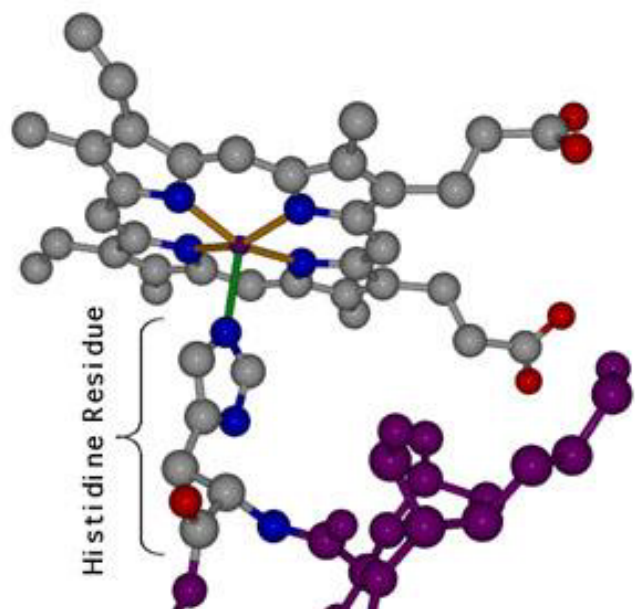
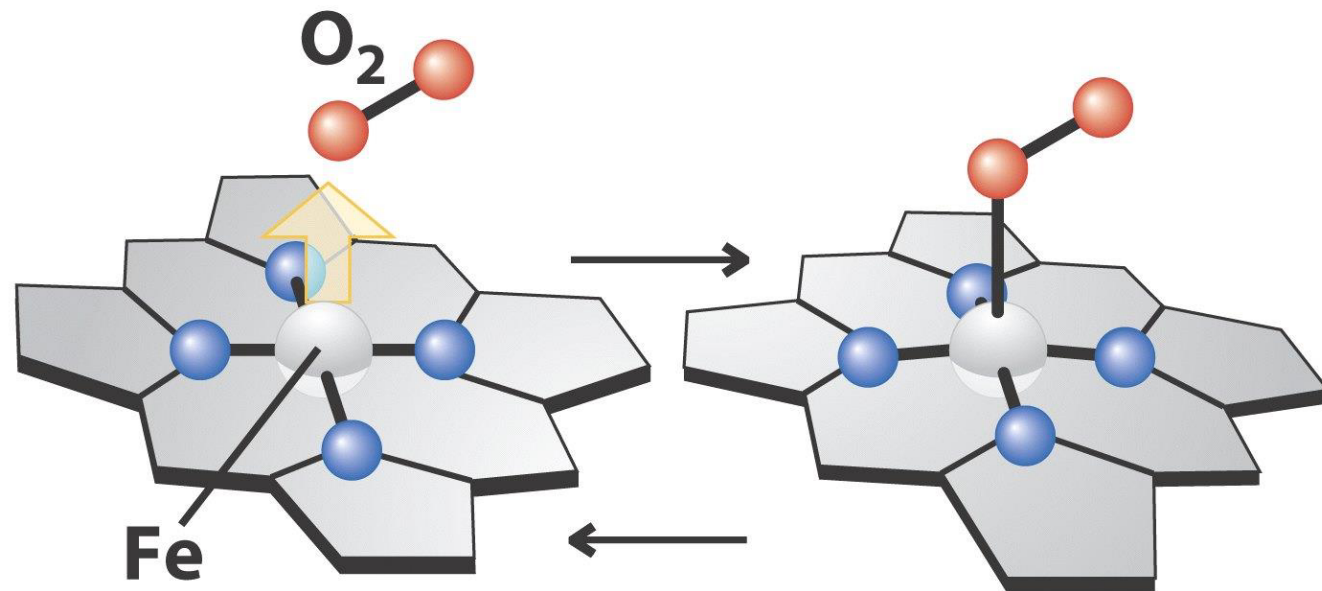


6-XIII

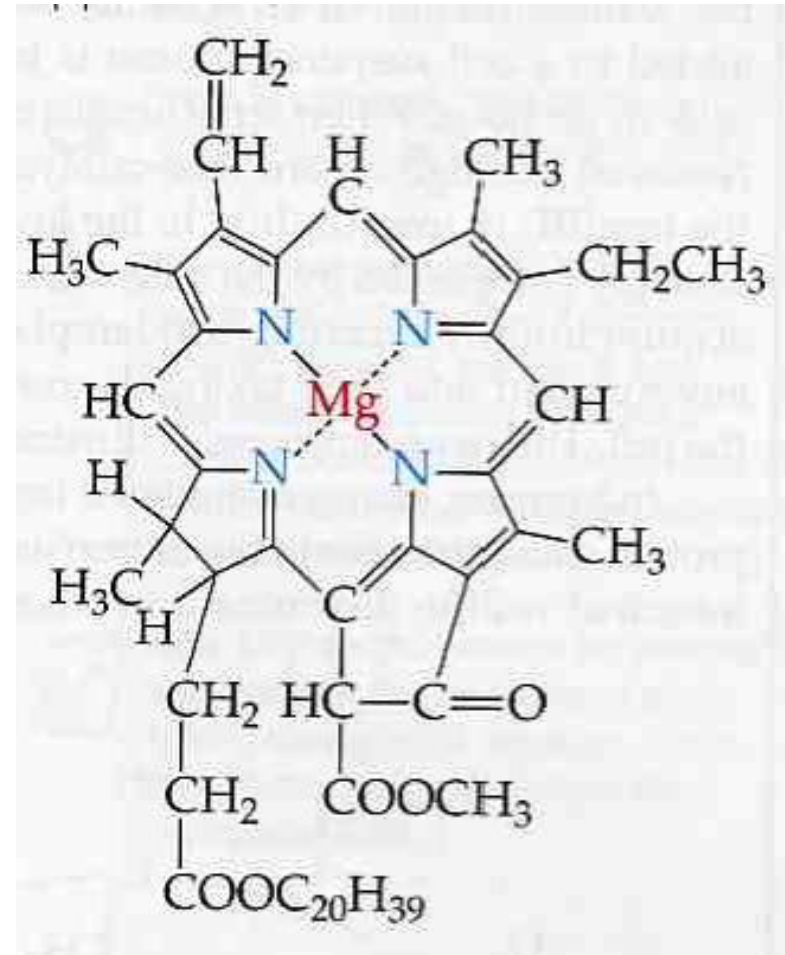
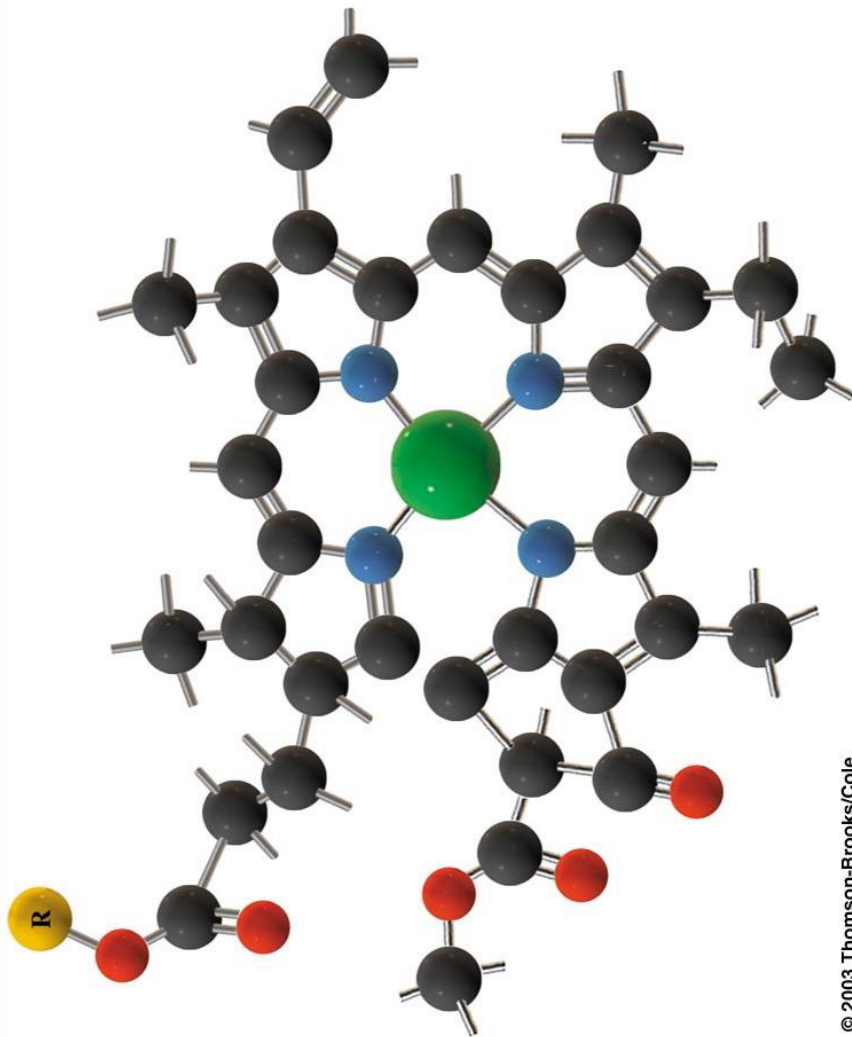


6-XIV





A very important porphine that converts solar photons into food energy: chlorophyll

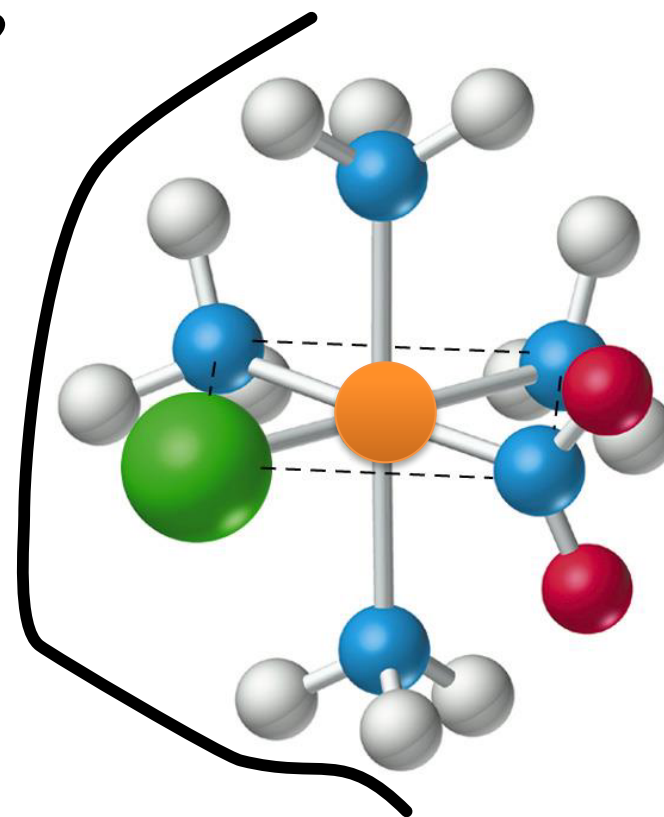
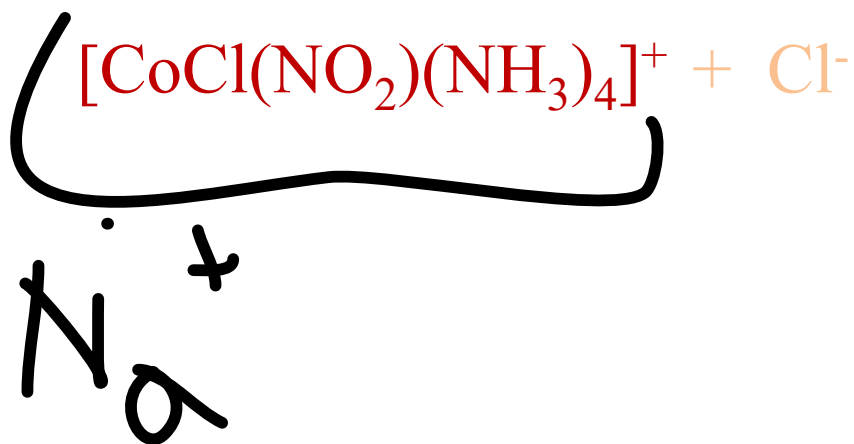


Chlorophyll ($\text{C}_{55}\text{H}_{72}\text{N}_4\text{O}_5\text{Mg}$)

Question

Relating the Formula of a Complex to the Coordination Number and Oxidation State of the Central Metal.

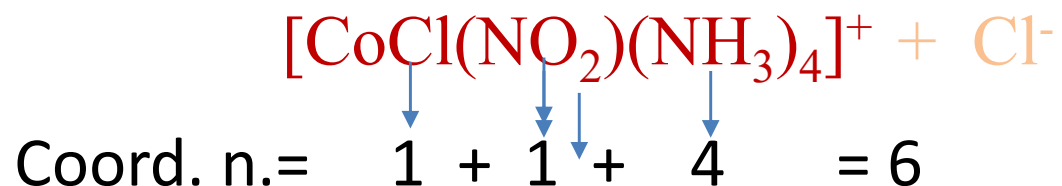
What are the coordination number and oxidation state of Co in the complex ion $[\text{CoCl}(\text{NO}_2)(\text{NH}_3)_4]\text{Cl}$?



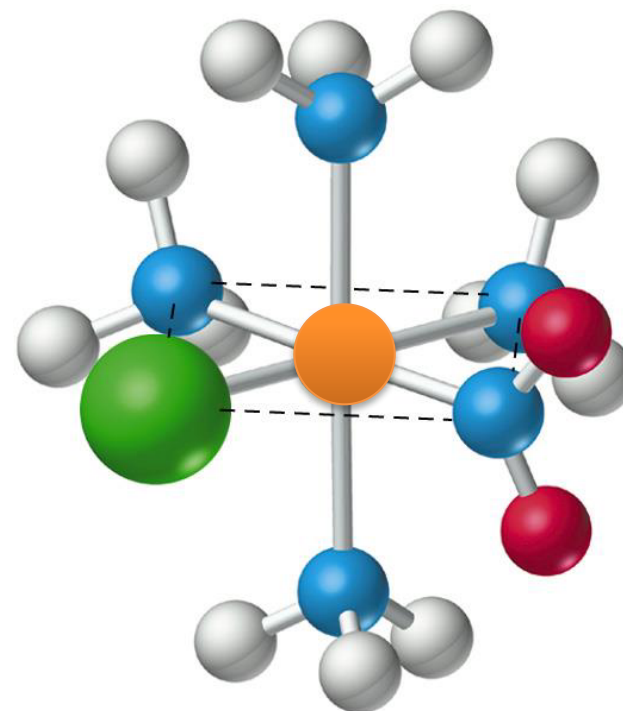
Question

Relating the Formula of a Complex to the Coordination Number and Oxidation State of the Central Metal.

What are the coordination number and oxidation state of Co in the complex ion $[\text{CoCl}(\text{NO}_2)(\text{NH}_3)_4]\text{Cl}$?



Octahedral geometry



Question

Relating the Formula of a Complex to the Coordination Number and Oxidation State of the Central Metal.

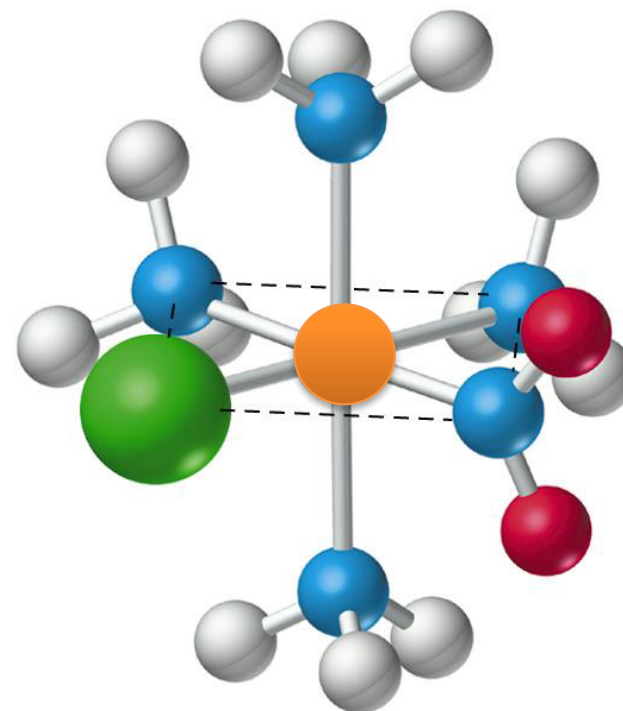
What are the coordination number and oxidation state of Co in the complex ion $[\text{CoCl}(\text{NO}_2)(\text{NH}_3)_4]\text{Cl}$?



$$\text{Ox. state.} \quad x \quad -1 \quad -1 \quad -4 \quad \cdot \emptyset = +1$$

$$x \quad -2 \quad \quad \quad = +1$$

$$\text{Ox. state} = x = +3$$

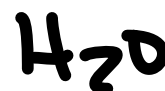
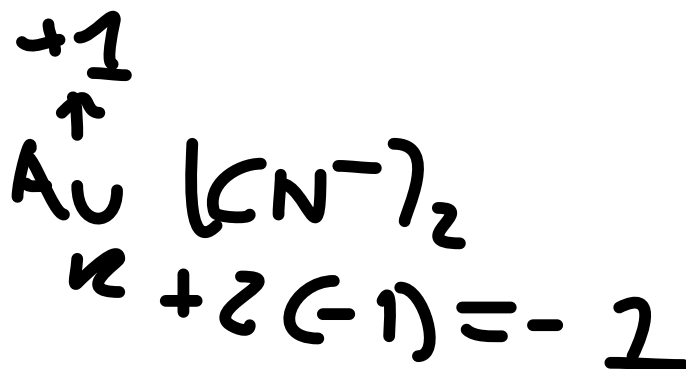
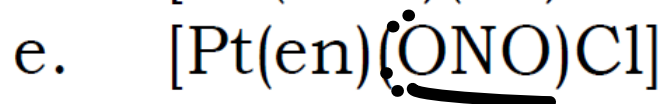
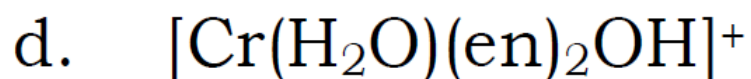
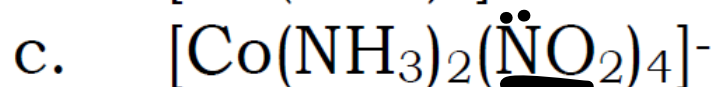
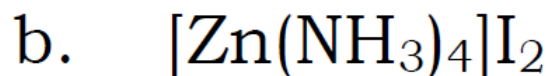
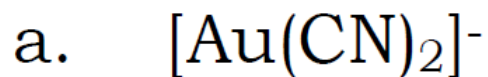


Exercise 1

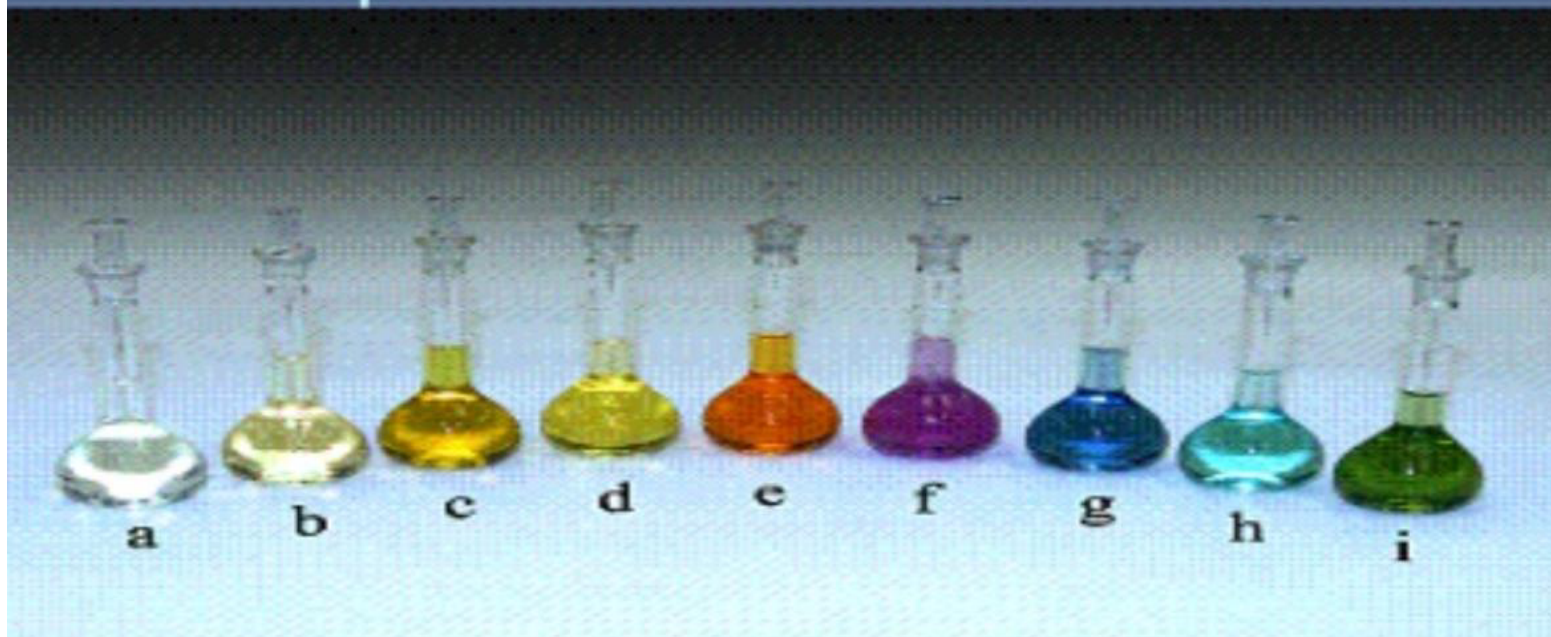


5 minutes

What are oxidation state and coordination number of the transition metal in the following coordination compounds
(en = ethylenediamine; while NO_2^- and $(\text{ONO})^-$ act as monodentate ligands)



The Spectrochemical Series



The complexes of cobalt (III) show the shift in color due to the ligand.

(a) CN^-

(b) NO_2^-

(c) phen

(d) en

(e) NH_3

(f) gly

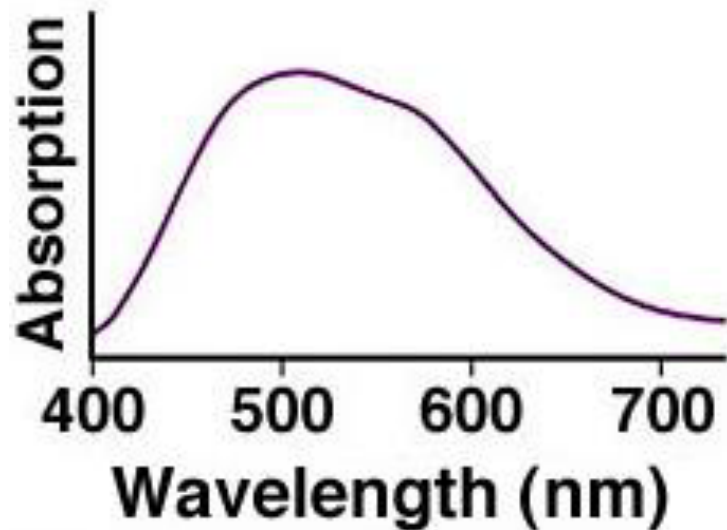
(g) H_2O

(h) ox^{2-}

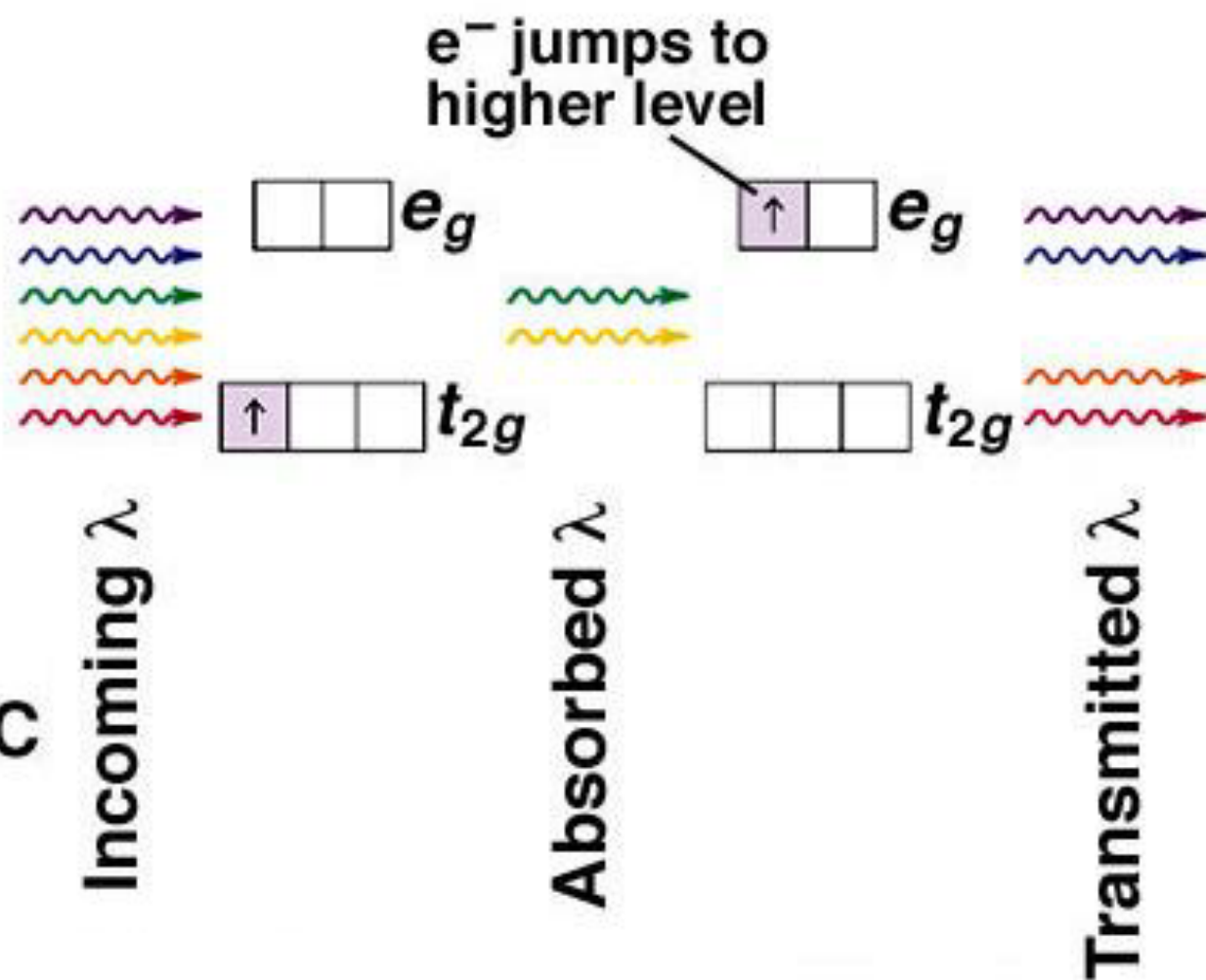
(i) CO_3^{2-}

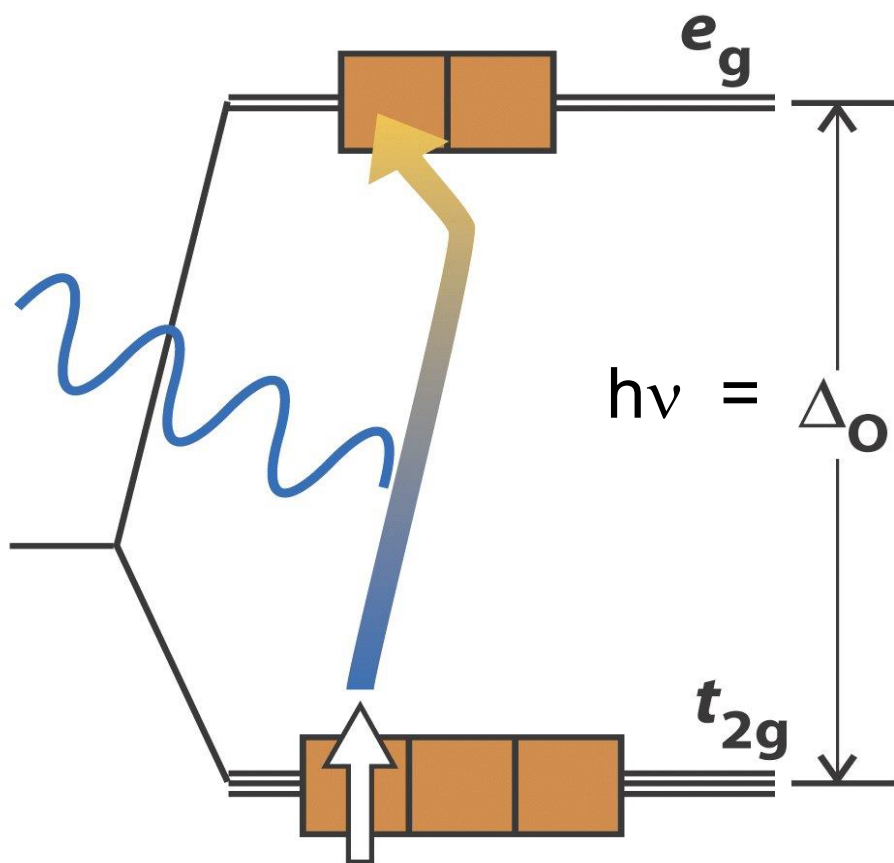
The Color of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$

A



C





The energy gap between the e_g and t_{2g} orbitals, Δ_o , (the crystal field splitting) equals the energy of a photon:

$$\Delta_o = h\nu = \Delta E$$

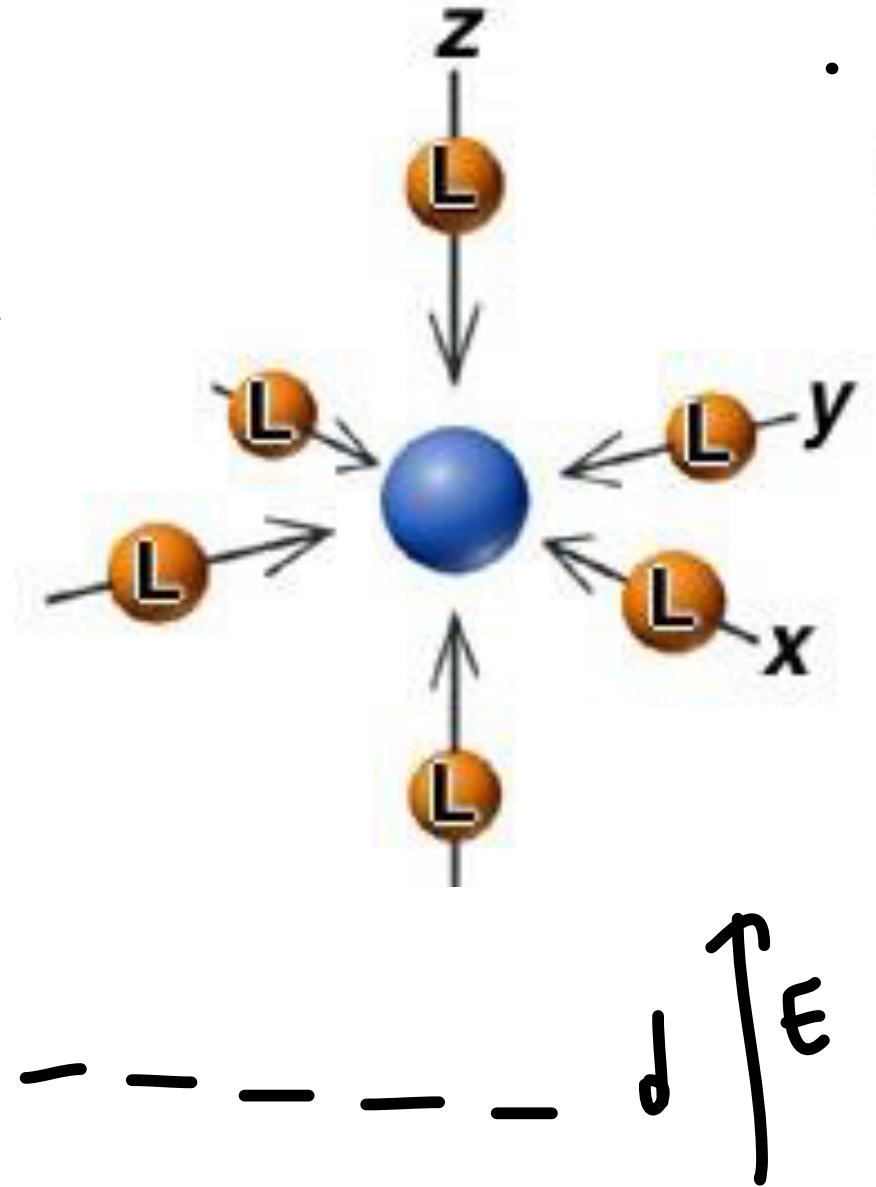
As Δ_o varies, $h\nu$ will also vary and the color of the compound will change.

Absorption of a photon causes a jump from an occupied orbital to an unoccupied orbital

Crystal field theory

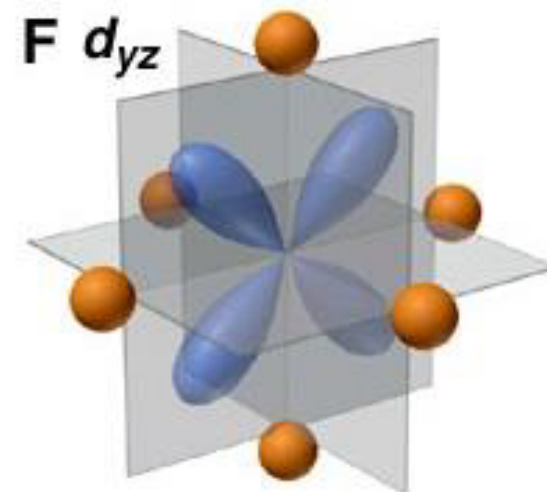
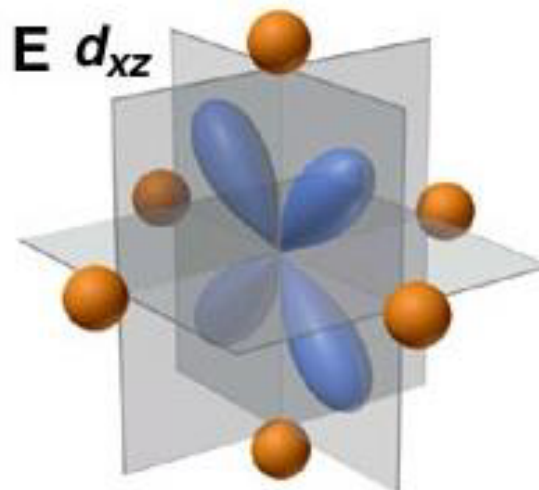
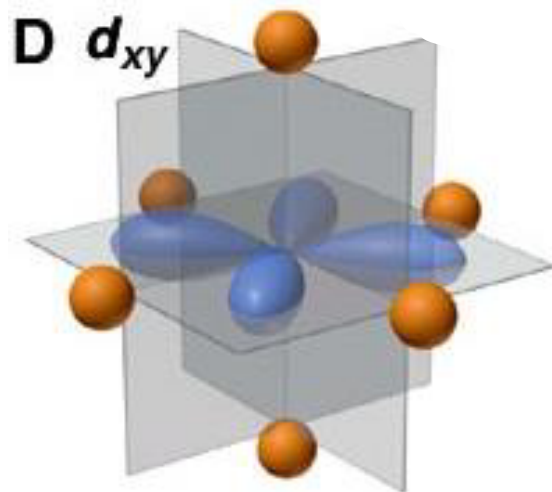
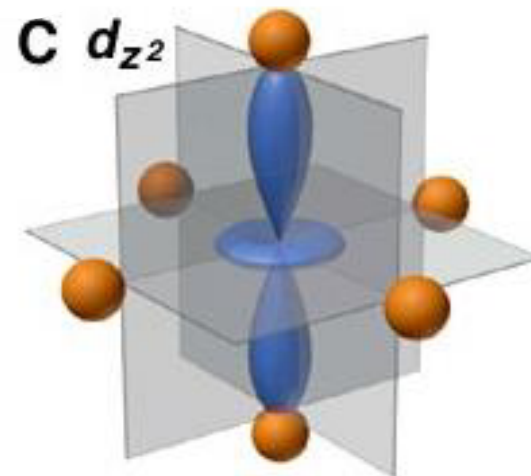
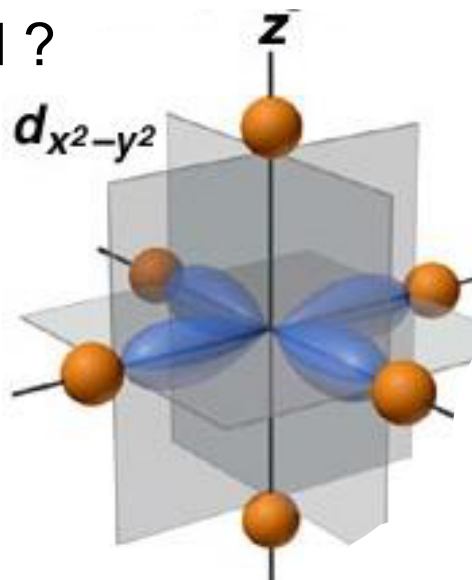
Bonds in a complex are purely electrostatic attraction between a positively charged nucleus and the electrons of the ligands.

- *Electrons on metal orbitals repel electrons on ligands.*
- *Focus particularly on the d-electrons on the metal ion.*

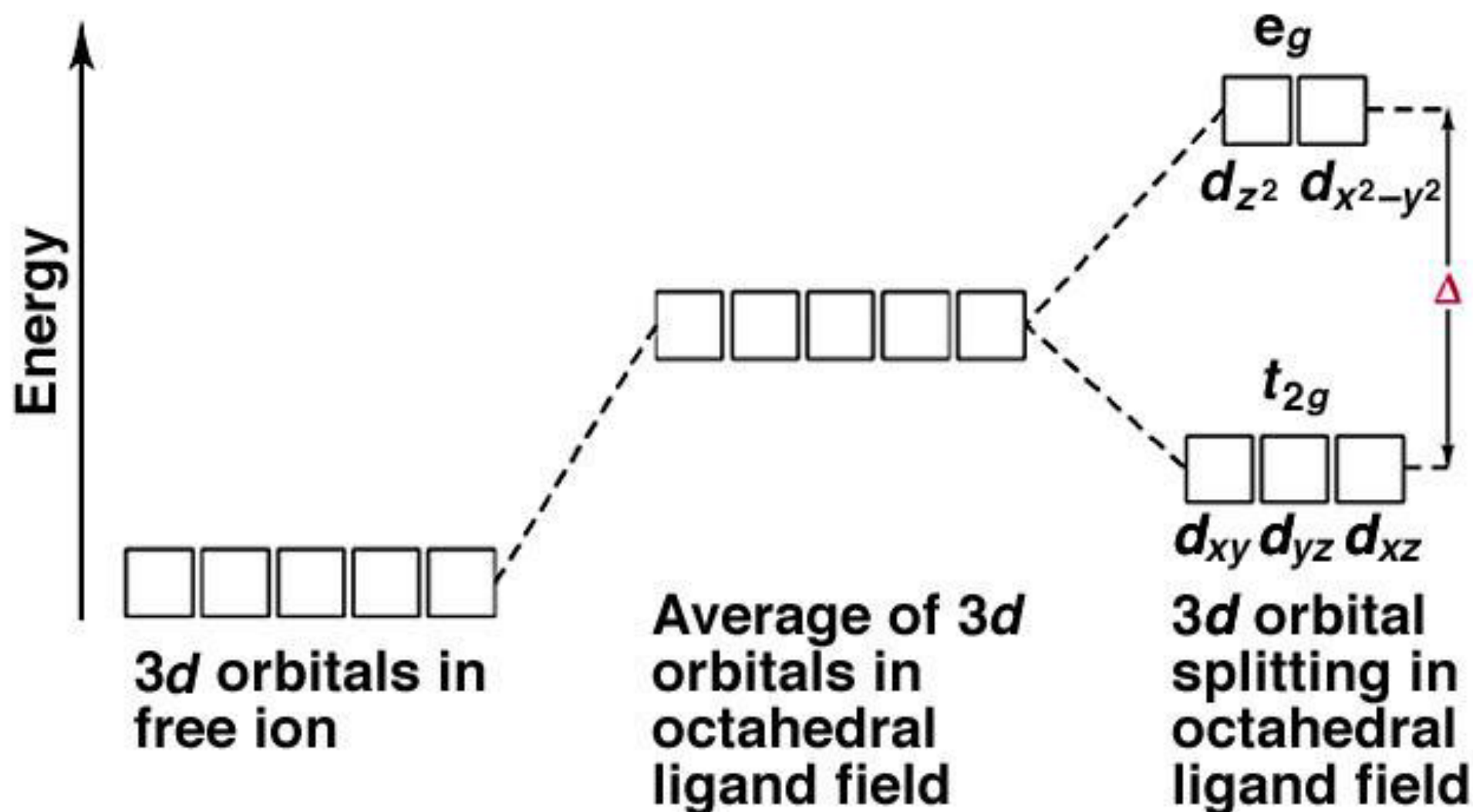


Question 1

Which are the d orbitals which have the strongest interaction with ligands in the octahedral field ?



Splitting of *d*-Orbital Energies by an Octahedral Field of Ligands



Exercise 1:



7 minutes

3	Na SODIUM	Mg MAGNESIUM	3	IIIB	4	IVB	5	VB	6	VIB	7	VIIIB	8	9	VIIIB	10	11	IB	12	IIIB	Al ALUMINIUM	Si SILICON	P PHOSPHORUS	S SULPHUR	Cl CHLORINE	Ar ARGON
	19 39.098	20 40.078	21 44.956	22 47.867	23 50.942	24 51.996	25 54.938	26 55.845	27 58.933	28 58.693	29 63.546	30 65.39	31 69.723	32 72.64	33 74.922	34 78.96	35 79.904	36 83.80								
4	K POTASSIUM	Ca CALCIUM	Sc SCANDIUM	Ti TITANIUM	V VANADIUM	Cr CHROMIUM	Mn MANGANESE	Fe IRON	Co COBALT	Ni NICKEL	Cu COPPER	Zn ZINC	Ga GALLIUM	Ge GERMANIUM	As ARSENIC	Se SELENIUM	Br BROMINE	Kr KRYPTON								

- Represent the 3d electron configuration of the following ions

Ti⁰ [Ar] 4s² 3d²

Ti³⁺ [Ar] 3d¹

Ti³⁺ _ _

1

Ti²⁺ _ _

_ _

V²⁺ _ _

_ _

Cr²⁺ _ _

_ _

Mn²⁺, Fe³⁺ _ _

_ _

Fe²⁺, Co³⁺ _ _

_ _

Co²⁺

_ _

_ _

Ni²⁺

_ _

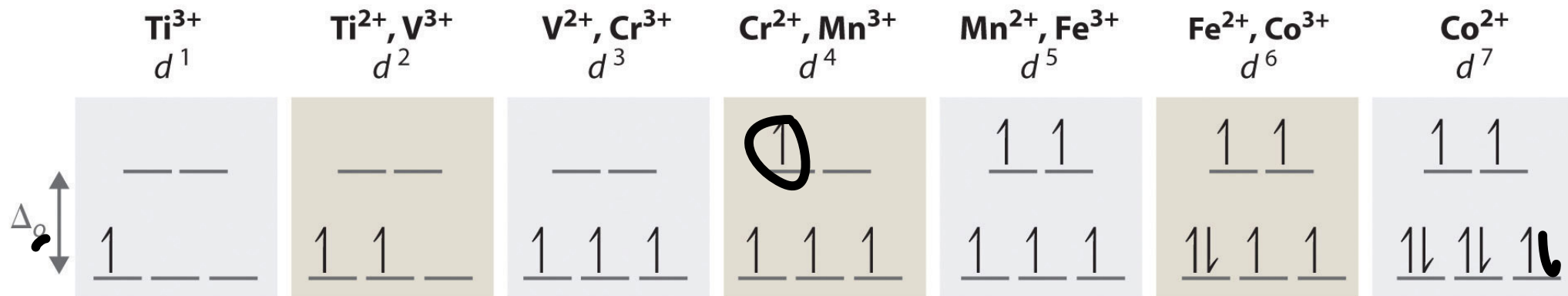
_ _

Cu²⁺ _ _

_ _

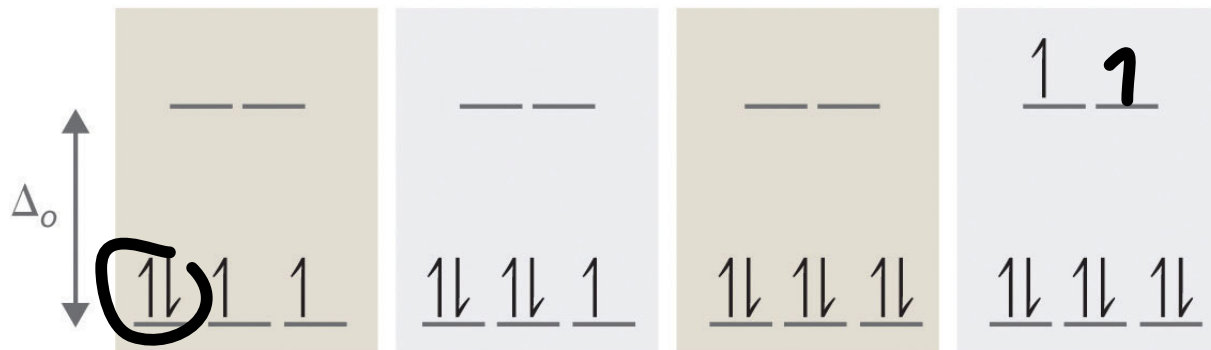
Zn²⁺ _ _

_ _



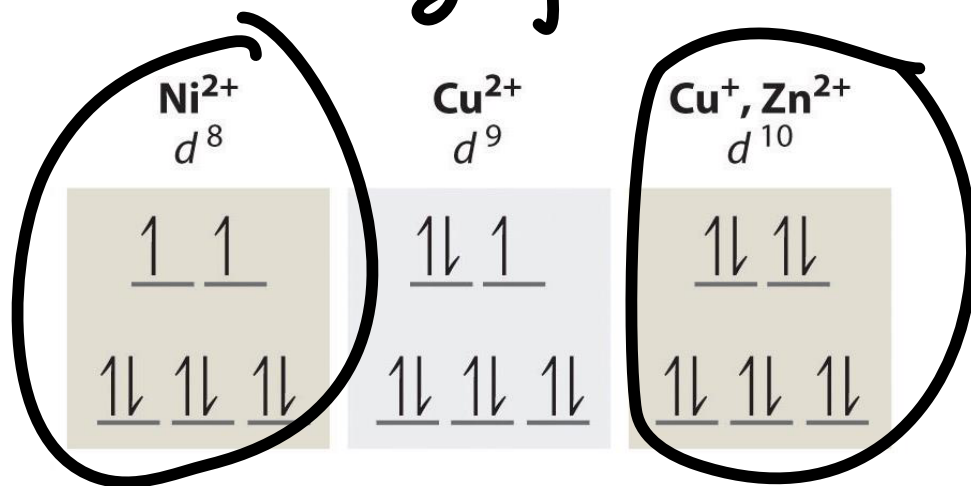
High spin

weak f



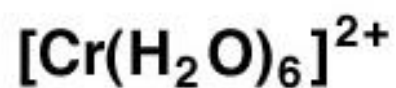
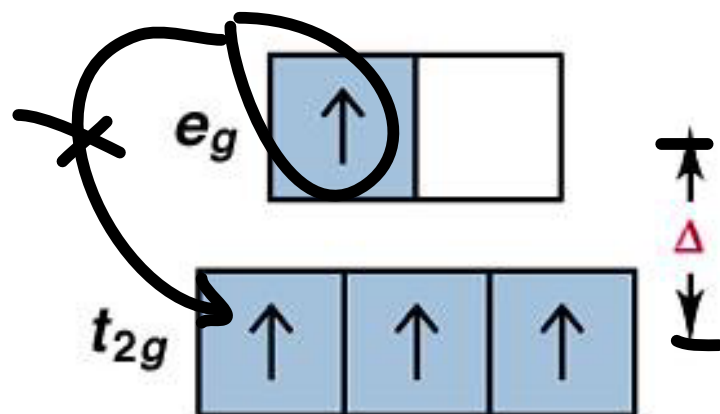
Low spin

strong f



Effect of the Ligand on Splitting Energy

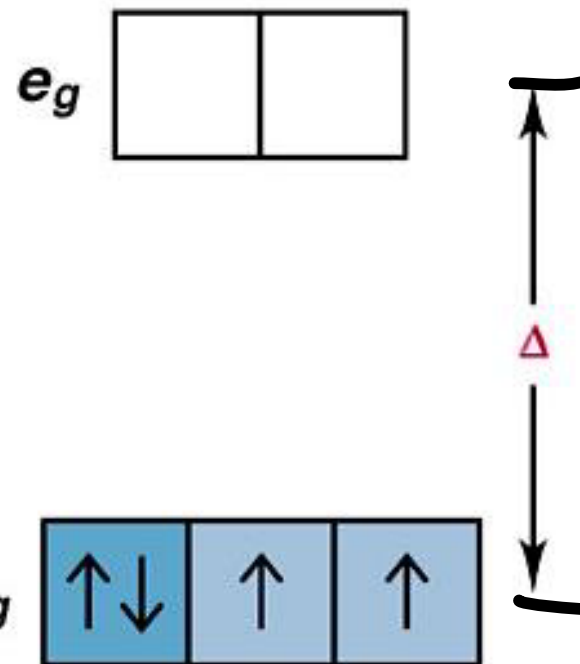
Weak-field ligands



$$\Delta_0 < P$$

High spin

Strong-field ligands



$$\Delta_0 > P$$

Low spin

Δ_o vs Pairing Energy

Complex	Config.	Δ_o, cm^{-1}	P, cm^{-1}	spin-state
$[\text{Fe}(\text{OH}_2)_6]^{2+}$	d^6	<u>10,400</u>	<u>17,600</u>	high-spin
$[\text{Fe}(\text{CN})_6]^{4-}$	d^6	<u>32,850</u>	17,600	low-spin

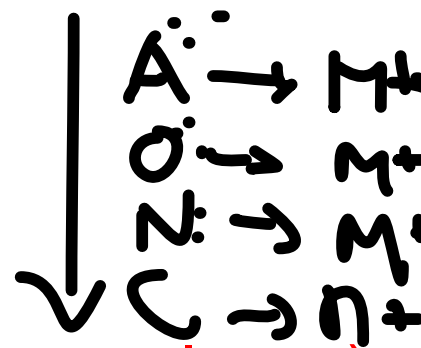
Ions		Ligands				
		Cl^-	H_2O	NH_3	en	CN^-
d^3	Cr^{3+}	13 700	17 400	21 500	21 900	26 600
d^5	Mn^{2+}	7500	8500		10 100	30 000
	Fe^{3+}	11 000	14 300			(35 000)
d^6	Fe^{2+}		10 400			(32 800)
	Co^{3+}		(20 700)	(22 900)	(23 200)	(34 800)
	Rh^{3+}		(20 400)	(27 000)	(34 000)	(34 600)
d^8	Ni^{2+}	7500	8500	10 800	11 500	

* Values are in cm^{-1} ; entries in parentheses are for low-spin complexes.

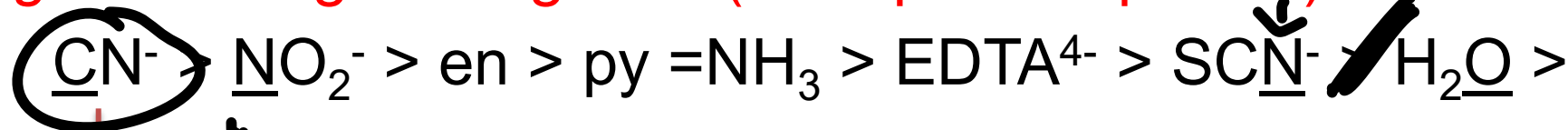
Source: H.B. Gray, *Electrons and chemical bonding*. Benjamin, Menlo Park (1965).

Strength of the Crystal field

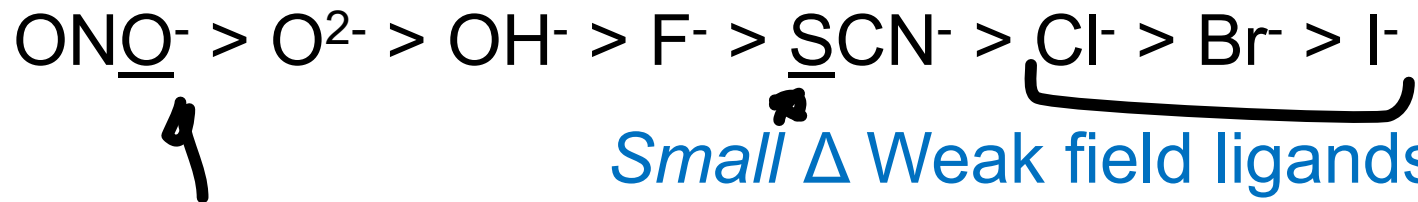
- Nature of the ligand



Large Δ Strong field ligands (low-spin complexes)



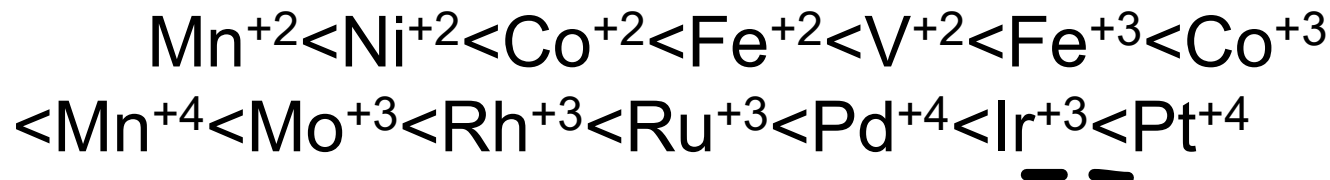
π -acceptor



Small Δ Weak field ligands (high-spin complexes)

π -donor

- Δ_o increases with increasing oxidation state on the metal.



- Δ_o increases with increases going down a group of metals.

The Spectrochemical Series



The complexes of cobalt (III) show the shift in color due to the ligand.

(a) CN^-

(b) NO_2^-

(c) phen

(d) en

(e) NH_3

(f) gly

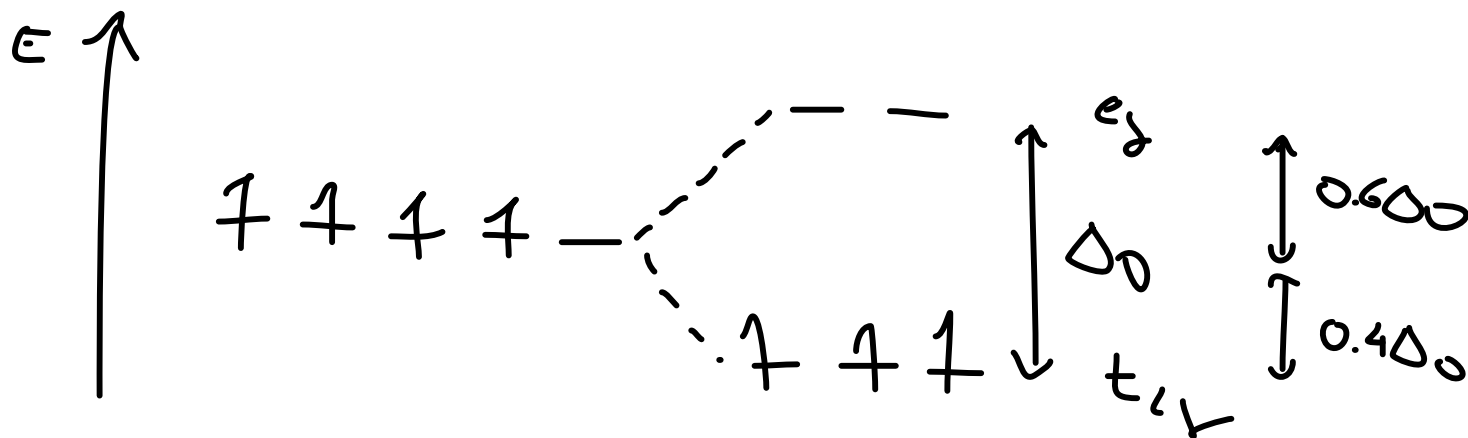
(g) H_2O

(h) ox^{2-}

(i) CO_3^{2-}

Worked exercises

Calculate the crystal field stabilization energy (CFSE) for complexes with a number of d electron from 1 to 10. Consider both the high-spin and low-spin case.

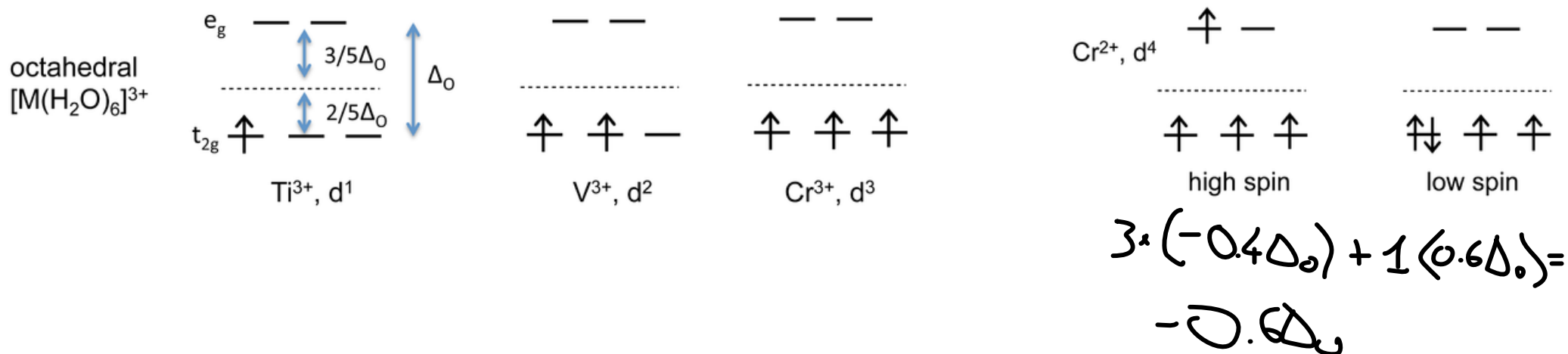


$$3 \times (-0.4\Delta_o) = -1.2\Delta_o$$

Worked exercises

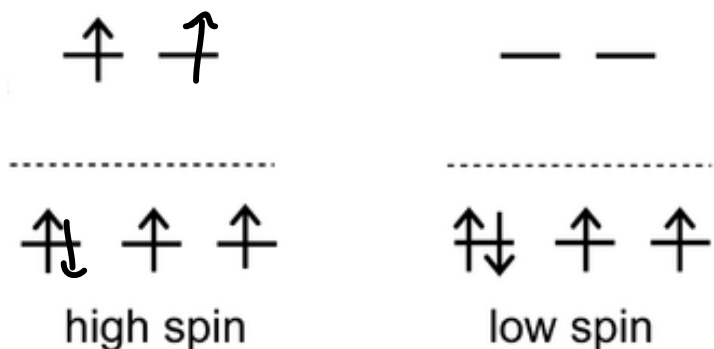
Calculate the crystal field stabilization energy (CFSE) for complexes with a number of *d* electron from 1 to 10. Consider both the high-spin and low-spin case.

	High-spin	Low-spin
d^1	$-0.4 \Delta_0$	$-0.4 \Delta_0$
d^2	$-0.8 \Delta_0$	$-0.8 \Delta_0$
d^3	$-1.2 \Delta_0$	$-1.2 \Delta_0$
d^4	$-0.6 \Delta_0$	$-1.6 \Delta_0 + P$



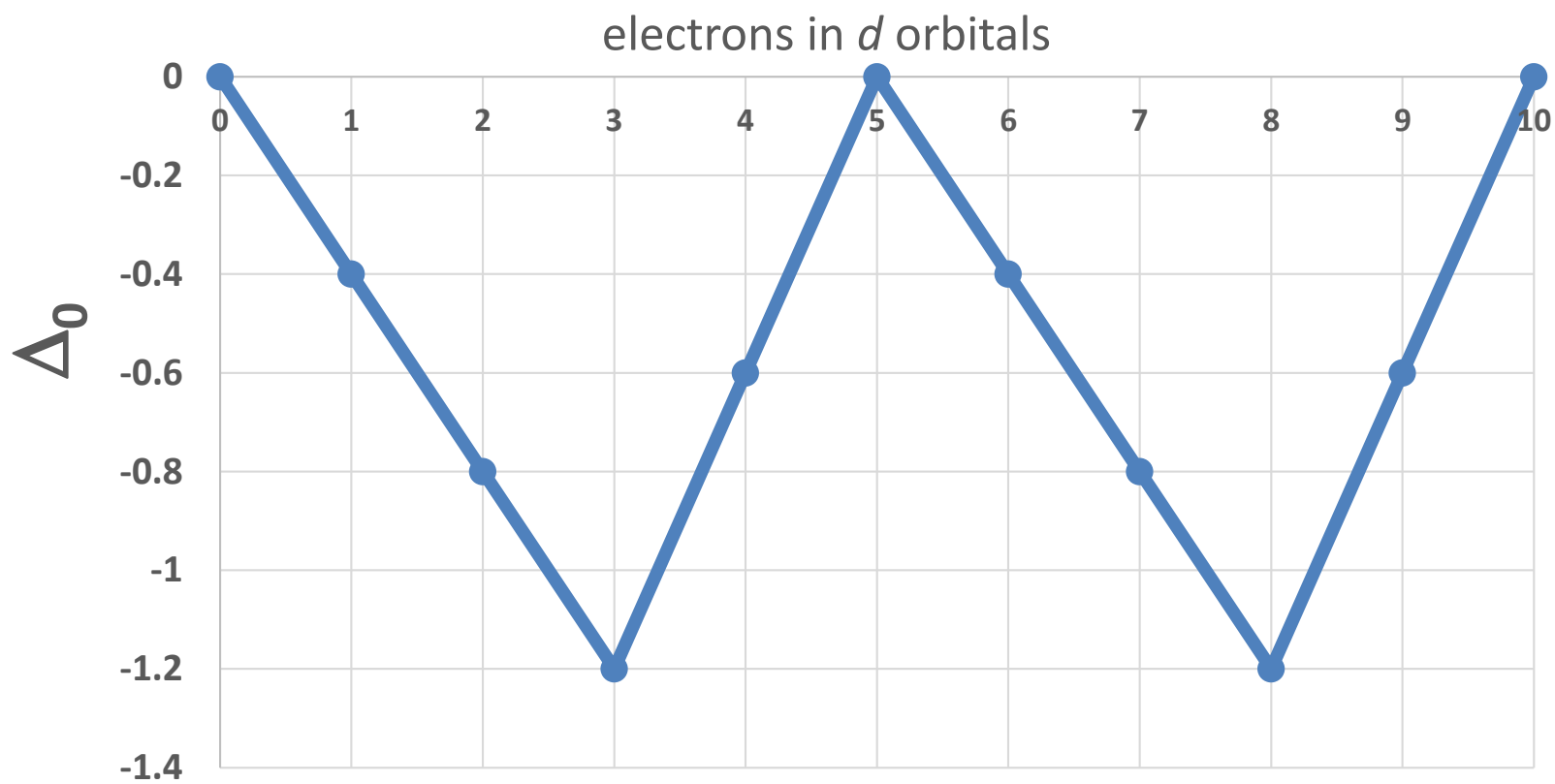
Worked exercises

Calculate the crystal field stabilization energy (CFSE) for complexes with a number of *d* electron from 1 to 10. Consider both the high-spin and low-spin case.



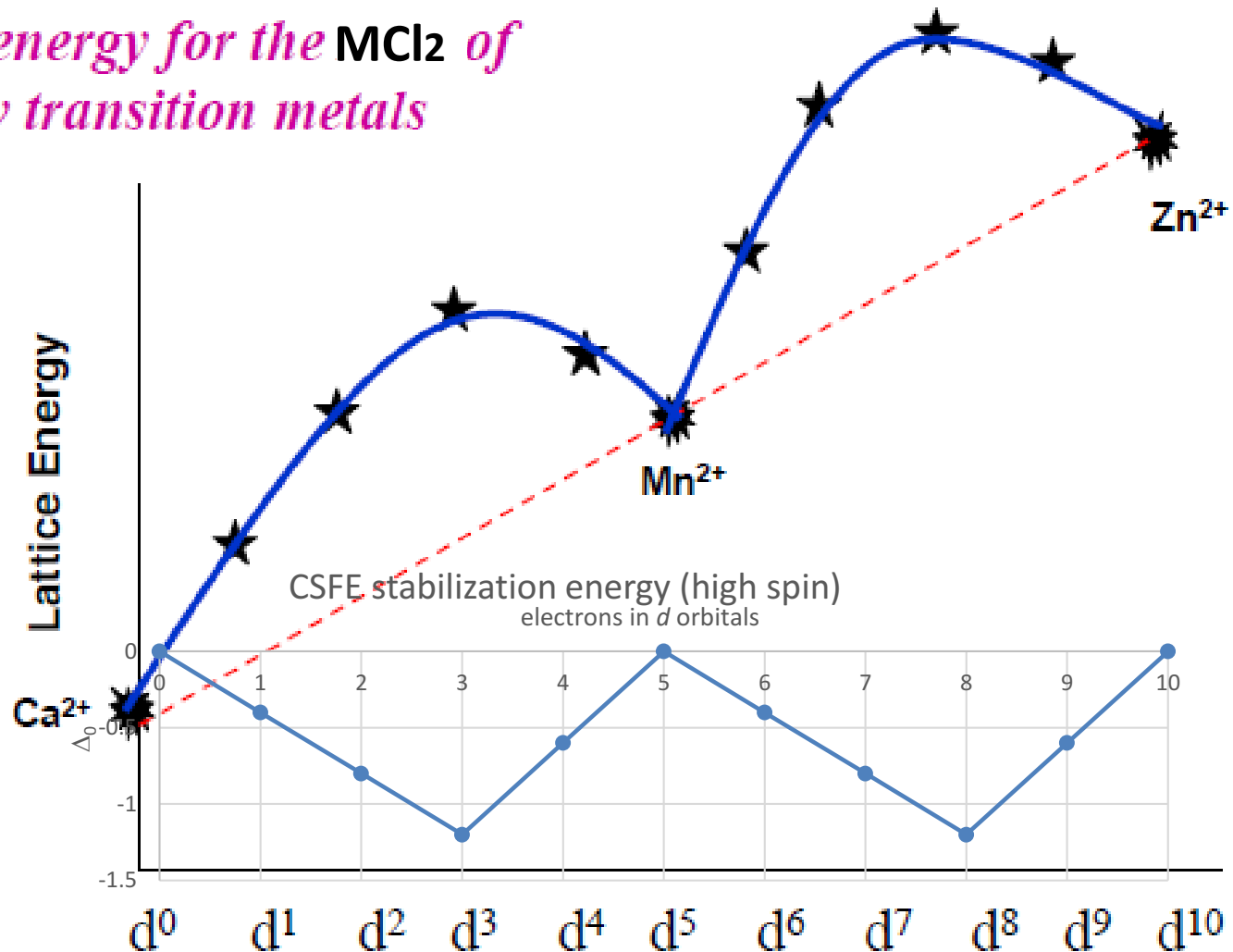
	High-spin	Low-spin
d^1	$-0.4 \Delta_0$	$-0.4 \Delta_0$
d^2	$-0.8 \Delta_0$	$-0.8 \Delta_0$
d^3	$-1.2 \Delta_0$	$-1.2 \Delta_0$
d^4	$-0.6 \Delta_0$	$-1.6 \Delta_0 + P$
d^5	$-0 \Delta_0$	$-2.0 \Delta_0 + 2P$
d^6	$-0.4 \Delta_0$	$-2.4 \Delta_0 + 2P$
d^7	$-0.8 \Delta_0$	$-1.8 \Delta_0 + P$
d^8	$-1.2 \Delta_0$	$-1.2 \Delta_0$
d^9	$-0.6 \Delta_0$	$-0.6 \Delta_0$
d^{10}	$-0 \Delta_0$	$-0 \Delta_0$

CSFE stabilization energy (high spin)



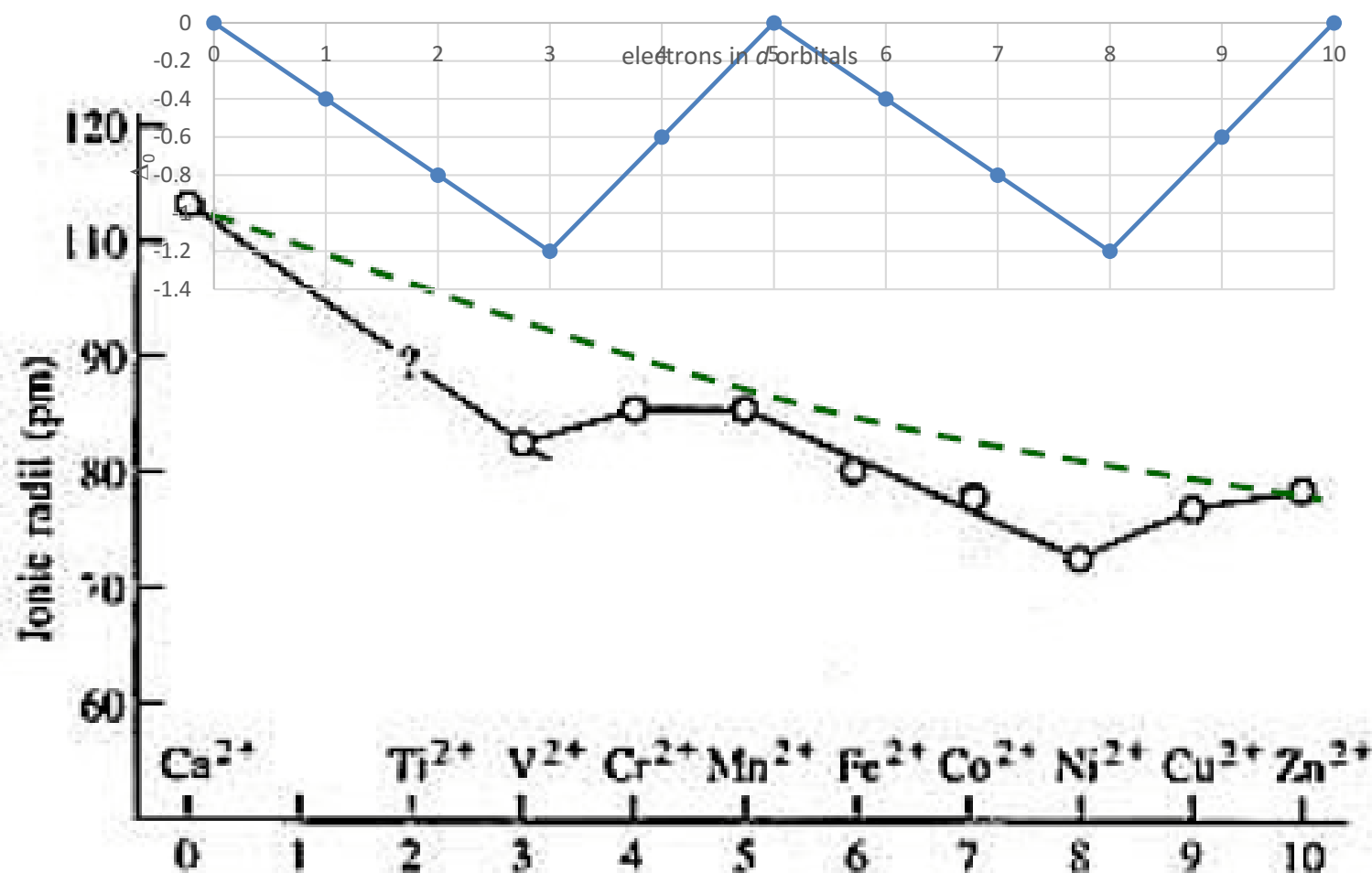
Applications of CFT

Lattice energy for the MCl_2 of first row transition metals

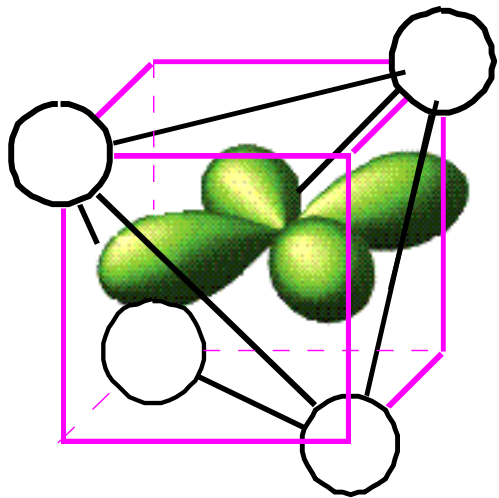


Applications of CFT

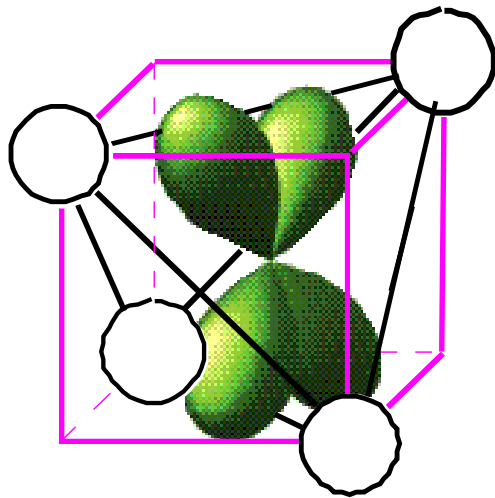
Ionic Radii. For a given oxidation state, the ionic radius decreases steadily on going from left to right in a transition series (dotted line).



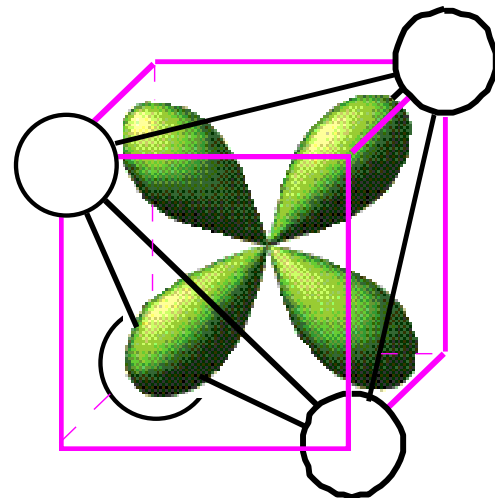
Tetrahedral Complexes



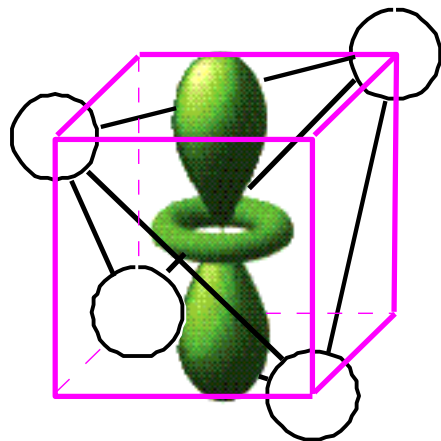
d_{xy}



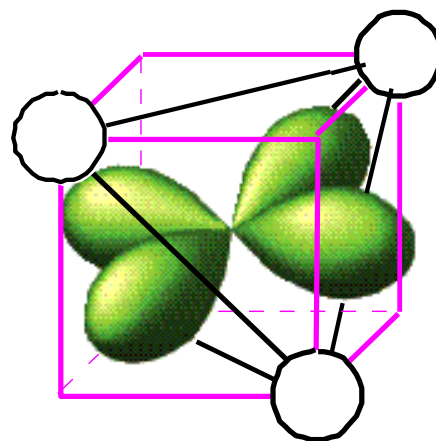
d_{xz}



d_{yz}

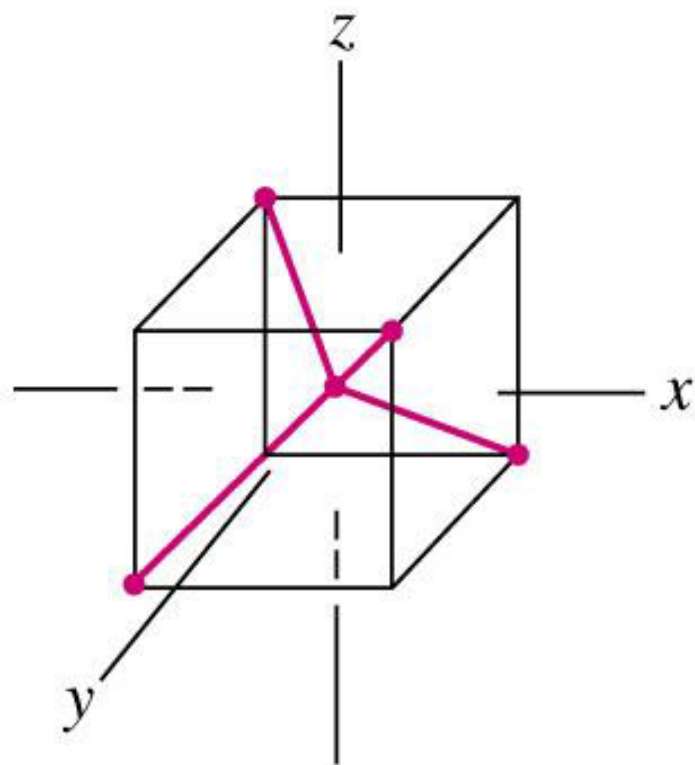


d_{z^2}

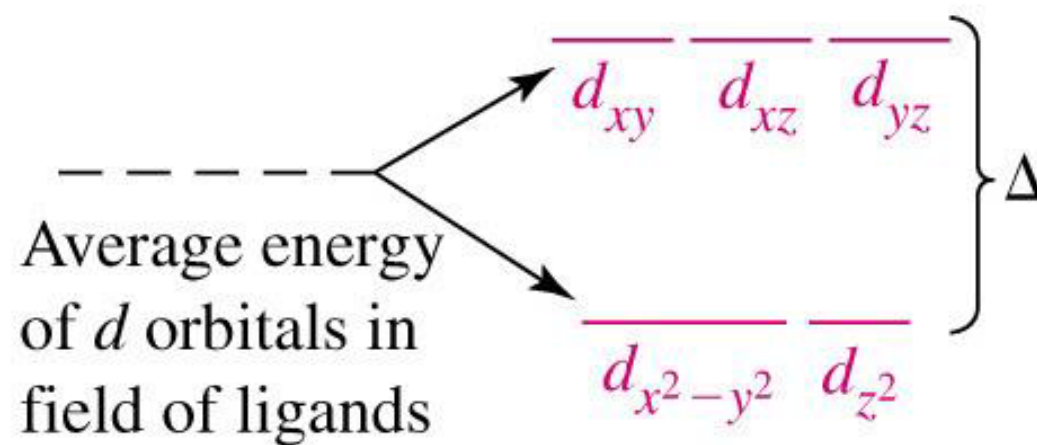


$d_{x^2-y^2}$

Tetrahedral Crystal Field



(a)



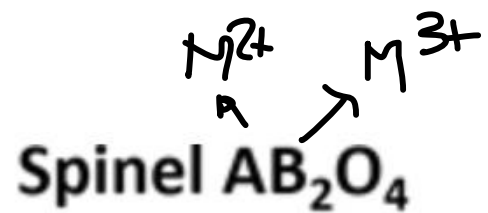
(b)

$$\Delta_t = 4/9 \Delta_o$$

There are only 4 ligands in the tetrahedral complex, and hence the ligand field is roughly 2/3 of the octahedral field.

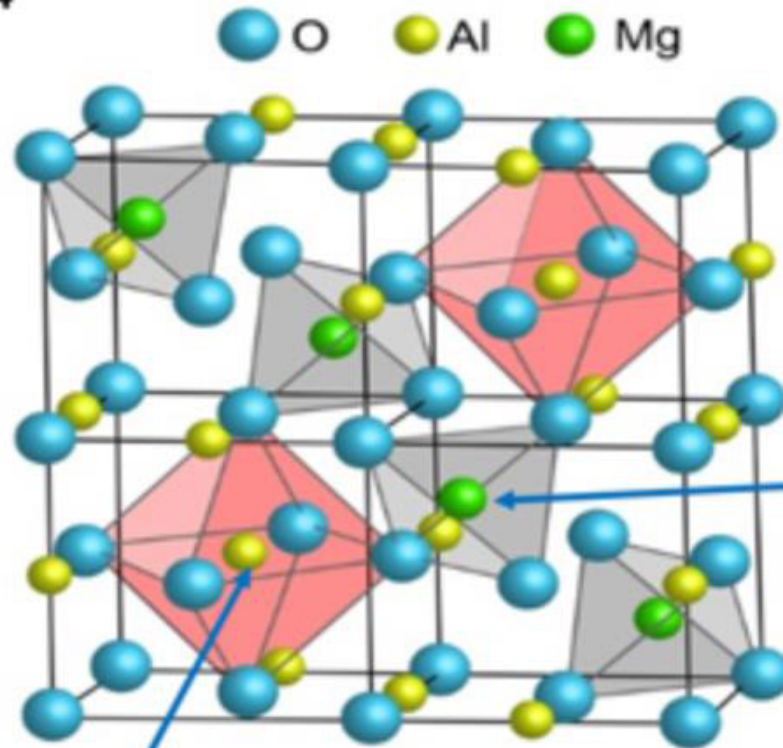
The direction of ligand approach in tetrahedral complex does not coincide with the d-orbitals. This reduces the field by a factor of 2/3. Therefore Δ_t is roughly $2/3 \times 2/3 = 4/9$ of Δ_o .

Do you expect that tetrahedral compounds are in the high-spin or low spin-state?



Normal $A_t B_o$

Inverse $M^{2+} \rightarrow o$
 $\frac{1}{2} M^{3+} \rightarrow t$



A Site – one metal with four nearest-neighbor oxygens.
Tetrahedral site

B site – one metal with six nearest-neighbor oxygens.
Octahedral site

Lecture 8

Coordination compounds (part B)

Chapter 19

Content:

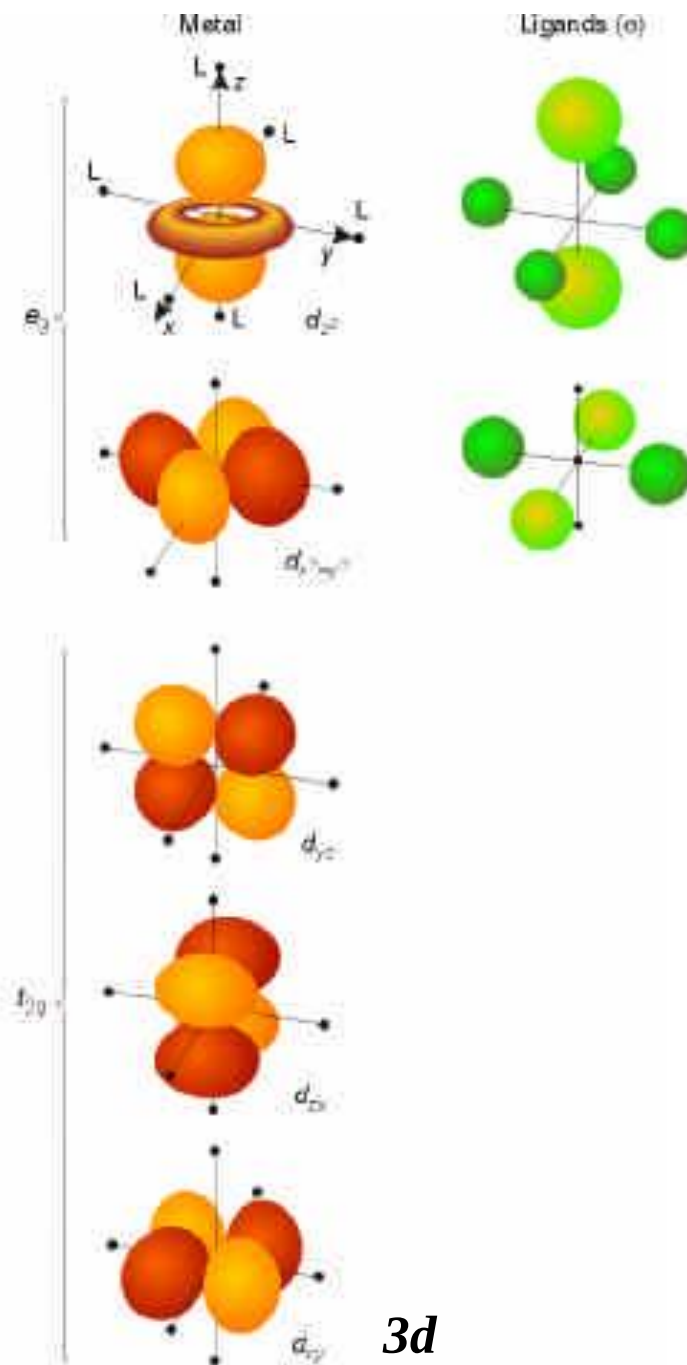
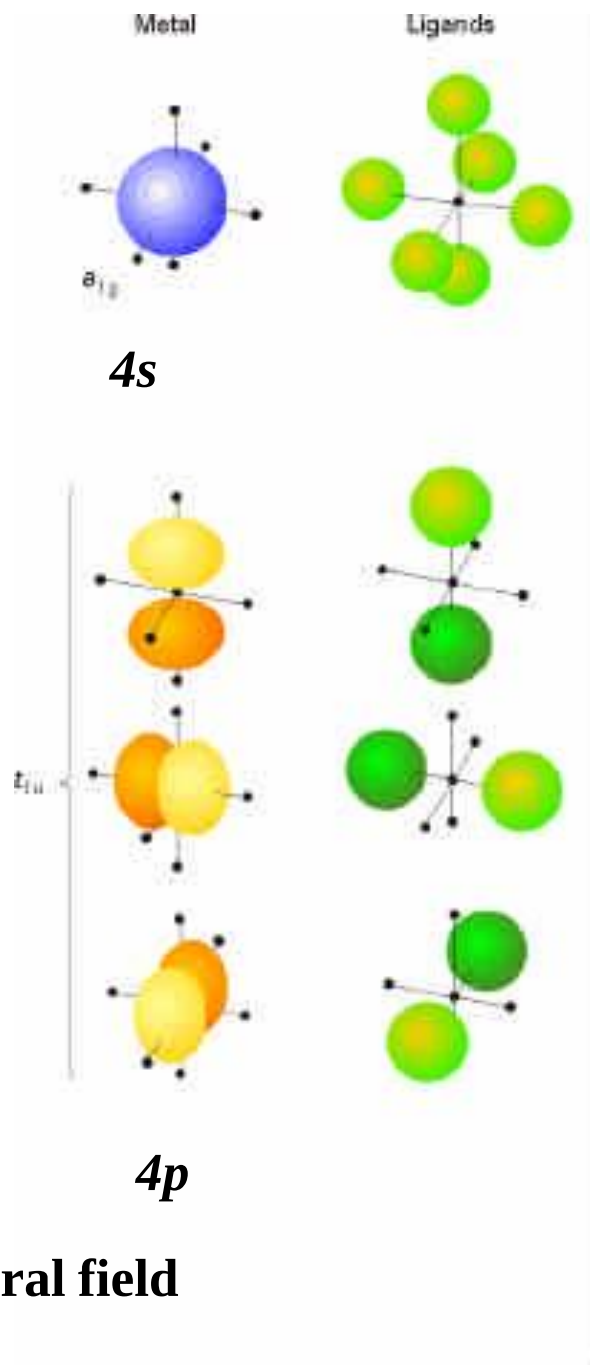
- **Ligand field theory**
 - **Absorption spectra**
 - **Isomerism**
 - **Reactivity**
- 

Crystal field theory:

treats the ligands as point charges and ignores overlap between orbitals on the ligands with orbitals on the metal

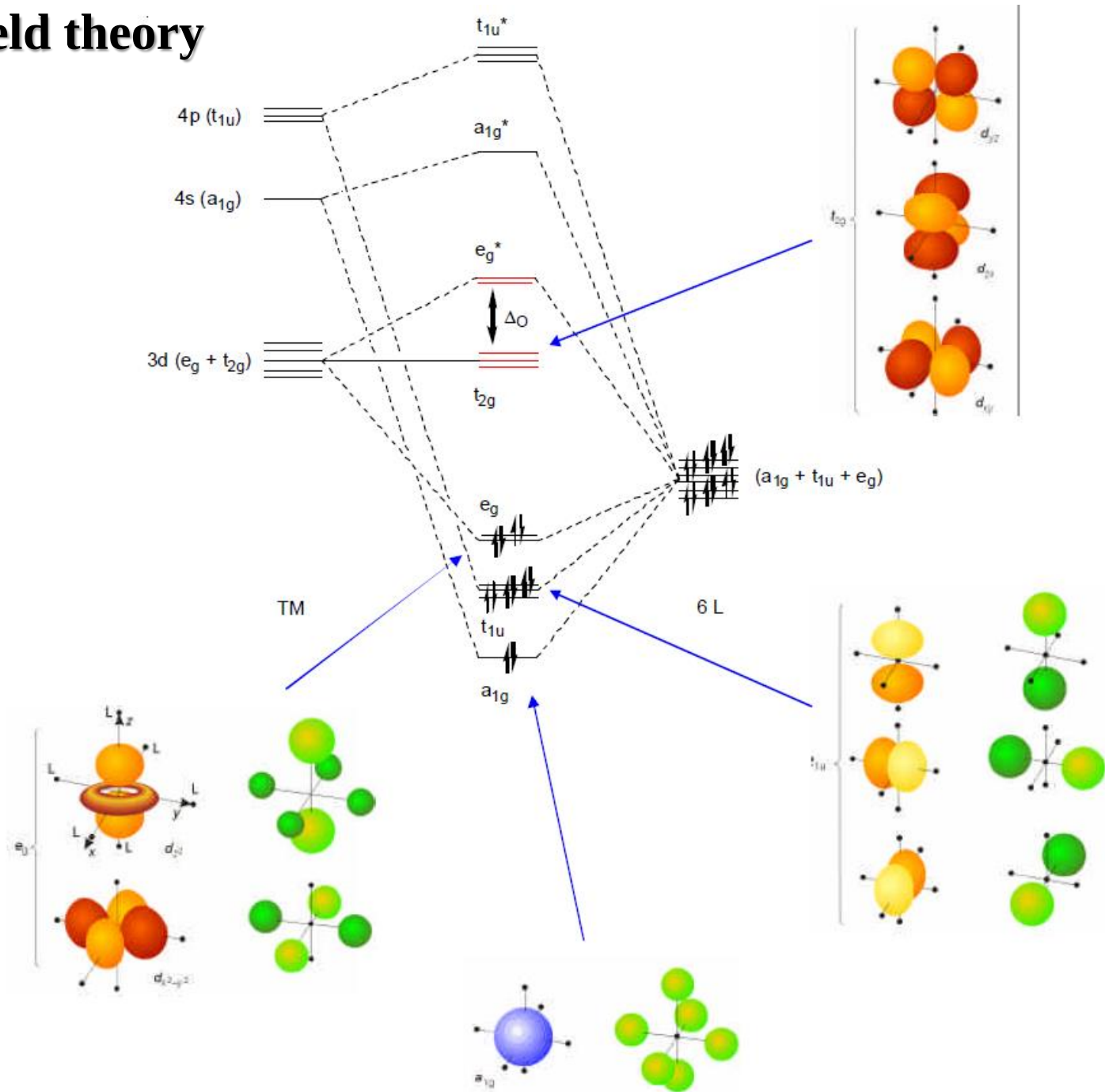
Ligand field theory:

- Metal orbitals are divided according to the symmetry;
- The orbitals of the ligands are combined to give linear combinations, which are of the correct symmetry to interact with the s, p and some of the d orbitals of the central metal;
- This is a much more correct approach than crystal field theory, but is not as easy to use.



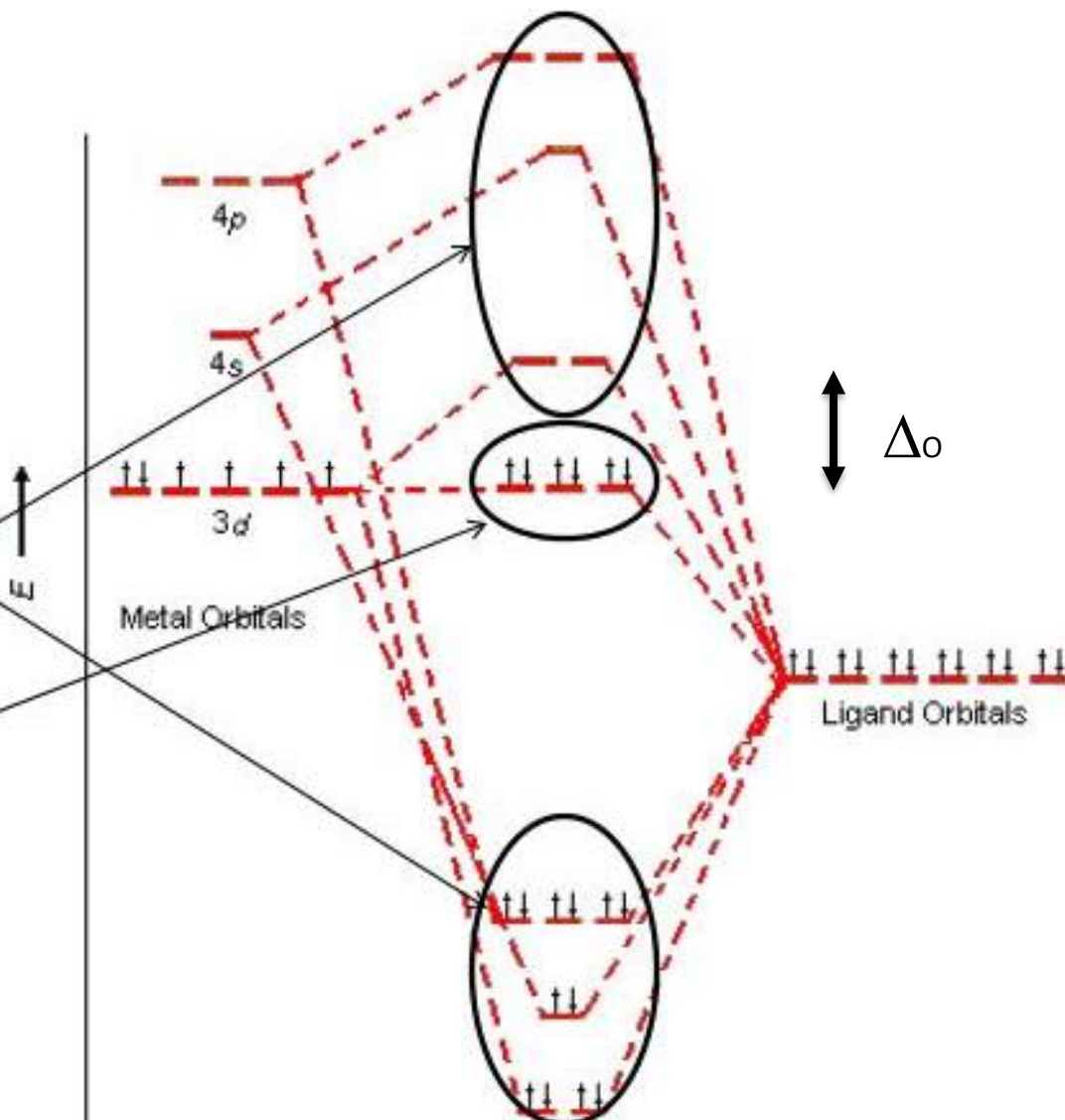
Octahedral field

Ligand field theory

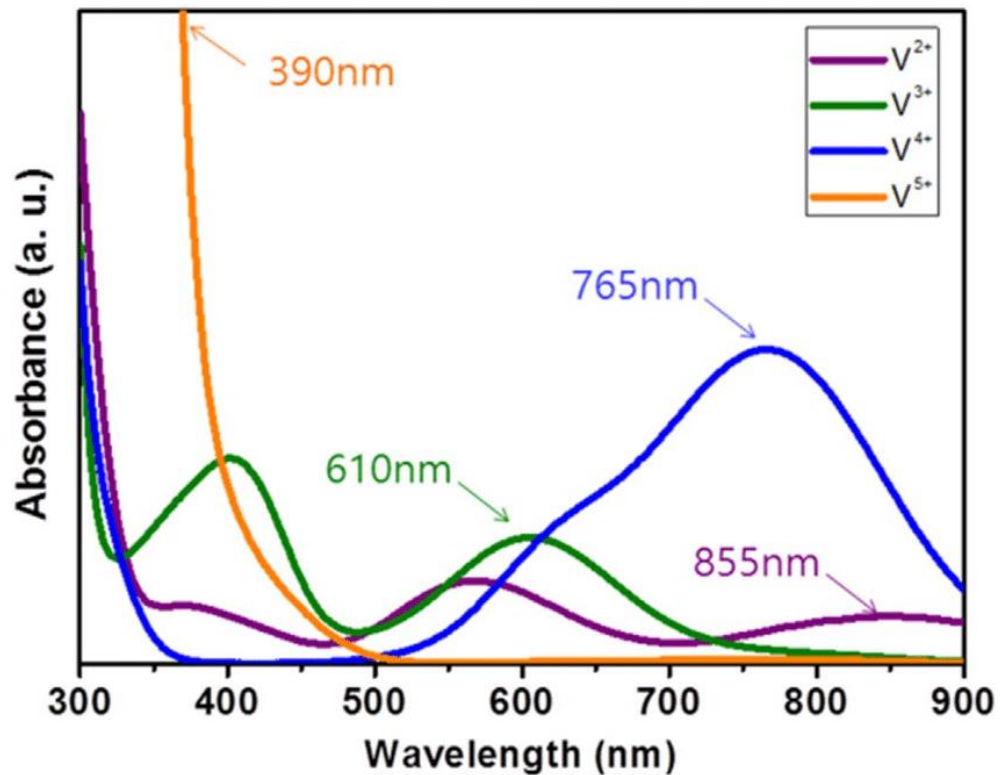
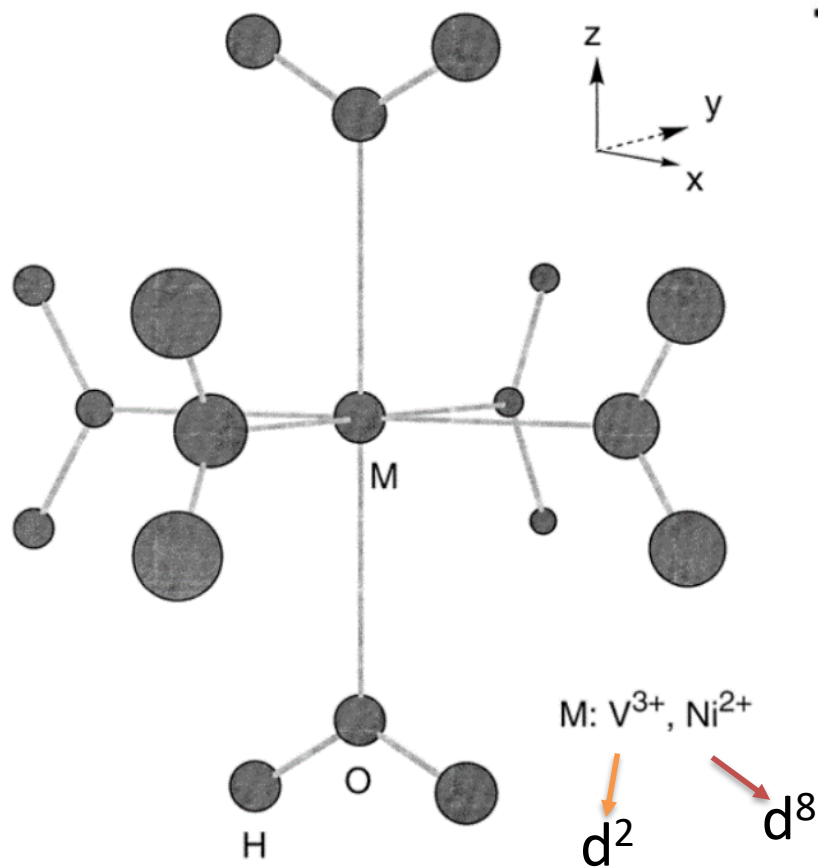


Ligand field theory

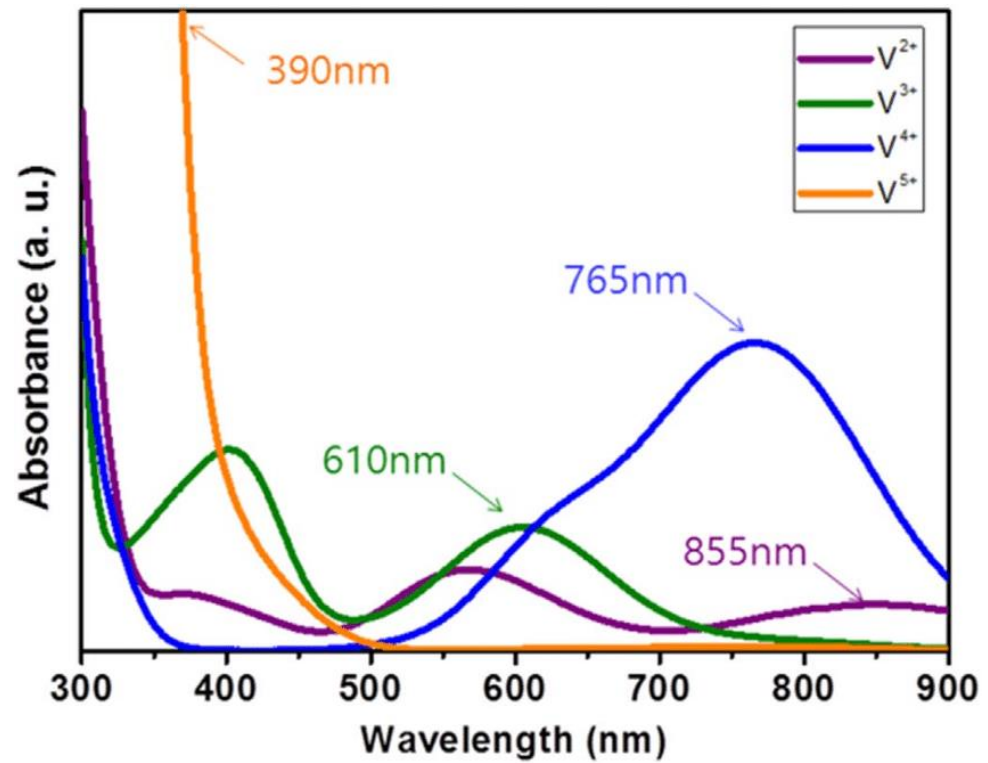
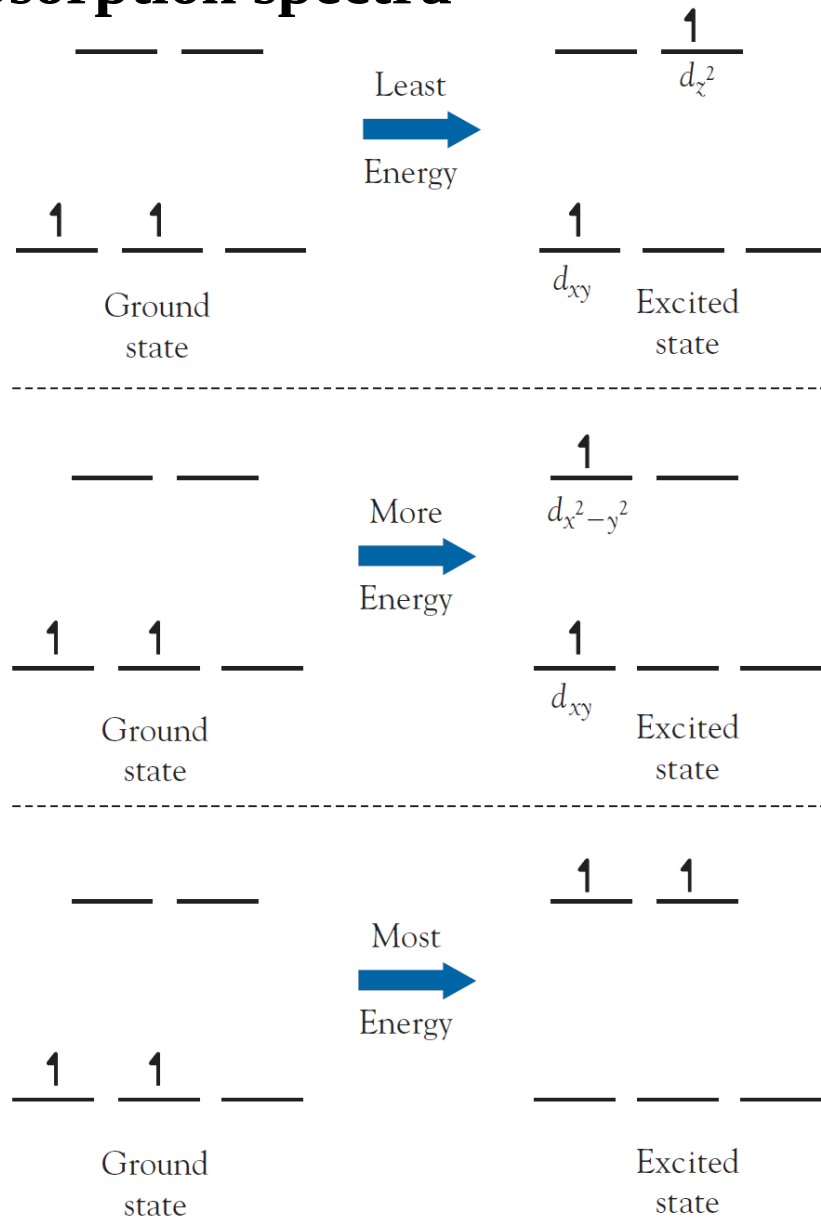
- nd , $n+1s$, and $n+1p$ orbitals on the metal overlap with one orbital on each of the six ligands
 - forms 15 molecular orbitals
- Six are bonding
 - energies are lower than original atomic orbitals
- Six are antibonding with higher energy
- Three are nonbonding
- Ligand-field theory describes how s , p , and d orbitals on the metal to overlap with orbitals on the ligand



Complexity of the absorption spectra



Complexity of the absorption spectra



Intensity of the absorption spectra

Transition type	Example	Comparative intensity
Spin forbidden Laporte forbidden	$[\text{Mn}(\text{OH}_2)_6]^{2+}$	1
Spin allowed (center of symmetry) Laporte forbidden	$[\text{Co}(\text{OH}_2)_6]^{2+}$	10^1
Spin allowed (no center of symmetry) Laporte allowed	$[\text{CoCl}_4]^{2-}$	5×10^2
Spin allowed Laporte allowed (charge transfer)	$[\text{MnO}_4]^-$	10^4

Spin Rule: transitions must involve the promotion of electrons without a change in their spin

Laporte rule: transitions between states of the same symmetry with respect to the inversion center are forbidden.

Isomers
Compounds with same formula
but different arrangement of atoms

Constitutional isomers
Compounds with different
connections between atoms

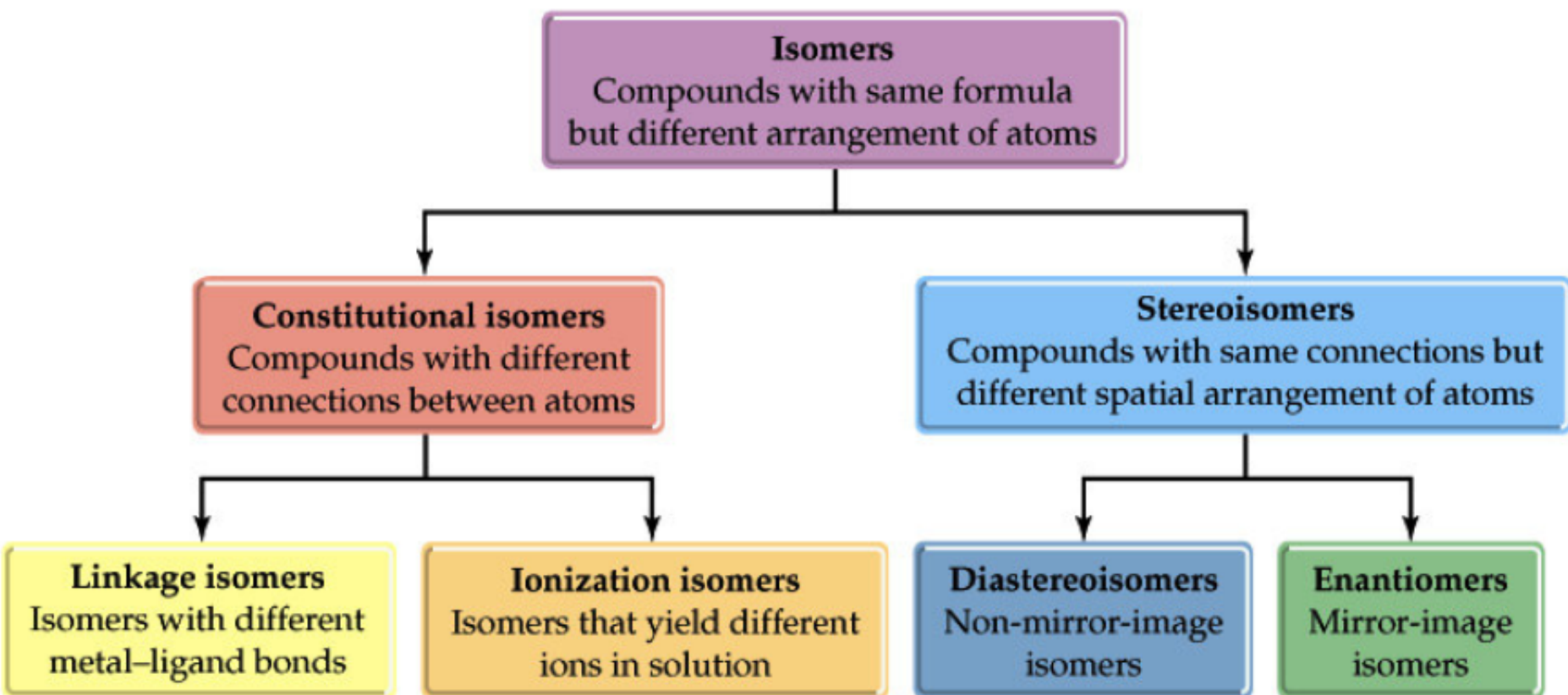
Stereoisomers
Compounds with same connections but
different spatial arrangement of atoms

Linkage isomers
Isomers with different
metal–ligand bonds

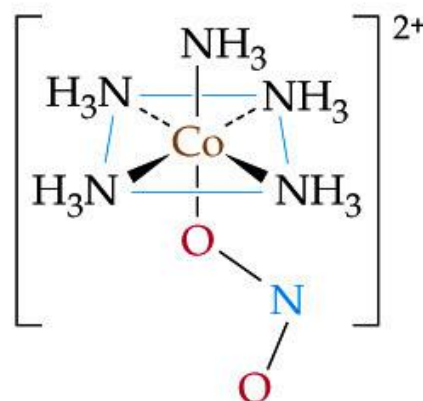
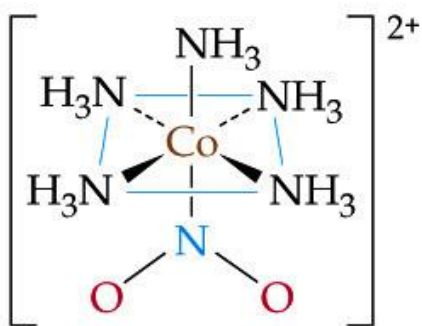
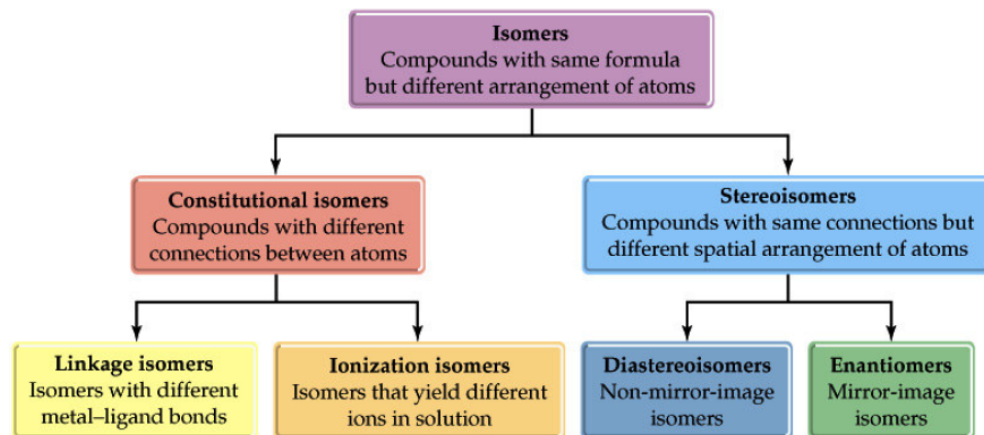
Ionization isomers
Isomers that yield different
ions in solution

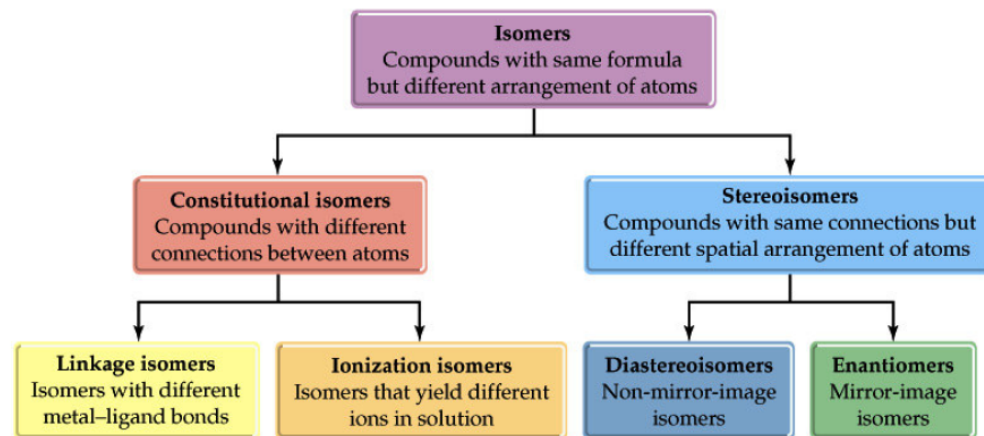
Diastereoisomers
Non-mirror-image
isomers

Enantiomers
Mirror-image
isomers



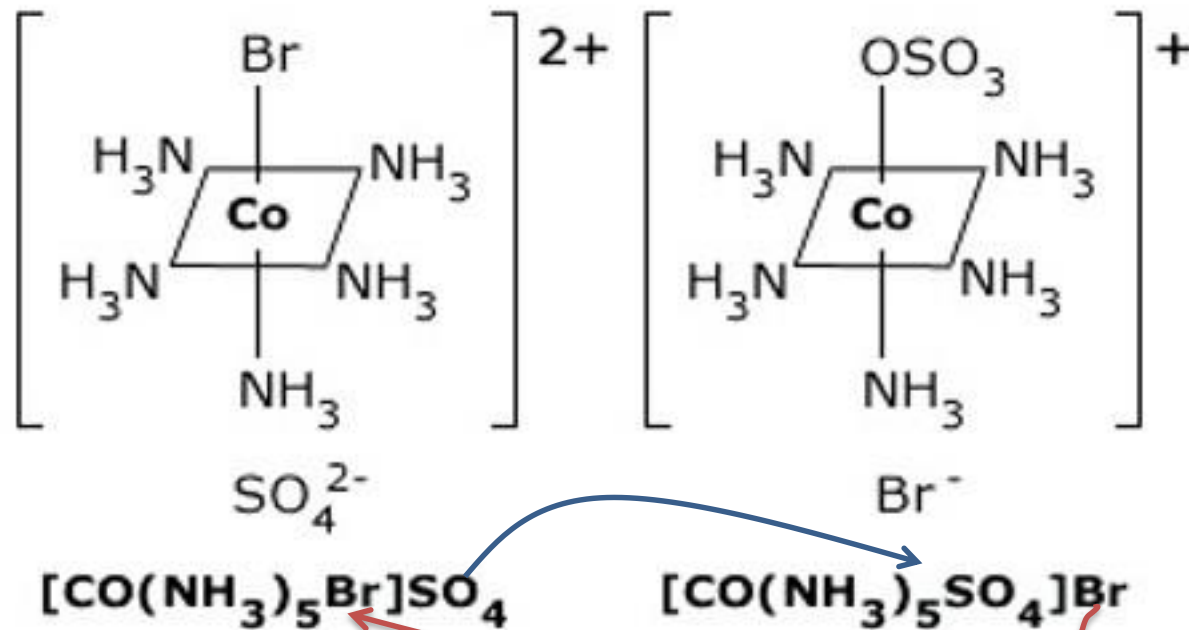
Linkage Isomerism

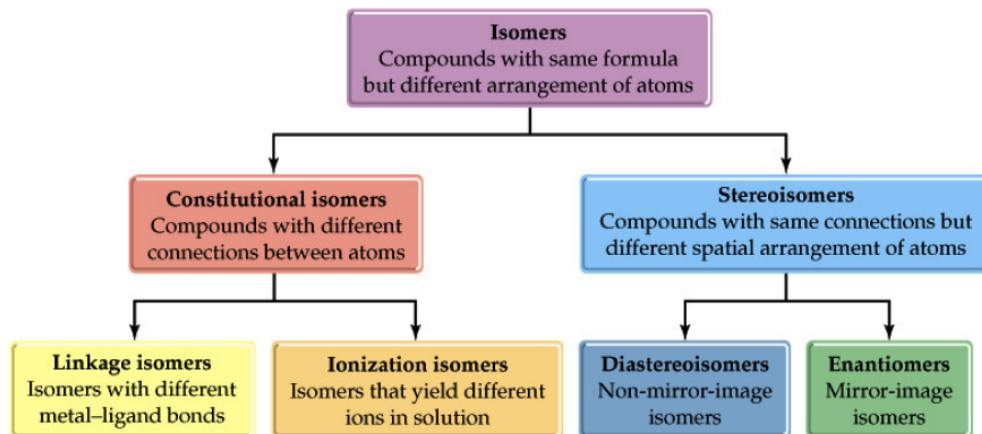




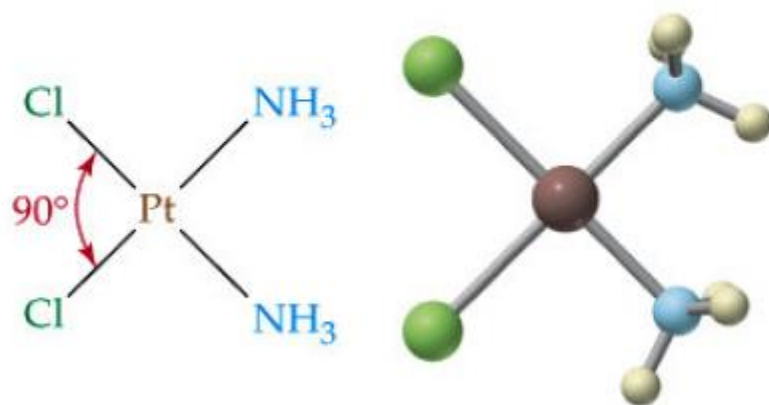
Ionization Isomerism

Compounds that have the same molecular formula but differ in which anions are coordinated yield different ions when in solution. For instance:

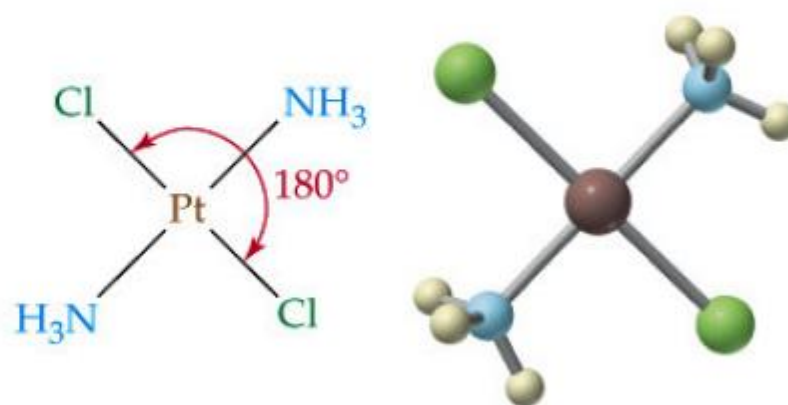




- **Geometric Isomers of $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$:** In the *cis* isomer, atoms are on the *same* side. In the *trans* isomer, atoms are on *opposite* sides.

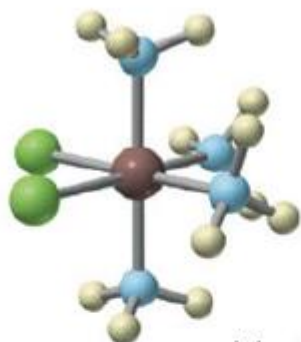


(a) cis

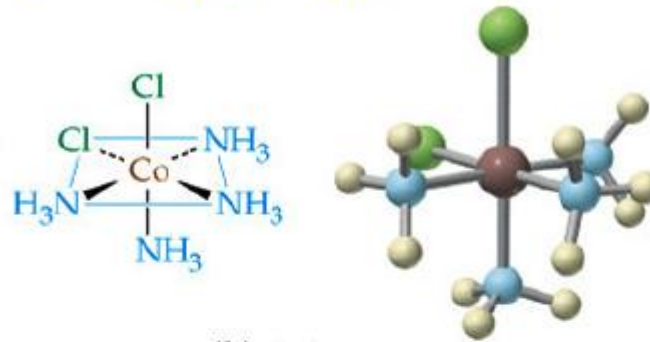
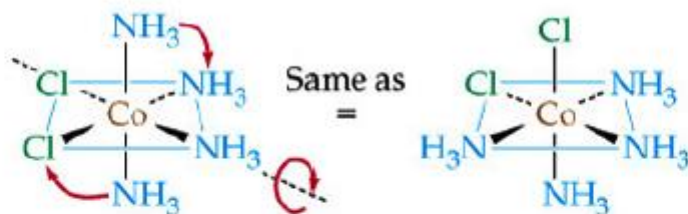


(b) trans

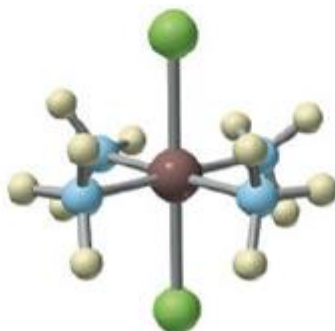
Geometric Isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$:



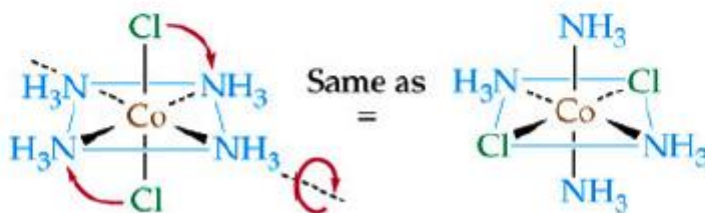
(a) cis isomer



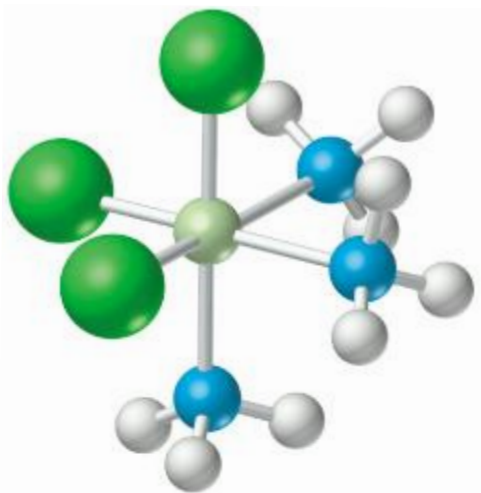
(b) cis isomer



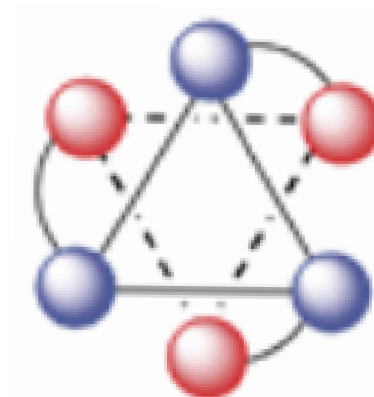
(c) trans isomer



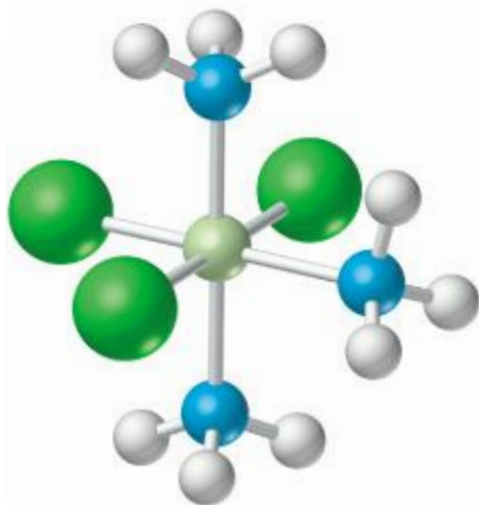
(d) trans isomer



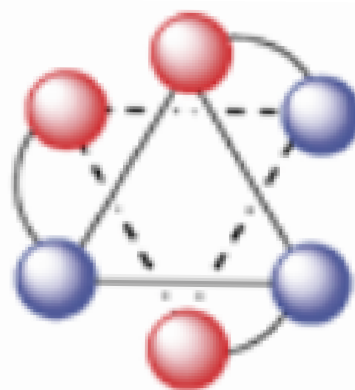
fac-[CoCl₃(NH₃)₃]



fac



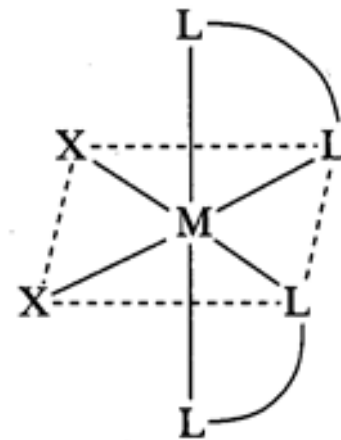
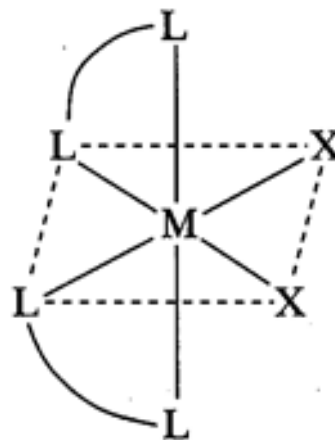
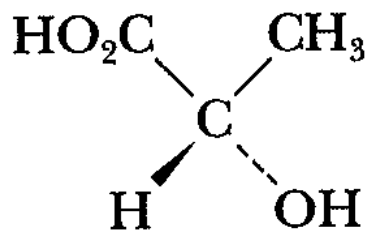
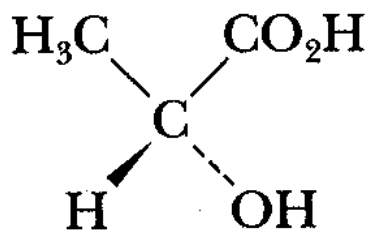
mer-[CoCl₃(NH₃)₃]



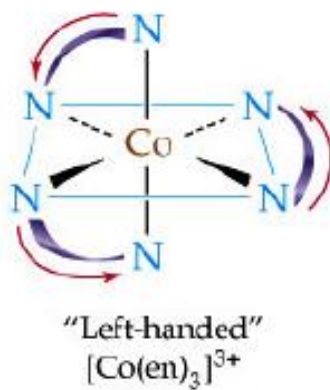
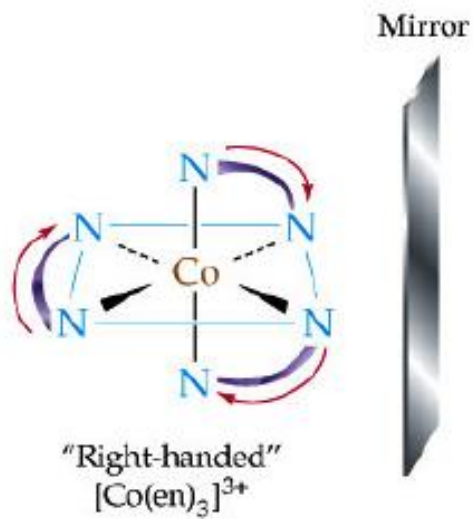
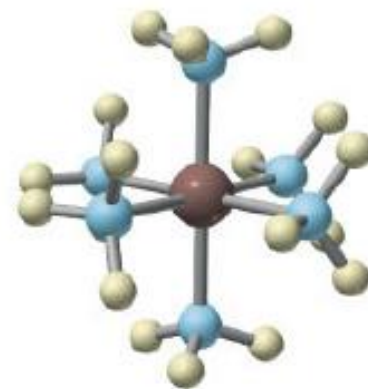
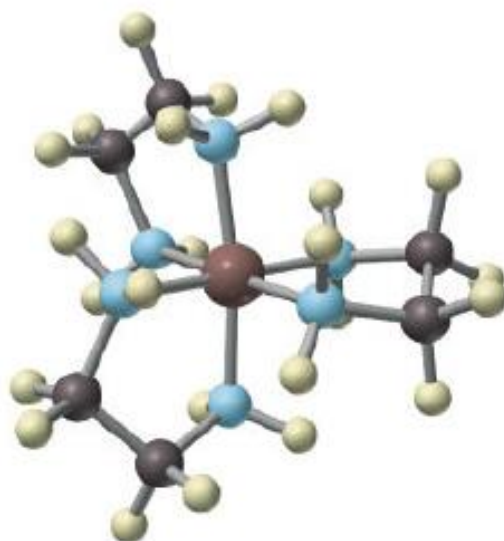
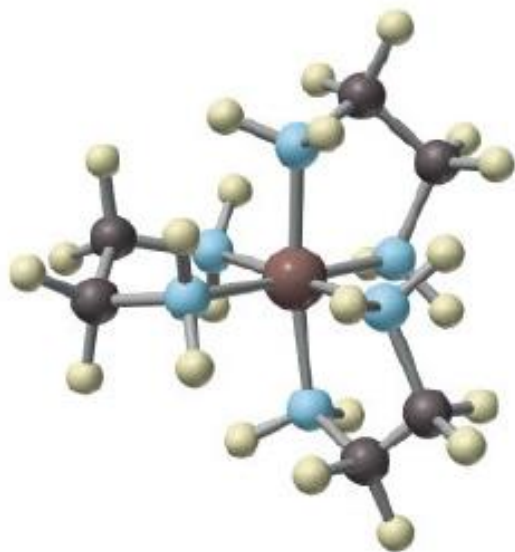
mer

3) optical isomers

molecules that are mirror images of each other that cannot be superimposed. They are also called enantiomorphs:

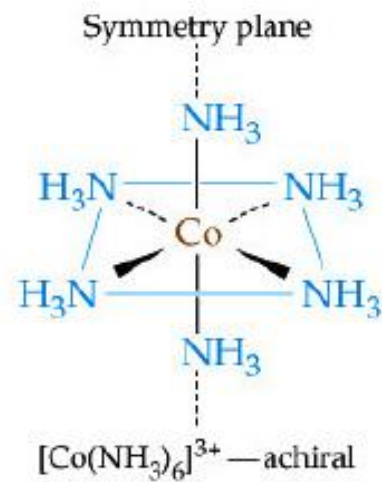


Trischelate:



Δ

Λ

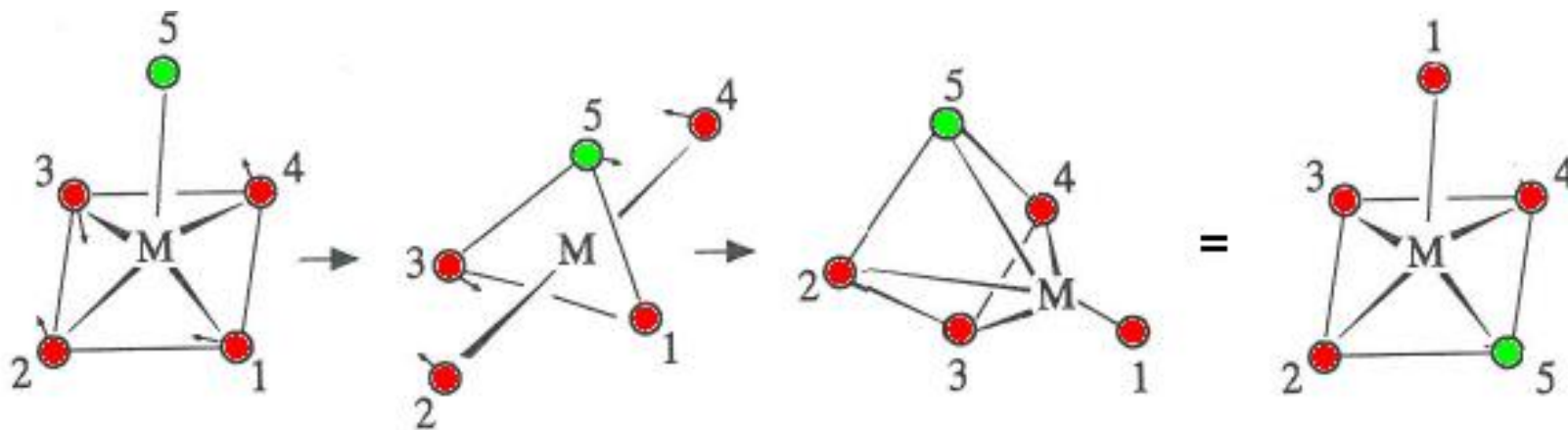


Stereochemical Nonrigidity

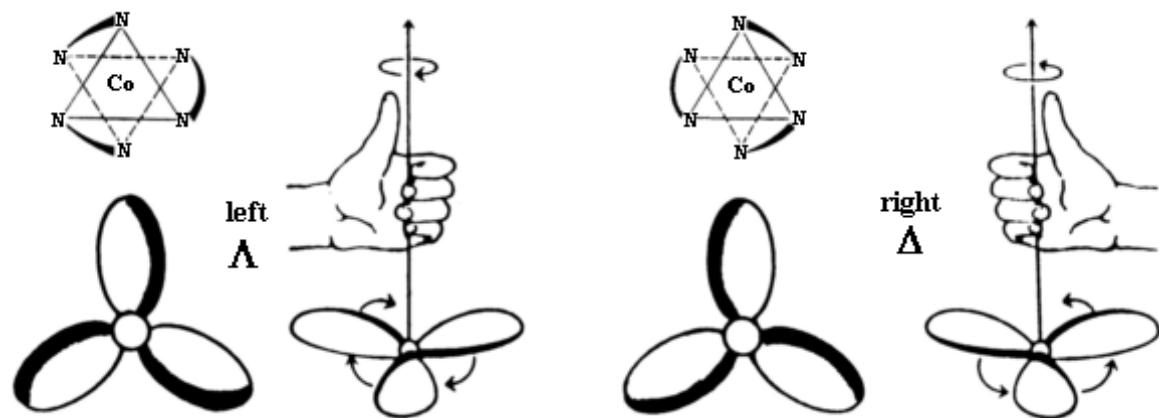
Basic idea: coordination compounds are not rigid. They undergo rapid deformational rearrangements of a much greater amplitude, in which atoms actually change places with each other, e.g. in PF_5 .

Five-Coordinate Complex: PF_5

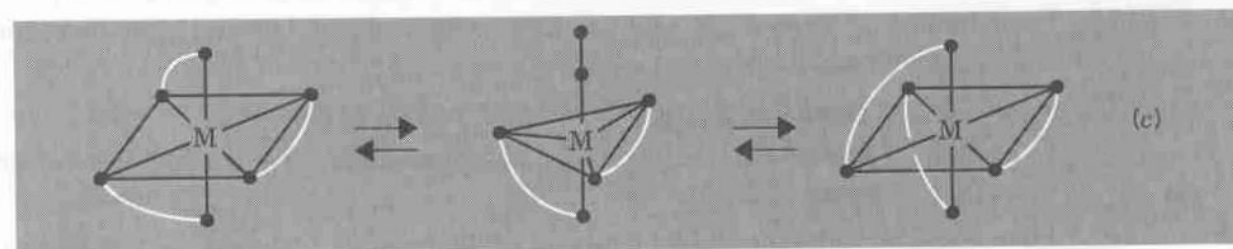
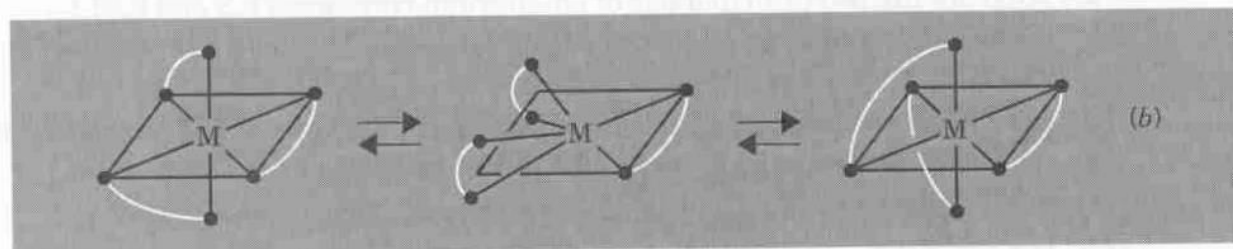
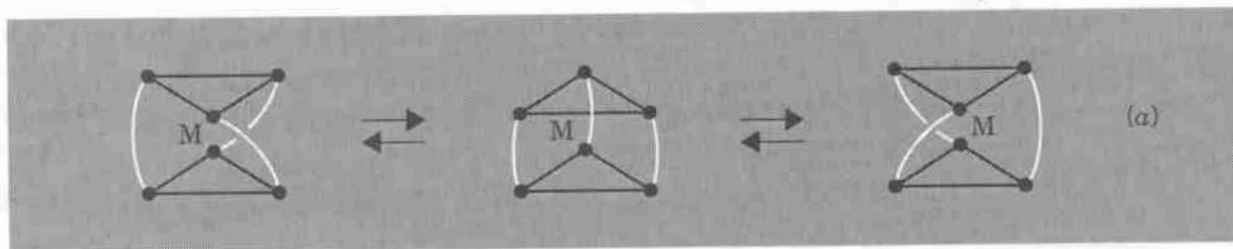
The exchange of axial and equatorial fluorine atoms in PF_5 is shown in the following figure:



Stereochemical Nonrigidity

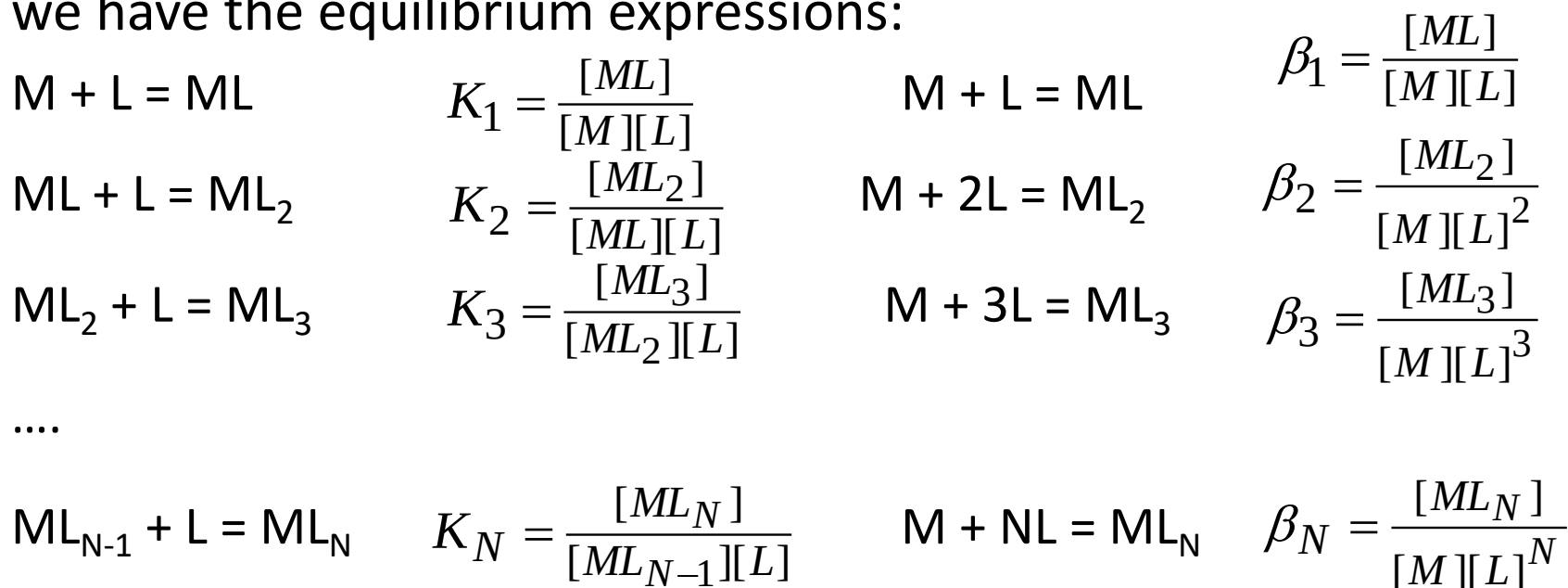


Racemization of tris-chelate Complexes



Equilibrium Constants for Formation of Complexes in solution

Suppose we put a metal M and some monodentate ligand, L, together in solution. Assuming that no insoluble products are formed, nor any species containing more than one metal ion, we have the equilibrium expressions:

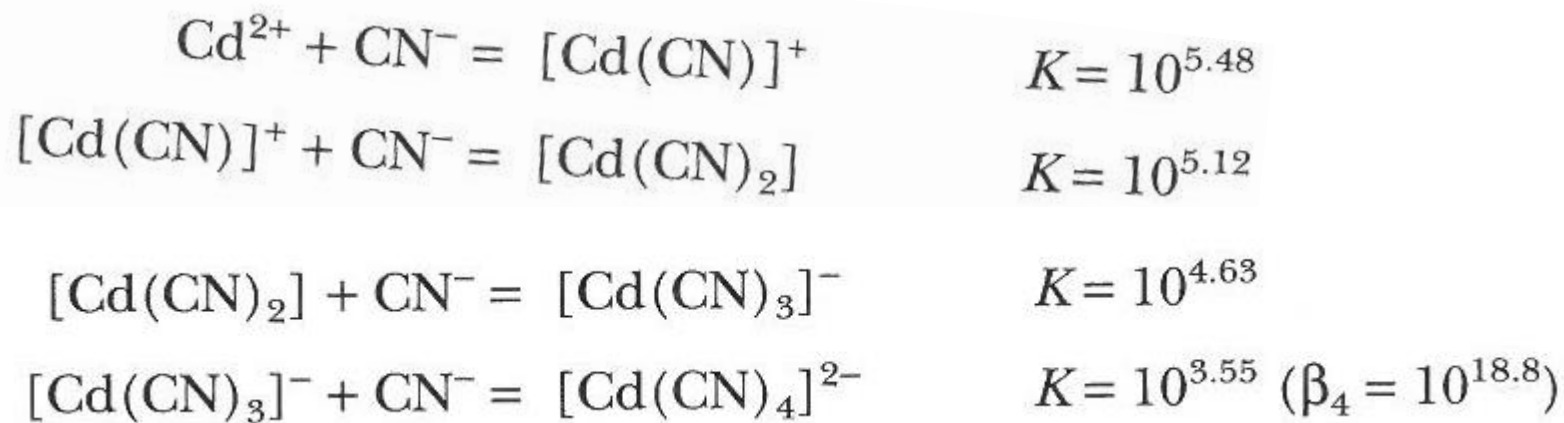
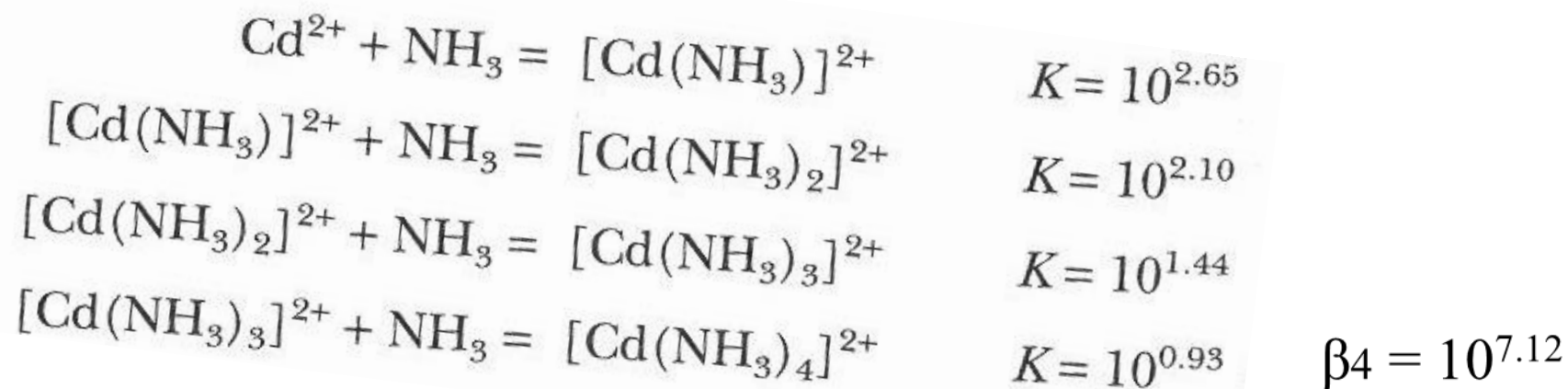


Multiplying both numerator and denominator by $[ML][ML_2]$ gives:

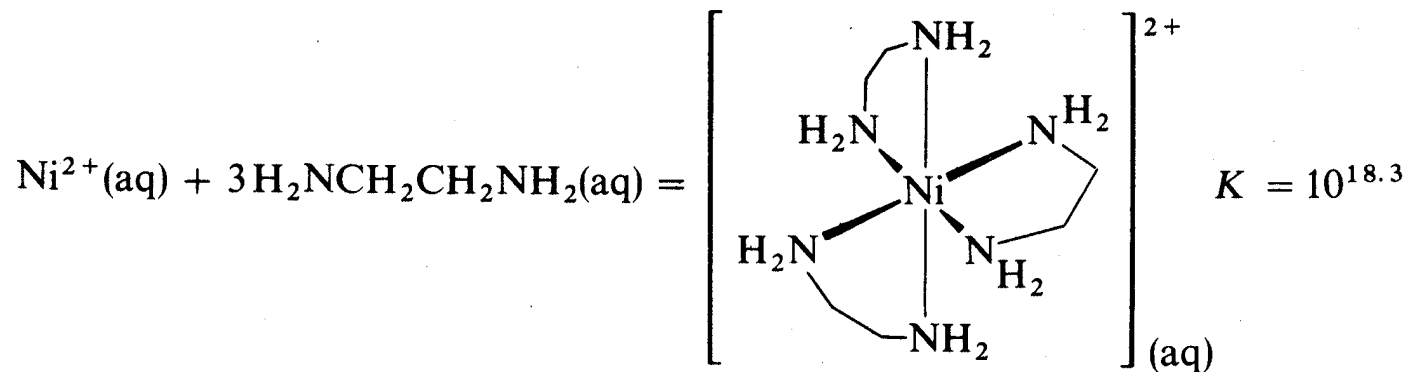
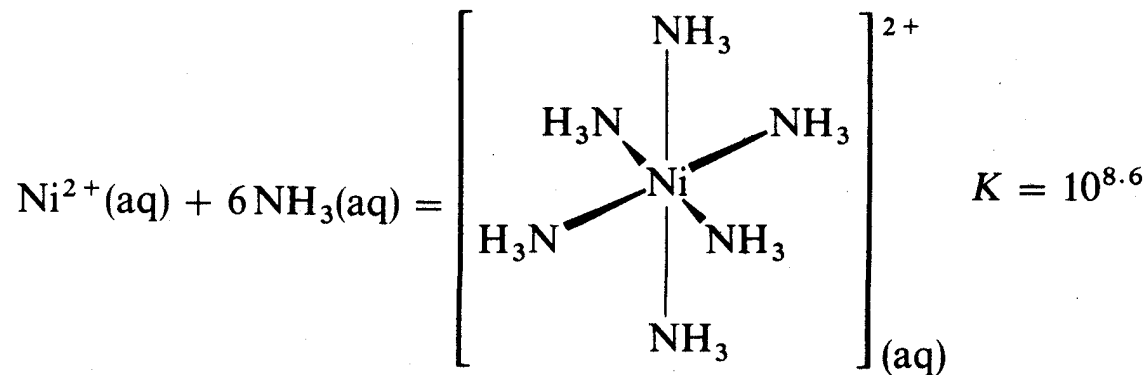
$$\begin{aligned}\beta_3 &= \frac{[ML_3]}{[M][L]^3} \cdot \frac{[ML][ML_2]}{[ML][ML_2]} \\ &= \frac{[ML]}{[M][L]} \cdot \frac{[ML_2]}{[ML][L]} \cdot \frac{[ML_3]}{[ML_2][L]} \\ &= K_1 K_2 K_3\end{aligned}$$

Thus, it is easy to get the general expression:

$$\beta_k = K_1 K_2 K_3 K_4 \dots K_k = \prod_{i=1}^{i=k} K_i$$



The Chelate Effect



Exercise 1

Why the formation constant is higher for *en* than for ammonia?

Reaction of Complexes

This is a big topic.

- 1) Ligand replacement reactions
- 2) Redox reactions
- 3) Partial decomposition reactions

One concept should be known:

lability: the ability of a complex ion to engage in reactions that result in replacing one or more ligands in its coordination sphere by others. Those complexes for which reactions of this type are very rapid are called **labile**, whereas those for which such reactions proceed only slowly or not at all are called **inert**.

It is important to know:

“Inert” and *“labile”* are different from *“stable”* and *“unstable”*.

The former refer to the kinetic aspect; the latter refer to the thermodynamic aspect.

Two examples:

$[\text{Co}(\text{NH}_3)_6]^{3+} + \text{H}_3\text{O}^+ = [\text{Co}(\text{H}_2\text{O})_6]^{3+} + 6\text{NH}_4^+$ kinetically $\beta \sim 10^{25}$
inert but thermodynamically unstable.

$[\text{Ni}(\text{CN})_4]^{2-} = \text{Ni}^{2+} + 4\text{CN}^-$ $\beta \sim 10^{-22}$

kinetically labile but thermodynamically stable.

There is no clear relation between thermodynamic stability and kinetic lability.

A practical definition of the terms labile and inert:

Inert complex: ligand-replacement reactions are slow enough to be studied in “classical” kinetic experiments by measuring changes in optical density, pH, gas evolution

Labile complex: the reaction is so fast that the modern method must be used to follow them. Such as flow systems, rapidly responding electronic recorders, and the like.

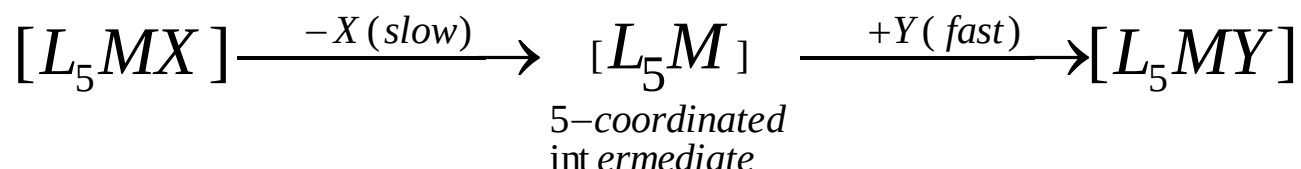
Two extreme mechanistic possibilities for ligand-replacement reactions:

- dissociative (D)
- associative (A)

Dissociative (D) mechanism: (*substitution and unimolecular*)

The complex dissociates, losing the ligand to be replaced, the vacancy in the coordination shell then being taken by the new ligand.

This path may be represented as follows:



The first step determines the complete process.

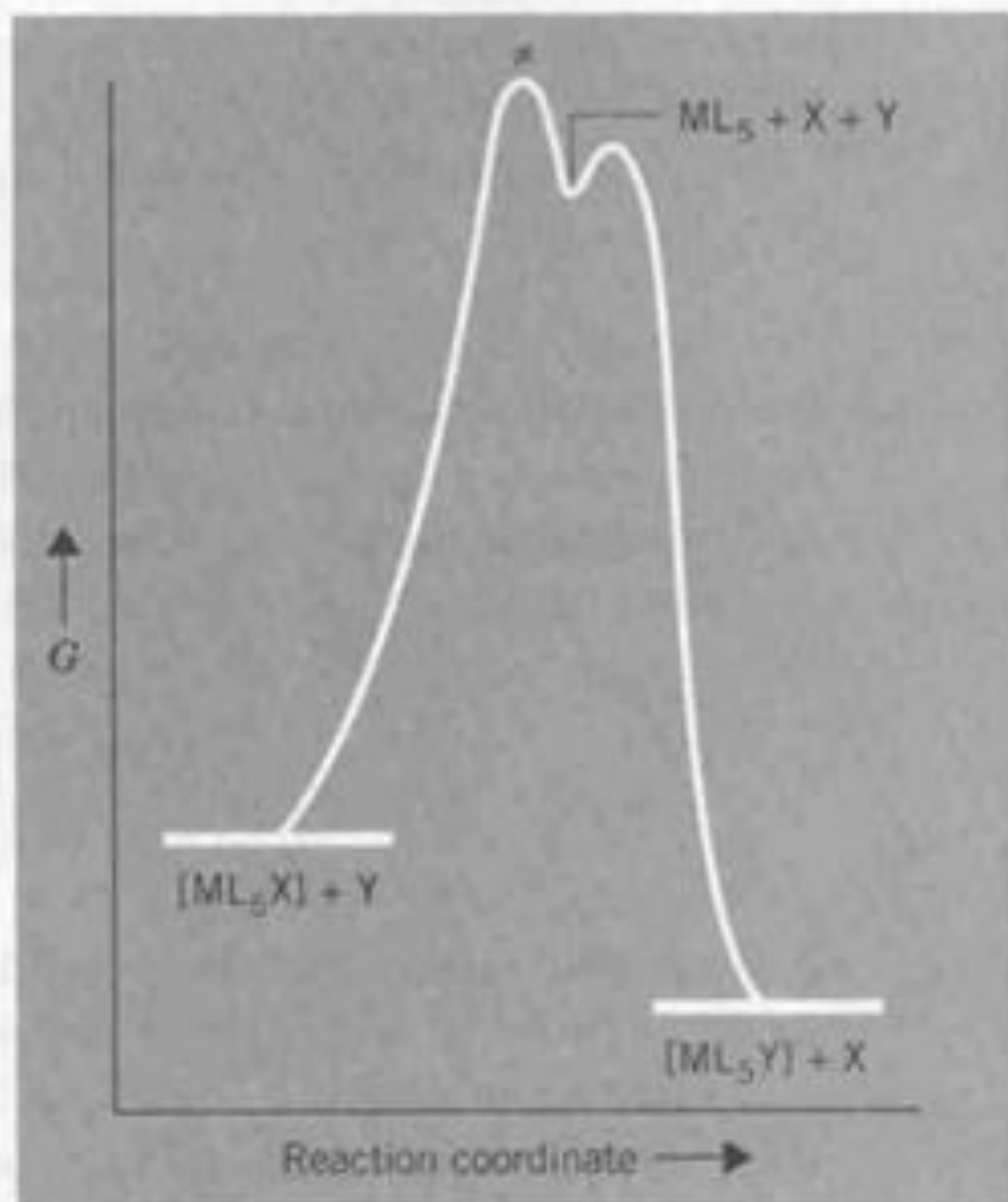
Then the rate law for such a process is

$$v = k[L_5MX]$$

which is proportional to the concentration of $[L_5MX]$

- but is independent of the concentration and the nature of the new ligand, Y.
- Common with high coordination numbers.

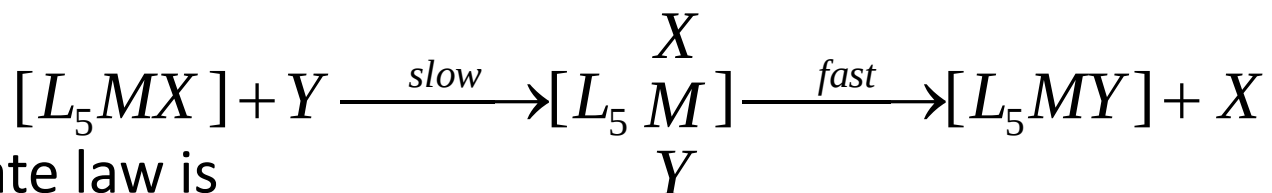
(b)



(d)

Associativ (A) mechanism: (*substitution and bimolecular*)

The new ligand attacks the original complex directly to form a 7-coordinated activated complex which then ejects the displaced ligand, as shown in this scheme:

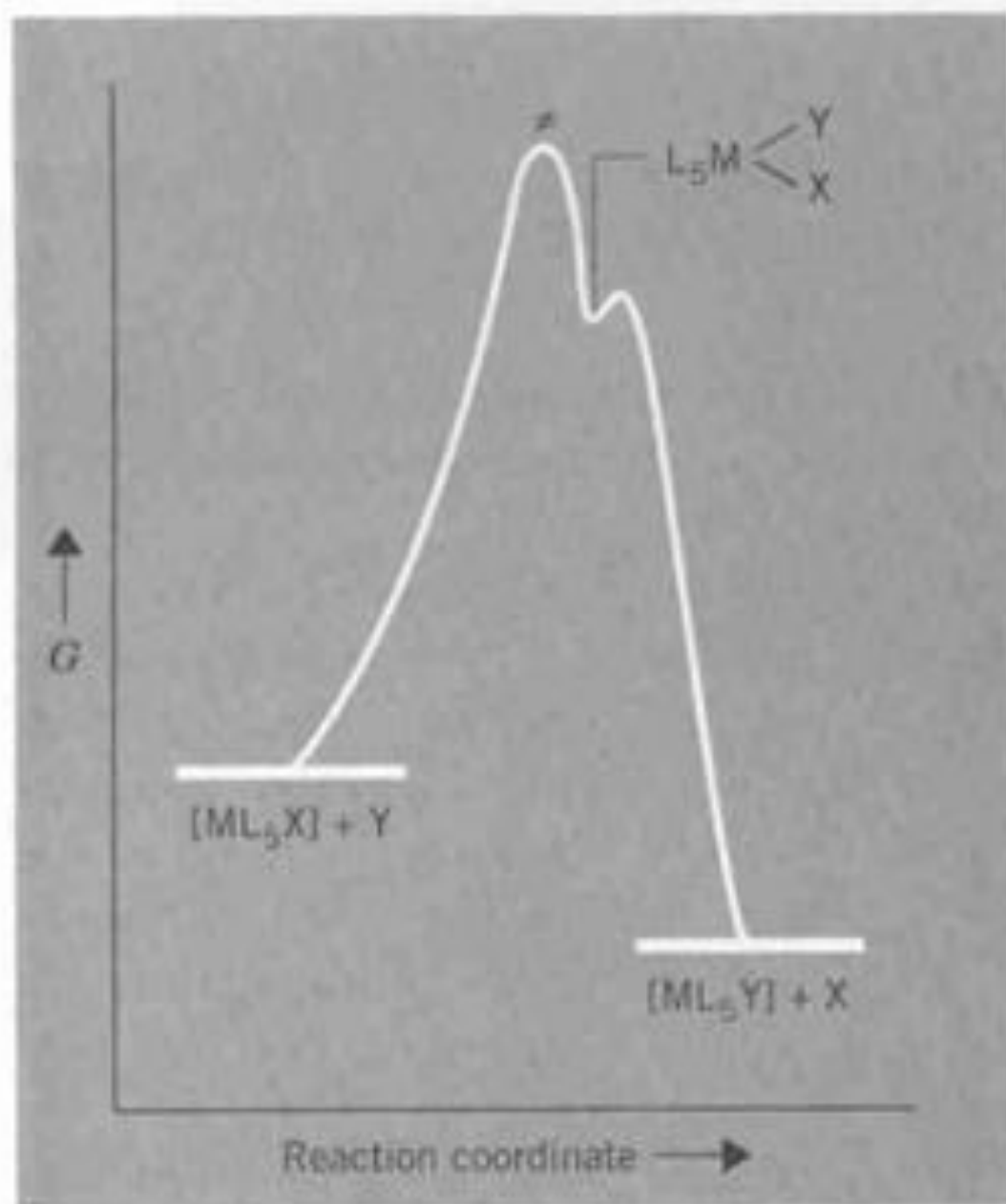


The rate law is

$$v = k[L_5MX][Y]$$

Sensitive to the nature of the incoming Y ligand.

Really common with compound with coordination $n < 6$



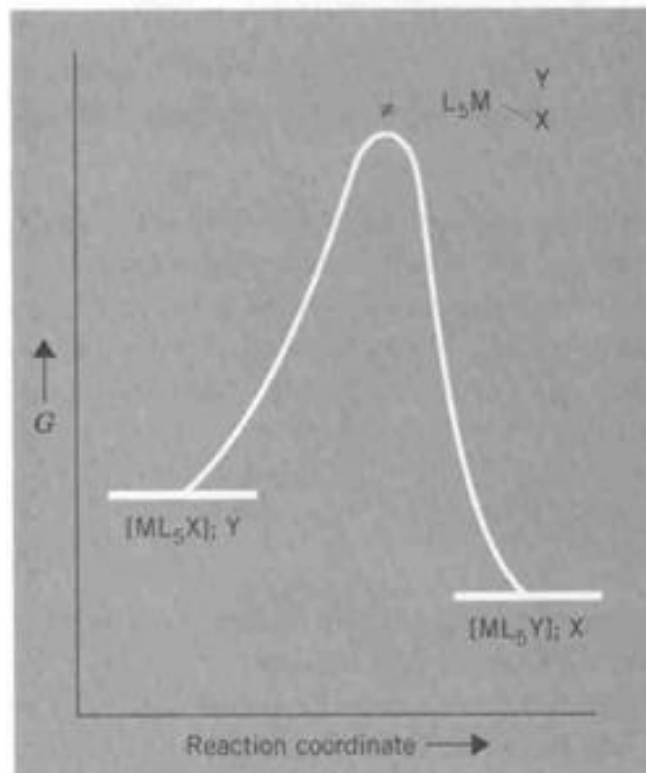
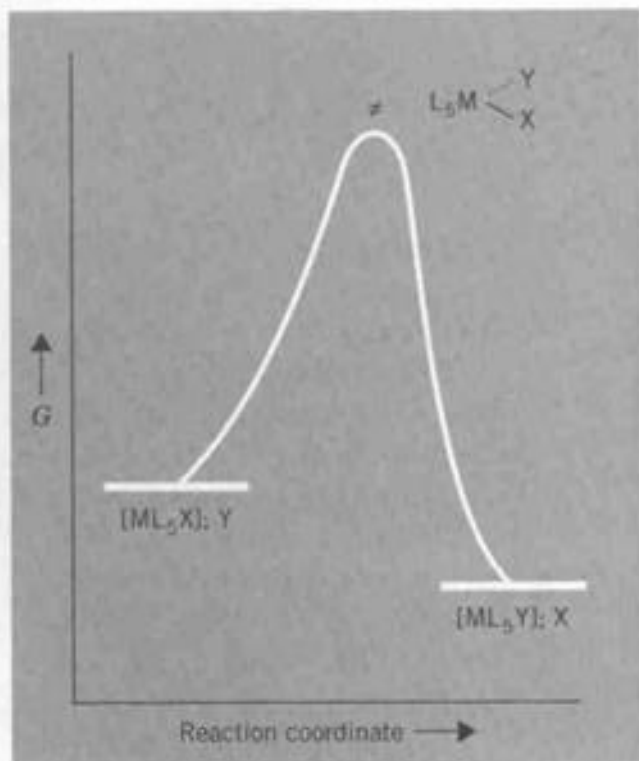
However, in reality, these two extreme mechanisms are seldom.

The real mechanism might be: some degree of bond making will occur before bond breaking is complete.

Therefore, the following two terms are used to describe mechanisms between the two extremes (**D** and **A**):

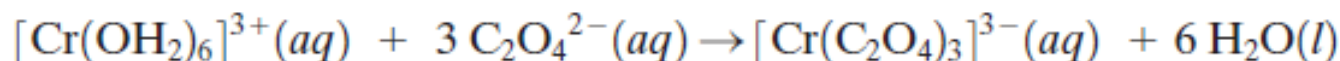
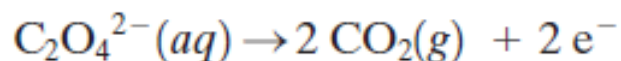
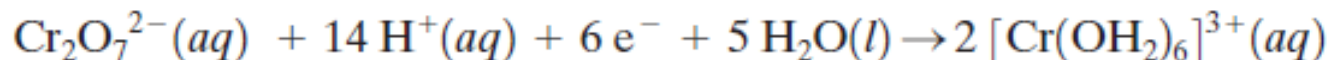
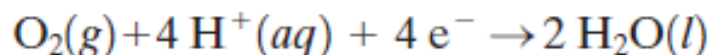
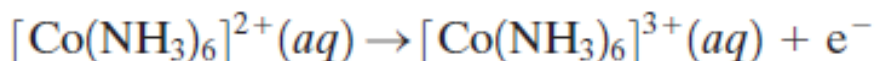
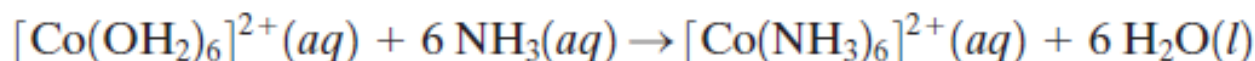
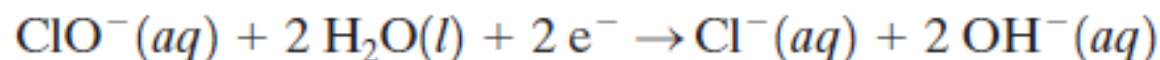
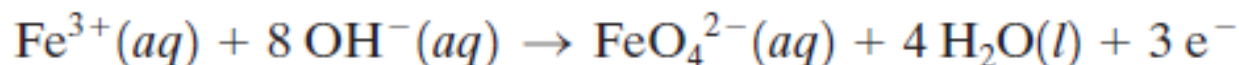
I_d - Interchange-dissociative

I_a - Interchange-associative



Redox reactions

examples:



Decomposition reactions

examples:

Vaporization of volatile ligands



Coordination Complexes and the HSAB Concept

Central metals

Hard	Borderline	Soft
Ti^{4+} , V^{4+} , Cr^{3+} , Cr^{6+} Mn^{2+} , Mn^{4+} , Mn^{7+} , Fe^{3+} , Co^{3+}	Fe^{2+} , Co^{2+} , Ni^{2+} , Cu^{2+}	Cu^{+} All metals with 0 or negative oxidation states

soft bases

hard bases

C	N	O	F
P	S	Cl	
As	Se	Br	
	Te	I	

Lesson 9

Iron Chemistry

Content

- **Iron smelting**
- **Iron in solution**
- **Rusting**
- **Ferrites**



Iron

Major component of Earth's core.

The most important material in our civilization, because:

1. Abundant and easy to mine.
2. The common ore can be easily and cheaply processed to obtain the metal.
3. The metal is malleable and ductile.
4. The melting point (1535°C) is low enough that the liquid phase can be handled without great difficulty.
5. By the addition of small quantities of other elements, alloys that have exactly the required combinations of strength, hardness, or ductility for very specific uses can be formed.



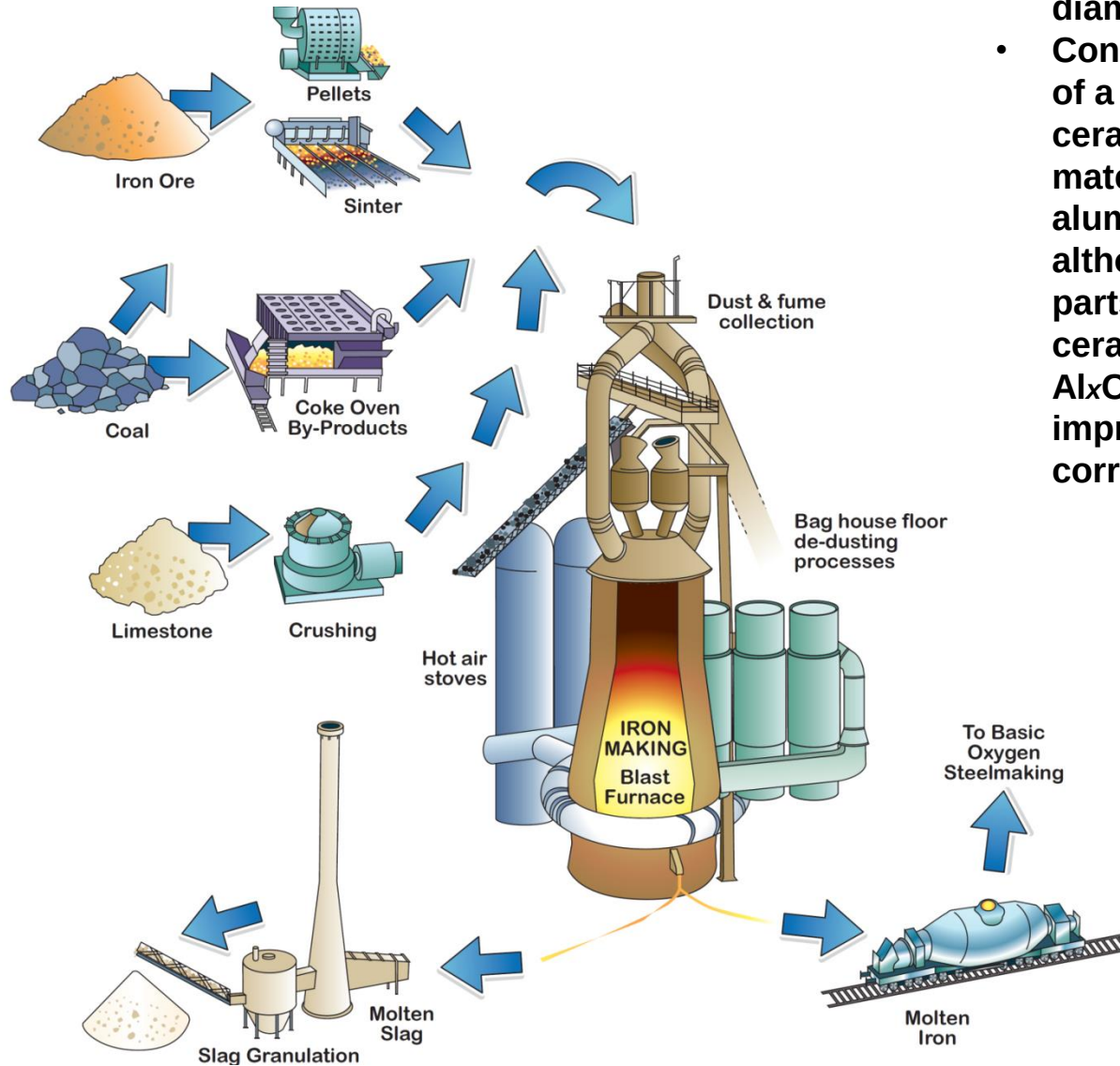
Chemical reactivity: **easy oxidation of iron is a major disadvantage**

Hematite Fe_2O_3 ,
Magnetite Fe_3O_4 , and
limonite
 $\text{FeO}(\text{OH}) \cdot n\text{H}_2\text{O}$

Iron production

Blast furnace

- 25 - 60 m in height, up to 14 m in diameter.
- Constructed of steel with a lining of a heat and corrosion-resistant ceramic bricks. The main ceramic material used for the lining is aluminum oxide (corundum), although the lining of the lower parts of the furnace consists of ceramic oxides of formula $\text{Al}_x\text{Cr}_{(2-x)}\text{O}_3$, which has an improved resistance to the corrosion.

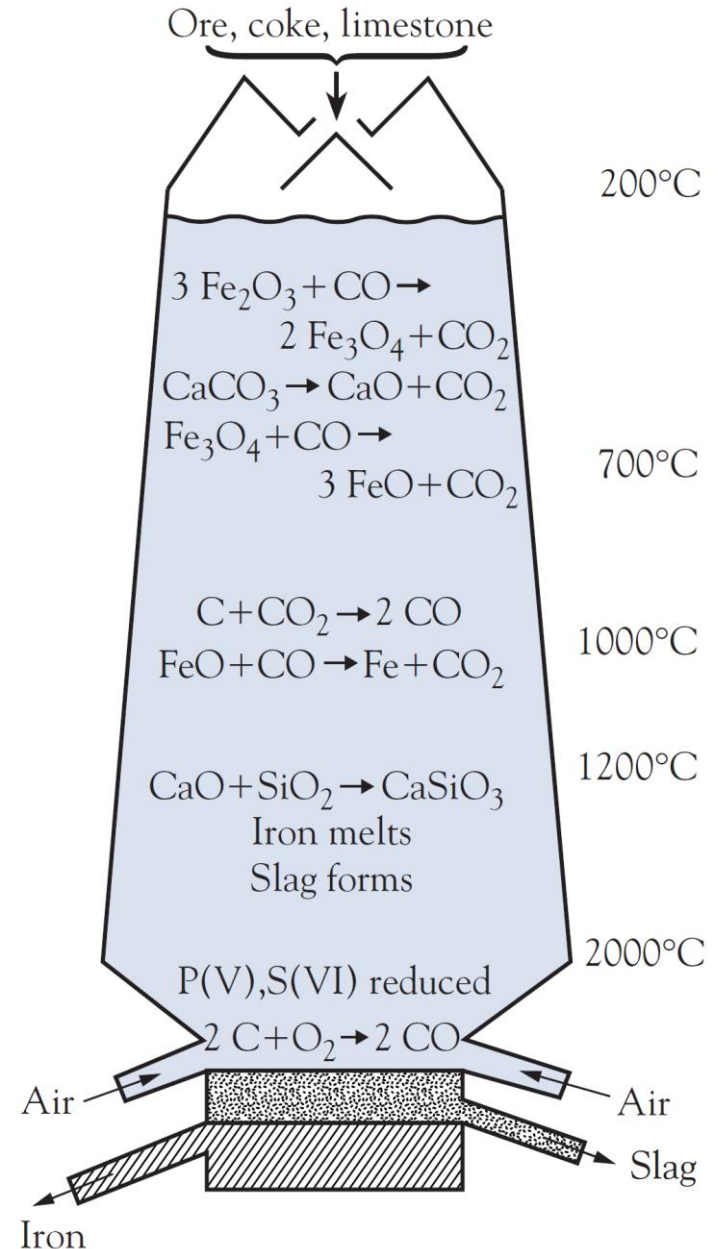


Blast furnace

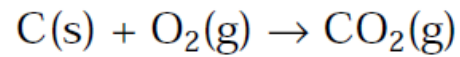
cone and hopper arrangement to hallow charge while the furnace is under operation.

Iron minerals
↓
Hot gases
↑

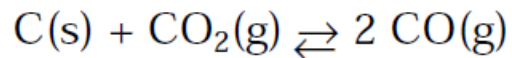
Air, preheated to 600 °C by combustion of the exhaust gases, is injected into the lower part of the furnace.



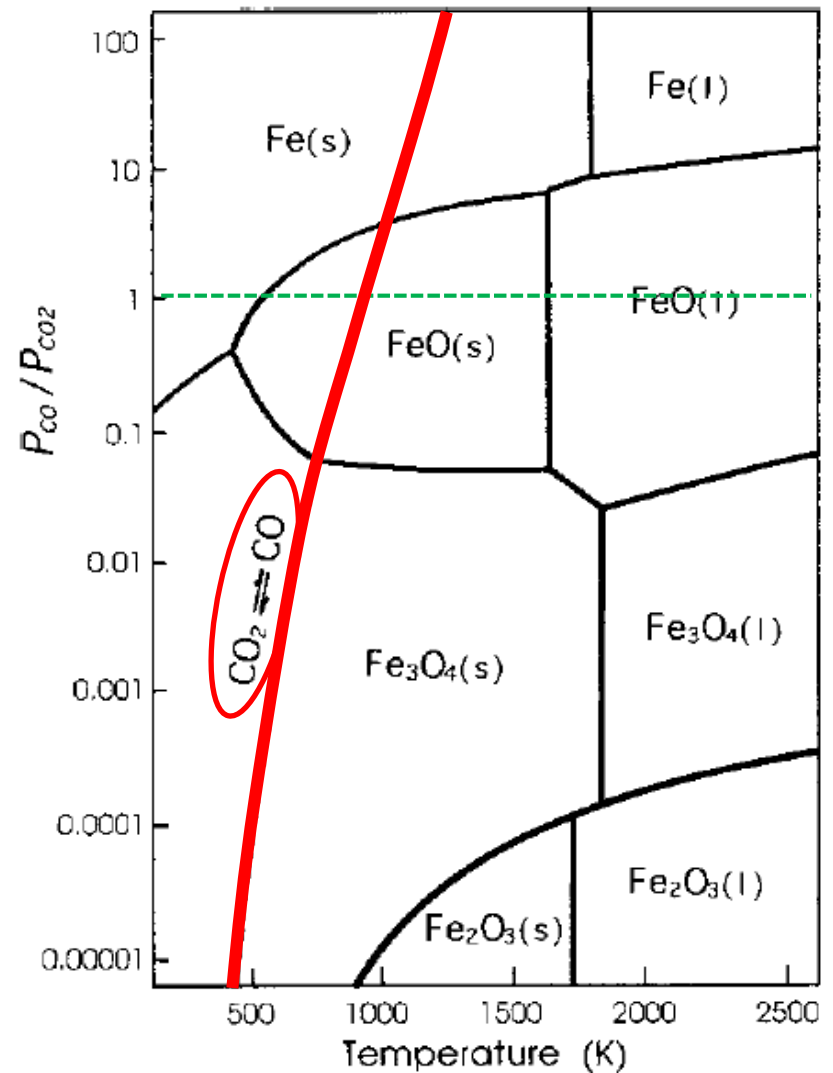
Combustion of coke



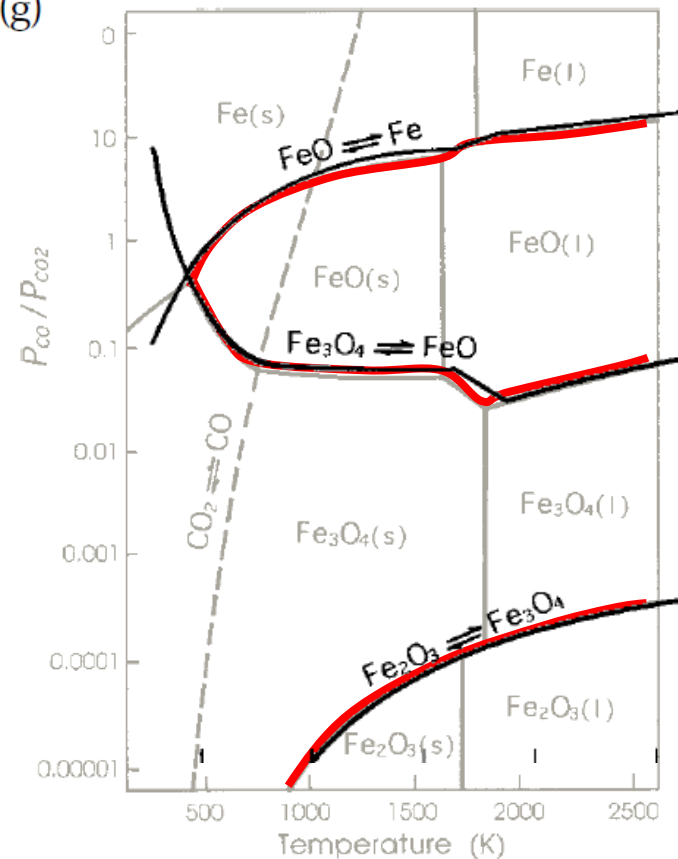
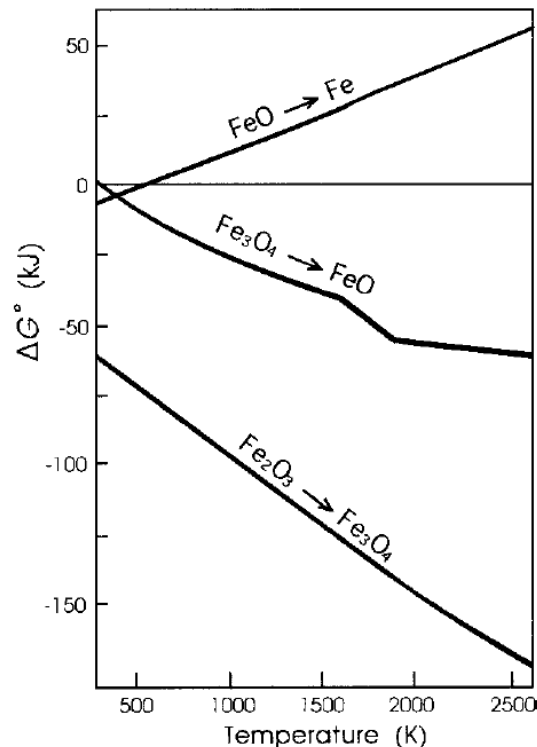
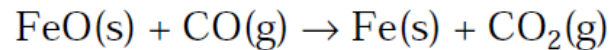
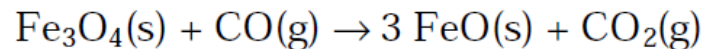
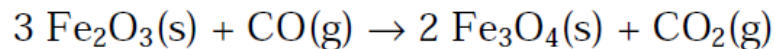
Production of CO



Temperature (K)	ΔH° (kJ)	ΔS° (J/K)	ΔG° (kJ)
$\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$			
298	-393.5	2.9	-394.4
1000	-394.6	1.3	-395.9
2000	-396.8	-0.2	-396.3
$\text{C(s)} + \text{CO}_2\text{(g)} \rightleftharpoons 2\text{CO(g)}$			
298	172.5	175.8	120.1
1000	170.7	175.3	-4.7
2000	159.0	167.4	-175.7



Iron reduction



At temperatures between 1200 °C and 1500 °C, the iron melts and sinks to the bottom of the furnace.

Two tap-holes that are plugged with clay:

- the lower one for the denser iron metal

- the upper one for the less dense slag.

- Blast furnaces are run 24/7

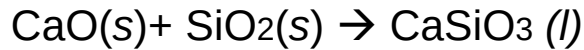
produce from 1000 to 10 000 tons of iron / day

The molten metal is usually conveyed directly in the liquid form to steelmaking plants.

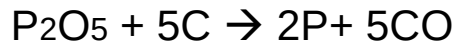
Iron reduction

On top: $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

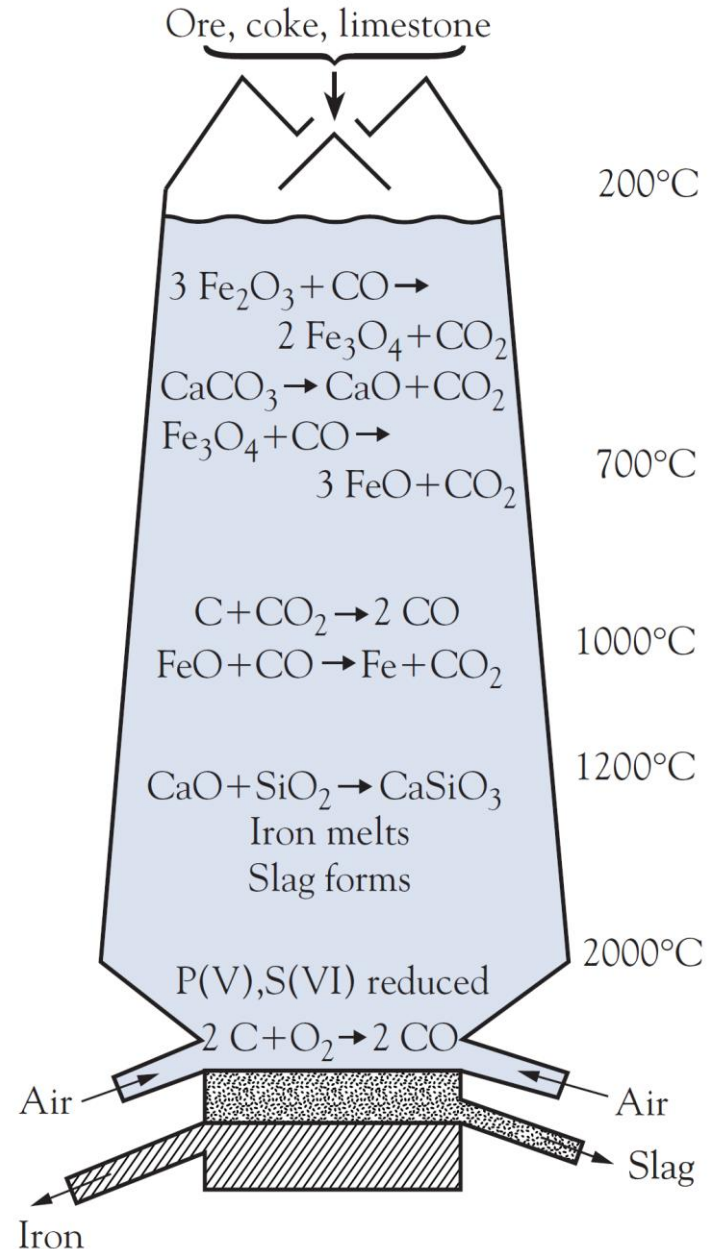
Dawn: Acid-base reaction between the basic metallic oxide, CaO, and the acidic nonmetal oxide, SiO₂:



Other minor components are also reduced



The slag can be either cooled to the solid phase, ground, and used in concrete manufacture or, while liquid, mixed with air and cooled into a “woolly” material that can be used for thermal insulation.



Exercise 1

Consider a blast furnace which is charged with iron ore coke and flux of the following composition:

Iron ore (weight %): $\text{Fe}_2\text{O}_3=78$, $\text{SiO}_2=8.4$, $\text{MnO} = 0.6$, $\text{Al}_2\text{O}_3=5.0$,
 $\text{P}_2\text{O}_5 = 1.7$ $\text{MgO} = 1.2$ and $\text{H}_2\text{O}=5.1$

Coke (weight %): $\text{C}=88$, $\text{SiO}_2 =9$, $\text{Al}_2\text{O}_3 =1$ and $\text{H}_2\text{O} = 2$

Flue : $\text{Ca CO}_3=96 \%$, $\text{Mg CO}_3 =2\%$ and $\text{SiO}_2 =2\%$

“Pig Iron” (iron produced by the blast furnace) has the following composition (w%):

$\text{Fe} = 92.7$, $\text{C} = 4$ $\text{Si} = 2$, $\text{P} = 0.9$ and $\text{Mn} = 0.4$

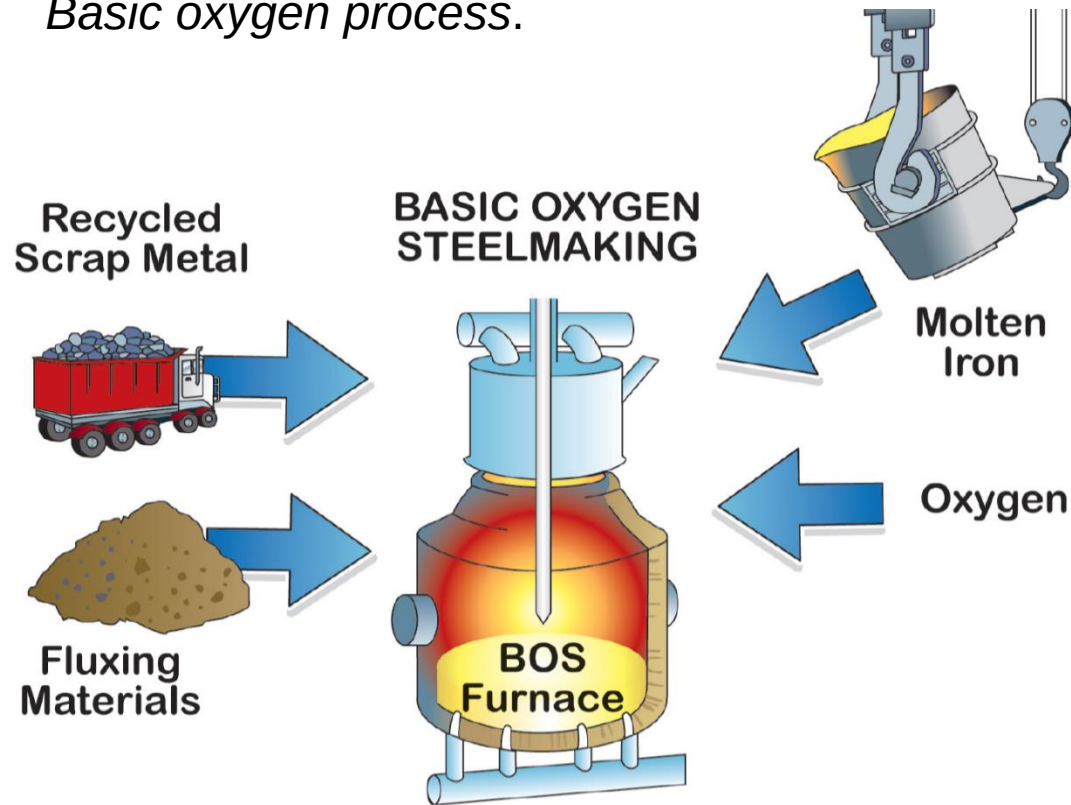
The furnace consume 900 Kg of coke and 450 Kg of flue for each ton of pig iron. During the process 99.5% of Fe mass is reduced and the 0.5% is lost in the slag. The CO/CO_2 ratio in the top gas is 2/1.

(i) Calculate the weight of the iron ore to produce a ton of pig Iron. (ii) Calculate the ratio between coke and flue over iron ore in the feed of the blast furnace

impurities, e.g. sulfur, phosphorus, carbon, and oxygen.

Carbon (4.5 %) is a particular contributor to the brittleness of the material.

Basic oxygen process.



Discontinuous process

Filled with about 60 ton of molten iron.

A blast of oxygen is blown through the converter.

Why Oxygen?

Oxygen reacts with the impurities and raises the temperature in the furnace to about 1900°C ,

Temperature control CO_2 and cold scrap metal

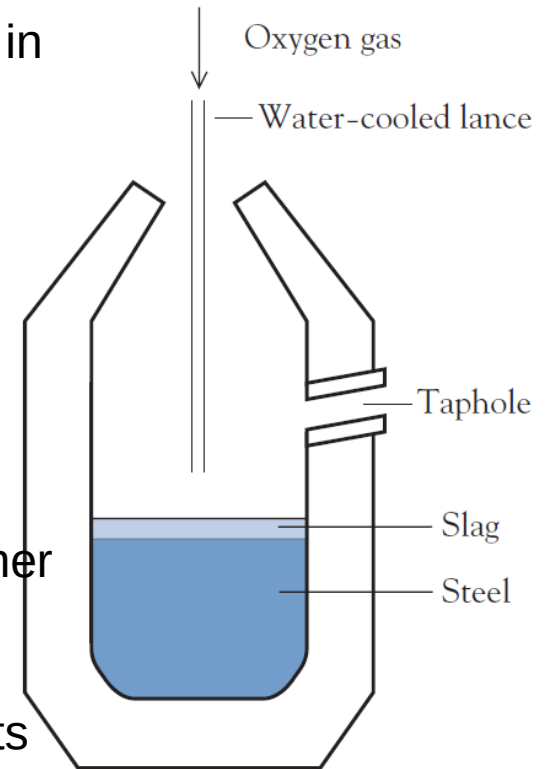
C is oxidized to CO, which burns to CO_2 at the top of the converter. After several minutes, the flame at the top of the converter sinks, indicating that the conversion occurred.

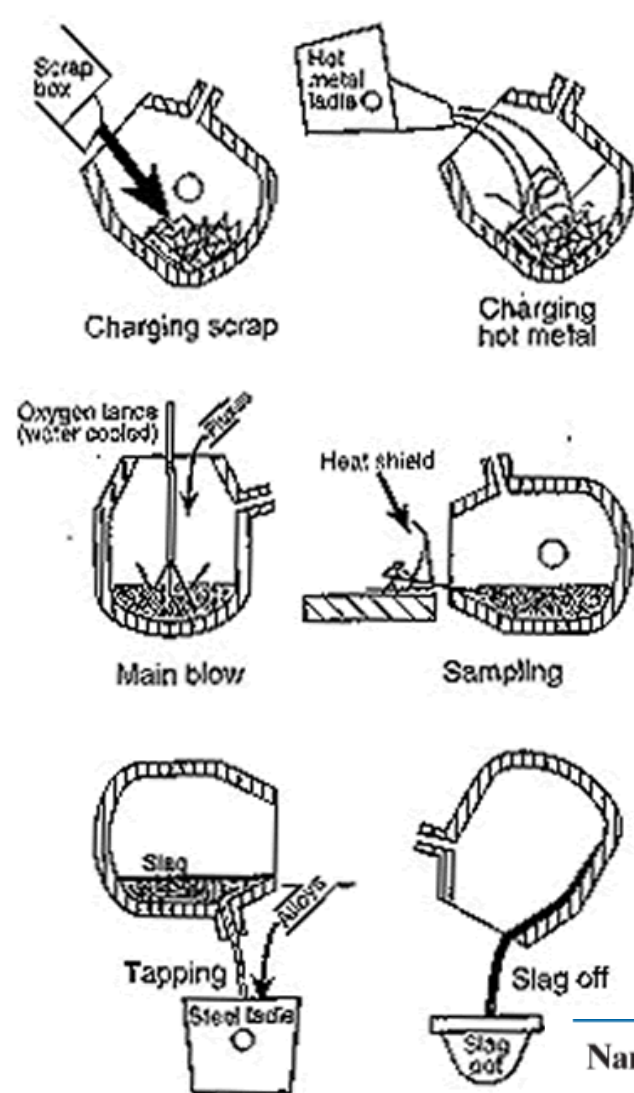
Si is oxidized to SiO_2 , which then reacts with the oxides of other elements to form a slag.

The furnace also is lined with limestone (CaCO_3), which reacts with acidic phosphorus-containing impurities.

At the end, the slag is poured off, and any required trace elements are added to the molten iron.

For normal steel, between 0.1 and 1.5 percent carbon is required.





Molten Steel
 98% Fe
 0.04 - 1.5% C
 1% Mn
 various alloys
 3000°F
 Fluidity ~ 6 cp

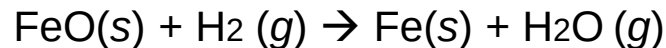
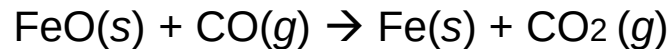
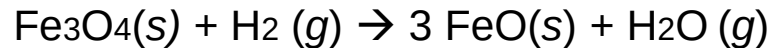
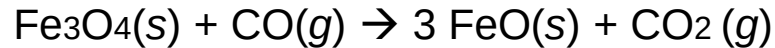
Name	Approximate composition	Properties and uses
Stainless steel	73% Fe, 18% Cr, 8% Ni	Corrosion resistant (tableware, cookware)
Tungsten steel	94% Fe, 5% W	Hard (high-speed cutting tools)
Manganese steel	86% Fe, 13% Mn	Tough (rock drill bits)
Permalloy	78% Ni, 21% Fe	Magnetic (electromagnets)

Direct Reduction Iron (DRI)

An increasing proportion of iron is produced by direct reduction from the ore in the solid phase.

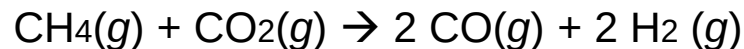
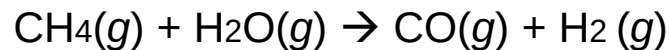
high-purity iron ore must be used.

Carbon monoxide and hydrogen, the reductants, are passed over heated iron ore.



The reducing gases are obtained by methane re-forming.

Thus, the economics of the process depend on the price of natural gas.

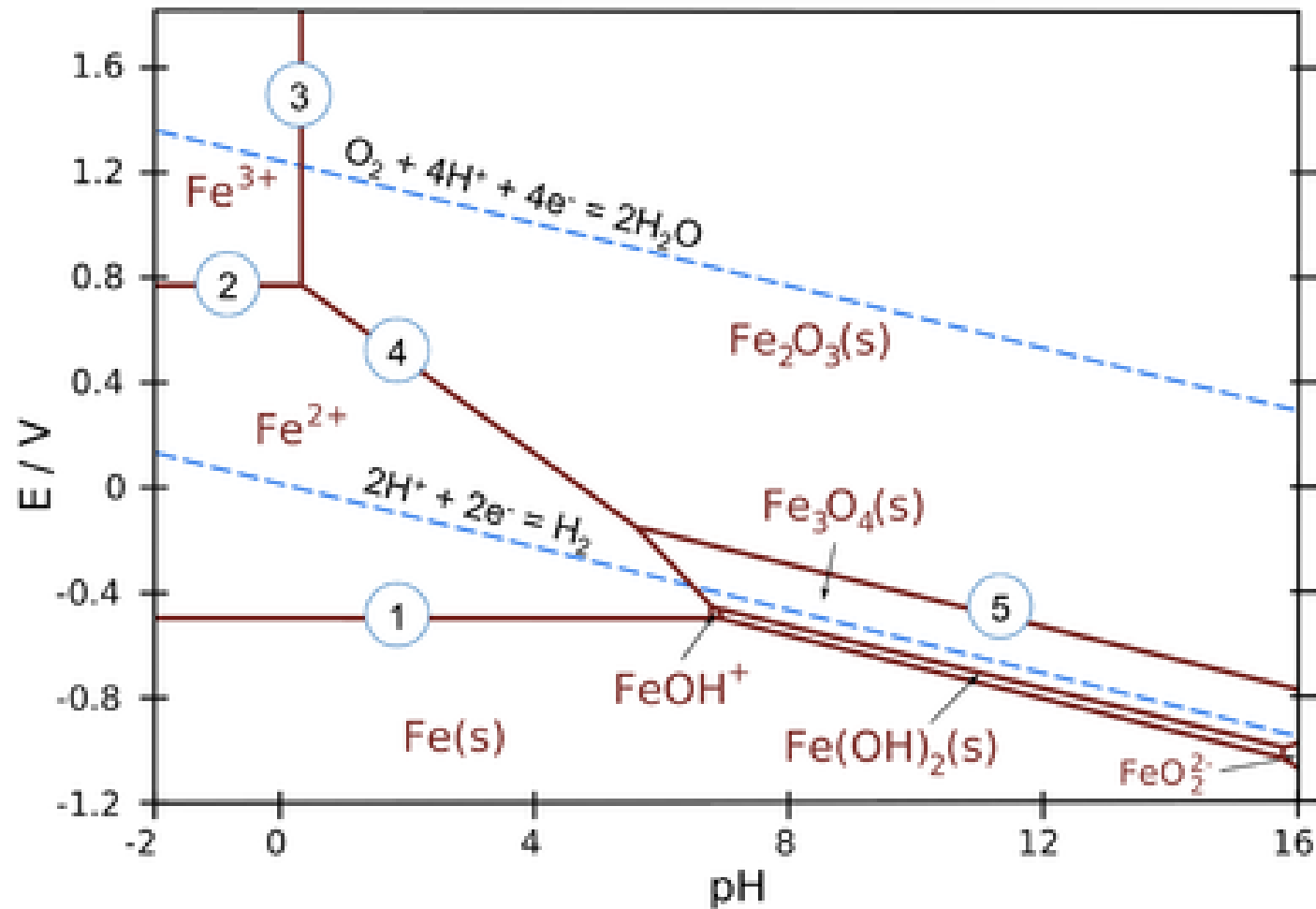


Advantage: economic, no further purification

Disadvantage: product contains most element impurities from the natural ore.

Chemistry of Iron

The Pourbaix diagram



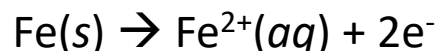
The Rusting Process

requires the presence of both dioxygen and water.

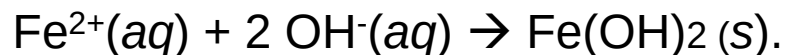
On iron surface:



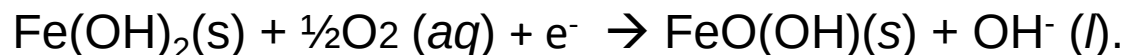
The bulk iron acts like a wire connected to a battery, conveying electrons from another point on the surface that has a lower oxygen concentration, a point at which the iron metal is oxidized to iron(II) ions:



These two ions diffuse through the solution, and where they meet, insoluble iron(II) hydroxide is formed:



And thus:



Iron Oxides

There are three common oxides of iron:

iron(II) oxide, FeO;

Wüstite

iron(III) oxide, Fe₂O₃;

Hematite

iron(II) iron(III) oxide, Fe₃O₄.

Magnetite

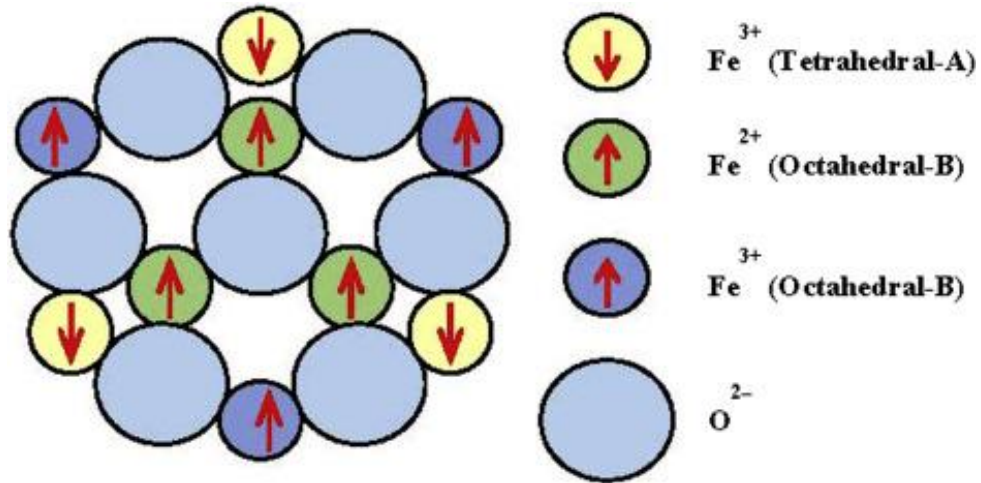
The iron(II) oxide is basic, dissolving in acids: $\text{FeO(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Fe}^{2+}(\text{aq}) + \text{H}_2\text{O(l)}$

Iron(III) oxide, or hematite, is found in large underground deposits, the oldest being about 2 billion years old and can exist in different crystalline forms.

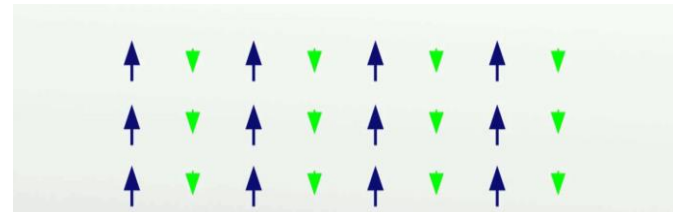
Exercise 2

Does Magnetite, Fe₃O₄, has normal or inverse spinel structure?

Magnetite, Fe_3O_4 , has commonly inverse spinel structure and is ferrimagnetic. Pieces of this magnetic compound were used as a primitive compass.



ferrimagnetic



Ferrites

mixed-metal oxides, one metal being iron (III), often have spinel structures

“soft” ferrites: can be magnetized rapidly and efficiently by an electromagnet, but they lose their magnetism as soon as the current is discontinued.

Structure: MFe_2O_4 , $M = Mn^{2+}, Ni^{2+}, Co^{2+}, \text{ or } Mg^{2+}$

Applications: Transformers and Antennas

“hard” ferrites: the hard ferrites retain their magnetic properties constantly

Structure: $MFe_{12}O_{19}$, $M = Ba^{2+}, Sr^{2+}$

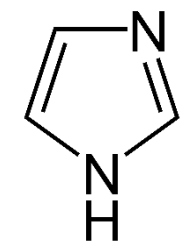
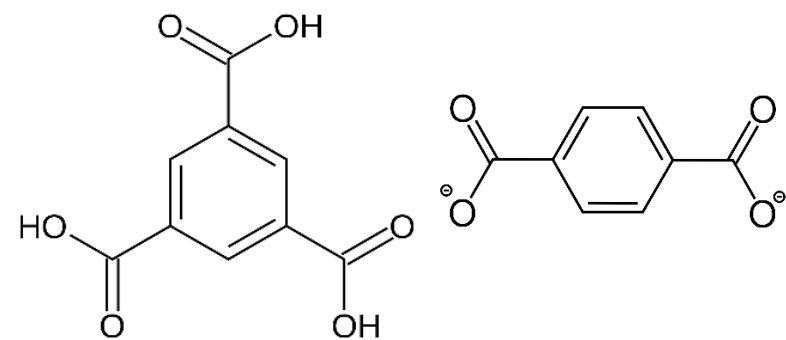
Applications: permanent magnets, e.g. refrigerator magnets.

Metal-organic frameworks (MOFs)

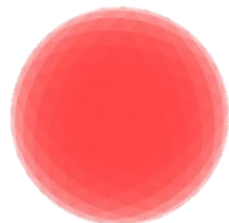
Xinxin Chen

What are Metal-Organic Frameworks (MOFs)?

Metal-Organic Frameworks (MOFs) are crystalline, periodic, highly porous (up to 94% empty space) frameworks



+



Organic
ligands

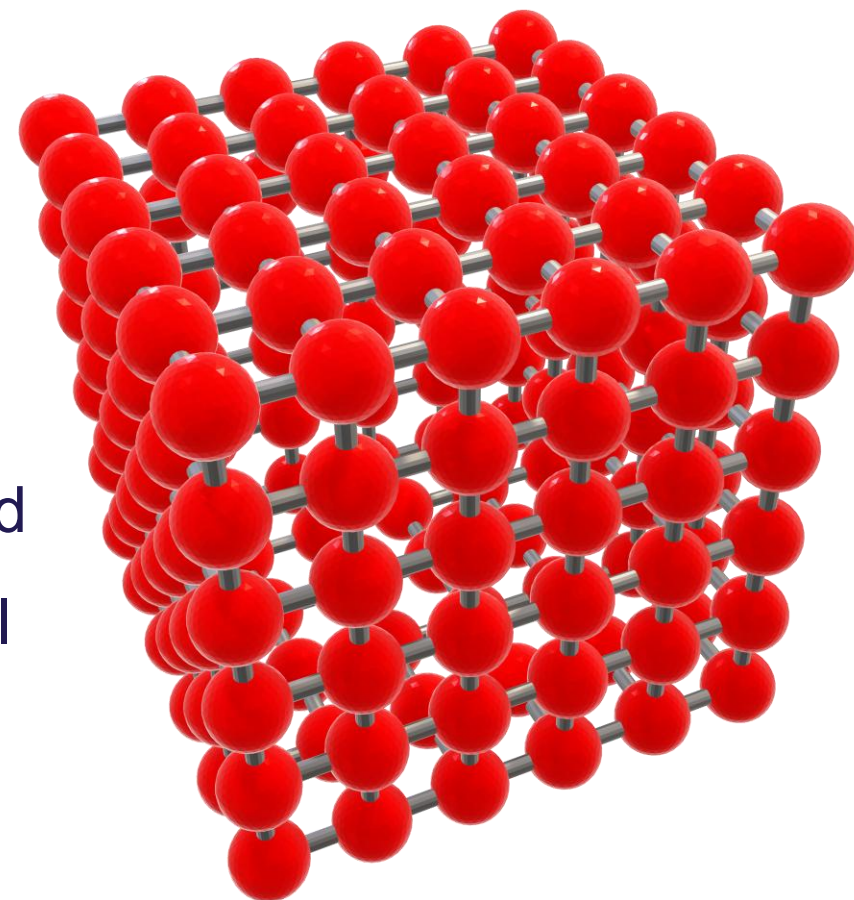
Metal ions

Co, Ni, Zr, Cr,
Cu, Fe, Al, etc.

solvothermal
sonochemical

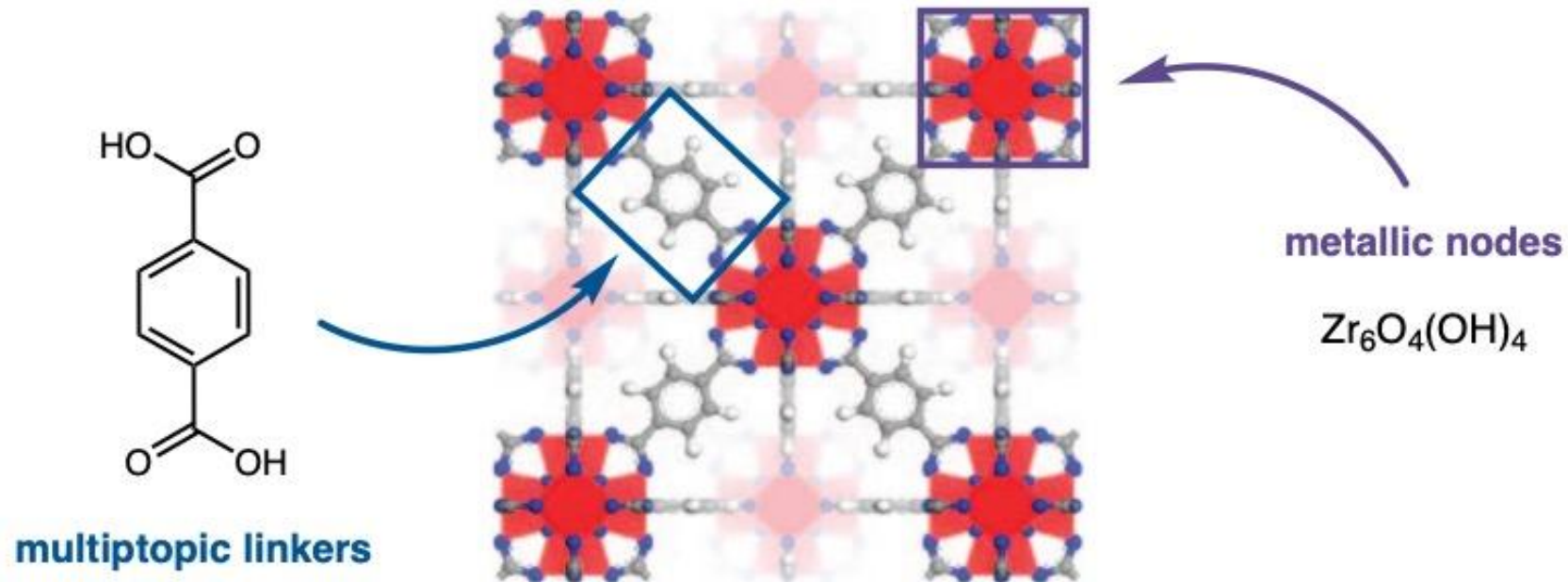


microwave-assisted
Mechano-chemical



What are Metal-Organic Frameworks (MOFs)?

Metal-Organic Frameworks (MOFs) are crystalline, periodic, highly porous (up to 94% empty space) frameworks



Framework self-assemble to form coordination bonds between transition metal cations and carboxylate anions

The Evolution of Metal-Organic Frameworks

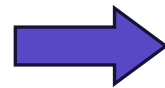
Reticular chemistry

a. MOFs & Porous materials

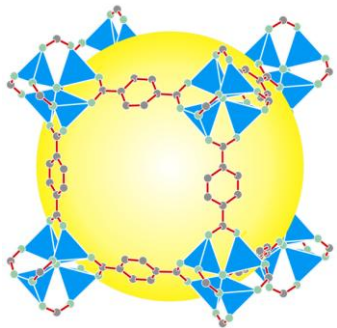
Molecules



Frameworks

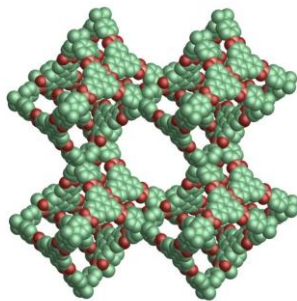


Applications



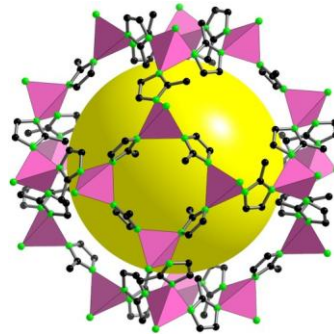
MOFs

[Nature](#) 1995, [JACS](#) 1998,
[Nature](#) 1999



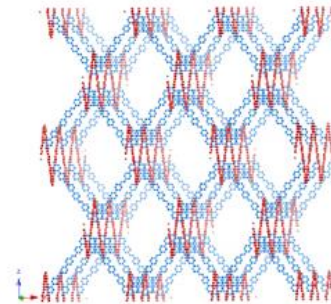
COFs

[Science](#) 2005, [2007](#)



ZIFs

[PNAS](#) 2006



Molecular Weaving

[Science](#) 2016

[Nobel Prize in Chemistry 2025](#)



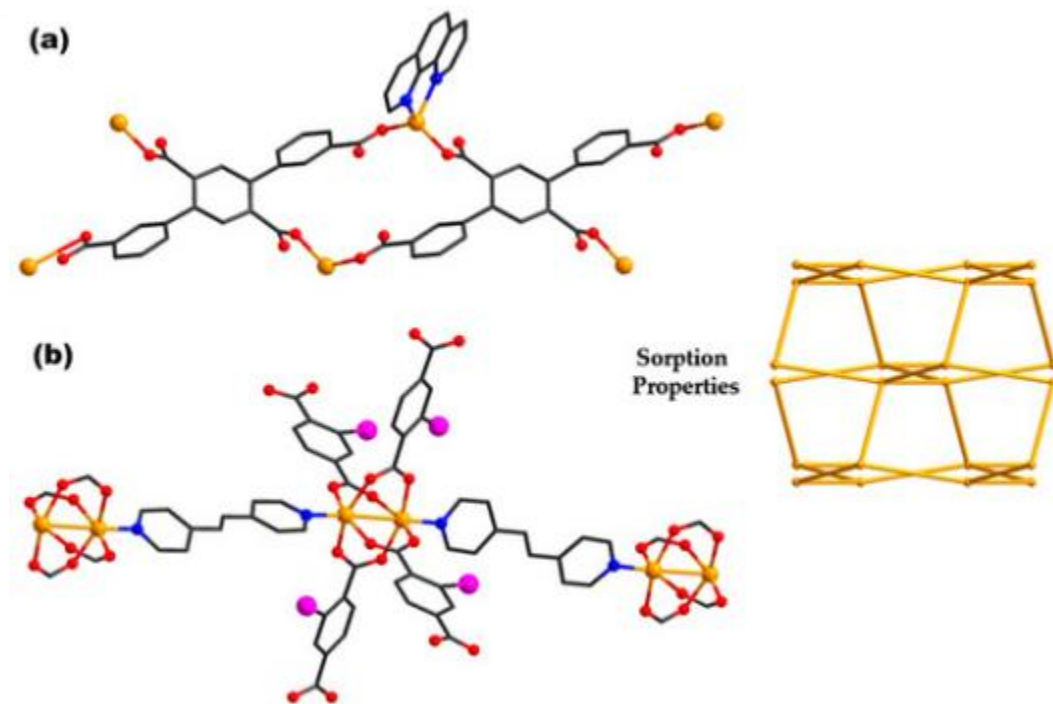
Prof. Omar M. Yaghi

1965: The Early Foundations

Zn-Coordination Polymers

Researchers at **DuPont** synthesized thermally stable coordination polymers using Zinc nodes.

While these materials showed the potential for repetitive bonding between metals and ligands, they were non-porous and lacked the crystalline order of modern frameworks.



1990: The Conceptual Breakthrough

Hoskins & Robson

Proposed the revolutionary design of **3D scaffold-like frameworks**. This theoretical shift moved the field from simple polymers to engineered architectures.

Infinite Lattices

They envisioned a material where metal atoms act as "joints" and organic ligands act as "struts," creating an infinite, open molecular lattice.

1995: The Birth of MOF Synthesis + 1999: MOF-5 – The Landmark

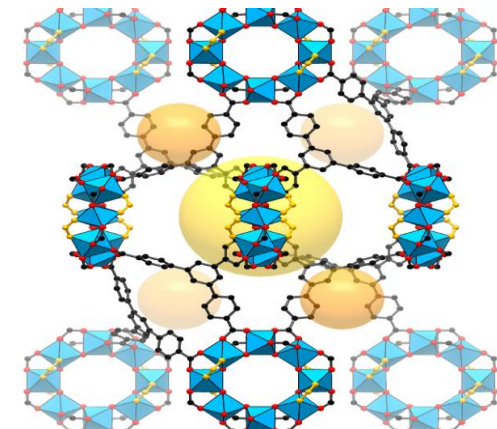
Nobel Prize in Chemistry 2025



Omar Yaghi introduces hydrothermal methods to reliably engineer metal-organic bonds.

In 1999, Omar Yaghi and colleagues published the synthesis of **MOF-5**. This structure proved that high porosity and structural integrity could coexist without the framework collapsing upon solvent removal. MOF-5 effectively launched the modern era of porous materials research.

MOF-5 effectively launched the modern era of porous materials research.

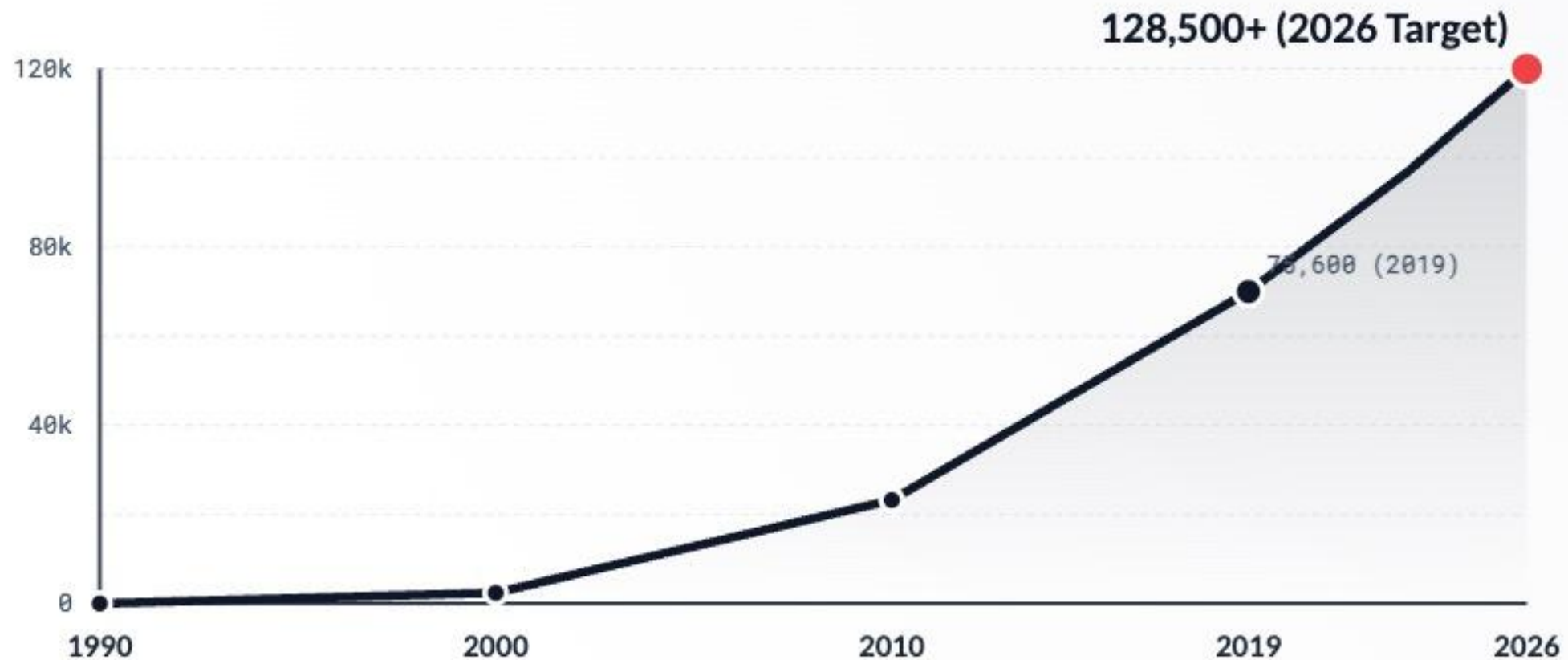


Performance: Surface Area Comparison



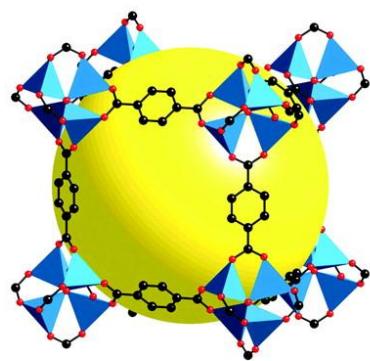
MOFs achieved surface areas that far exceeded traditional porous materials, enabling new heights in gas adsorption.

The "Cambrian Explosion" of MOFs



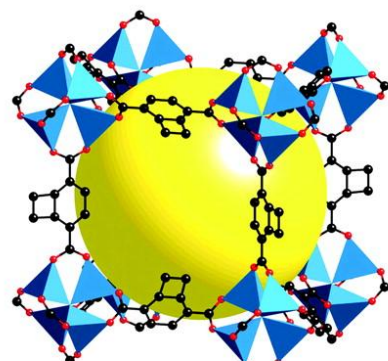
Registered structures in the Cambridge Structural Database (CSD).

Examples of MOF Structures



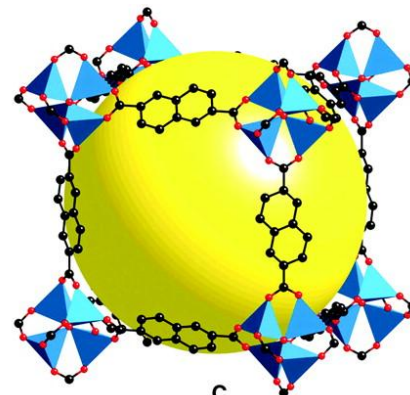
A

MOF-5



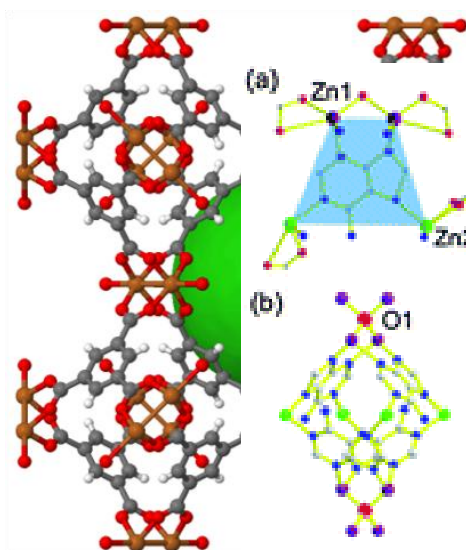
B

MOF-6

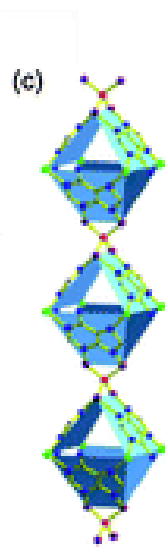


C

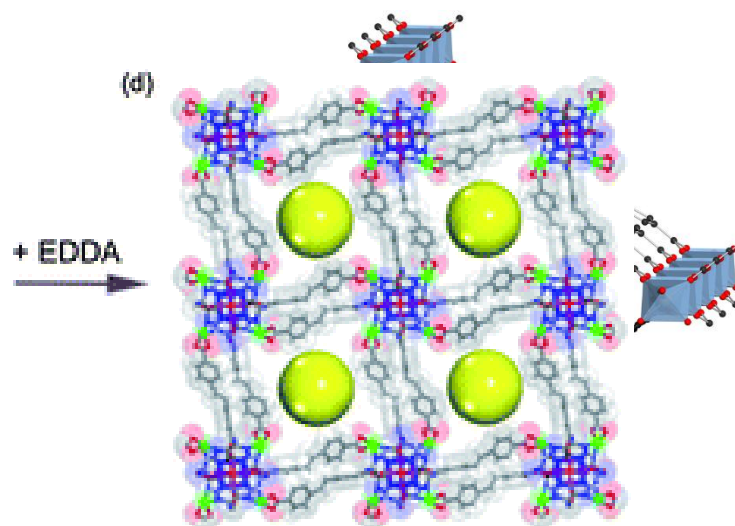
MOF-8



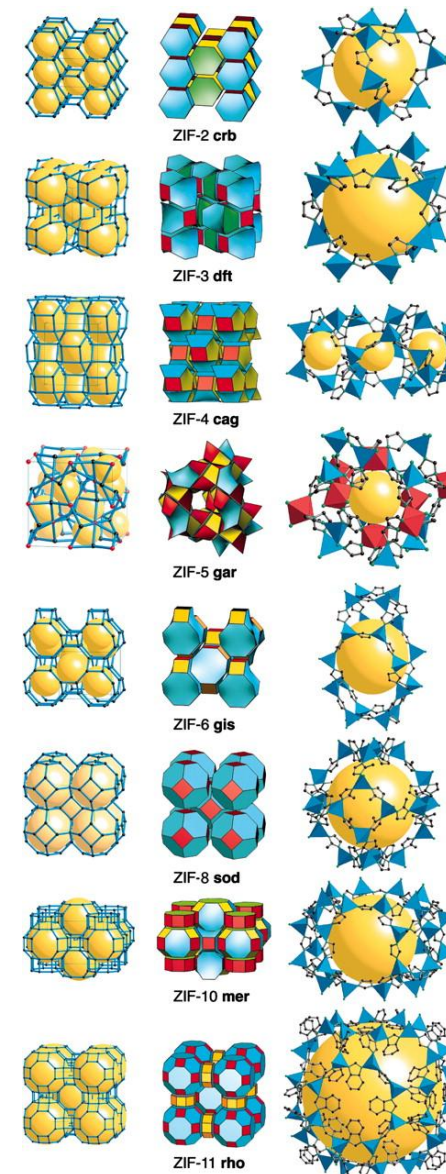
HKUST



ZJU

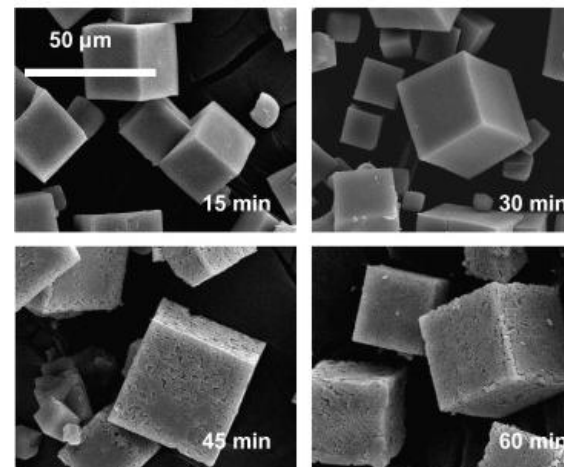
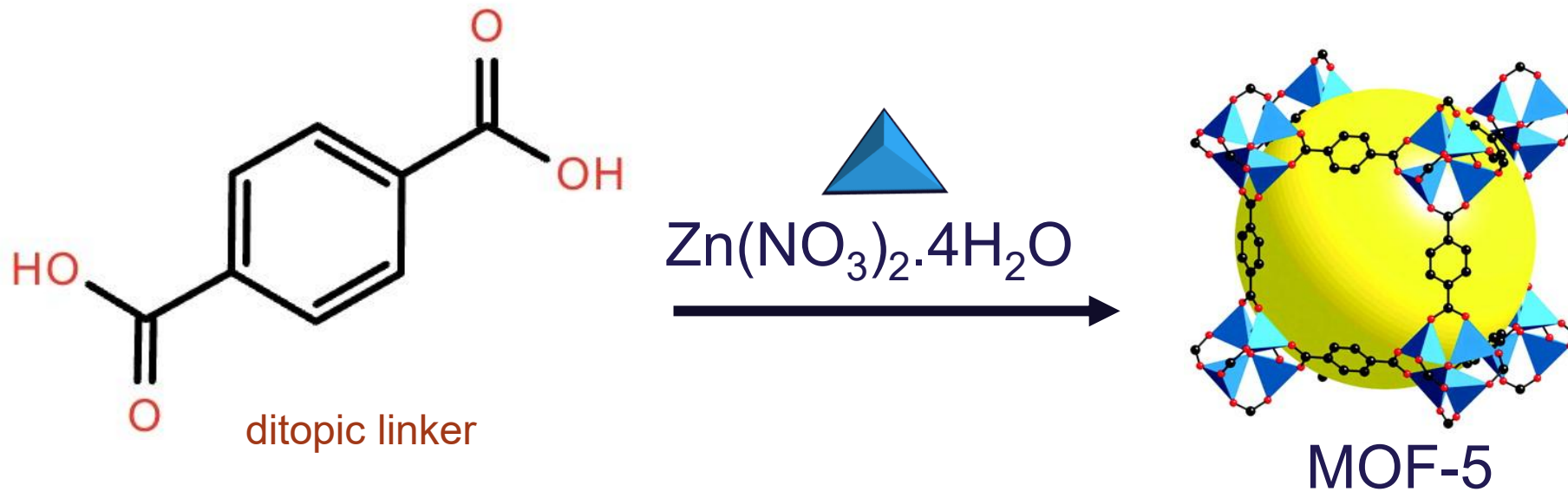


MIL

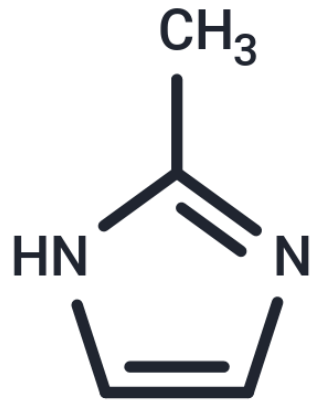


ZIF

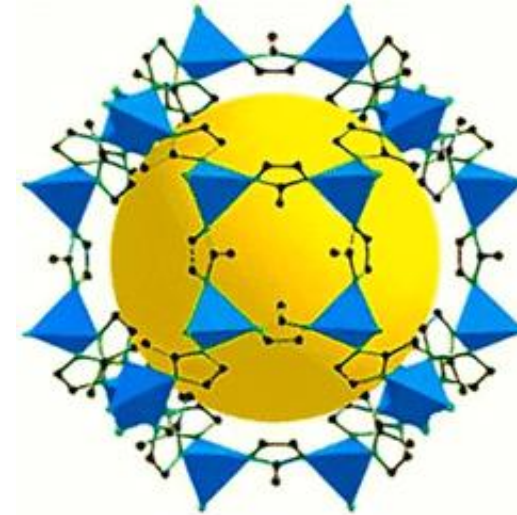
Examples of MOF Structures



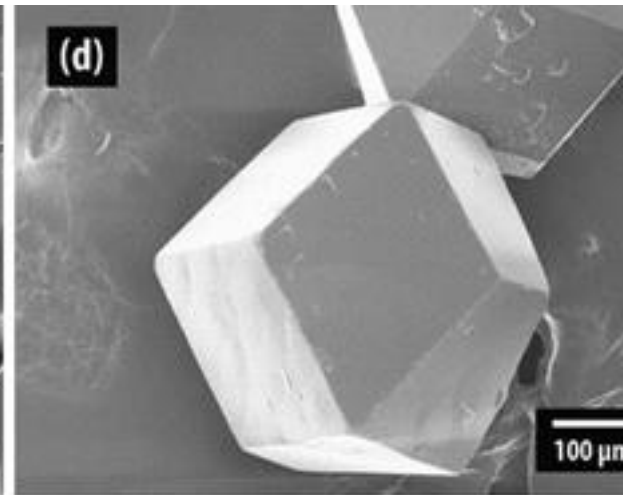
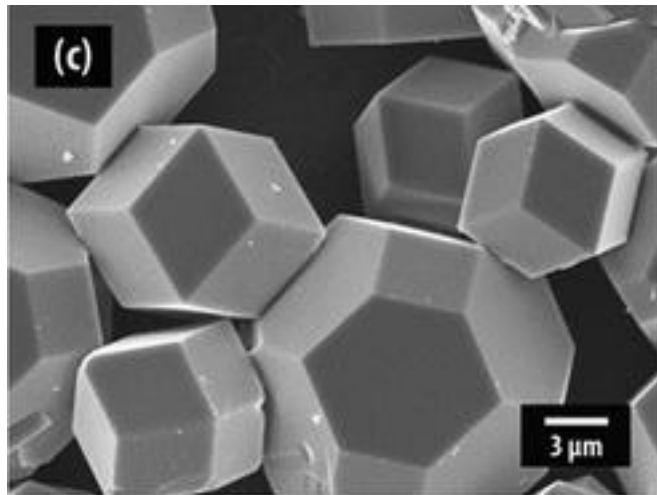
Examples of MOF Structures



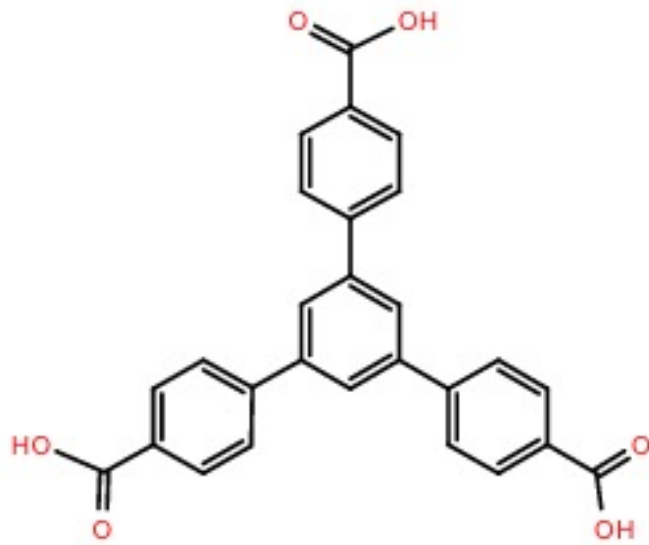
ditopic linker



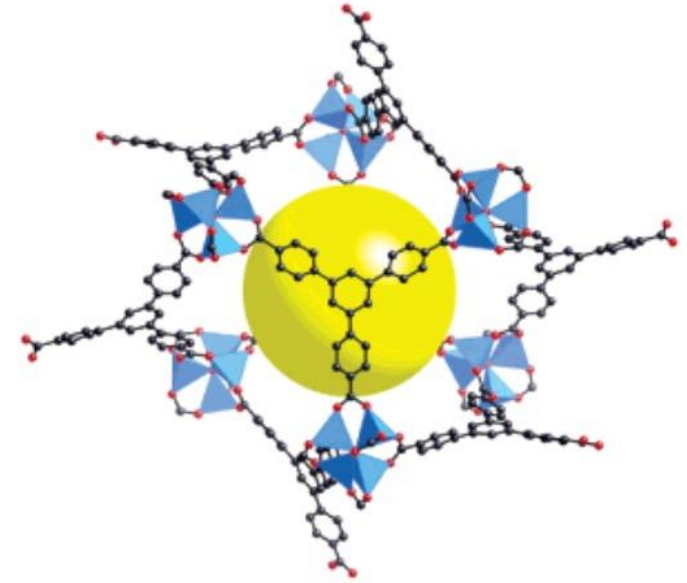
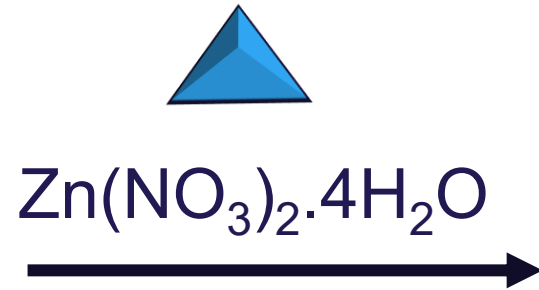
ZIF-8



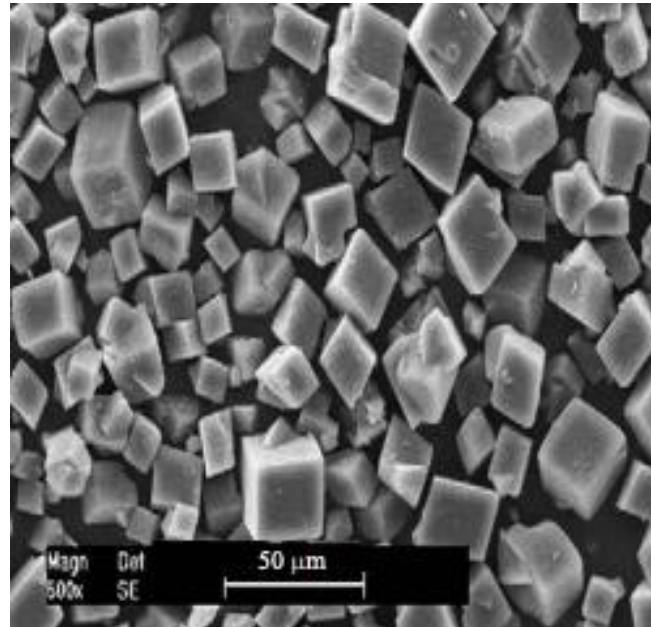
Examples of MOF Structures



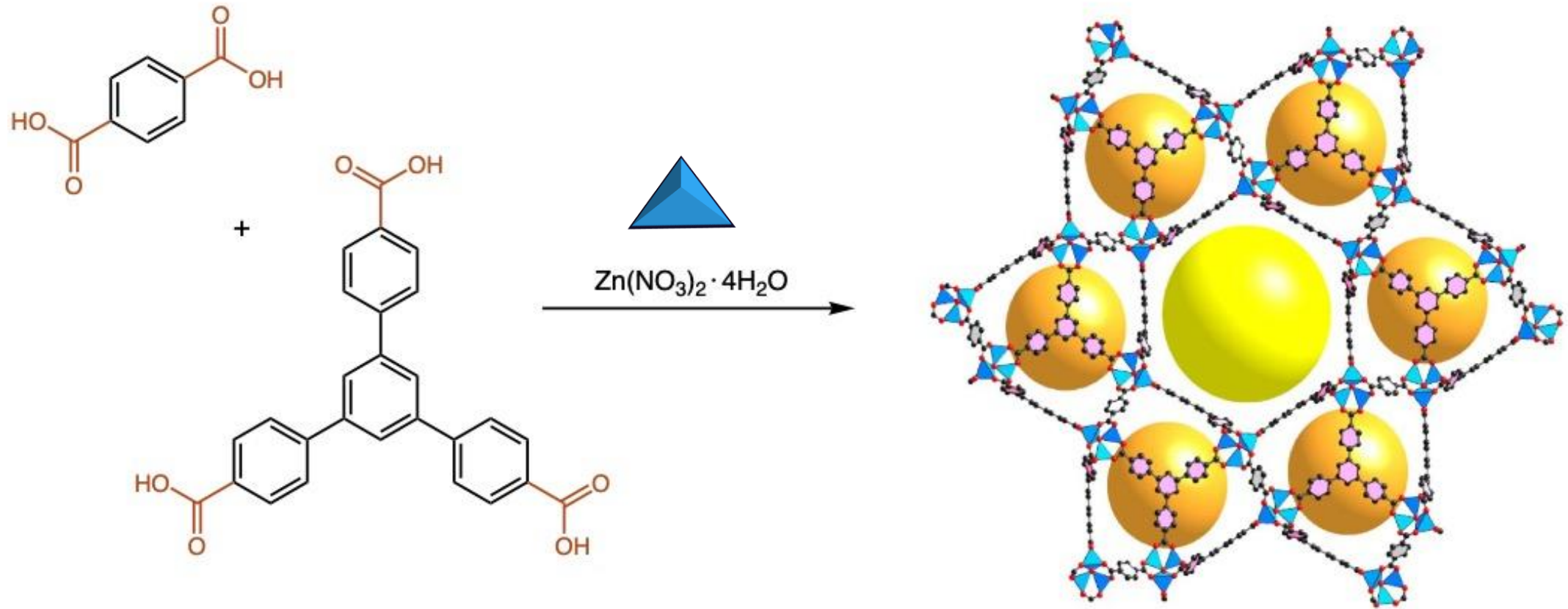
tritopic linker



MOF-177



Examples of Mixed-Linker MOF Structures



Due to the modularity of MOF design, a nearly infinite number of topographies can be imagined.

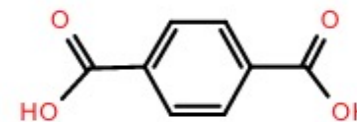
- **Synthesis through self-assembly**
- **Activation by removal of solvent**
- **Characterization**

Synthesis and Activation

The Conventional Route: Solvothermal Synthesis

ZnCl₄

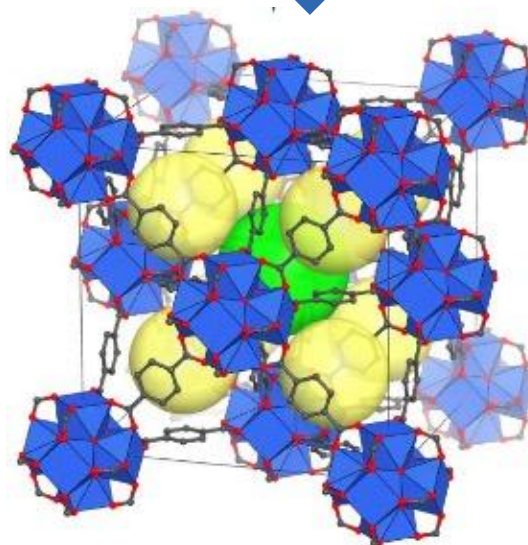
Simple metal salt



Organic linker

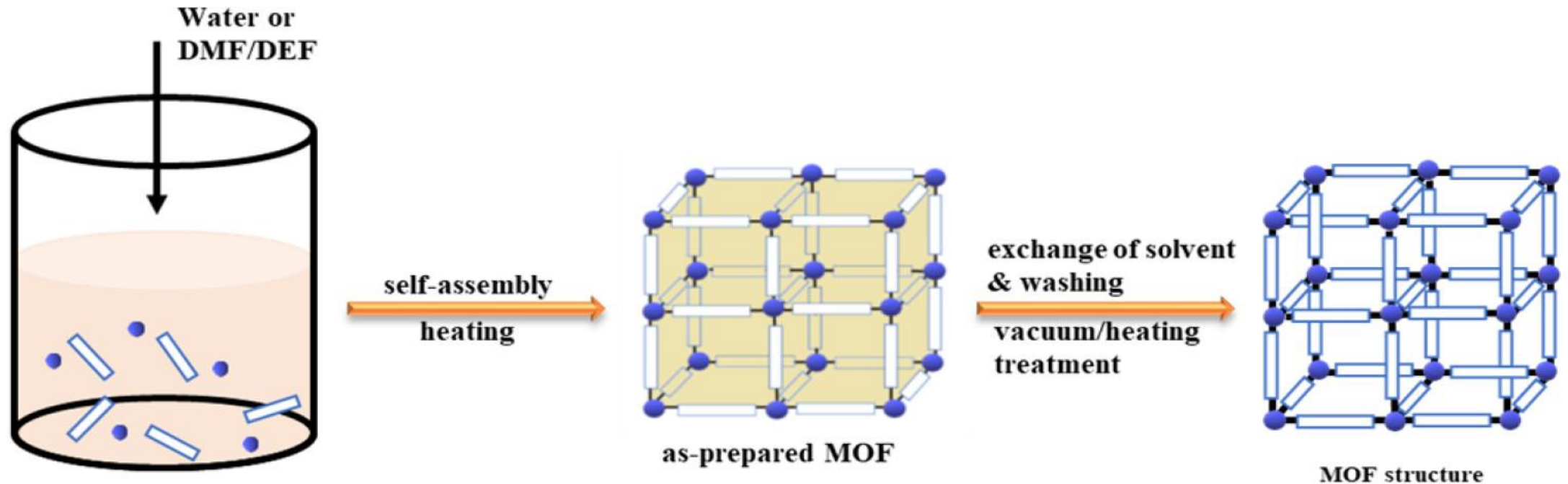
1. Sonicate in high-boiling solvents (e.g., DMF).
2. Heating for hours/ days
3. Centrifuge and filter

Optimization: Controlling temperature, order of addition, and metal source.



The Role of Modulators: Benzoic or acetic acids slow nucleation to promote ideal framework assembly.

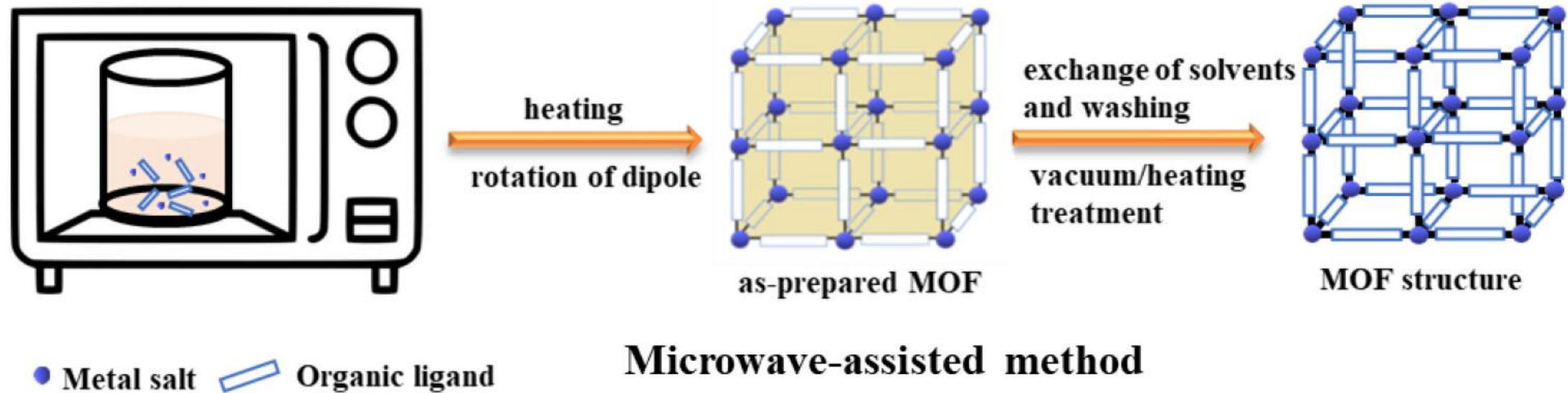
The Conventional Route: Solvothermal Synthesis



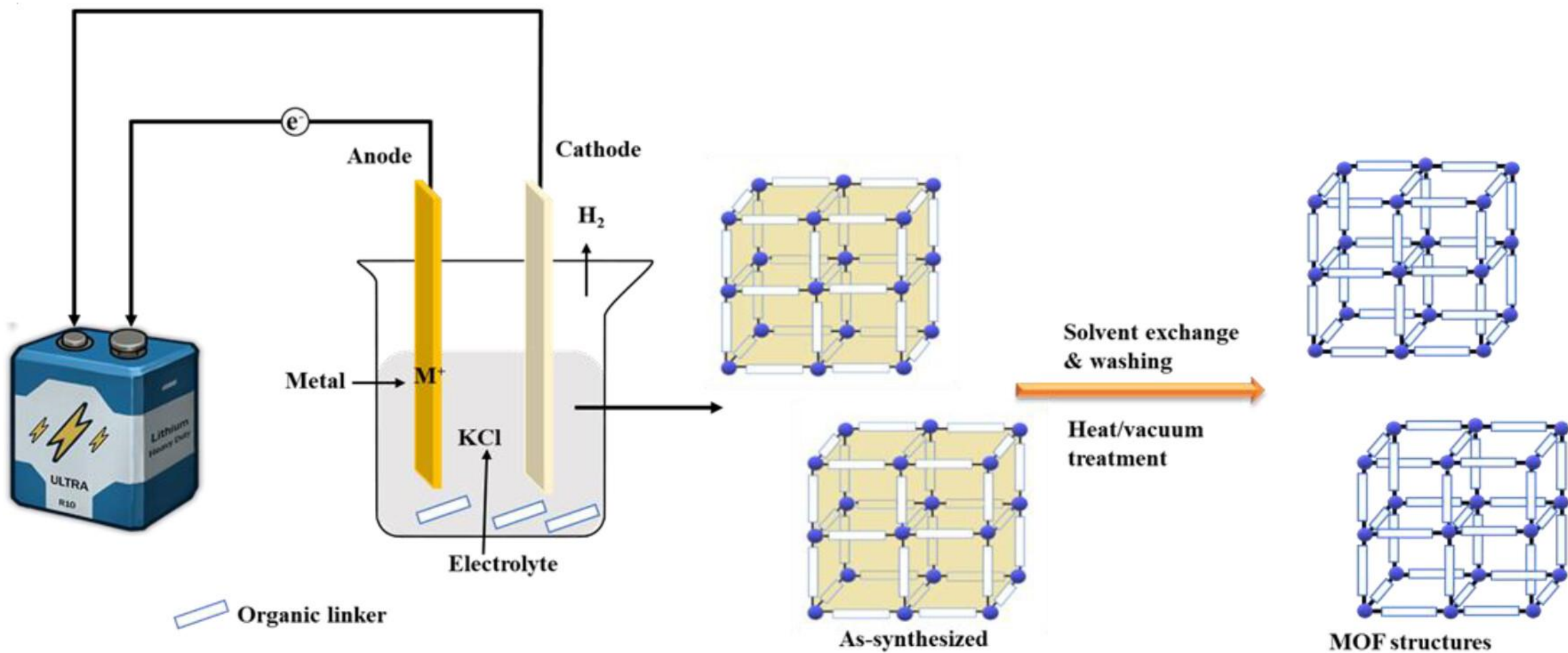
● Metal salt ▭ Organic ligand

Solvothermal method

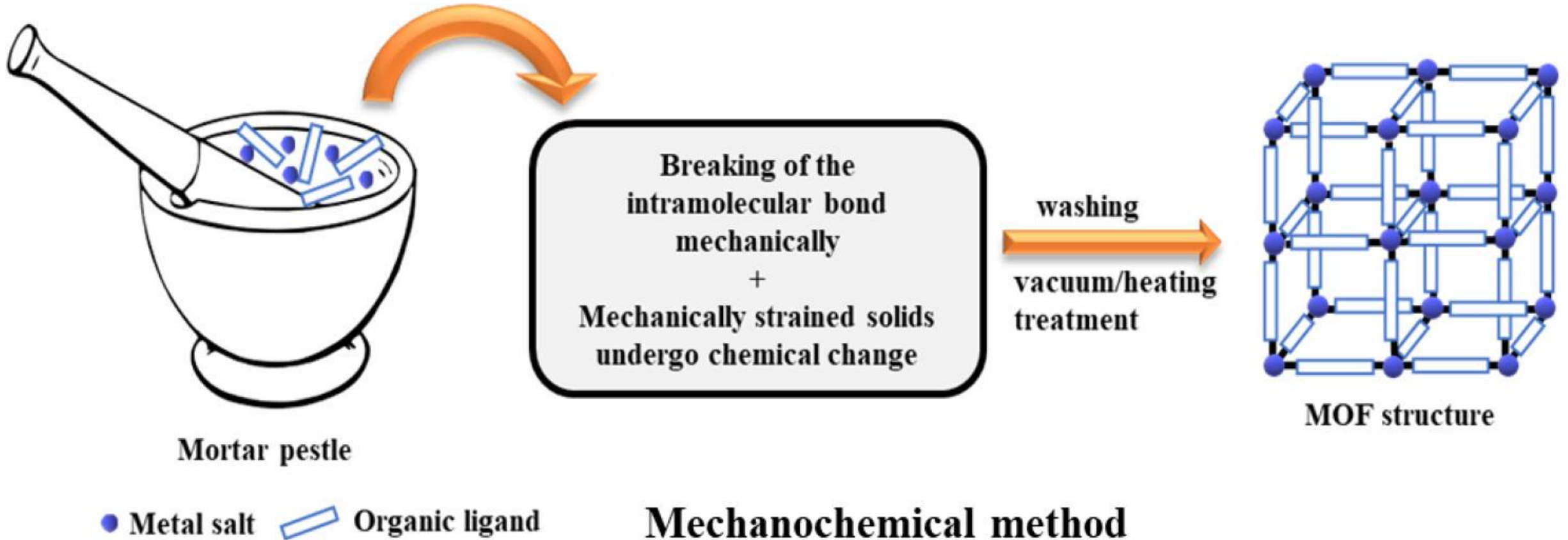
The Conventional Route: Microwave-Assisted Method



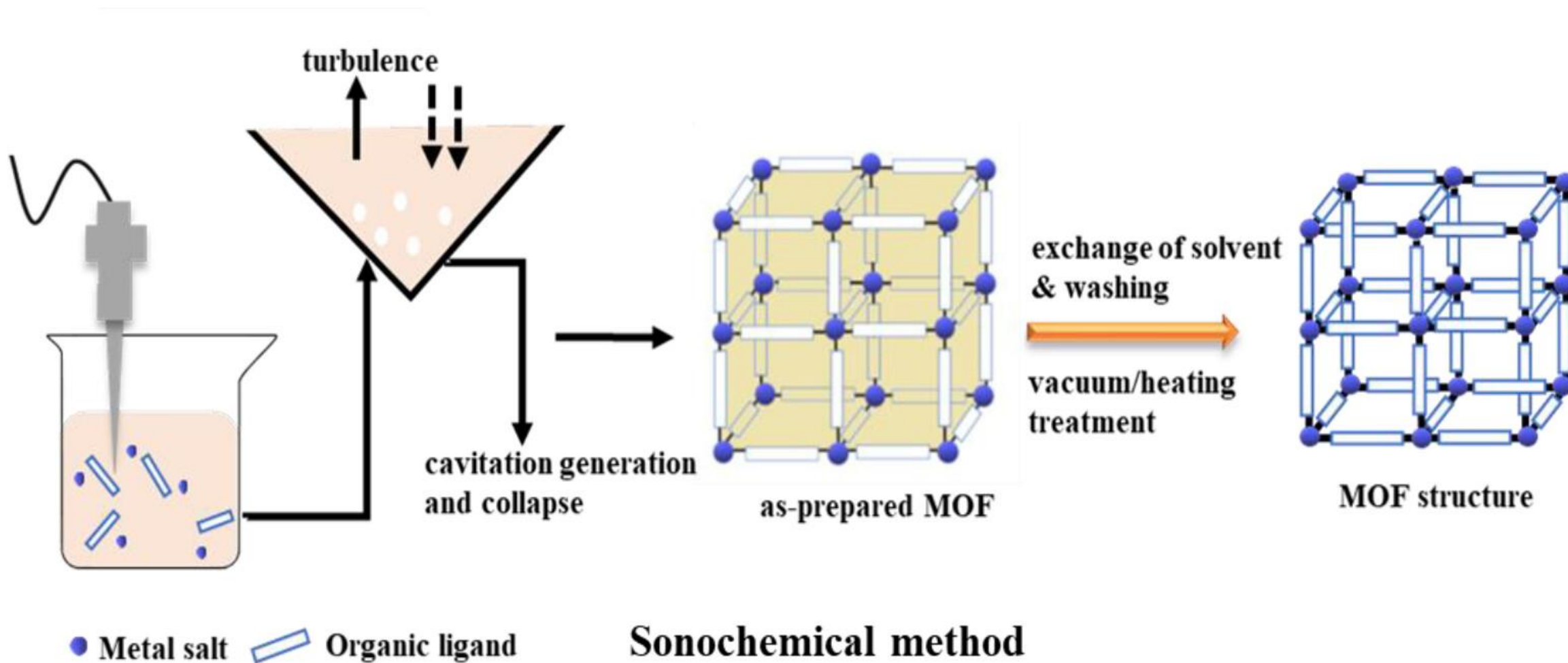
Electrochemical Method



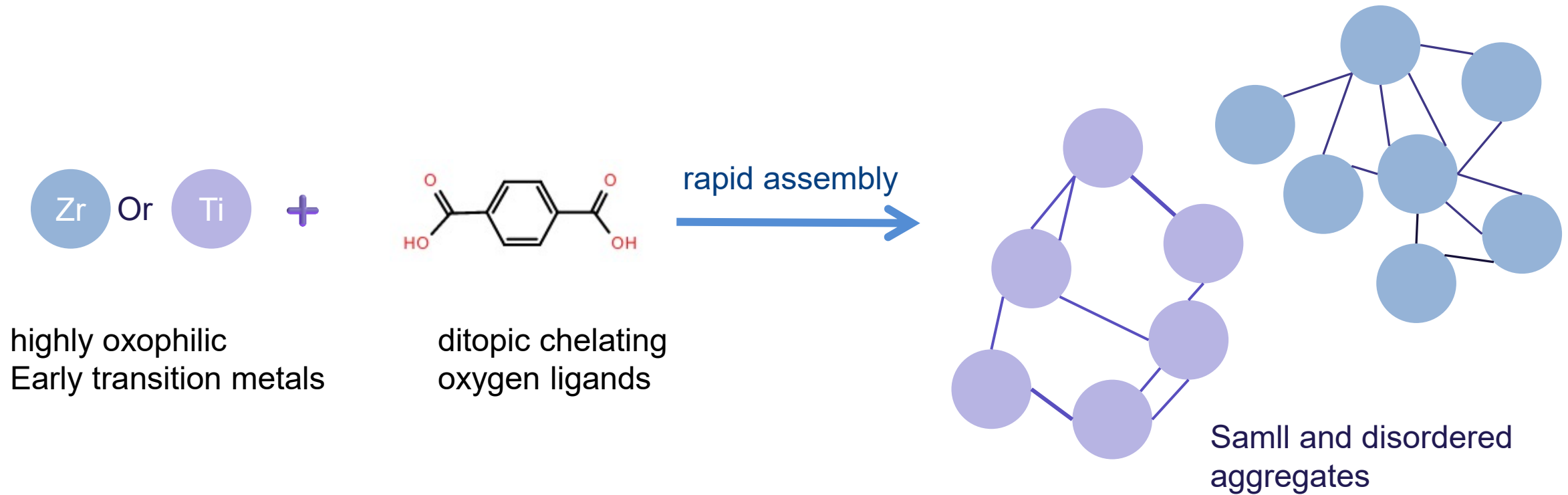
Mechanochemical Method



Sonochemical Method



Pitfalls of Uncontrolled Self-Assembly



Task 1 (10min)

How do we slow down self-assembly and control lattice quality?

Why does uncontrolled self-assembly lead to poor crystallinity?

Modulators as Small Molecule Regulators of MOF Growth

Mechanisms of Action

1. Reversibly occupying coordination sites on metal clusters

The modulator (monocarboxylic acid) occupies the Zr site first, but it is a "weak competitor". So it can be changed by the linker

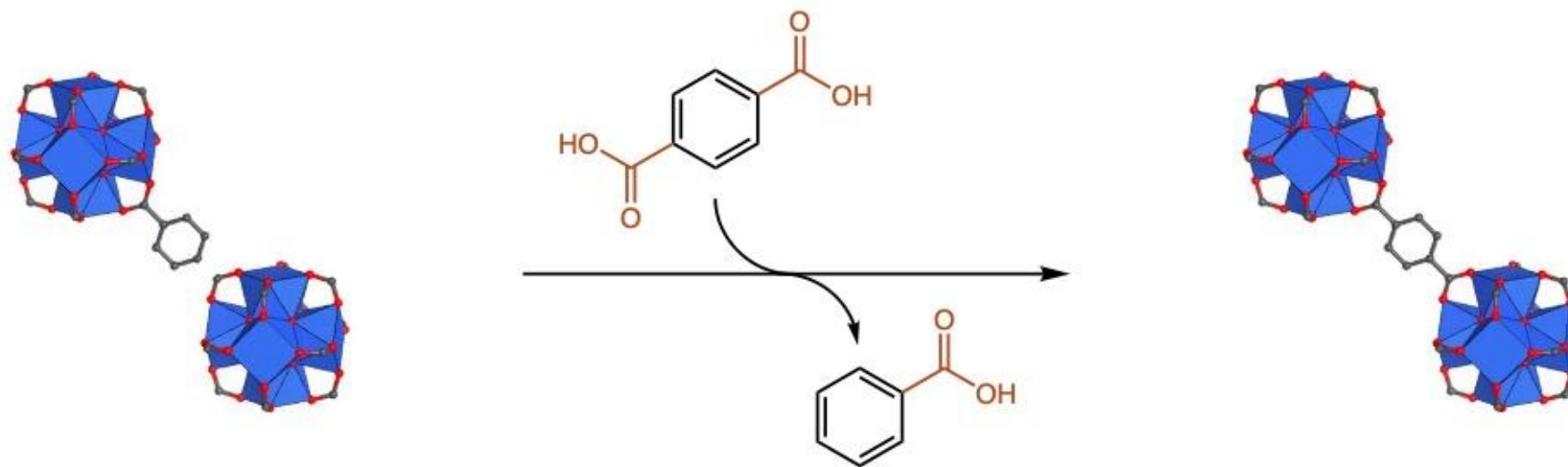
2. Accelerating metal cluster formation

Accelerating: $\text{Zr}_6\text{O}_4(\text{OH})_4$ cluster formation

But slower : Framework extension

Modulators as Small Molecule Regulators of MOF Growth

Case 1: Benzoic and acetic acids



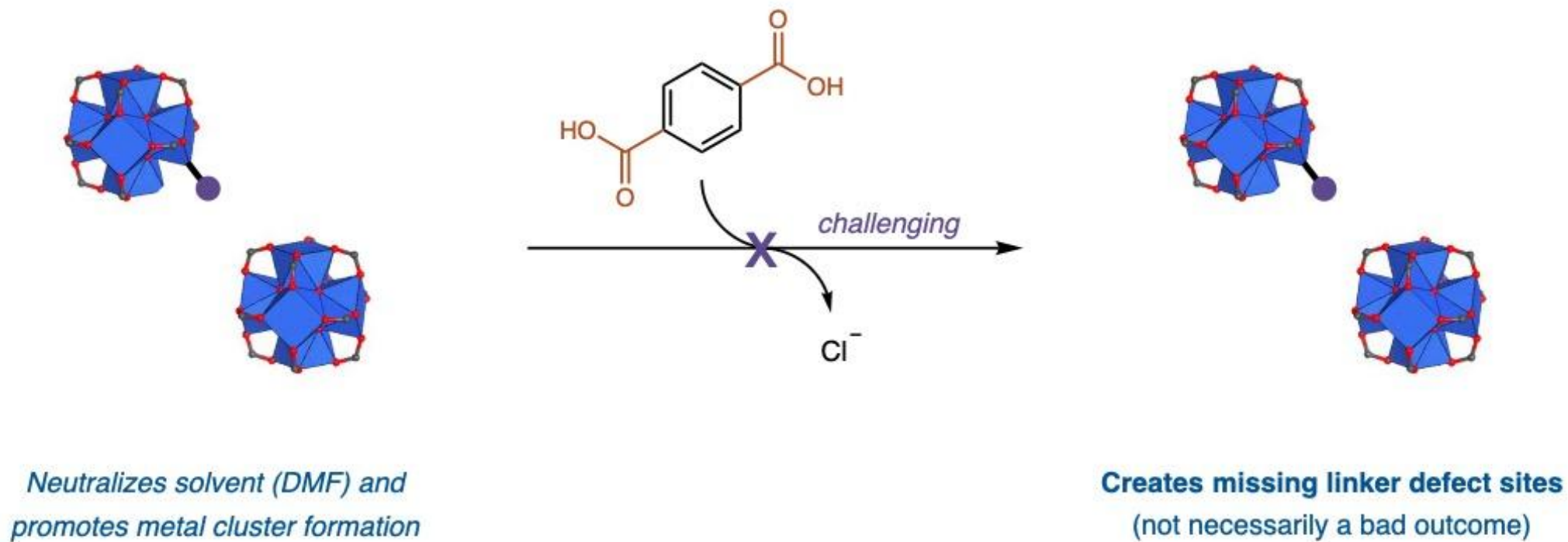
*Monotopic coordination to metal clusters
slows nucleation rate*

Promotes ideal framework assembly

Shaate, A.; Roy, P.; Godt, A.; Lippke, J.; Waltz, F.; Wiebcke, M.; Behrens, P. *Chem. Eur. J.* **2011**, *17*, 6643-6651.

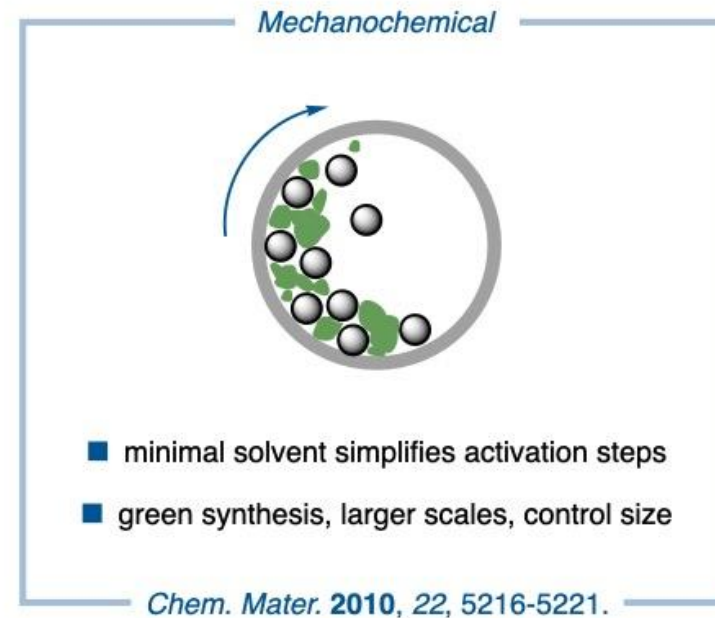
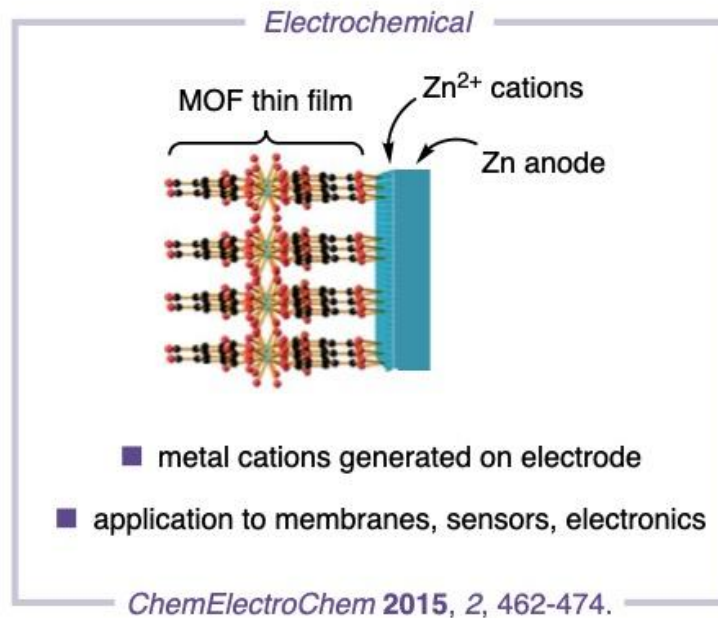
Modulators as Small Molecule Regulators of MOF Growth

Case 2: Hydrochloric Acid



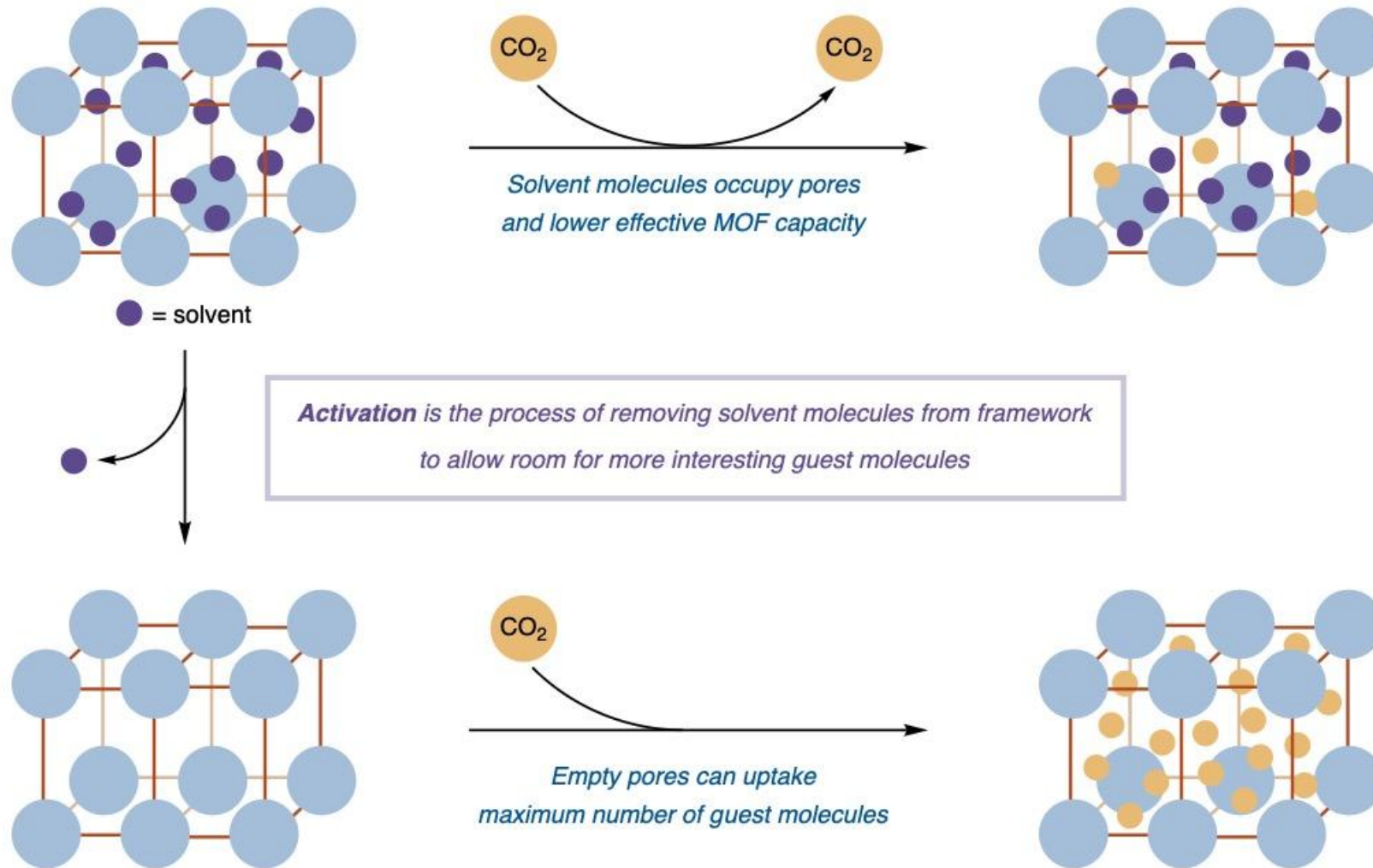
Katz, M. J.; Brown, Z. J.; Colon, Y. J.; Siu, P. W.; Scheidt, K. A.; Snurr, R. Q.; Hupp, J. T.; Farha, O. K. *Chem. Commun.* **2013**, 49, 9449-9451.

Alternatives to Solvothermal MOF Synthesis



Activation by removal of solvent

Activation as the Key to Unlocking a MOF's Potential



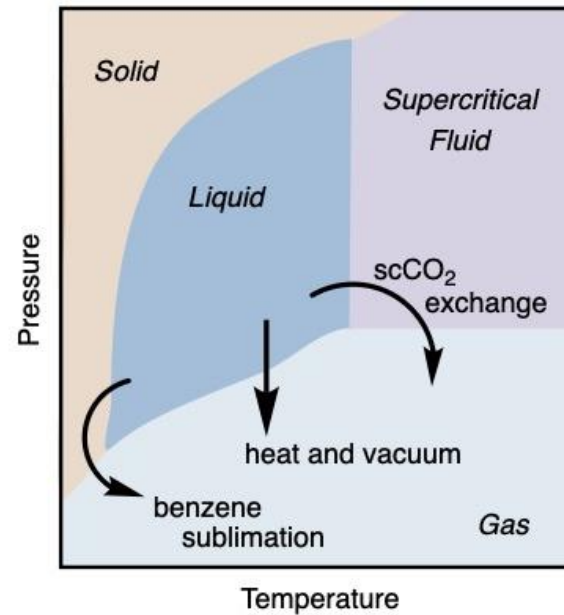
Activation as the Key to Unlocking a MOF's Potential

Activation can be performed by simply heating and applying vacuum

*Removing high boiling, high surface tension solvents (like DMF and DMSO)
can cause **framework collapse** due to capillary forces*

Alternative Strategy #1

*Soak MOF in lower boiling solvent
then heat under vacuum*



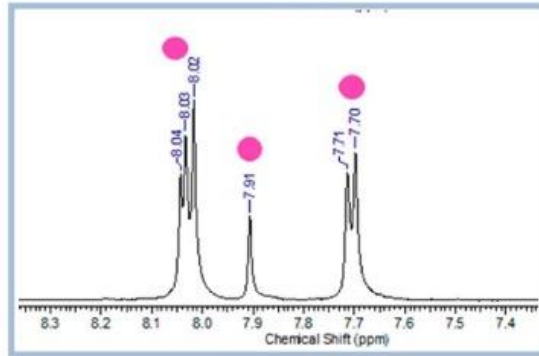
Alternative Strategy #2

Avoid the liquid-gas phase transition

- “green” activation method
- easily scalable

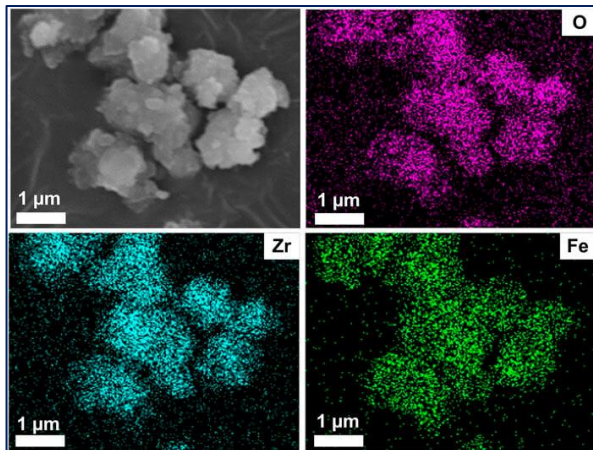
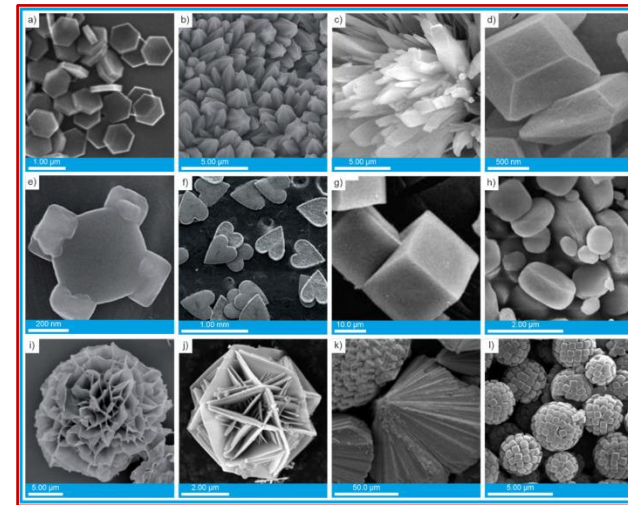
Characterization to assess purity and crystallinity

MOF Characterization Techniques



^1H NMR Spectroscopy assays the bulk purity of the sample by quantifying the relative amounts of incorporated and free organic linker

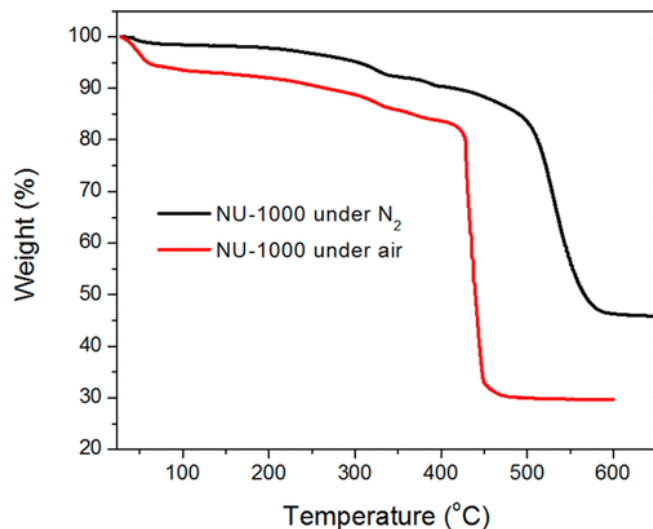
Scanning Electron Microscopy (**SEM**) allows the chemist to observe crystal size, morphology, and uniformity



SEM in conjunction with Electron Diffraction X-Ray (**EDX**) Spectroscopy maps the elemental composition throughout the framework

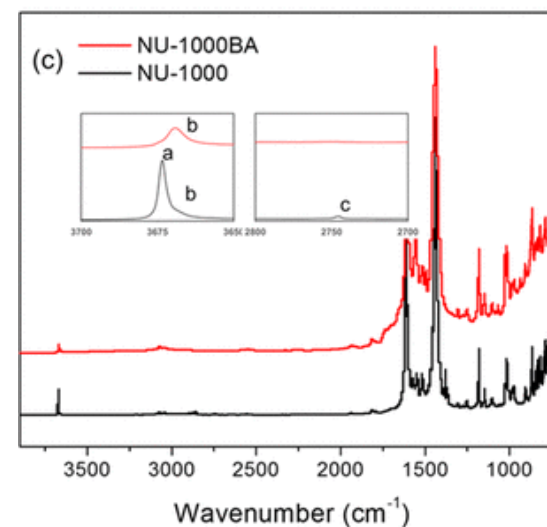
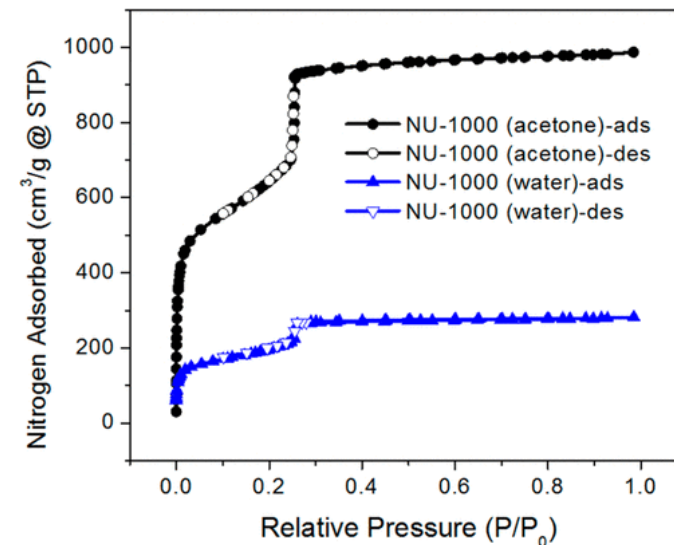
MOF Characterization Techniques

N₂ Adsorption/Desorption Isotherms measure the apparent surface area of the MOF (i.e., the capacity of the MOF to store guest molecules)



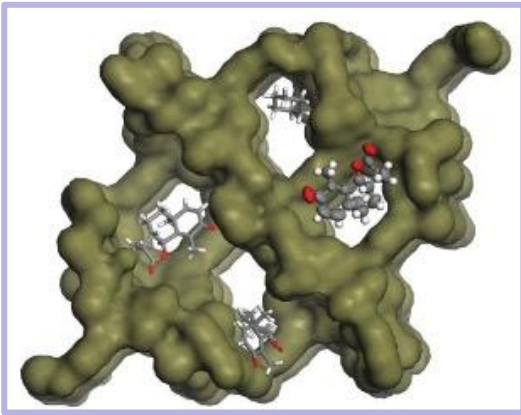
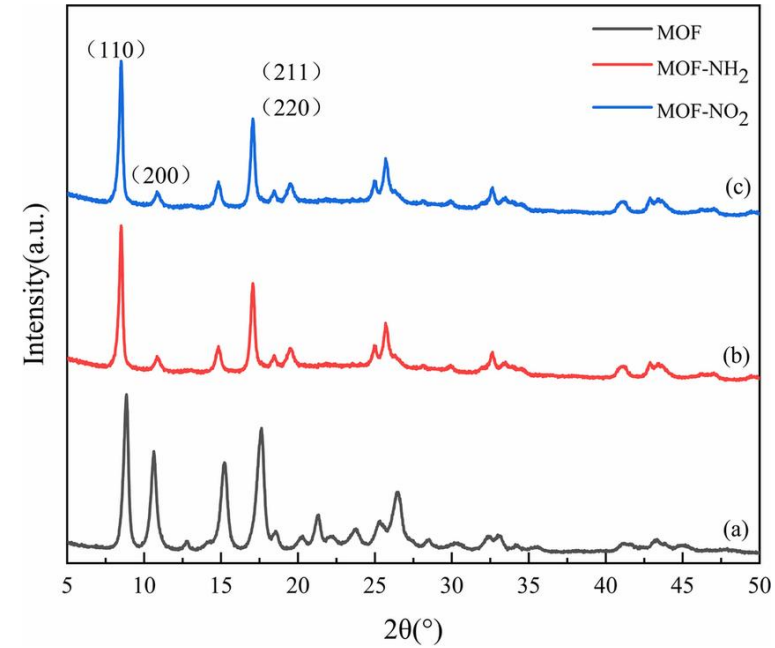
Stability tests assess the framework under relevant conditions (e.g.; aqueous, pH, photochemical, and thermal stability)

Diffuse Reflectance Infrared Fourier Transform Spectroscopy (DRIFTS) can show the interaction of the framework with guest molecules



MOF Characterization Techniques

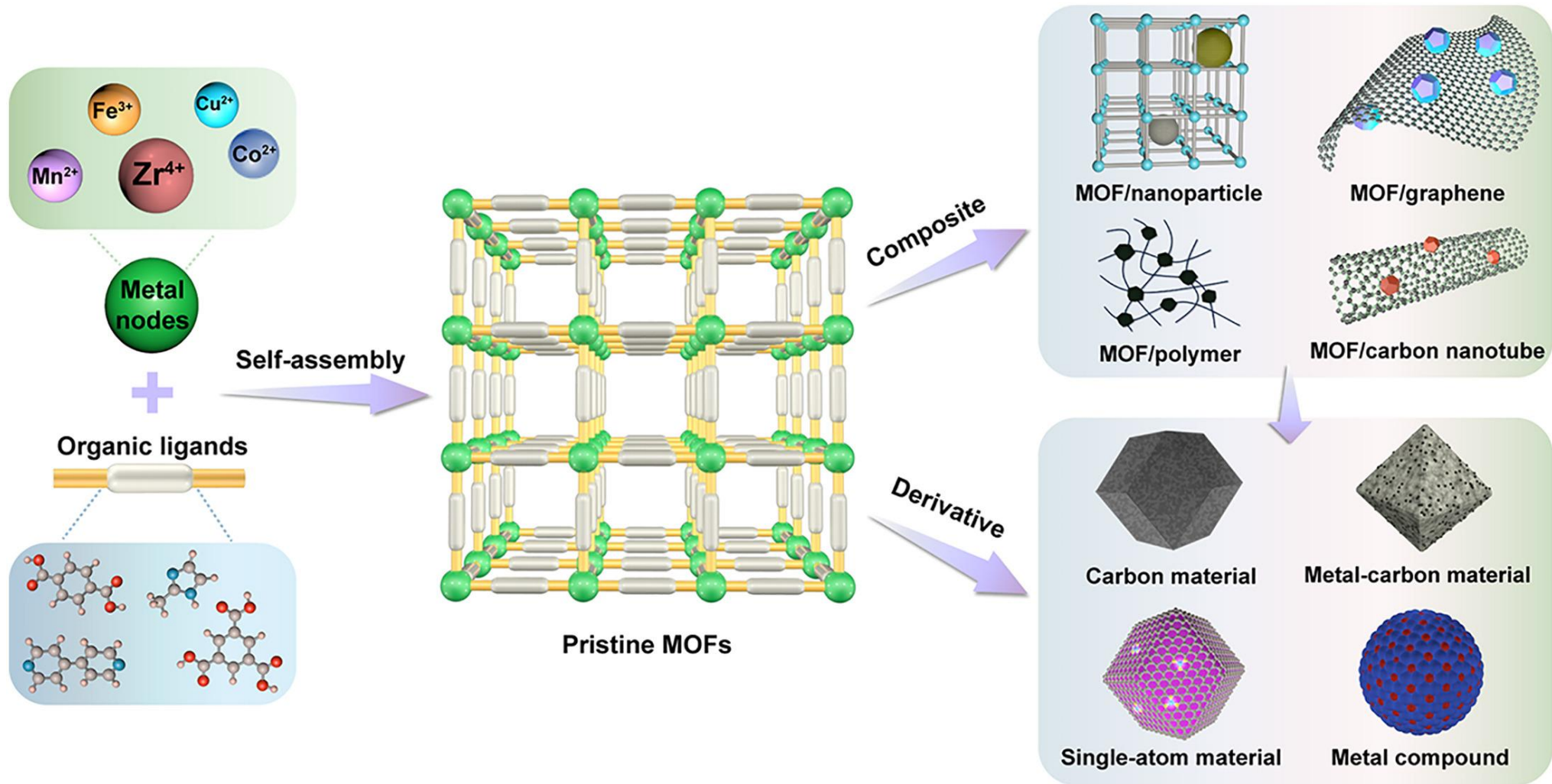
X-Ray Diffraction (XRD) confirms bulk crystallinity of the sample and unit cell size can also be extrapolated



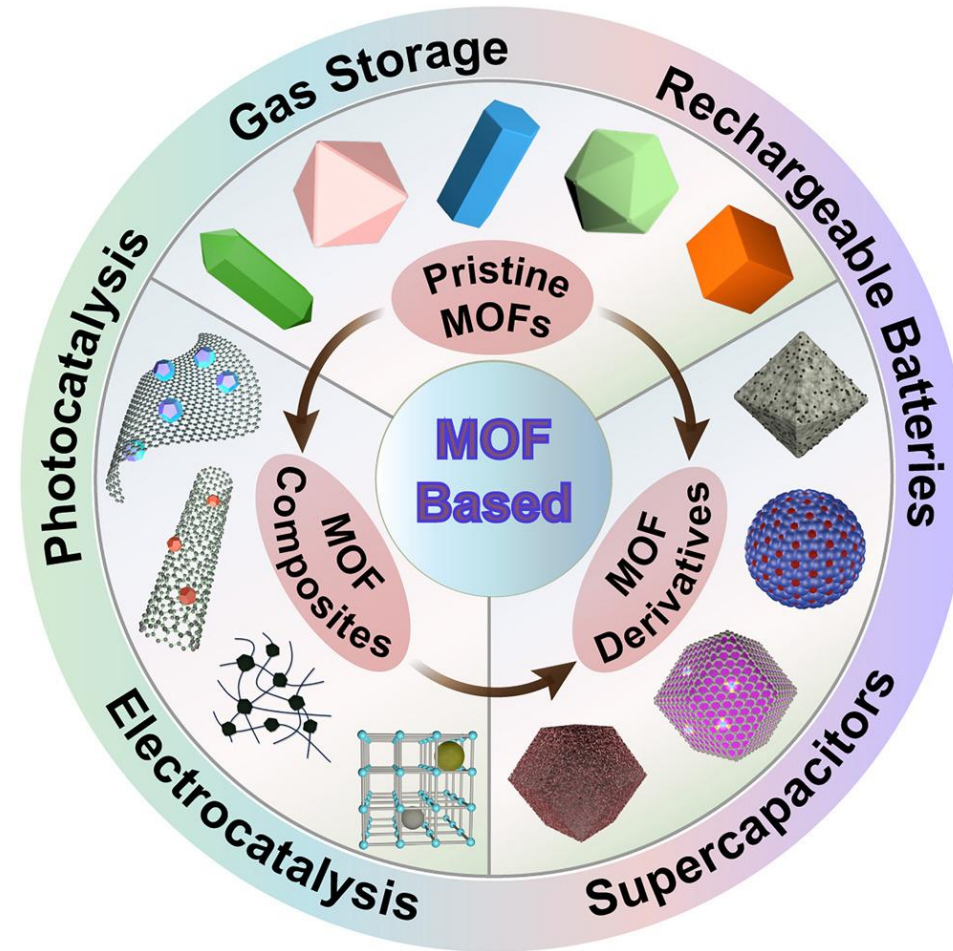
Single Crystal X-Ray Diffraction gives absolute structural information but is limited by the ability to grow single crystals of sufficient size (5-10 μm)

MOF Broad Applications

MOF's Applications

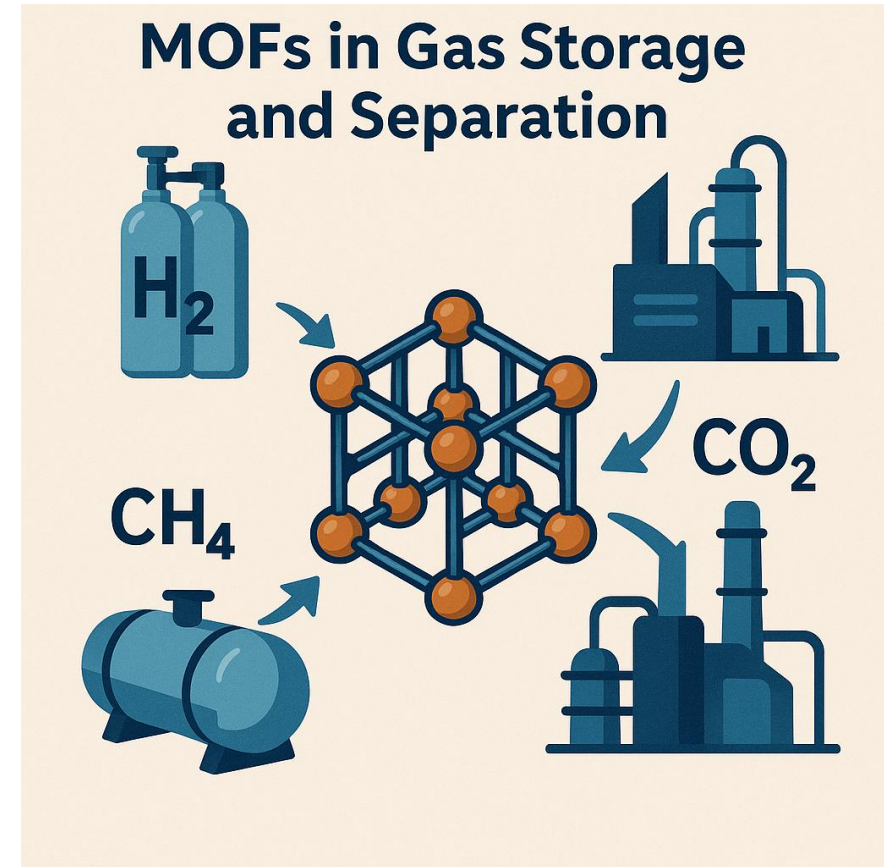
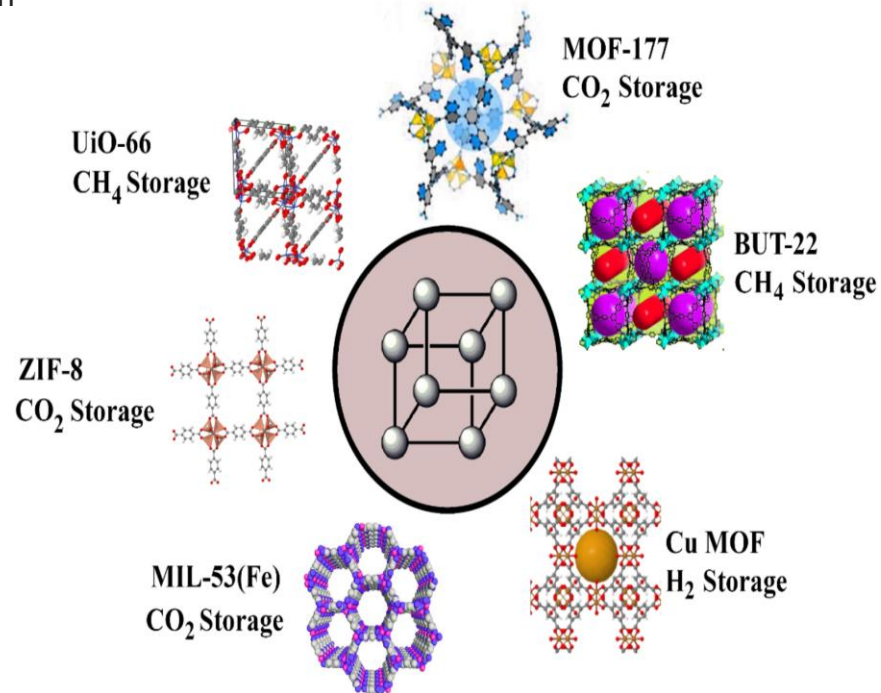


MOF's Applications



MOF's Applications : Gas storage and Separation

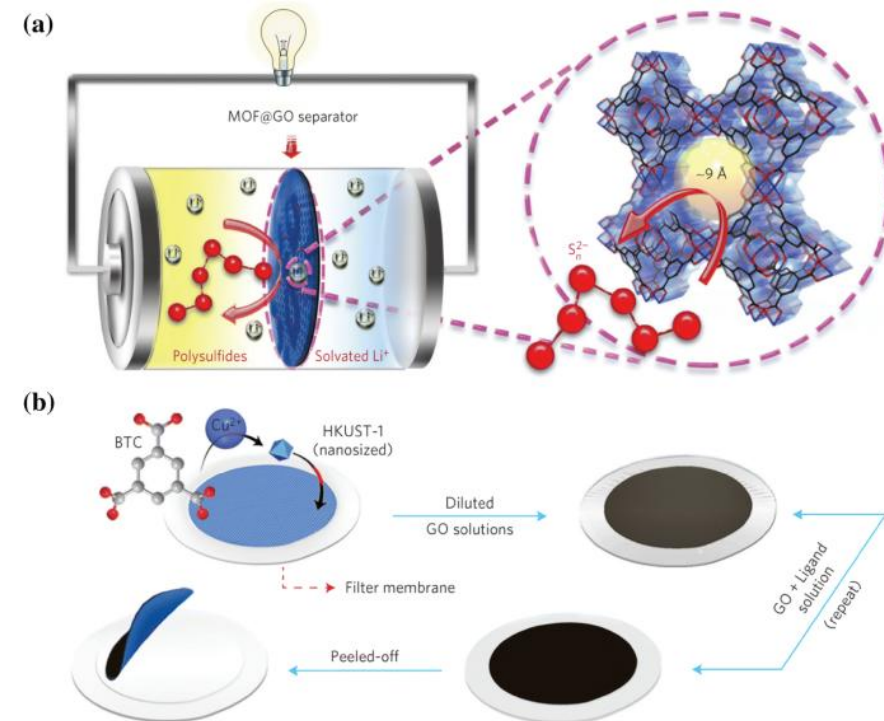
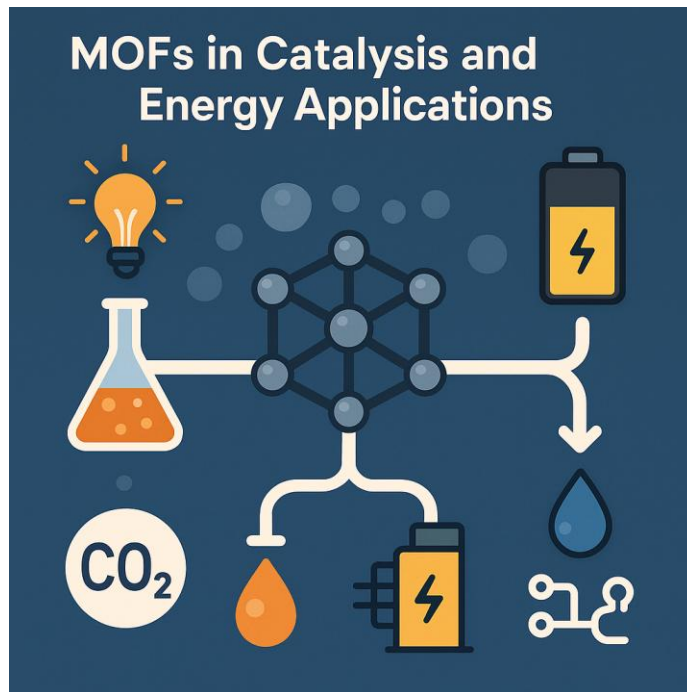
MOFs high porosity and tunable pore environments provide significant advantages over traditional adsorbents. Their internal surface areas can reach over 7,000 m²/g, making them highly effective for capturing and storing various gases — from energy carriers like hydrogen (H₂) and methane (CH₄) to greenhouse gases like carbon dioxide (CO₂).



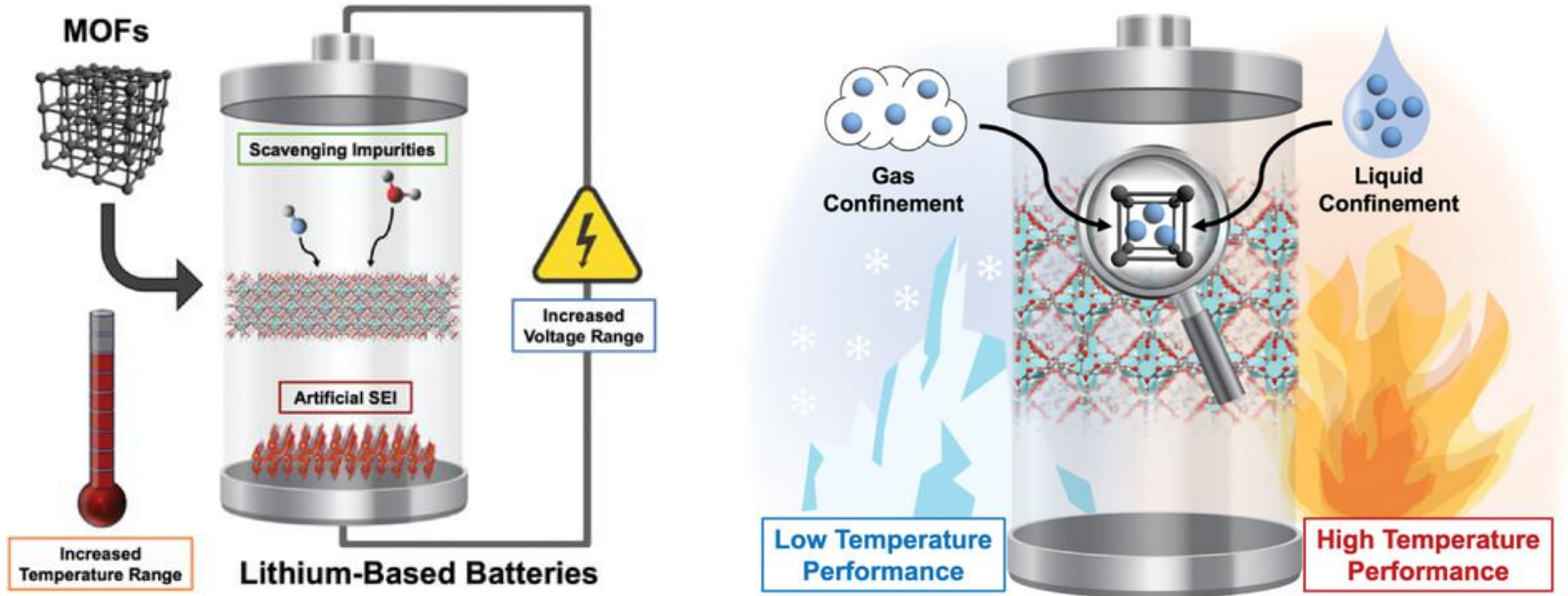
Different MOFs for gas storage applications.

MOF's Applications: Electrochemical Application

MOFs energy-conversion reactions owing to the distinct advantages of structural diversity, porous structures, and tailorable catalytic sites. They are highlighted as appropriate electrocatalysts with high activity and stability for various frontier reactions, including oxygen evolution reaction (OER), hydrogen evolution reaction (HER), oxygen reduction reaction (ORR), and carbon dioxide reduction reaction (CO₂RR).

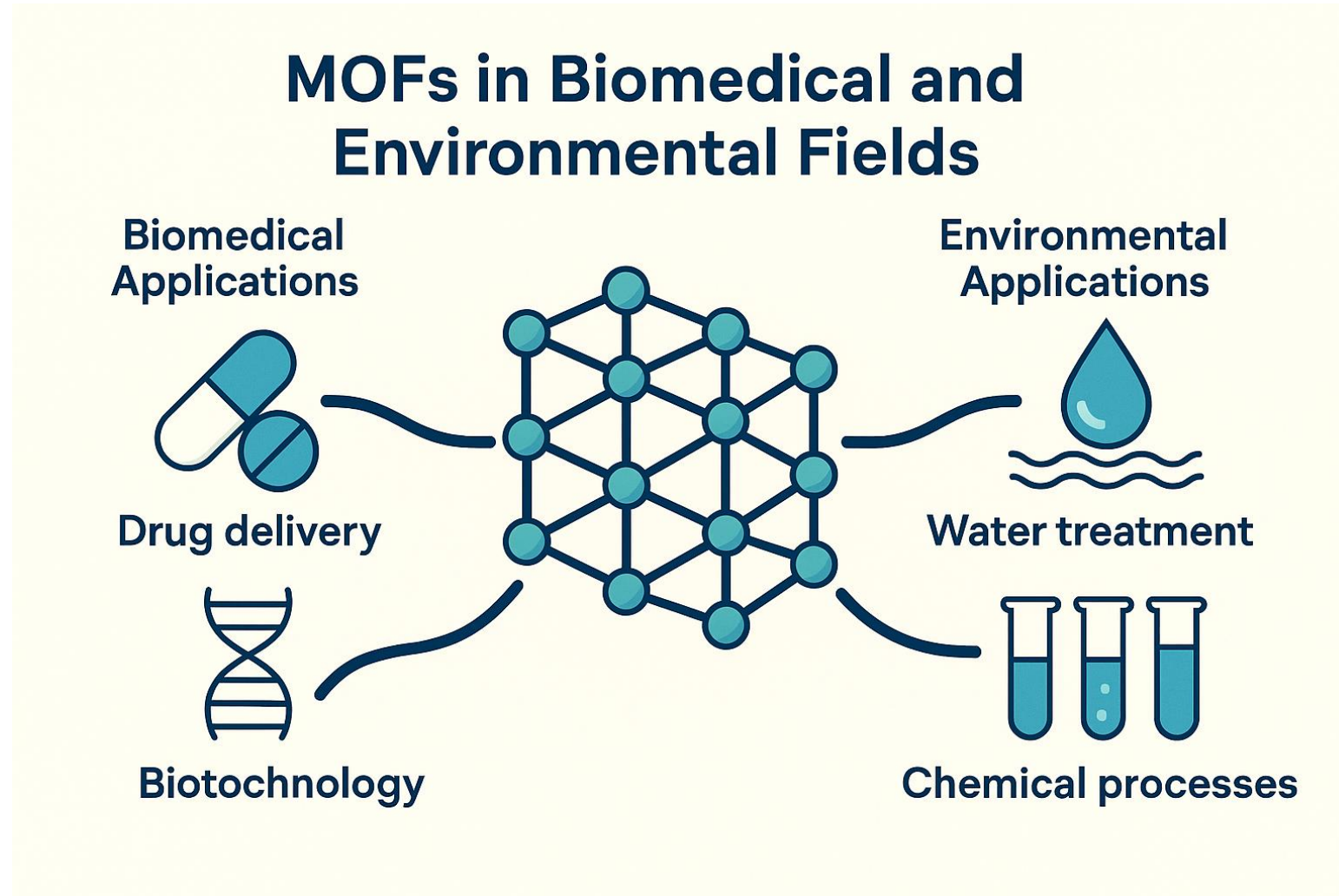


MOF's Applications: Electrochemical Application



Schematic showing the MOF-based electrolytes for batteries operating under extreme temperatures.

MOF's Applications: Biomedical and Environmental Fields



MOF Membrane for Wastewater Treatment

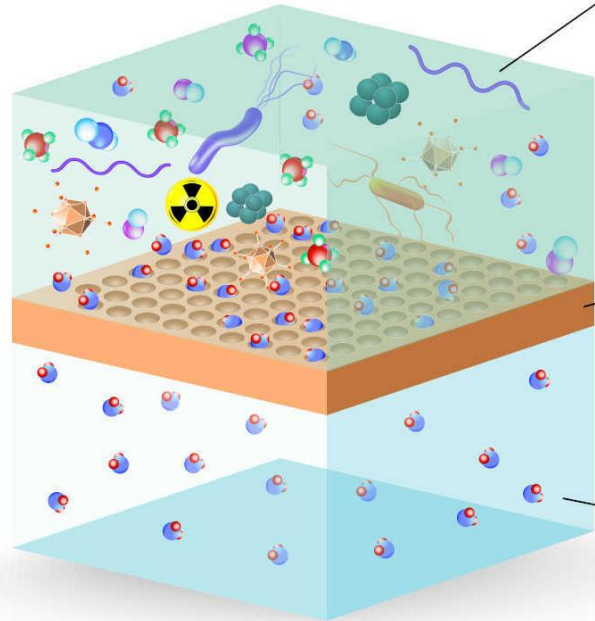
Membrane technology

Pressure

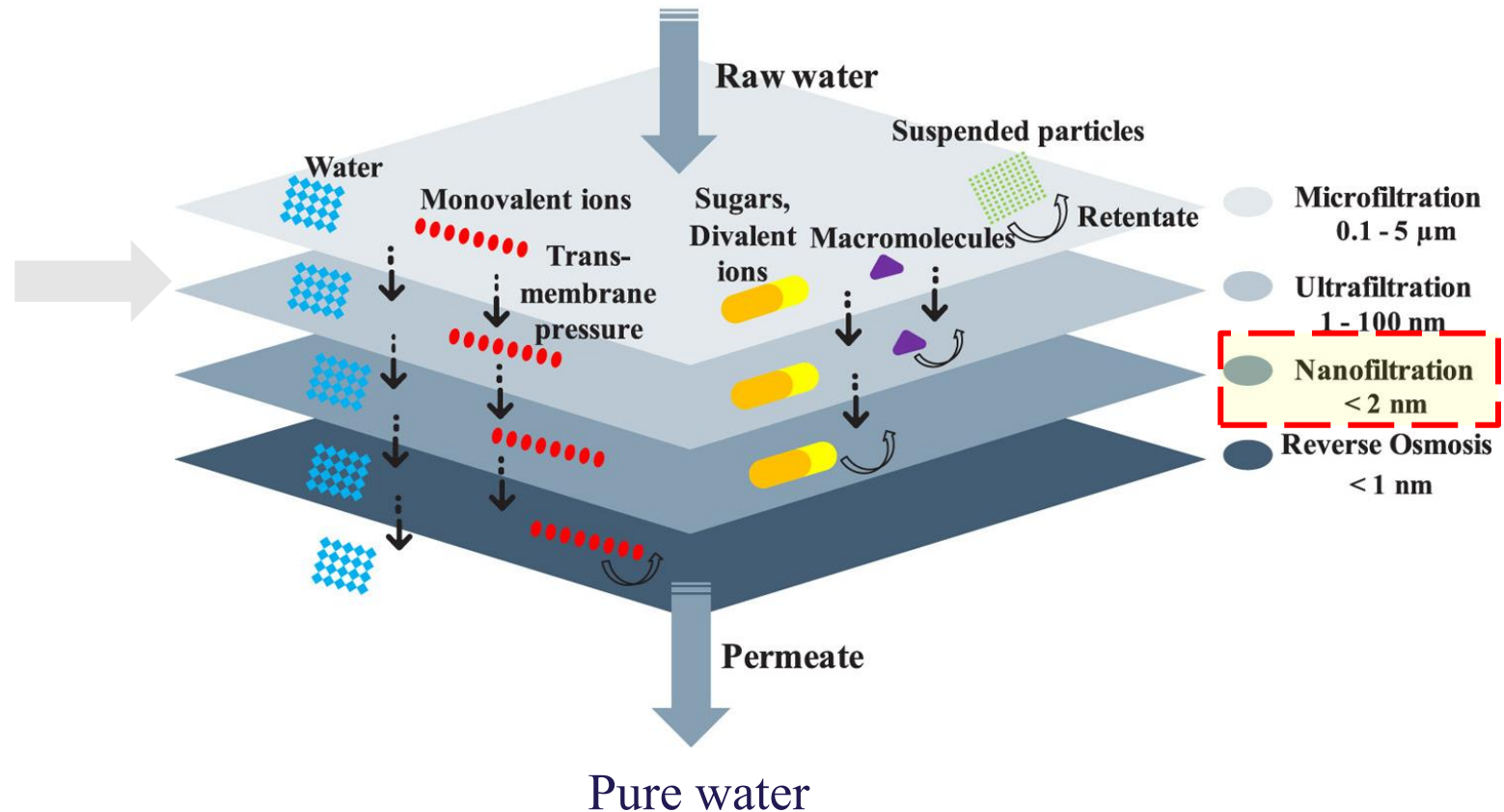
Higher
contaminant
concentration

Semipermeable
membrane

Lower
contaminant
concentration

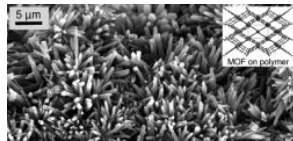


Membrane filtration Classification



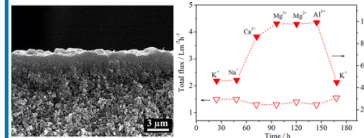
MOF Membrane for Wastewater Treatment

First MOF membranes prepared by microwave on polymer substrate



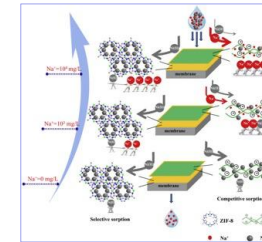
J. Am. Chem. Soc., 2010, 132:15687-15691

UIO-66 fibers for desalination



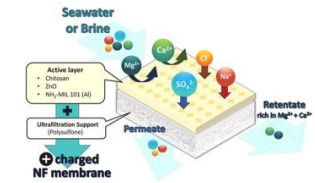
I. Am. Chem. Soc. 2015, 137(22), 6999-7002.

MOF hybrid membrane firstly used for removing trace Ni(II)



Water Res., 2018, 143:87-98

NH2-MIL-101(Al) nanofiltration membrane



J. Membr. 2023, 685, 121941

2005

2010

2012

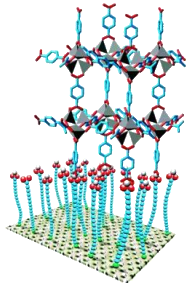
2015

2018

2021

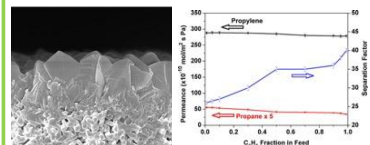
2023

First bare MOF (MOF-5 membranes)



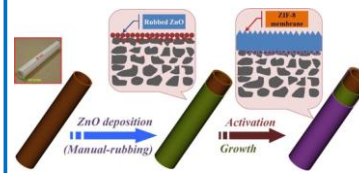
J. Am. Chem. Soc. 2005, 127, 40.13744-13745.

ZIF8 membrane with long-term stability for separation



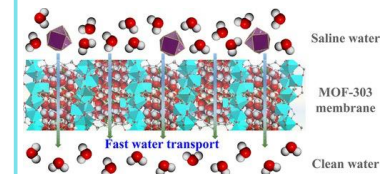
J. Membr. 2015, 391(3), 93-98

Mechanical stability ZIF-8 membrane for separation



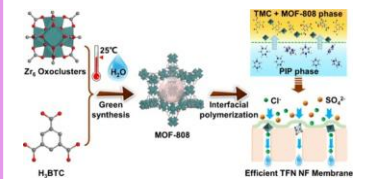
J. Membr. 2015, 490, 354-363

MOF-303 Membranes for Desalination



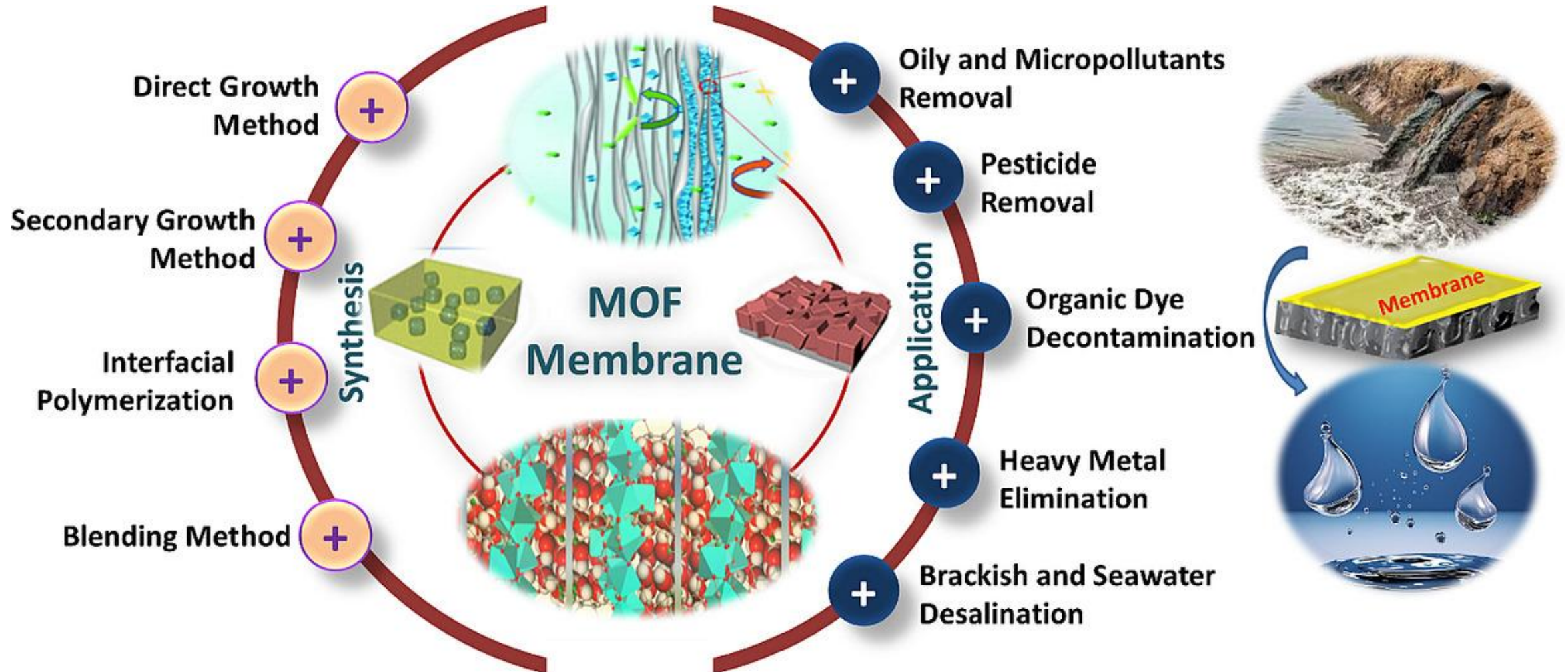
J. Am. Chem. Soc. 2021, 143 (48), 20055-20058

MOF-808/Polyamide Nanofiltration Membranes

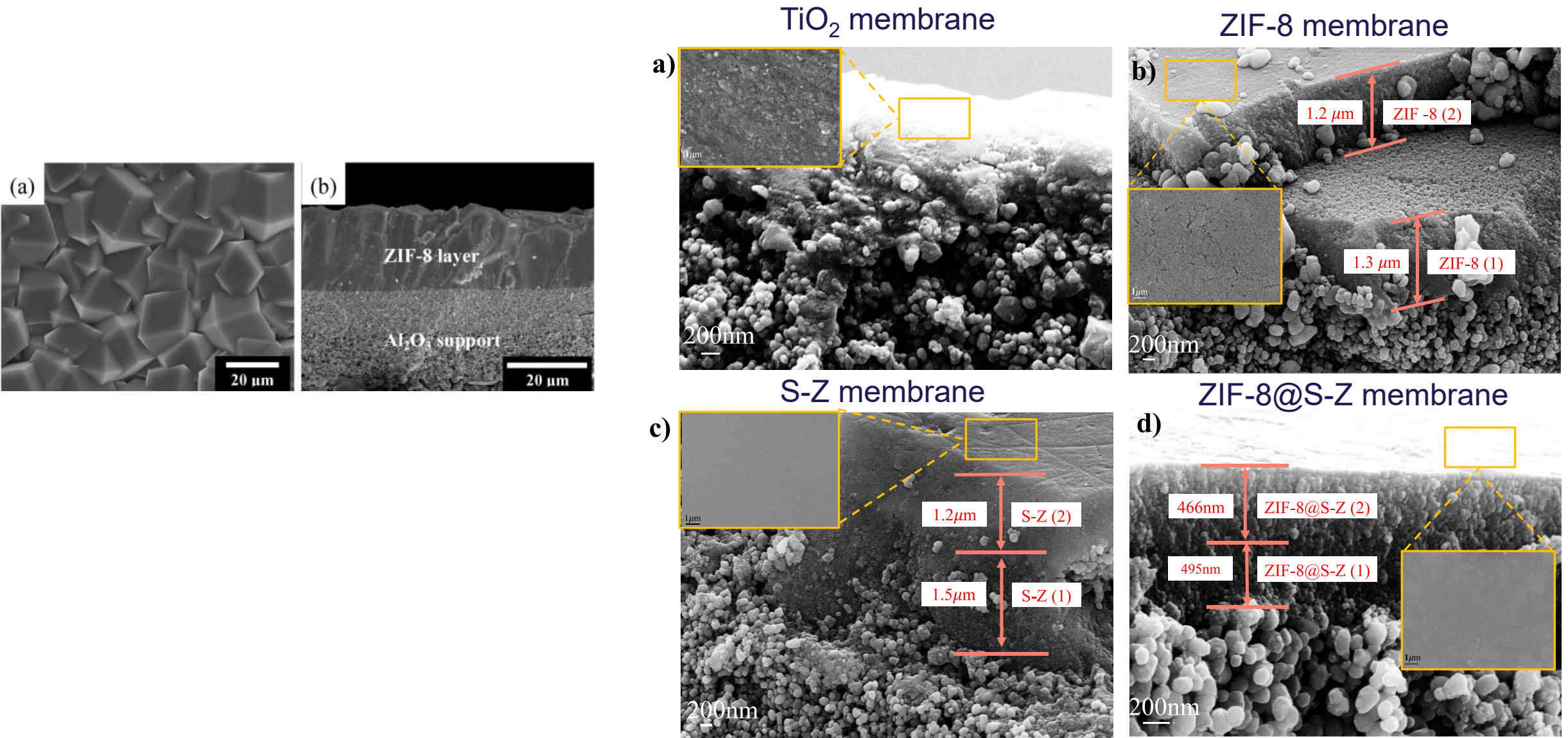


ACS Appl. Nano Mater. 2023, 6 (19), 17615-17625

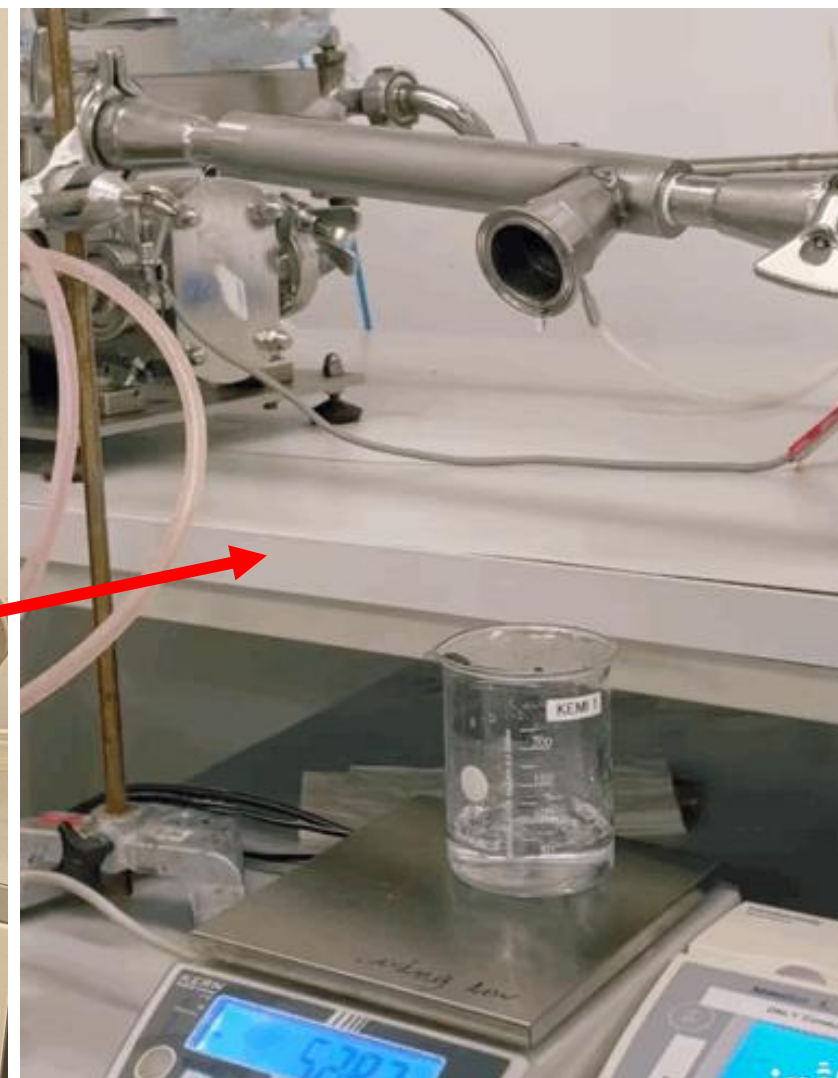
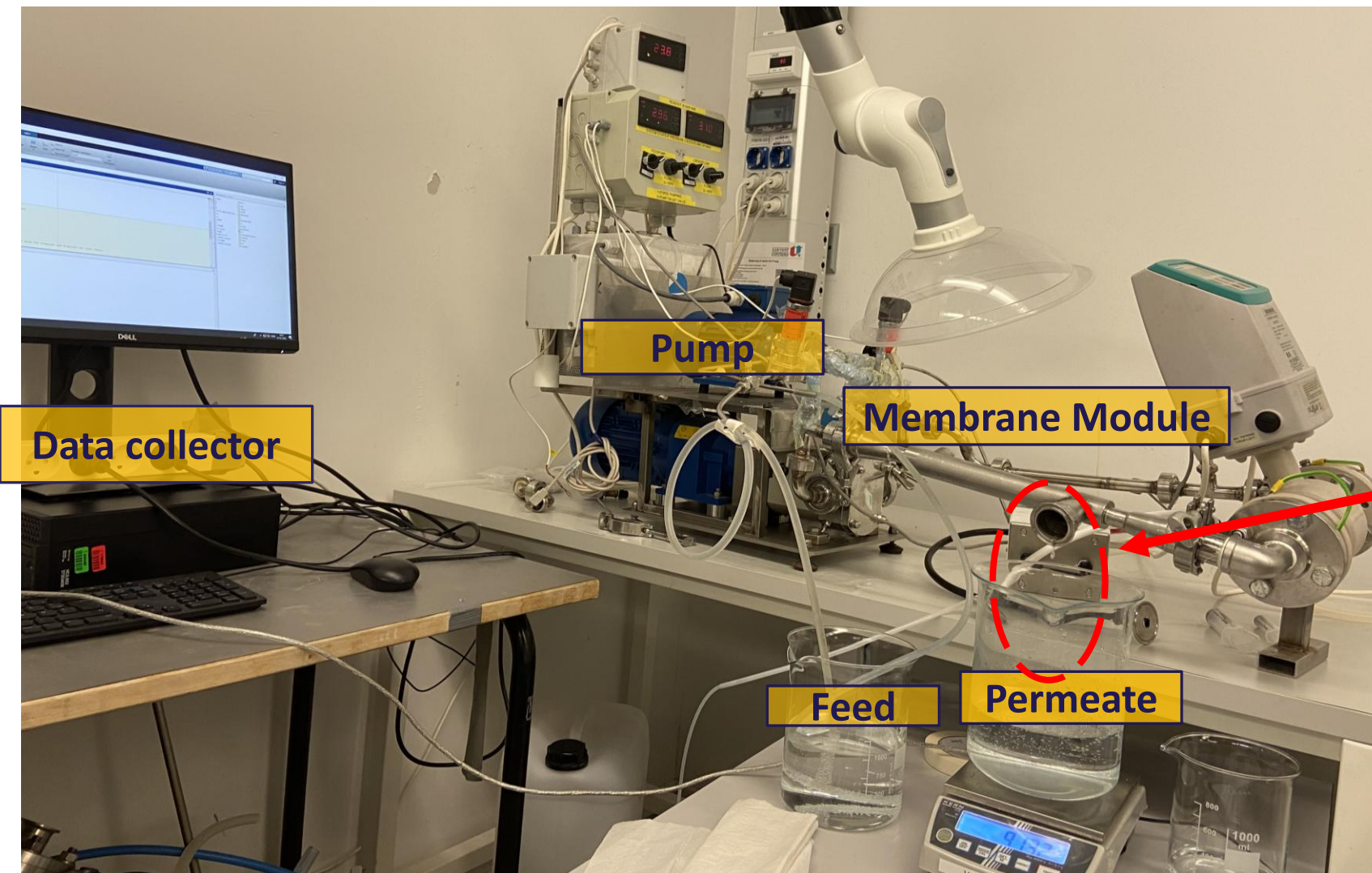
MOF Membrane for Wastewater Treatment



MOF Membrane for Wastewater Treatment

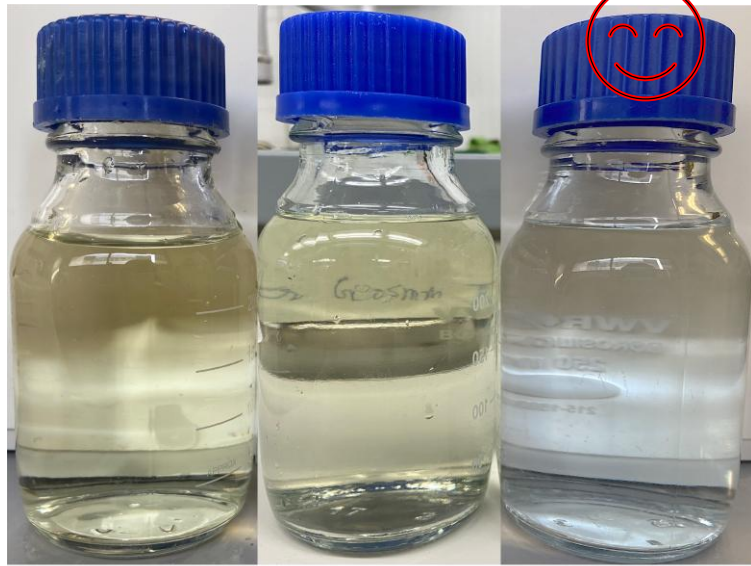


MOF Membrane for Wastewater Treatment



MOF Membrane for Wastewater Treatment

Danish Fish Farm Samples



Aquaculture
effluent (feed)

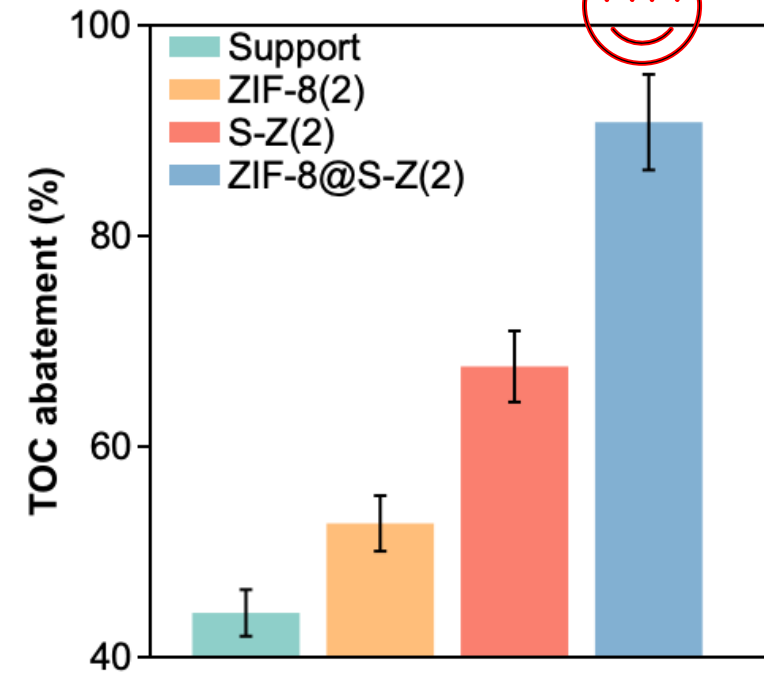
Support
permeate

ZIF-8@S-Z-2
permeate

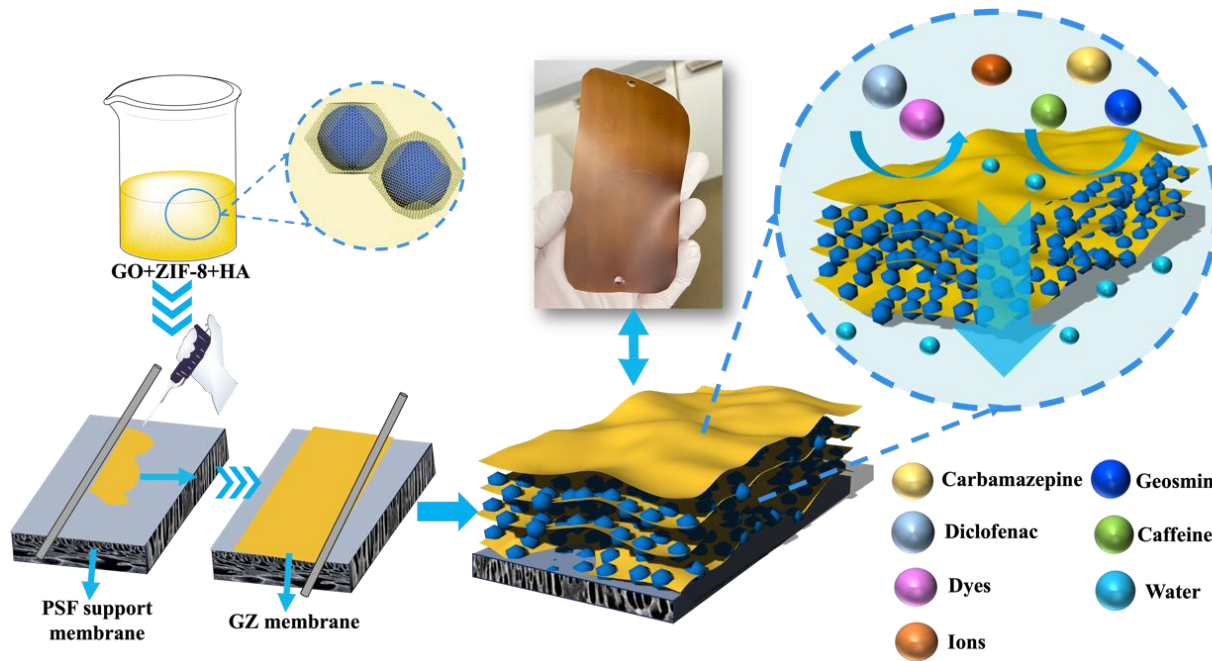
yellowish
turbid

Colorless
clear

TOC abatement



MOF Membrane for Wastewater Treatment



Feed Support Permeate GZ7-3 Permeate

Challenges and Future Outlook for MOFs

Current Challenges:

Stability Issues

Many MOFs suffer from poor water and thermal stability, especially those with labile metal-ligand bonds.

Scalability and Cost

MOFs are typically synthesized under solvothermal conditions using expensive solvents and linkers

Integration into Devices

Translating MOFs from powder form to functional devices—such as membranes, electrodes, or sensors—requires innovations in materials processing, shaping, and coating techniques.

Challenges and Future Outlook for MOFs

Future Directions:

AI-Assisted MOF Design

With thousands of known MOF structures and even more possible combinations, machine learning and computational modeling are being used to predict optimal frameworks for specific tasks.

Stimuli-Responsive (Smart) MOFs

MOFs that respond to pH, light, heat, or specific chemicals could revolutionize fields like targeted drug delivery, environmental sensing, and self-healing materials.

Green and Circular MOFs

Sustainable synthesis, recyclable frameworks, and bio-based linkers for MOFs

Industrial Deployment

Areas like carbon capture, clean hydrogen storage, and selective gas separation are already seeing pilot-scale testing, signaling the dawn of MOF-based industrial applications.

Assignment

1: MOF Design Challenge

Task:

Design a MOF for a specific application.

Choose ONE application:

- CO₂ capture
- Water purification
- Catalysis
- Drug delivery

You need to answer:

- 1.What metal node would you choose? Why?
- 2.What type of organic linker? (ditopic / tritopic / functionalized)
- 3.What pore size / property is needed?
- 4.What challenge might your MOF face?

2: Controlling MOF Growth

Task:

Based on what you learned about self-assembly and modulators:

👉 Propose a strategy to improve MOF crystallinity.

Discuss:

- 1.What problem occurs in uncontrolled nucleation?
- 2.How do modulators (e.g., acetic acid) help?
- 3.Could you design a new modulator?
- 4.What parameters would you optimize? (temperature, solvent, concentration, etc.)

Thank you for your attention!



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DENMARK